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Abstracts for

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PAPERS PRESENTED AT MEETINGS

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Fall Southeastern Virtual Sectional Meeting, October 10–11, 2020

Abstracts of the 1161st Meeting.

00 ► General

1161-00-239 **Q. S. Baldwin*** (qbaldwin8382@tuskegee.edu) and **Eleni Panagiotou** (eleni-panagiotou@utc.edu). *The local topological free energy of viral glycoproteins.*

Many viruses infect cells by using a mechanism that involves binding of a viral protein to the host cell. During this process, the three-dimensional conformation of the viral binding protein changes significantly and disruption of this process could be achieved by targeting key locations in the viral protein that are essential in this rearrangement. In this manuscript we propose to use the local geometry/topology of the crystal structure of the viral protein backbone alone to identify these essential locations. Our results show that the local Writhe, local Average Crossing Number and the local Torsion alone can identify "exotic" amino acids that are essential in the viral protein infection mechanism. We apply this method to the SARS-Cov-2 Spike protein to propose target amino-acids for drug discovery. (Received August 17, 2020)

05 ► Combinatorics

1161-05-15 **Asaf Ferber** and **Matthew Kwan***, Department of Mathematics, 450 Jane Stanford Way, Stanford, CA 94305. *Dirac-type theorems in random hypergraphs.*

For positive integers $d < k$ and n divisible by k, let $m_d(k, n)$ be the minimum d-degree ensuring the existence of a perfect matching in a k-uniform hypergraph. In the graph case (where $k = 2$), a classical theorem of Dirac says that $m_1(2, n) = \lceil n/2 \rceil$. However, in general, our understanding of the values of $m_d(k, n)$ is still very limited, and it is an active topic of research to determine or approximate these values. In this talk, we discuss a new "transference" theorem for Dirac-type results relative to random hypergraphs. Specifically, for any $d < k$, any $\varepsilon > 0$ and any "not too small" p, we prove that a random k-uniform hypergraph G with n vertices and edge

probability p typically has the property that every spanning subgraph of G with minimum d-degree at least $(1+\varepsilon)m_d(k,n)p$ has a perfect matching. One interesting aspect of our proof is a "non-constructive" application of the absorbing method, which allows us to prove a bound in terms of $m_d(k,n)$ without actually knowing its value. (Received July 19, 2020)

1161-05-16 **Sergey Norin** (sergey.norin@mcgill.ca), **Luke Postle** (lpostle@uwaterloo.ca) and **Zi-Xia Song*** (zixia.song@ucf.edu), Department of Mathematics, University of Central Florida, Orlando, FL 32816. *Breaking the degeneracy barrier for coloring graphs with no K_t minor.*

In 1943, Hadwiger conjectured that every graph with no K_t minor is $(t-1)$-colorable for every $t \geq 1$. In the 1980s, Kostochka and Thomason independently proved that every graph with no K_t minor has average degree $O(t\sqrt{\log t})$ and hence is $O(t\sqrt{\log t})$-colorable. In this talk we present our recent breakthrough on making the first improvement on the order of magnitude of the Kostochka-Thomason bound: we show that every graph with no K_t minor is $O(t(\log t)^\beta)$-colorable for every $\beta > 1/4$. (Received July 20, 2020)

1161-05-27 **Jeff Kahn** and **Jinyoung Park*** (jinyoungpark0@gmail.com). *Tuza's Conjecture for random graphs.*

A celebrated conjecture of Zs. Tuza says that in any (finite) graph, the minimum size of a cover of triangles by edges is at most twice the maximum size of a set of edge-disjoint triangles. Resolving a recent question of Bennett, Dudek, and Zerbib, we show that this is true for random graphs; more precisely:

$$\text{for any } p = p(n),\ \mathbb{P}(G_{n,p} \text{ satisfies Tuza's Conjecture}) \to 1 \ (\text{as } n \to \infty).$$

(Received July 24, 2020)

1161-05-39 **Tao Jiang**, **Zilin Jiang*** (zilinj@mit.edu) and **Jie Ma**. *Negligible obstructions and Turán exponents.*

The conjecture on the realizability of rational exponents states that for every rational number r in $(1,2)$ there is a graph F_r such that $\mathrm{ex}(n,F_r) = \Theta(n^r)$. In their beautiful work, Bukh and Conlon resolved a weaker version of the conjecture, where F_r is allowed to be a finite family of graphs. Subsequent work has been focusing on narrowing this family down to a single graph. We formulate a framework, that is taking shape in recent work, to attack the conjecture on the realizability of rational exponents. As an application of the framework, we show that for every rational number $r \in (1,2)$ of the form $2 - a/b$, where $a, b \in \mathbb{N}^+$ satisfy $\lfloor a/b \rfloor^3 \leq a \leq b/(\lfloor b/a \rfloor + 1) + 1$, there exists a graph F_r such that the Turán number $\mathrm{ex}(n,F_r) = \Theta(n^r)$. Joint work with Tao Jiang and Jie Ma. (Received August 02, 2020)

1161-05-40 **Lingjuan Shi*** (shilj18@nwpu.edu.cn), **Hongwei Wang** and **Heping Zhang**. *The resonant faces of $(4,6)$-fullerenes.* Preliminary report.

A $(4,6)$-fullerene graph G is a plane cubic graph whose faces are squares and hexagons. A resonant set of G is a set of pairwise disjoint faces of G such that the boundaries of such faces are M-alternating cycles for a perfect matching M of G. A resonant set of G is referred to as sextet pattern whenever it only includes hexagonal faces. We get the following conclusions. i) The Clar (resp. Fries) number of G is equal to its maximum forcing (resp. anti-forcing) number, which extends some known results for hexagonal systems with a perfect matching. Moreover, two formulas dependent only on the order of G are obtained, which count the Clar number and Fries number of G respectively. Hence we can compute the maximum forcing number of a $(4,6)$-fullerene graph in linear time. This answers an open problem proposed by Afshani et al. (2004) in the case of $(4,6)$-fullerene graphs. ii) All the maximum sextet patterns of G are characterized. Via such characterizations, a formula only depending on the order of G and the number of a fixed subgraph of G is obtained for counting the maximum sextet patterns of G, where the count equals the coefficient of the term with largest degree in its sextet polynomial. (Received August 03, 2020)

1161-05-48 **Jiaao Li*** (lijiaao@nankai.edu.cn), Nankai University, 94 Weijin Road, Tianjin, 300071, Peoples Rep of China. *Integer Flows and Modulo Flows of Signed Graphs.*

It was conjectured by Bouchet in 1983 that every flow-admissible signed graph admits a nowhere-zero integer 6-flow. Recently, we prove that every flow-admissible signed graph admits a balanced $Z_2 \times Z_3$-flow, and apply it to obtain a nowhere-zero integer 11-flow for every such signed graph. For a better understanding of Bouchet's conjecture, it is crucial to study on how to convert modulo flows to better integer flows. In this talk, we will show how to convert Z_2-, Z_3-flow to integer flows in the 11-flow theorem proof, and will also discuss some new results on converting Z_5-flow and unitary Z_{2k+1}-flows to certain integer flows. It seems much more difficult to convert Z_{2k}-flows to (desired) integer flows in signed graphs. In particular, is it possible to covert Z_6-flows to

integer k-flows for some small value k (which would be able to improve the current integer 11-flow theorem)? (Received August 04, 2020)

1161-05-49 **Alex Cameron** and **Gregory J. Puleo*** (gjp0007@auburn.edu). *A lower bound on the saturation number, and graphs for which it is sharp.*

Let H be a fixed graph. We say that a graph G is H-saturated if it has no subgraph isomorphic to H, but the addition of any edge to G results in an H-subgraph. The *saturation number* $\mathrm{sat}(H, n)$ is the minimum number of edges in an H-saturated graph on n vertices. Kászonyi and Tuza, in 1986, gave a general upper bound on the saturation number of a graph H, but a nontrivial lower bound has remained elusive. We give a general lower bound on $\mathrm{sat}(H, n)$ and prove that it is asymptotically sharp (up to an additive constant) on a large class of graphs. This class includes all threshold graphs and many graphs for which the saturation number was previously determined exactly. Our work thus gives an asymptotic common generalization of several earlier results. The class also includes disjoint unions of cliques, allowing us to address an open problem of Faudree, Ferrara, Gould, and Jacobson. (Received August 04, 2020)

1161-05-54 **Bing Wei*** (bwei@olemiss.edu). *Gallai-Ramsey numbers for some graphs with chromatic number three.*

A Gallai coloring of a complete graph is an edge-coloring such that no triangle has all its edges colored differently. A Gallai k-coloring is a Gallai coloring that uses k colors. Given a graph H and an integer $k \geq 1$, the Gallai-Ramsey number $GR_k(H)$ is defined to be the minimum integer n such that every Gallai k-coloring of the edges of K_n contains a monochromatic copy of H. In this talk, we investigate Gallai-Ramsey numbers for some graphs with chromatic number three such as $\widehat{K}_m$ for $m \geq 2$, where $\widehat{K}_m$ is a kipas with $m+1$ vertices obtained from the join of K_1 and P_m, and a class of graphs with five vertices, denoted by $\mathcal{H}$. We first provide a general lower bound of such graphs and propose a conjecture for the exact value of $GR_k(\widehat{K}_m)$. Then we determine the Gallai-Ramsey numbers for many graphs in $\mathcal{H}$ and obtain the exact value of $GR_k(\widehat{K}_4)$ for $k \geq 1$. Our outcomes not only indicate that the conjecture on $GR_k(\widehat{K}_m)$ is true for $m = 4$, but also imply several results on $GR_k(H)$ for some $H \in \mathcal{H}$ which are proved individually in different papers.

This is a joint work with Qinghong Zhao. (Received August 05, 2020)

1161-05-57 **Wesley Hamilton*** (wham@live.unc.edu). *A Graph Spectral Flow for Computing Nodal Deficiencies.*

In this paper we propose a spectral flow for graph Laplacians, and prove that it counts the number of nodal domains for a given Laplace eigenvector. This extends work done for Laplacians on $\mathbb{R}^n$ to the graph setting. We mention some open problems relating the topology of a graph to the analytic behaviour of its Laplace eigenvectors, and include numerical examples illustrating our flow. (Received August 06, 2020)

1161-05-67 **Guantao Chen*** (gchen@gsu.edu), Department of Mathematics and Statistics, Georgia State University, Atlanta, GA 30303, and **Xiaowei Yu** and **Yi Zhao**. *Ramsey number of fans.* Preliminary report.

For a given graph H, the Ramsey number $r(H)$ is the minimum N such that any 2-edge-coloring of the complete graph K_N yields a monochromatic copy of H. Given a positive integer n, let nK_3, F_n and B_n be three graphs formed by n triangles that share zero, one, and two common vertices, respectively. Burr, Erdős and Spencer in 1975 showed that $r(nK_3) = 5n$ for $n \geq 2$. Rousseau and Sheehan in 1978 showed that $r(B_n) \leq 4n + 2$ and equality holds for infinitely many values of n. We believe that $r(B_n) \leq r(F_n) \leq r(nK_3)$ for sufficiently large n. We confirm the first inequality by showing that $9n/2 - 5 \leq r(F_n) \leq 11n/2 + 6$ for any n. This improves previously known bounds $4n + 2 \leq r(F_n) \leq 6n$. (Received August 09, 2020)

1161-05-72 **Xiaoya Zha*** (xzha@mtsu.edu), Department of Mathematical Scienced, Middle Tennessee State University, Murfreesboro, TN 37132. *Edge Partition of Graphs Embeddable in the Projective Plane and the Klein Bottle.*

In a previous paper we show that every non-planar toroidal graph can be edge partitioned into a planar graph and an outerplanar graph. This edge partition then implies some results in thickness and outerthickness of toroidal graphs. In particular, if each planar graph has outerthickness at most 2 (conjectured by Chartrand, Geller and Hedetniemi in 1971 and the confirmation of the conjecture was announced by Gonçalves in 2005), then the outerthickness of toroidal graphs is at most 3 which is the best possible due to K_7.

In this talk we continue to study the edge partition for projective planar graphs and Klein bottle embeddable graphs. We show that (1) every non-planar but projective planar graph can be edge partitioned into a planar graph and a union of caterpillar trees; and (2) every non-planar Klein bottle embeddable graph can be edge partitioned into a planar graph and a subgraph of two vertex amalgamation of a caterpillar tree with a cycle with

pendant edges. As consequences, the thinkness of projective planar graphs and Klein bottle embeddabe graphs are at most 2, which are the best possible, and the outerthickness of these graphs are at most 3. (Received August 10, 2020)

1161-05-76 **Abigail Raz*** (araz2@unl.edu) and **Xavier Perez-Gimenez**. *Perfect matchings in the random bipartite geometric graph.* Preliminary report.

We consider the standard random bipartite geometric graph process in which n red vertices and n blue vertices are placed at random on the unit d-dimensional cube and edges are added sequentially, between vertices of different colors, in increasing order of edge-length. A natural question is to ask whether the first edge in the process that results in the minimum degree being at least one coincides, with high probability, with the first edge that creates a perfect matching. While this was already known to be false when $d = 2$, as the thresholds are not even of the same order, we are able to positively answer it for $d \geq 3$.
Joint work with Xavier Pérez-Giménez (Received August 10, 2020)

1161-05-82 **Debarun Ghosh**, **Ervin Győri** and **Ryan R. Martin***, Department of Mathematics, 428 Carver Hall, Iowa State University, Ames, IA 50011, and **Addisu Paulos** and **Chuanqi Xiao**. *Planar Turán number of the* 6*-Cycle.*

Let $\mathrm{ex}_{\mathcal{P}}(n, T, H)$ denote the maximum number of copies of T in an n-vertex planar graph which does not contain H as a subgraph. When $T = K_2$, $\mathrm{ex}_{\mathcal{P}}(n, T, H)$ is the well studied function, the planar Turán number of H, denoted by $\mathrm{ex}_{\mathcal{P}}(n, H)$. The topic of extremal planar graphs was initiated by Dowden (2016). He obtained sharp upper bound for both $\mathrm{ex}_{\mathcal{P}}(n, C_4)$ and $\mathrm{ex}_{\mathcal{P}}(n, C_5)$. Later on, Y. Lan, et al. continued this topic and proved that $\mathrm{ex}_{\mathcal{P}}(n, C_6) \leq \frac{18(n-2)}{7}$. In this paper, we give a sharp upper bound $\mathrm{ex}_{\mathcal{P}}(n, C_6) \leq \frac{5}{2}n - 7$, for all $n \geq 18$, which improves Lan's result. We also pose a conjecture on $\mathrm{ex}_{\mathcal{P}}(n, C_k)$, for $k \geq 7$. (Received August 10, 2020)

1161-05-95 **Mark Ellingham*** (mark.ellingham@vanderbilt.edu) and **Joanna A. Ellis-Monaghan**. *Orientable embeddings with two euler circuit faces.*

In a *directed embedding* of a digraph each face is bounded by a directed walk; if the digraph is connected it must be eulerian. An orientable directed embedding has at least two faces, and it has exactly two faces if and only if it is an *orientable two-euler-face*, or *OTEF*, directed embedding, with each face bounded by a directed euler circuit. Such an embedding is a directed embedding of maximum genus. There is an obvious necessary condition that a digraph with an OTEF directed embedding must satisfy. In 2002 Bonnington, Conder, Morton and McKenna showed that every regular tournament satisfying this condition has an OTEF directed embedding. We consider generalizations of their result. We show that the necessary condition is not in general sufficient. In fact, we show the stronger result that there are 4-edge-connected eulerian undirected graphs with no OTEF embedding (defined analogously for undirected graphs). But we also show that a very large class of eulerian digraphs do have OTEF directed embeddings, namely those in which every vertex has degree congruent to 2 modulo 4. (Received August 11, 2020)

1161-05-99 **Michael D Plummer*** (michael.d.plummer@vanderbilt.edu), TN 37215, and **Art Finbow** and **Bert Hartnell**. *On the structure of planar well-covered graphs.*

A graph is *well-covered* if all its maximal independent sets of vertices have the same cardinality. The property of being well-covered is known to be co-NP-complete and hence, not surprisingly, the study of such graphs has proven to be difficult. Hence the authors and others have focused on the structure of various special subclasses of these graphs. In the present paper the authors undertake the study of those well-covered graphs which are 4-regular and planar. (Received August 12, 2020)

1161-05-103 **Matthew Kwan**, **Lisa Sauermann*** (lsauerma@mit.edu) and **Yufei Zhao**. *Extension complexity of low-dimensional polytopes.*

Sometimes, it is possible to represent a complicated polytope as a projection of a much simpler polytope. To quantify this phenomenon, the extension complexity of a polytope P is defined to be the minimum number of facets in a (possibly higher-dimensional) polytope from which P can be obtained as a (linear) projection. This notion has been studied for several decades, motivated by its relevance for combinatorial optimisation problems. It is an important question to understand the extent to which the extension complexity of a polytope is controlled by its dimension, and in this paper we prove three different results along these lines. First, we prove that for a fixed dimension d, the extension complexity of a random d-dimensional polytope (obtained as the convex hull of random points in a ball or on a sphere) is typically on the order of the square root of its number of vertices. Second, we prove that any cyclic n-vertex polygon (whose vertices lie on a circle) has extension complexity at most $24\sqrt{n}$. This bound is tight up to the constant factor 24. Finally, we show that there exists

an $n^{o(1)}$-dimensional polytope with at most n facets and extension complexity $n^{1-o(1)}$. (Received August 13, 2020)

1161-05-117 **Chun-Hung Liu*** (chliu@math.tamu.edu). *Asymptotic dimension of minor-closed families and beyond.*

The asymptotic dimension of metric spaces is an important notion in geometric group theory introduced by Gromov. The metric spaces considered in this talk are the ones whose underlying spaces are the vertex-sets of graphs and whose metrics are the distance functions in graphs. A standard compactness argument shows that it suffices to consider the asymptotic dimension of classes of finite graphs.

In this talk we prove that the asymptotic dimension of any minor-closed family, any class of graphs of bounded tree-width, and any class of graphs of bounded layered tree-width are at most 2, 1, and 2, respectively. The first result solves a question of Fujiwara and Papasoglu; the second and third results solve a number of questions of Bonamy, Bousquet, Esperet, Groenland, Pirot and Scott. These bounds for asymptotic dimension are optimal and improve a number of results in the literature. (Received August 14, 2020)

1161-05-118 **Chun-Hung Liu*** (chliu@math.tamu.edu). *Immersion and clustered coloring.*

Hadwiger and Hajós conjectured that for every positive integer t, K_{t+1}-minor free graphs and K_{t+1}-topological minor free graphs are properly t-colorable, respectively. Clustered coloring version of these two conjectures which only require monochromatic components to have bounded size has been extensively studied. In this talk we consider the clustered coloring version of the immersion-variant of Hadwiger's and Hajós' conjecture proposed by Lescure and Meyniel and independently by Abu-Khzam and Langston. We determine the minimum number of required colors for H-immersion free graphs, for any fixed graph H, up to a small additive absolute constant. Our result is tight for infinitely many graphs H. (Received August 14, 2020)

1161-05-125 **Andrzej Dudek*** (andrzej.dudek@wmich.edu), Department of Mathematics, Western Michigan University, Kalamazoo, MI 49008. *Variations on twins in permutations.*

Let π be a permutation of the set $[n] = \{1, 2, \ldots, n\}$. Two disjoint order-isomorphic subsequences of π are called *twins*. How long twins are contained in every permutation? The well known Erdős-Szekeres theorem implies that there is always a pair of twins of length $\Omega(\sqrt{n})$. On the other hand, by a simple probabilistic argument Gawron proved that for every $n \geqslant 1$ there exist permutations with no twins of length greater than $O(n^{2/3})$. His conjecture states that the latter bound is the correct size of the longest twins guaranteed in every permutation. In this talk we show that asymptotically almost surely a random permutation contains twins of length at least $\Omega(n^{2/3})$, which supports this conjecture. (This was also proved recently by Bukh and Rudenko.) We also discuss several variants of the problem with diverse restrictions imposed on the twins.

This is a joint work with Jarosław Grytczuk and Andrzej Ruciński. (Received August 14, 2020)

1161-05-132 **Songling Shan*** (sshan12@ilstu.edu). *Antimagic orientation of graphs with minimum degree at least 33.* Preliminary report.

An antimagic labeling of a directed graph D with n vertices and m arcs is a bijection from the set of arcs of D to the integers $\{1, \cdots, m\}$ such that all n oriented vertex sums are pairwise distinct, where an oriented vertex sum is the sum of labels of all arcs entering that vertex minus the sum of labels of all arcs leaving it. A graph G has an antimagic orientation if it has an orientation which admits an antimagic labeling. Hefetz, Mütze, and Schwartz conjectured that every connected graph admits an antimagic orientation. We show that every bipartite graph with no vertex of degree 0 or 2 admits an antimagic orientation and every graph G with $\delta(G) \geq 33$ admits an antimagic orientation. Our proof relies on a newly developed structural property of bipartite graphs, which might be of independent interest. (Received August 14, 2020)

1161-05-136 **Gweneth McKinley*** (gmckinley@ucsd.edu), **Asaf Ferber** and **Kyle Luh**. *Resilience of the rank of random matrices.*

For an $n \times m$ matrix M of independent Rademacher (± 1) random variables, it is well known that if $n \leq m$, then M is of full rank with high probability. We show that this property is resilient to adversarial changes to M. More precisely, if $m \geq n + n^{1-\varepsilon/6}$, then even after changing the sign of $(1-\varepsilon)m/2$ entries, M is still of full rank with high probability. Note that this is asymptotically best possible as one can easily make any two rows proportional with at most $m/2$ changes. Moreover, this theorem gives an asymptotic solution to a slightly weakened version of a conjecture made by Van Vu in 2008. (Received August 14, 2020)

1161-05-138 **Alan M Frieze*** (alan@random.math.cmu.edu), 5000 Forbes Avenue, Pittsburgh, PA 15213. *The effect of adding randomly weighted edges.*

We consider the following question. We have a dense regular graph G with degree αn, where $\alpha > 0$ is a constant. We add $m = o(n^2)$ random edges. The edges of the augmented graph $G(m)$ are given independent edge weights $X(e), e \in E(G(m))$. We estimate the minimum weight of some specified combinatorial structures. We show that in certain cases, we can obtain the same estimate as is known for the complete graph, but scaled by a factor α^{-1}. We consider spanning trees, shortest paths and perfect matchings in (pseudo-random) bipartite graphs. (Received August 14, 2020)

1161-05-146 **Yifan Jing***, 273 Altgeld Hall 1409 W. Green Street, Urbana, IL 61801, and **Bojan Mohar**. *Approximate the genus of dense graphs.*

The genus $g(G)$ of a graph G is defined as the minimum genus of a surface in which G can be embedded (drawn without crossings). Thomassen proved that it is NP-hard to determine whether $g(G) < k$, when the graph G and an integer k are given to us as the input. Robertson and Seymour, and later Mohar, proved that this problem is FPT (fixed-parameter tractable). However, it is wide open whether the value of $g(G)$ can be approximated. The speaker will give an overview of this problem, describe underlying conjectures, and present a complete solution for the case when the graph is dense. The solution uses Szemeredi Regularity Lemma and a result on the genus of quasi-random graphs. This is joint work with Bojan Mohar. (Received August 15, 2020)

1161-05-155 **Yan Cao**, **Guantao Chen** and **Guangming Jing*** (gjing@augusta.edu), 2500 Walton Way, Augusta, GA 30904. *A note on Goldberg's conjecture of total chromatic numbers.*

Let $G = (V(G), E(G))$ be a multigraph with maximum degree $\Delta(G)$, chromatic index $\chi'(G)$ and total chromatic number $\chi''(G)$. The Total Coloring conjecture proposed by Behzad and Vizing, independently, states that $\chi''(G) \leq \Delta(G) + \mu(G) + 1$ for a multigraph G, where $\mu(G)$ is the multiplicity of G. Moreover, Goldberg conjectured that $\chi''(G) = \chi'(G)$ if $\chi'(G) \geq \Delta(G) + 3$ and noticed the conjecture holds when G is an edge-chromatic critical graph. In this note we show that $\chi''(G) = \chi'(G)$ if $\chi'(G) \geq \max\{\Delta(G) + 2, |V(G)| + 1\}$. Consequently, $\chi''(G) = \chi'(G)$ if $\chi'(G) \geq \Delta(G) + 2$ and G has a spanning edge-chromatic critical subgraph. (Received August 15, 2020)

1161-05-163 **M N Ellingham**, **Linyuan Lu** and **Zhiyu Wang*** (zwang672@gatech.edu). *Maximum spectral radius of outerplanar 3-uniform hypergraphs.* Preliminary report.

A 3-uniform hypergraph $\mathcal{H}$ is called *outerplanar* if its shadow has an outerplanar embedding such that every hyperedge of $\mathcal{H}$ is the vertex set of an interior triangular face of the shadow. Cvetković and Rowlinson [*Linear and Multilinear Algebra, 1990*] conjectured that among all outerplanar graphs on n vertices, the graph $K_1 + P_{n-1}$ attains the maximum spectral radius. We show a hypergraph analogue of the Cvetković-Rowlinson conjecture. In particular, we show that for sufficiently large n, the n-vertex outerplanar 3-uniform hypergraph of maximum spectral radius is the unique 3-uniform hypergraph whose shadow is $K_1 + P_{n-1}$. (Received August 16, 2020)

1161-05-164 **Srilakshmi Pattabiraman**, **Ryan Gabrys** and **Olgica Milenkovic*** (milenkov@illinois.edu), Coordinated Science Lab, Department of Electrical and Computer Enginee, Urbana, IL 61801. *Reconstruction and Error-Correction Codes for Polymer-Based Data Storage.*

Motivated by polymer-based data-storage platforms that use chains of binary synthetic polymers as the recording media and read the content via tandem mass spectrometers, we propose a new family of codes that allows for unique string reconstruction and correction of one mass error. Our approach is based on introducing redundancy that scales logarithmically with the length of the string and allows for the string to be uniquely reconstructed based only on its erroneous substring composition multiset. The key idea behind our unique reconstruction approach is to interleave Catalan-type paths with arbitrary binary strings and "reflect" them so as to allow prefixes and suffixes of the same length to have different weights. For error correction, we add a constant number of bits that provides information about the weights of reflected pairs of bits and hence enable recovery from a single mass error. The asymptotic code rate of the scheme is one, and decoding is accomplished via a simplified version of the backtracking algorithm used for the Turnpike problem. (Received August 16, 2020)

1161-05-166 **Matthew Kahle** and **Elliot Paquette*** (elliot.paquette@gmail.com), , Canada, and **Erika Roldan**. *The threshold for simple–connectedness in hypercube percolation.*

We study the fundamental group of certain random 2-dimensional cubical complexes. We show a 2-dimensional generalization of a theorem of Burtin and Erdos-Spencer on the connectivity threshold for bond percolation. In the 2-dimensional setting, the natural analogue is a transition for the simple–connectivity of the space. This is in contrast to the 2-dimensional analogue of simplicial complexes, in which the natural analogue of the Erdos-Renyi

theorem is the threshold for homological connectivity of the space (due to Linial–Meshulam). We also show that below the connectivity threshold, the fundamental group factors as a product of a finitely generated pieces, and that as the density parameter goes to 0, every finitely generated group appears. (Received August 16, 2020)

1161-05-167 **Hongqiu Chen*** (hchen1@memphis.edu), Department of Mathematical Sci, 365 Dunn Hall, University of Memphis, Memphis, TN 38152, and **Jerry Bona** and **Colette Guillope**.
Further Theory for a Higher-order Water Wave Model. Preliminary report.

A class of higher-order models for unidirectional water wave of the form

$$\eta_t + \eta_x - \gamma_1\beta\eta_{xxt} + \gamma_2\beta\eta_{xxx} + \delta_1\beta^2\eta_{xxxxt} + \delta_2\beta^2\eta_{xxxxx} + \frac{3}{4}\alpha(\eta^2)_x + \alpha\beta\Big(\gamma(\eta^2)_{xx} - \frac{7}{48}\eta_x^2\Big)_x - \frac{1}{8}\alpha^2(\eta^3)_x = 0$$

was derived by Bona, Carvajal, Panthee and Scialom. With appropriate choices of $\gamma_1, \gamma_2, \delta_1, \delta_2$ and γ, this equation serves as a model for the propagation of small-amplitude, long-crested surface waves moving to the direction of increasing values of the spatial variable x. Where $\eta = \eta(x,t)$ is a real-valued function of $x \in (-\infty, \infty), t \geq 0$ representing the deviation of the free surface from its undisturbed position at the point corresponding to x at time t. Moreover, γ_1 and γ_2 are restricted by $\gamma_1 + \gamma_2 = \frac{1}{6}$. The new result is that when $\gamma = \frac{7}{48}$, $\delta_2 > \delta_1 > 0$ and the initial data

$$\eta(x,0) = \eta_0(x,0)$$

lies in H^1 and not too big, then the initial-value problem is globally well posed and the H^1-norm of the solution is uniformly bounded for $t \geq 0$. (Received August 16, 2020)

1161-05-169 **Yue Wang*** (15064013175@163.com) and **Gexin Yu** (gyu@wm.edu). .
Let $s \geq 2$ and $t \geq 2$ be integers. A graph G is (s,t)-*splittable* if $V(G)$ can be partitioned into two sets S and T such that $\chi(G[S]) \geq s$ and $\chi(G[T]) \geq t$. The well-known Erdős-Lovász Tihany Conjecture from 1968 states that every graph G whose chromatic number $\chi(G)$ is more than its clique number $\omega(G)$ is (s,t)-splittable. We prove an enhanced version of the Erdős-Lovász Tihany Conjecture for the following two classes of graphs.

For every line graph of multigraph G with $\chi(G) > \omega(G)$ and for nonnegative integer ℓ, any two integers $s, t \geq 3.5\ell + 2$ with $s + t = \chi(G) + 1$, G is $(s, t+\ell)$-splittable. In particular, when $\ell = 1$, we can obtain the same result just for any $s, t \geq 4$. The Erdős-Lovász Tihany conjecture is a special case when $\ell = 0$.

For every graph G with independence number two with $\chi(G) > \omega(G) + 1$ and any two integers $s, t \geq 2$ with $s + t = \chi(G) + 1$, G is $(s, t+1)$-splittable. We found examples showing that this result is best possible.
(Received August 16, 2020)

1161-05-174 **Hoi Nguyen***, 231 W 18th Ave, Columbus, OH 43210. *On roots of random trigonometric polynomials.*

We show that for various general ensembles of random trigonometric polynomials with iid coefficients the variance of the number of roots is asymptotically linear in terms of the expectation; furthermore the multiplicative constant in this linear relationship depends only on the kurtosis of the common distribution of the polynomial's coefficients. We will also show that the number of roots satisfies an exponential concentration of measure.

Based on joint work with Y. Do and O. Nguyen, and with O. Zeitouni (Received August 16, 2020)

1161-05-175 **Martin Rolek*** (mrolek1@kennesaw.edu), Department of Mathematics, 850 Polytechnic Lane, MD #9085, Marietta, GA 30060, and **Paul Scemama**. *A fractional look at Steinberg's conjecture.*

Steinberg's conjecture claims that every planar graph with no 4- or 5-cycles is 3-colorable. After significant attention in recent decades, the conjecture was finally disproved in 2017. Hope remains, however, for a fractional version of Steinberg's conjecture, which claims that every planar graph with no 4- or 5-cycles has fractional chromatic number at most 3. A graph G is $(s:t)$-*colorable* if there is a function ϕ which assigns to each vertex of G a t-element subset of $\{1, 2, \ldots, s\}$ such that $\phi(u) \cap \phi(v) = \emptyset$ whenever $uv \in E(G)$, and the *fractional chromatic number* of a graph G is $\min\{s/t : G \text{ is } (s:t)\text{-colorable}\}$. In this talk, we will discuss some relaxations of Steinberg's conjecture that have been investigated over the years, and we will look at some very recent results on fractional coloring which relate to these relaxations of Steinberg's conjecture. In particular, the authors have shown that any planar graph with no 4- or 5-cycles and no incident triangles has fractional chromatic number at most 7/2. (Received August 16, 2020)

1161-05-178 **Yuanan Diao**, **Gabor Hetyei** and **Van Anh Pham***, 9201 University City Blvd, Department of Mathematics and Statistics, Charlotte, NC 28223. *Writhe-like invariants for alternating link diagrams derived from Seifert graphs.* Preliminary report.

In this paper, we introduce quantities that are derived from Seifert graphs of reduced alternating link diagrams. We prove that they are "writhe-like" invariants; that is to say, they are invariant among reduced alternating link

diagrams. The determination of these invariants are elementary and non-recursive, so they are easy to calculate. We demonstrate that different alternating links can be easily distinguished by one of these new invariants. As an application, we also derive an if-and-only-if condition for a strongly invertible rational link. (Received August 16, 2020)

1161-05-182 **Keith Frankston*** (keith.frankston@math.rutgers.edu), **Jeff Kahn**, **Bhargav Narayanan** and **Jinyoung Park**. *Thresholds Versus Fractional Expectation-Thresholds.*

Given an increasing family F of subsets of a finite set X, its measure according to μ_p increases and often exhibits a threshold behavior, growing quickly as p increases from near 0 to near 1 around a specific value p_c. These thresholds have been a central focus of the study of random discrete structures, with estimation of thresholds for specific properties the subject of some of the most challenging work in the area.

In 2006, Kahn and Kalai conjectured that a natural (and often easy to calculate) lower bound $q(F)$ (which we call the "expectation-threshold") for p_c is never far from its actual value. We prove a fractional version (proposed by Talagrand) of this so called "expectation-threshold" conjecture showing that for any increasing family we have $p_c(F) = O(q_f(F)\log \ell(F))$, where q_f is the "fractional expectation-threshold" and $\ell(F)$ is the maximum size of a minimal element of F.

This result easily implies previously difficult results in probabilistic combinatorics such as thresholds for perfect hypergraph matchings (Johansson-Kahn-Vu) and bounded-degree spanning trees (Montgomery). Our approach builds on a recent breakthrough of Alweiss, Lovett, Wu and Zhang on the Erdős-Rado "Sunflower Conjecture." (Received August 16, 2020)

1161-05-185 **Linyuan Lu*** (lu@math.sc.edu), Department of Mathemetics, University of South Carolina, 1523 Greene Street, Columbia, SC 29208, and **Zhiyu Wang**, School of Mathematics, Georgia Institute of Technology, Atlanta, GA 30332. *On the size of planar graphs with positive Lin-Lu-Yau Ricci curvature.*

Let G be a planar graph embedded into a sphere. The combinatorial curvature $\phi(v)$ at a vertex v is defined by $\phi(v) = 1 - \frac{\deg(v)}{2} + \sum_{\sigma \in F(v)} \frac{1}{|\sigma|}$, where $F(v)$ is the set of faces touching v and $|\sigma|$ is the number of edges bounding a face σ. G is said to be positively curved if $\phi(v) > 0$ for every $v \in V(G)$. Higuchi [*J. Graph Theory, 2001*] conjectured that if G is a simple connected positively curved graph embedded into a 2-sphere and with minimum degree at least 3, then G is finite. DeVos and Mohar [*Trans. Amer. Math. Soc., 2007*] showed that if G is not a prism, and an antiprism, then $|V(G)| \leq 3444$, resolving Higuchi's conjecture. In this talk, we show an analogue of Higuchi's conjecture in the context of the Lin-Lu-Yau Ricci curvature. In particular, we show that if a planar graph G with minimum degree at least 3 has positive Lin-Lu-Yau Ricci curvature on every edge, then G is finite. (Received August 16, 2020)

1161-05-194 **Paul Balister*** (paul.balister@maths.ox.ac.uk), Mathematical Institute, Radcliffe Observatory Quarter, Woodstock Road, Oxford, OX2 6GG, United Kingdom, and **Bela Bollobas**, **Robert Morris**, **Julian Sahasrabudhe** and **Marius Tiba**. *Flat Littlewood polynomials exist.*

A Littlewood polynomial is a polynomial with all coefficients equal to $+1$ or -1. Erdős (1957) and Littlewood (1966) asked if one could find $\delta, \Delta > 0$ such that for all large n there exist Littlewood polynomials $P(z)$ of degree n such that $\delta\sqrt{n} \leq |P(z)| \leq \Delta\sqrt{n}$ for all complex z with $|z| = 1$. In this talk I will describe our proof of this conjecture, which uses recent results in discrepancy theory. (Received August 17, 2020)

1161-05-196 **Alexander Riasanovsky*** (awnr@iastate.edu) and **Ryan Martin** (rymartin@iastate.edu). *On the edit distance function of the random graph.*

Given a hereditary property of graphs $\mathcal{H}$ and a $p \in [0,1]$, the edit distance function $\mathrm{ed}_{\mathcal{H}}(p)$ is asymptotically the maximum proportion of edge-additions plus edge-deletions applied to a graph of edge density p sufficient to ensure that the resulting graph satisfies $\mathcal{H}$. The edit distance function is directly related to other well-studied quantities such as the speed function for $\mathcal{H}$ and the $\mathcal{H}$-chromatic number of a random graph.

Let $\mathcal{H}$ be the property of forbidding an Erdős-Rényi random graph $F \sim \mathbb{G}(n_0, p_0)$, and let φ represent the golden ratio. In this paper, we show that if $p_0 \in [1 - 1/\varphi, 1/\varphi]$, then a.a.s. as $n_0 \to \infty$,

$$\mathrm{ed}_{\mathcal{H}}(p) = (1 + o(1))\,\frac{2\log n_0}{n_0} \cdot \min\left\{\frac{p}{-\log(1-p_0)}, \frac{1-p}{-\log p_0}\right\}.$$

Moreover, this holds for $p \in [1/3, 2/3]$ for any $p_0 \in (0,1)$. (Received August 17, 2020)

1161-05-198 **Pawel Pralat*** (pralat@ryerson.ca), 350 Victoria Str., Toronto, ON M5B 2K3, Canada.
A variant of the Erdős-Rényi random graph process.

We consider a natural variant of the Erdős-Rényi random graph process in which k vertices are special and are never put into the same connected component. The model is natural and interesting on its own, but is actually inspired by the combinatorial data fusion problem that itself is connected to a number of important problems in graph theory. We will show that a phase transition occurs when the number of special vertices is roughly $n^{1/3}$, where n is the number of vertices. (Joing work with Adam Logan and Mike Molloy.) (Received August 17, 2020)

1161-05-200 **R. Kirsch*** (kirsch@iastate.edu) and **A. J. Radcliffe**. *Maximizing the density of K_t's in graphs of bounded degree and clique number.*

Zykov showed in 1949 that among graphs on n vertices with clique number $\omega(G) \leq \omega$, the Turán graph $T_\omega(n)$ maximizes not only the number of edges but also the number of copies of K_t for each size t. The problem of maximizing the number of copies of K_t has also been studied within other classes of graphs, such as those on n vertices with maximum degree $\Delta(G) \leq \Delta$.

We combine these restrictions and investigate which graphs with $\Delta(G) \leq \Delta$ and $\omega(G) \leq \omega$ maximize the number of copies of K_t per vertex. We define $f_t(\Delta, \omega)$ as the supremum of ρ_t, the number of copies of K_t per vertex, among such graphs, and show for fixed t and ω that $f_t(\Delta, \omega) = (1 + o(1))\rho_t(T_\omega(\Delta + \lfloor \frac{\Delta}{\omega - 1} \rfloor))$. For two infinite families of pairs (Δ, ω), we determine $f_t(\Delta, \omega)$ exactly for all $t \geq 3$. For another we determine $f_t(\Delta, \omega)$ exactly for the two largest possible clique sizes. Finally, we demonstrate that not every pair (Δ, ω) has an extremal graph that simultaneously maximizes the number of copies of K_t per vertex for every size t. (Received August 17, 2020)

1161-05-201 **Pu Gao*** (pu.gao@uwaterloo.ca), Waterloo, Ontario N2L 3G1, Canada, and **Mikhail Isaev** and **Brendan McKay**. *Sandwiching random regular graphs by binomial random graphs.*

Random regular graphs are among the best known random graph models. However, they are hard to analyse, especially when the degree grows with the number of vertices. In 2004 Kim and Vu conjectured that if $d \gg \log n$ then the random d-regular graph G can be coupled with two binomial random graphs $G(n, p_1)$ and $G(n, p_2)$ such that both p_1 and p_2 are asymptotic to d/n, and with high probability, $G(n, p_1) \subseteq G \subseteq G(n, p_2)$. We prove this conjecture for all $d \gg n/\sqrt{\log n}$, and we prove an approximate version of the conjecture for smaller d. This is joint work with Mikhail Isaev and Brendan McKay. (Received August 17, 2020)

1161-05-205 **Alan Frieze** and **Tomasz Tkocz***, Carnegie Mellon University, Pittsburgh, PA 15213.
Shortest paths with a cost constraint: a probabilistic analysis.

We consider a constrained version of the shortest path problem on the complete graphs whose edges have independent random lengths and costs. We shall sketch how to establish the asymptotic value of the minimum length as a function of the cost-budget within a wide range. (Received August 17, 2020)

1161-05-209 **Joe Briggs** and **Chris Cox*** (cocox@iastate.edu). *Restricted online Ramsey numbers for matchings.*

Consider a two-player game between players Builder and Painter. Painter begins the game by picking a coloring of the edges of K_n, which is hidden from Builder. In each round, Builder points to an edge and Painter reveals its color. Builder's goal is to locate a large monochromatic matching in Painter's coloring by revealing the color of as few edges as possible. We show that if Painter has access to t colors, then Builder can locate a monochromatic matching on at least $\frac{n-t+1}{t+1}$ edges by revealing at most $O(n \log t)$ edges. Time permitting, we will discuss results pertaining to the game wherin Builder is instead tasked with locating a large tree. (Received August 17, 2020)

1161-05-213 **Victor Falgas-Ravry**, **Klas Markström**, **Andrew Treglown** and **Yi Zhao*** (yzhao6@gsu.edu). *Existence thresholds and Ramsey properties of random posets.*

Let P_n denote the power set of $[n]$, ordered by inclusion, and let $P(n, p)$ denote the random poset obtained from P_n by retaining each element from P_n independently at random with probability p and discarding it otherwise.

Given *any* fixed poset F, we determine the threshold for the property that $P(n, p)$ contains F as an induced subposet. We also asymptotically determine the number of copies of a fixed poset F in P_n. Finally, we obtain a number of results on the Ramsey properties of the random poset $P(n, p)$. (Received August 17, 2020)

1161-05-215 **Deepak Bal***, deepak.bal@montclair.edu. *Size Ramsey numbers of paths and cycles.* Preliminary report.

Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two families of graphs. The size Ramsey number $\hat{R}(\mathcal{F}_1, \mathcal{F}_2)$ is the smallest number m such that there exists a graph G with m edges in which every red-blue coloring of $E(G)$ contains either a red graph from $\mathcal{F}_1$ or a blue graph from $\mathcal{F}_2$. Let P_n represent the path on n vertices and let $\mathcal{C}$ represent the family of all cycles. We discuss some recent progress on size Ramsey numbers in the cases where $\mathcal{F}_1 = \mathcal{F}_2 = \{P_n\}$ (improvement of lower bound) and where $\mathcal{F}_1 = \{P_n\}$, $\mathcal{F}_2 = \mathcal{C}$ (improvement of lower and upper bounds). These result are joint with Louis DeBiasio and Ely Schudrich respectively. (Received August 17, 2020)

1161-05-216 **Ross Berkowitz** and **Pat Devlin*** (patrick.devlin@yale.edu). *Central limit laws in majority dynamics (and a proof that bribing three voters suffices).* Preliminary report.

Let V be a set of N vertices, half of which are initially colored red and half are colored blue. At each time step, all of the vertices simultaneously update their color to match the majority opinion among their neighbors. We examine the evolution of this process when the underlying graph is chosen from G(N, p) and show that giving one color (say blue) a lead of size O(1/p) can lead to every vertex being blue within four steps. We also prove a central limit theorem for the number of vertices of each color after the first step. (Received August 17, 2020)

1161-05-222 **JD Nir*** (nir.justin@gmail.com) and **Xavier Perez**. *The Chromatic Number of Random Lifts of Regular Graphs.*

Graph coloring is one of several constraint satisfaction problems studied on random structures. The problem at the heart of this talk is to bound the chromatic number of a random d-regular graph. Rather than use the uniform model, we take inspiration from a question of Linial and generate our regular graph by taking random lifts of a regular host graph. This method produces a family of noncontiguous models of random graphs. We provide an upper bound for the chromatic number of such lifts. In the case where the host graph is complete, we also provide a nearly matching lower bound. Our proof contains several techniques with broad applications in random structures, including Laplace summation, the saddle point method, and the small subgraph conditioning method. (Received August 17, 2020)

1161-05-223 **Zdeněk Dvořák**, **Bernard Lidický*** (lidicky@iastate.edu) and **Luke Postle**. *11/4-colorability of subcubic triangle-free graphs.* Preliminary report.

We prove that every connected subcubic triangle-free graph except for two exceptional graphs on 14 vertices has fractional chromatic number at most 11/4. (Received August 17, 2020)

1161-05-224 **Patrick Bennett*** (patrick.bennett@wmich.edu), **Ryan Cushman** and **Andrzej Dudek**. *Tuza's conjecture in random graphs.*

A long-standing conjecture of Zsolt Tuza asserts that the triangle covering number $\tau(G)$ is at most twice the triangle packing number $\nu(G)$, where the *triangle packing number* $\nu(G)$ is the maximum size of a set of edge-disjoint triangles in G and the *triangle covering number* $\tau(G)$ is the minimal size of a set of edges intersecting all triangles. In this paper, we prove that Tuza's conjecture holds in the Erdős-Rényi random graph $G(n, m)$ for all range of m, closing the "gap" in what was previously known. (Recently, this result was also independently proved by Jeff Kahn and Jinyoung Park.) We employ a random greedy process called the *online triangle packing process* to produce a triangle packing in $G(n, m)$ and analyze this process by using the differential equations method. (Received August 17, 2020)

1161-05-238 **Anton Bernshteyn*** (bahtoh@gatech.edu) and **Clinton T. Conley** (clintonc@andrew.cmu.edu). *Equitable colourings of infinite graphs.*

A proper k-coloring of a finite graph G is called equitable if every two color classes differ in size at most by one. In particular, if G has n vertices and k divides n, then in an equitable k-coloring of G every color class has size exactly n/k. There is a natural way to extend this definition to infinite graphs on probability spaces. Namely, if G is a graph whose vertex set $V(G)$ is a probability space, then a proper k-coloring of G is equitable when every color class has measure $1/k$. In this talk I will discuss extensions of some classical results about equitable colourings to this setting, including an infinite version of the Hajnal–Szemerédi theorem on equitable k-colorings for $k \geq \Delta(G) + 1$. This is joint work with Clinton T. Conley. (Received August 17, 2020)

1161-05-240 **Beka Ergemlidze** and **Theodore Molla*** (molla@usf.edu). *Transversal C_k-factors in subgraphs of the balanced blowup of C_k.* Preliminary report.

Call a blowup of a graph F an n-blowup if each part has size n. For a subgraph of a blowup of F, we define the minimum partial degree of G to be the smallest minimum degree over all of the bipartite subgraphs of G that correspond to edges of F. Johannson proved that when the minimum partial degree of a spanning subgraph

of the n-blowup of a triangle is $2n/3 + n^{1/2}$, then G can be tiled with n vertex disjoint triangles. Fischer's Conjecture, which was proved by Keevash and Mycroft in 2015, is a generalization of this result to complete graphs larger than the triangle. Another generalization, conjectured independently by Fischer and Häggkvist, is the following: If G is a spanning subgraph of the n-blowup of C_k with minimum partial degree $(1+1/k)n/2+1$, then G contains n vertex disjoint copies of C_k that each intersect all of the k parts. In this talk, we will discuss a proof of an asymptotic version of this conjecture. We will also discuss some related conjectures and results. (Received August 17, 2020)

1161-05-252 **Michael Anastos*** (manastos@zedat.fu-berlin.de). *On a k-matching algorithm and finding k-factors in random graphs with minimum degree $k+1$ in linear time.*

We show that for $k+1 \geq 3$ and $c > (k+1)/2$ w.h.p. the random graph on n vertices, cn edges and minimum degree $k+1$ contains a (near) perfect k-matching. As an immediate consequence we get that w.h.p. the $(k+1)$-core of $G_{n,p}$, if non empty, contains a (near) spanning k-regular subgraph. This improves upon a result of Chan and Molloy and completely resolves a conjecture of Bollobás, Kim and Verstraëte. In addition, we show that such a subgraph can be found in a linear time w.h.p. A substantial element of the proof is the analysis of a randomized algorithm for finding k-matchings in random graphs with minimum degree $k+1$. (Received August 18, 2020)

1161-05-280 **He Guo*** (he.guo@gatech.edu), 686 Cherry Street, School of Mathematics, Georgia Institute of Technology, Atlanta, GA 30332, and **Lutz Warnke**. *On the power of random greedy algorithms.*

In recent years, as part of the maturation of the probabilistic method, random greedy algorithms have been successfully used to show the existence of hard-to-construct combinatorial objects. In particular, some of the best-known Ramsey and Turán bounds are obtained via the graphs produced by the H-free process. In this talk we explore the random greedy paradigm in the context of additive combinatorics. We improve the best-known lower bound on the van der Waerden numbers $W(r,t)$, by analyzing the r-term arithmetic progression free process (which proceeds by step-by-step adding random integers from $[n]$ that avoid the creation of r-term arithmetic progressions).

Based on joint work with Lutz Warnke. (Received August 18, 2020)

1161-05-283 **Ruth Luo*** (ruluo@ucsd.edu), **Zoltan Furedi** and **Sam Spiro**. *Traces of graphs in hypergraphs.*

Let F be a graph. We say a hypergraph H is a trace of F if there exists a bijection f from the edges of F to the hyperedges of H such that for all $xy \in E(F)$, $f(xy) \cap V(F) = \{x, y\}$. In this talk, we show asymptotics for the maximum number of edges in an r-uniform hypergraph that does not contain a trace of F. We also obtain better bounds in the case $F = K_{2,t}$. This is joint work with Zoltan Furedi and Sam Spiro. (Received August 18, 2020)

1161-05-289 **Lutz Warnke*** (warnke@math.gatech.edu), Georgia Institute of Technology. *Counting extensions revisited.*

We consider rooted subgraphs in random graphs, i.e., extension counts such as (i) the number of triangles containing a given 'root' vertex or (ii) the number of paths of length three connecting two given 'root' vertices. In 1989, Spencer gave sufficient conditions for the event that, with high probability, these extension counts are asymptotically equal for all choices of the root vertices (for vertex degrees this is true when np grows faster than log n). For the important strictly balanced case, Spencer also raised the fundamental question whether these conditions are necessary. We answer this question by a careful second moment argument, and discuss some intriguing problems that remain open.

Based on joint work with Matas Sileikis, see arXiv:1911.03012 (Received August 18, 2020)

1161-05-306 **Runrun Liu**, **Martin Rolek** and **Chris Stephens*** (chris.stephens@mtsu.edu), Box 34, Department of Mathematical Sciences, 130 E. Main St., Murfreesboro, TN 37132, and **Dong Ye** and **Gexin Yu**. *Connectivity for kite-linked graphs.*

For a given graph H, a graph G is H-linked if, for every injection $\phi : V(H) \to V(G)$, the graph G contains a subdivision of H with $\phi(v)$ corresponding to v, for each v in $V(H)$. Let $f(H)$ be the minimum integer k such that every k-connected graph is H-linked. Among connected simple graphs H with at least four vertices, the exact value $f(H)$ is only know when H is a star, or a path with four vertices, or a cycle with four vertices. A kite is graph obtained from K_4 by deleting two adjacent edges, i.e., a triangle together with a pendant edge. In

this project, we find the exact value for $f(H)$ when H is the kite by showing that every 7-connected graph is kite-linked. (Received August 19, 2020)

11 ► Number theory

1161-11-83 **A.C. Cojocaru*** (cojocaru@uic.edu), 851 S Morgan St, SEO 415, Chicago, IL 60607, and **Matthew Fitzpatrick**. *The growth of the absolute discriminant of the ring of endomorphisms of the reduction of an elliptic curve.*

Let $E/\mathbb{Q}$ be a non-CM elliptic curve. Assuming GRH, we prove that, for a set of primes p of density 1, the absolute discriminant of the $\mathbb{F}_p$-endomorphism ring of the reduction of E modulo p is close to maximal. (Received August 11, 2020)

1161-11-87 **Kirsten Eisentraeger***, Department of Mathematics, Penn State University, University Park, PA 16802, and **Sean Hallgren**, **Chris Leonardi**, **Travis Morrison** and **Jennifer Park**. *Computing endomorphism rings of supersingular elliptic curves.*

Computing endomorphism rings of supersingular elliptic curves is an important problem in computational number theory, and it is also closely connected to the security of some of the recently proposed isogeny-based cryptosystems. In this talk we give a new algorithm for computing the endomorphism ring of a supersingular elliptic curve E defined over a finite field. The algorithm works by first computing two cycles in the ℓ-isogeny graph that create an order in the endomorphism ring of E. Then we determine which maximal order containing this order is the endomorphism ring of E. (Received August 11, 2020)

1161-11-176 **Abbey Bourdon*** (bourdoam@wfu.edu), **David R. Gill**, **Jeremy Rouse** and **Lori D. Watson**. *On Isolated Points of Odd Degree.*

Let C be a curve defined over a number field k, and suppose $C(k)$ is nonempty. We say a closed point x on C of degree d is isolated if it does not belong to an infinite family of degree d points parametrized by the projective line or a positive rank abelian subvariety of the curve's Jacobian. In this talk we will identify the non-CM elliptic curves with rational j-invariant which give rise to an isolated point of odd degree on some modular curve $X_1(N)$, where N is a positive integer. (Received August 16, 2020)

1161-11-207 **Giacomo Micheli***, University of South Florida, Department of Mathematics, Tampa, FL , and **Andrea Ferraguti** (and.ferraguti@gmail.com), Università degli Studi di Torino, Department of Mathematics, Torino, Italy. *An Equivariant Isomorphism Theorem for Arboreal Galois Representations.*

In this talk we first recall the notion of arboreal Galois representation and then we develop a method to effectively determine the set of primes p for which certain arboreal Galois representations are surjective modulo p. Our method is based on a combination of height bounds on integral points on elliptic curves over function fields in positive characteristic and the ABC theorem for function fields.

Using this technique we prove Jones' conjecture on the surjectivity of the arboreal Galois representation attached to $f = x^2 + t$ [Conjecture 6.7, Compositio Math. 43 (5) 1108–1126 (2007)].

This is a recent joint work with Andrea Ferraguti. (Received August 17, 2020)

1161-11-211 **Trevor Griffin**, **Nathan Kenshur**, **Abigail Price**, **Bradshaw Vandenberg-Daves** and **Hui Xue*** (huixue@clemson.edu), Clemson University, Clemson, SC 29634, and **Daozhou Zhu**. *Zeros of Eisenstein Series.*

We present some recent progress on the locations of zeros of Eisenstein series of level one. As applications we also discuss the monomial relations between Eisenstein series and the separation property of zeros of Eisenstein series. (Received August 17, 2020)

1161-11-232 **Paul Pollack*** (pollack@uga.edu). *Thoughts on the order of a mod p.*

I will discuss some old problems concerning the distribution of the order of a mod p, for a fixed integer a and varying primes p. The emphasis will be on connections between these problems and recent work of the speaker with K. Agrawal (UGA) — on finite sets of integers at least one of which usually has a large order mod p — and Z. Engberg (Wasatch Academy, UT) — on arithmetic properties of Mersenne numbers $2^n - 1$. (Received August 17, 2020)

1161-11-237 **Larry Rolen*** (larry.rolen@vanderbilt.edu), **Zack Tripp** and **Ian Wagner**. *Cranks for partition function congruences.*

Dyson famously defined his partition rank function, which he conjectured "explains" Ramanujan's congruences mod 5,7. Building on his speculation, Garvan and Andrews-Garvan later discovered a "crank" function which explains the Ramanujan congruences mod 5,7, and 11. Here, we describe how the family of colored partitions and their Ramanujan-type congruences fit into the recent theory of theta blocks, due to Gritsenko, Skoruppa, and Zagier. We describe a method for producing new families of crank functions. This method should be useful for further studies of partition functions. (Received August 17, 2020)

1161-11-260 **Edoardo Persichetti***, 777 Glades Road, Boca Raton, FL 33431. *LESS: designing code-based signatures without decoding.*

Code-based cryptography was born in 1978 with the intuition of Robert J. McEliece, who designed a cryptosystem based on the hardness of the Decoding Problem for random linear codes. Since then, many variants of McEliece's cryptosystem were devised, primarily in the context of encryption, always relying on the hardness of decoding. These schemes are all indirectly based on the Code Equivalence problem: in fact, the private code used in the system is hidden, by transforming it to an equivalent code, which is then released as the public code. A crucial point is that this public code is indistinguishable from random, which allows to leverage the well-known result about NP-completeness of decoding a random linear code. However, to date, no system was known to rely on Code Equivalence as a standalone problem. In this talk, we show how to build such a system, basing the security entirely on the difficulty of finding a linear isometry between codes. We then explain why this approach is promising, in particular with regards to building signature schemes, where the traditional methods have so far proved to be far from optimal. (Received August 18, 2020)

1161-11-265 **Christelle Vincent***, 82 University Place, Burlington, VT 05405. *Computing hyperelliptic invariants from period matrices.* Preliminary report.

In this talk we present an obstacle to computing invariants of curves whose Jacobian has CM (complex multiplication), when the genus of the curve is greater than 1. The problem is essentially that while the Jacobian has everywhere potential good reduction, the curve does not. We show the connection between this obstacle and a certain embedding problem which we define in the talk, and present our progress on analyzing the embedding problem. This is joint work with Ionica, Kilicer, Lauter, Lorenzo Garcia and Manzateanu. (Received August 18, 2020)

1161-11-266 **Agbolade Olakunle Akande*** (agbolade.akande@uga.edu). *Generalization of Euler's recurrence of the partition function and other recursive formulas for partition-based functions.*

Euler introduced the recursive formula for the partition function, $p(n)$:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + \cdots$$

In this talk, we will prove a generalization of this recurrence: we will give a family of functions $h_{r,l}(n)$ defined on a subset N of $\mathbb{N}$, where $h_{r,l}(n) = p(n)$ in N, and then investigate what occurs as $l \to \infty$. Also using generating functions, similar to Euler's recurrence, other recurrence can be produced for other partition-based functions, for example, a recursive formula for the number of partition with distinct parts. (Received August 18, 2020)

1161-11-270 **Michael A. Freeze*** (freezem@uncw.edu), Department of Mathematics and Statistics, 601 South College Road, Wilmington, NC 28403. *A Modified Erdős-Ginzburg-Ziv Constant for Cyclic Groups.* Preliminary report.

A classical result of Erdős, Ginzburg, and Ziv states that any sequence of $2n-1$ elements in the cyclic group $\mathbb{Z}_n$ contains a subsequence of length n that sums to the group identity 0 (that is, a zero-sum subsequence). Let G be an additive abelian group and let $\mathcal{L} \subseteq \mathbb{N}$.Define the *modified* generalized Erdős-Ginzburg-Ziv constant $s'_{\mathcal{L}}(G)$ to be the minimum length ℓ such that any zero-sum sequence of length at least ℓ in G contains a subsequence that sums to 0, with the length of the subsequence being in $\mathcal{L}$. Note that the notation for this constant follows that of A. Berger's recent work considering the case when $\mathcal{L}$ is singleton with $G = \mathbb{Z}$, among other restrictions. We discuss the value of $s'_{\mathcal{L}}(\mathbb{Z}_n)$ for several choices of length set $\mathcal{L}$. (Received August 18, 2020)

1161-11-296 **Kristin E Lauter*** (klauter@microsoft.com), **Brooke Feigon**, **Anamaria Costache**, **Maike Massierer** and **Anna Puskas**. *Ramanujan Graphs in Cryptography.*

Quantum computers force us to consider what hard problems in mathematics our next generation of cryptographic systems can be based on. Supersingular Isogeny Graphs were proposed for use in cryptography in 2006 by Charles, Goren, and Lauter. Supersingular Isogeny Graphs are examples of Ramanujan graphs, which are

optimal expander graphs. These graphs have the property that relatively short walks on the graph approximate the uniform distribution, and for this reason, walks on expander graphs are often used as a good source of randomness in computer science. But the reason these graphs are important for cryptography is that finding paths in these graphs, i.e. routing, is hard: there are no known subexponential algorithms to solve this problem, either classically or on a quantum computer. For this reason, cryptosystems based on the hardness of problems on Supersingular Isogeny Graphs are currently under consideration for standardization in the NIST Post-Quantum Cryptography (PQC) Competition. This talk will introduce these graphs and explain how the security of the Key Exchange protocol depends on the hardness of the routing problem. It will also explain an alternate description of these graphs in terms of quaternion algebras, and in terms of Bruhat-Tits trees. (Received August 18, 2020)

12 ► Field theory and polynomials

1161-12-50 **Daniel Panario*** (daniel@math.carleton.ca). *Construction of Irreducible Polynomials through Rational Transformations.*

Let $\mathbb{F}_q$ be the finite field with q elements, where q is a power of a prime. We discuss recursive methods for constructing irreducible polynomials over $\mathbb{F}_q$ of high degree using rational transformations. In particular, given a divisor $D > 2$ of $q+1$ and an irreducible polynomial $f \in \mathbb{F}_q[x]$ of degree n such that n is even or $D \not\equiv 2 \pmod 4$, we show how to obtain from f a sequence $\{f_i\}_{i\geq 0}$ of irreducible polynomials over $\mathbb{F}_q$ with $\deg(f_i) = n \cdot D^i$. (Received August 04, 2020)

13 ► Commutative rings and algebras

1161-13-32 **Todd Morra***, tmorra@clemson.edu. *Canonical modules of Stanley-Reisner rings of certain independence complexes and related free resolutions.* Preliminary report.

We use results of D'Alì, Fløystad, and Nematbakhsh [DFN] to explicitly describe the canonical modules of Stanley-Reisner rings of certain Cohen-Macaulay simplicial complexes. We will also report on progress towards finding a DG-algebra structure for a resolution given in [DFN] when restricting to such complexes. (Received July 30, 2020)

1161-13-33 **Saeed Nasseh*** (snasseh@georgiasouthern.edu), **Maiko Ono** and **Yuji Yoshino**. *Weak liftings of DG modules.* Preliminary report.

We discuss some of the recent progress on the problems of liftings and weak liftings of DG modules. We then consider some applications of these notions to the self-vanishing of Ext for finitely generated modules over commutative noetherian rings.

This talk is based on joint works with Maiko Ono and Yuji Yoshino. (Received July 30, 2020)

1161-13-36 **Zhibek Kadyrsizova**, **Jennifer Kenkel**, **Janet Page**, **Jyoti Singh**, **Karen E Smith**, **Adela Vraciu*** (vraciu@math.sc.edu) and **Emily Witt**. *Bounds for F-pure threshold and extremal singularitites in positive characteristic.*

For a hypersurface over a field in positive characteristic, the F-pure threshold can be viewed as a measure of the singularity. More singular hypersurfaces have lower F-pure thresholds. The degree also gives a measure of the singularity and it provides a bound for the F-pure threshold. We describe and classify hypersurfaces that have the lowest possible F-pure threshold, taking the degree into consideration. (Received August 01, 2020)

1161-13-56 **Susan Loepp*** (sloepp@williams.edu), 33 Stetson Ct., Williams College, Williamstown, MA 01267, and **Teresa Yu**. *Completions of Countable Excellent Local Domains.*

The main focus of this talk is to present necessary and sufficient conditions for a complete local (Noetherian) ring containing the rationals to be the completion of a countable excellent local domain. Time permitting, we also present necessary and sufficient conditions for a complete local ring to be the completion of a countable noncatenary local domain, as well as necessary and sufficient conditions for it to be the completion of a countable noncatenary local unique factorization domain. (Received August 06, 2020)

1161-13-77 **Justin Lyle*** (justin.lyle@ku.edu), **Jonathan Montaño** and **Sean K. Sather-Wagstaff**. Tor*-persistence and related conjectures.*

A commutative Noetherian ring R is said to be Tor-persistent if, for any finitely generated R-module M, the vanishing of $\mathrm{Tor}_i^R(M, M)$ for all $i \gg 0$ implies that M has finite projective dimension. An open question

of Avramov, Iyengar, Nasseh, and Sather-Wagstaff asks whether every commutative Noetherian ring is Tor-persistent. We show that any local ring $(R, \mathfrak{m})$ with $\mathfrak{m}^3 = 0$ is Tor-persistent, and we provide some partial results in the graded case. Our main approach is to exploit properties of exterior and symmetric squares of both modules and complexes. We discuss connections between these problems and a well-known open conjecture of Tachikawa. (Received August 10, 2020)

1161-13-94 **Evan Houston*** (eghousto@uncc.edu) and **Muhammad Zafrullah**. **-Super potent domains.*

For a finite-type star operation $\star$ on a domain R, we say that R is $\star$-super potent if each maximal $\star$-ideal of R contains a finitely generated ideal I such that (1) I is contained in no other maximal $\star$-ideal of R and (2) J is $\star$-invertible for every finitely generated ideal $J \supseteq I$. Examples of t-super potent domains include domains each of whose maximal t-ideals is t-invertible (e.g., Krull domains). We show that if the domain R is $\star$-super potent for some finite-type star operation $\star$, then R is t-super potent, we study t-super potency in polynomial rings and pullbacks, and we prove that a domain R is a generalized Krull domain if and only if it is t-super potent and has t-dimension one. (Received August 11, 2020)

1161-13-126 **Ela Celikbas*** (ela.celikbas@math.wvu.edu), Morgantown, WV 26506. *Weakly Arf Rings.* Preliminary report.

In 1971, Lipman proved that, if $(R, \mathfrak{m})$ is a complete, one-dimensional local domain with an algebraically closed field of characteristic zero, and R is saturated, then R has minimal multiplicity, that is, the embedding dimension of R is equal to the multiplicity of R. In the proof, Lipman used the fact that such a ring R is an Arf ring, i.e., R satisfies a certain condition that was studied by Arf in 1949 pertaining to a certain classification of curve singularities. The defining condition of an Arf ring is easy to state: if R is as above, then R is Arf provided, whenever $0 \neq x \in \mathfrak{m}$ and y/x, $z/x \in \mathrm{Frac}(R)$ are integral over R, one has that $yz/x \in R$.

In this talk, we introduce weakly Arf rings, which is a generalization of Arf rings. We give characterizations of weakly Arf rings, explore the relation between Arf and weakly Arf rings, and also give several examples. This talk is based on the recent joint work with O. Celikbas, C. Ciupercă, N. Endo, S. Goto, R. Isobe, and N. Matsuoka. (Received August 14, 2020)

1161-13-128 **Ela Celikbas*** (ela.celikbas@math.wvu.edu), Morgantown, WV 26506. *Arf and weakly Arf rings arising from fiber products.* Preliminary report.

In this talk, we discuss weakly Arf rings, which is a generalization of Arf rings, defined by J. Lipman in 1971. We investigate the Arf and weakly Arf properties of local rings obtained from fiber products. This talk is based on the recent joint work with O. Celikbas, C. Ciupercă, N. Endo, S. Goto, R. Isobe, and N. Matsuoka. (Received August 14, 2020)

1161-13-147 **Simplice Tchamna*** (simplice.tchamna@gcsu.edu), 231 W. Hancock Street, Box 017, Milledgeville, GA 31061. *Some results on the star-hash property for ring extensions.*

we give the definition of a ring extension having the star-hash property. We study the hash-star property in pullback diagrams of type $\square$. A characterization of Prüfer extensions having the star-hash property with respect to star operations of finite type is given We investigate condition under which the extension $R[X] \subseteq S[X]$ has the β-hash property given that the extension $R \subseteq S$ has the α-hash property, where X is an indeterminate over S, and α, β are strict star operations of finite type on $R \subseteq S$ and on $R[X] \subseteq S[X]$, respectively. (Received August 15, 2020)

1161-13-170 **Florian Enescu*** (fenescu@gsu.edu), Department of Mathematics and Statistics, 25 Park Place, Georgia State Univ, Atlanta, GA 30303, and **Yongwei Yao**. *Graded rings of rational twist in prime characteristic.* Preliminary report.

We study the generating function associated to complexity sequence of the twisted construction of a $\mathbb{N}$-graded ring. We regard this as an object reflecting the properties of the ring and its grading and perform a detailed analysis of the case of the polynomial ring with general $\mathbb{N}$-grading. Applications to the Frobenius complexity of determinantal rings are provided. (Received August 16, 2020)

1161-13-172 **Michael C. Loper*** (michael.loper@uwrf.edu). *What makes a complex virtual.*

Let S be the Cox ring of a smooth projective toric variety and B be the irrelevant ideal. In 2017, Berkesch, Erman, and Smith introduced virtual resolutions for toric varieties as an analogue of minimal free resolutions for projective varieties. Virtual resolutions are complexes of free S-modules that allow B-torsion homology. I will name two algebraic conditions that determine whether a bounded chain complex of free S-modules is a virtual

resolution. This theorem is similar to the depth criterion of exactness that Buchsbaum and Eisenbud published in 1973. (Received August 16, 2020)

1161-13-184 **Lokendra P Paudel***, 807 Hampton St, Walterboro, SC 29488, and **Simplice Tchamna**, Milledgeville, GA 31061. *Some results on star operations.* Preliminary report.

Let $\star$ be a star operation on a ring extension $R \subseteq S$. A ring extension $R \subseteq S$ is called Prüfer *star*-multiplication extension (P$\star$ME) if $(R_{[m]}, m_{[m]})$ is a Manis pair in S for every $\star$-maximal ideal m of R. In this talk, we discuss some results on star operations, and we show that under appropriate conditions, the extension $R \subseteq S$ is a P$\star_1$ME if and only if $R[X]_{[N(\star_1)]} \subseteq S[X]$ is P$\star_2$ME, where $\star_1$ and $\star_2$ are star operations on $R \subseteq S$ and $R[X] \subseteq S[X]$ respectively, and $N(\star_1) = \{f \in R[X] : c_R(f)^{\star_1} = R\}$. (Received August 16, 2020)

1161-13-195 **Ayah Almousa*** (aka66@cornell.edu) and **Keller VandeBogert**. *Resolutions of initial ideals of some determinantal facet ideals.*

We consider the class of determinantal facet ideals which form a Gröbner basis with respect to a diagonal term order. We show that the multigraded Betti numbers of initial ideals of such determinantal facet ideals are always either 0 or 1, and that the standard graded Betti numbers of these determinantal facet ideals and their initial ideals coincide when generators of the ideal come from maximal minors of a generic $n \times m$ matrix with $n > 2$. We give explicit differentials for the linear strand of initial ideals of determinantal facet ideals which form a Gröbner basis with respect to a diagonal term order. (Received August 17, 2020)

1161-13-197 **Keller VandeBogert*** (kellerlv@math.sc.edu). *Trimming Complexes and Their Applications.*

In this talk, I will introduce trimming complexes. In the most general situation, these complexes are applied to the following data: let R be a commutative ring, $I = (\phi_1, \dots, \phi_n) \subseteq R$ an ideal, and $K_1, \dots, K_\ell \subseteq R$ an arbitrary collection of ideals with $\ell \leq n$. Then the trimming complex associated to this data will yield a free resolution of the ideal $J = (\phi_1, \dots, \phi_{n-\ell}) + K_1\phi_{n-\ell+1} + \cdots + K_\ell\phi_n$. I will then discuss applications of these resolutions to the structure of certain classes of grade 3 homogeneous ideals defining compressed rings, and Betti tables of determinantal facet ideals. (Received August 17, 2020)

1161-13-220 **John Eagon**, **Ezra Miller** and **Erika Ordog*** (erika.ordog@tamu.edu). *Sylvan structures on near-cones.* Preliminary report.

Sylvan resolutions are minimal resolutions of monomial ideals whose differentials are sums over lattice paths of weights associated to higher dimensional analogues of spanning trees. The differentials arise from splitting the boundary maps of the simplicial complexes that record the local structure of the ideal near each lattice point. When resolving stable ideals, these simplicial complexes are near-cones. By selecting certain splittings of the boundary maps of near-cones, the resulting sylvan resolution is the Eliahou-Kervaire resolution. (Received August 17, 2020)

1161-13-226 **Eleonore Faber**, **Martina Juhnke-Kubitzke**, **Haydee Lindo**, **Claudia Miller**, **Rebecca R.G.*** (rrebhuhn@gmu.edu) and **Alexandra Seceleanu**. *Canonical resolutions over Koszul algebras.*

Koszul algebras are a class of graded algebras with a duality between commutative and anti-commutative algebras, much like the duality between the symmetric and exterior algebras. We find the minimal free resolutions of powers of the homogeneous maximal ideal over arbitrary Koszul algebras. (Received August 17, 2020)

1161-13-231 **Ben Briggs**, **Srikanth Iyengar**, **Janina Letz** and **Josh Pollitz*** (pollitz@math.utah.edu), University of Utah, 155 1400 E, JWB 130, Salt Lake City, UT 84112. *A characterization of relative complete intersections.*

Broadly speaking, the goal of this talk is to discuss how ring theoretic information is captured in terms of derived categories and illustrate applications from this perspective. The starting point is the celebrated Auslander-Buchsbaum-Serre theorem which can be phrased in the following way: A local ring is regular if and only if each object of its bounded derived category is a small object; this solved the localization problem for regular rings. An analogous characterization for complete intersections was established by Dwyer-Greenlees-Iyengar and P. for local complete intersections in terms of "proxy small objects;" similarly, this provided a solution to the localization problem for complete intersections. In this talk, we briefly review these points and discuss new work that characterizes when a surjective ring map is complete intersection (i.e., its kernel is generated by a regular sequence), which is in terms of how the proxy small objects ascend along the ring map. Applications of this characterization include a simpler proof of the factorization theorems for relative complete intersections. This talk is based on joint work with Ben Briggs, Srikanth Iyengar and Janina Letz. (Received August 17, 2020)

1161-13-236 **Oana Veliche*** (o.veliche@northeastern.edu), 360 Huntington Ave, Boston, MA 02115-5005, and **Pedro Macias Marques** and **Jerzy Weyman**. *On the Structure of Short Grade Four Artinian Gorenstein Algebras.* Preliminary report.

We consider homogeneous perfect ideals I of grade 4 in the polynomial ring $Q = k[x, y, z, w]$ such that Q/I defines a Gorenstein ring of socle degree 4. After a change of variables, we identify inside I a perfect ideal J of grade 3 such that Q/I is obtained as a quotient of Q/J by its the canonical module. (Received August 17, 2020)

1161-13-242 **Lars Winther Christensen*** (lars.w.christensen@ttu.edu), **Sergio Estrada** and **Peder Thompson**. *Singularity categories for schemes.*

In local algebra, the quotient of the bounded derived category by the subcategory of perfect complexes is often referred to as the *singularity category*. The quotient is trivial for a regular ring, and for a Gorenstein ring it is triangulated equivalent the stable category of maximal Cohen–Macaulay modules or, from our point of view, to the homotopy category of totally acyclic complexes of finitely generated projective modules. I will give an overview of a few recent papers in which we use the flat–cotorsion theory to extend these ideas to schemes (Received August 17, 2020)

1161-13-243 **J. R. Juett** and **Lois W. Ndungu*** (lwn8@txstate.edu). *Krull rings, regular π-rings, and generalizations.* Preliminary report.

Let $\star$ be a star operation on a commutative ring R. (We will review star operations and other relevant definitions.) We define R to be a *regular $\star$-π-ring* if every regular proper principal ideal is a $\star$-product of prime $\star$-ideals. We find this notion to hold interest because it simultaneously encapsulates two important generalizations of factoriality. That is, if R is an integral domain and $\star = d$ (resp., $\star = t$), then R is a regular $\star$-π-ring if and only if it is a π-domain (resp., Krull domain). Many characterizations of π-domains and Krull domains can be found in the literature, showcasing a rich interplay between the notions of (t-)products of prime (t-)ideals, (t-)invertibility, and various ideal arithmetic properties. In this talk we give several characterizations of regular $\star$-π-rings (in the case where $\star$ has finite character), simultaneously generalizing much of the known theory of π-domains and Krull domains under a unified framework, as well as obtaining a few results that are new even in these special cases. (Received August 17, 2020)

1161-13-247 **Rasoul Ahangari Maleki** (segal@umkc.edu) and **Liana M. Sega*** (segal@umkc.edu). *Resolutions over generic Gorenstein rings of socle degree 3.* Preliminary report.

Generic Gorenstein rings of socle degree 3 are known to be Koszul. We address the question of whether ideals generated by linear forms have a linear resolution over such rings. (Received August 18, 2020)

1161-13-263 **Claudia Miller*** (clamille@syr.edu) and **Sophia Vassiliadou**. *Exterior powers of differentials over complete intersection rings.*

In this talk, after giving some of the history of the classic Lipman-Zariski Conjecture and the generalized Lipman-Zariski questions of Graf, I'll relate some mostly known results on the torsion and cotorsion of exterior powers of the module of K'ahler differentials over complete intersection rings and say how these are used to prove a generalized Lipman-Zariski result under certain conditions. (Received August 18, 2020)

1161-13-276 **Philippe Gimenez** and **Hema Srinivasan*** (srinivasanh@missouri.edu), Department of Mathematics, University of Missouri, Columbia, MO 65211. *Semigroup Rings in Higher Dimensions.*

Let $\langle A\rangle$ be the semigroup finitely generated by a subset $A = \{\mathbf{a_1}, \ldots, \mathbf{a_p}\}$ of $\mathbb{N}^n$. Let k be an arbitrary field. The semigroup ring $k[A]$ over A is isomorphic to $k[x_1, \ldots, x_p]/I_A$, where I_A is a binomial ideal. We will also denote by A the $n \times p$ integer matrix whose columns are the elements in A. It is well known that the $\dim k[A] = RankA$. Let A and B be two $n \times p$ and $n \times q$ matrices over natural numbers. Two semigroups $\langle A\rangle$ and $\langle B\rangle$ are said to be glued to obtain the semigroup $\langle C\rangle$ if the ideal of I_C is generated by a single special element modulo $I_A + I_B$. When $n = 1$, the dimension of $k[A]$ is one, which is the case of the numerical semigroup rings. Two numerical semigroups can always be glued and this is not the true in higher dimensions. We will discuss the problem of when two semigroups can be glued and how it relates to various properties of semigroups such as depth, Cohen-Macaulay and symmetry. (Received August 18, 2020)

1161-13-279 **Jennifer Kenkel***, 530 Church Street, Ann Arbor, MI 48109, and **Uwe Nagel**. *A Construction of Local Cohomology Modules With an FI Structure.* Preliminary report.

Let R_n be a polynomial ring in n variables, and I_n an ideal of R_n. There are many ways to embed R_n into R_{n1}, and, depending on the situation, many ways to embed I_n compatibly into I_{n1}. Enforcing certain structure on

these embeddings allows us to consider this whole class of rings as one object, known as an FI-algebra. In this talk, I will explain this structure, with lots of examples, and explain how to construct local cohomology modules with an FI-Structure. (Received August 18, 2020)

1161-13-291 **Lance Edward Miller*** (lem016@uark.edu), SCEN, 850 W. Dickson St. #309, Fayetteville, 72701, and **Kyle Maddox**, 1460 Jayhawk Blvd, Lawrence, KS 66045.
F-nilpotent type singularities. Preliminary report.

Singularities of local and graded rings defined by the Frobenius have a rich history with F-rational singularities being among the best behaved. Takagi and Srinivas formalized a class of singularities considered without name by Blickle and Lyubeznik called F-nilpotent singularities, defined by considering when Frobenius actions on local cohomology is nilpotent. These have been the subject of significant recent study which generalized the notion in various ways. In this talk, we will review these singularity types, their relationships, and discuss work in progress with K. Maddox concerning sufficient conditions for rings to have these types of singularities. (Received August 18, 2020)

1161-13-292 **Haydee Lindo*** (hlindo@hmc.edu). *Trace Modules and Rigidity.* Preliminary report.

I will speak on some recent developments in the theory of trace modules over commutative Noetherian rings. This will include applications of trace modules in understanding endomorphism rings and a discussion of ongoing work examining the relationship between trace ideals and modules having no self-extensions. (Received August 18, 2020)

1161-13-297 **Ranthony A.C. Edmonds***, Edmonds.110@osu.edu, and **Austin Antoniou**, **Bethany Kubik**, **Christopher O'Neill** and **Shannon Talbott**. *On Atomic Density of Numerical Semigroup Algebras.*

A numerical semigroup S is a cofinite, additively-closed subset of the nonnegative integers that contains 0. In this talk, we discuss our initiation of the study of atomic density, an asymptotic measure of the proportion of irreducible elements in a given ring or semigroup, for semigroup algebras. For a fixed field $\mathbb{F}$ and a numerical semigroup S, the numerical semigroup algebra $\mathbb{F}[S]$ is the subring of $\mathbb{F}[x]$ consisting only of terms of the form x^a for $a \in S$.

It is known that the atomic density of the polynomial ring $\mathbb{F}_q[x]$ is zero for any finite field $\mathbb{F}_q$. We prove that the numerical semigroup algebra $\mathbb{F}_q[S]$ also has atomic density zero for any numerical semigroup S. We also examine the particular algebra $\mathbb{F}_2[x^2, x^3]$ in more detail, providing a bound on the rate of convergence of the atomic density as well as a counting formula for irreducible polynomials using Möbius inversion, comparable to the formula for irreducible polynomials over a finite field $\mathbb{F}_q$. (Received August 18, 2020)

14 ► Algebraic geometry

1161-14-25 **Douglas A Torrance*** (dtorrance@piedmont.edu) and **Nick Vannieuwenhoven** (nick.vannieuwenhoven@kuleuven.be). *All secant varieties of the Chow variety are nondefective for cubics and quaternary forms.*

The Chow rank of a form is the length of its smallest decomposition into a sum of products of linear forms. For a generic form, this corresponds to finding the smallest secant variety of the Chow variety which fills the ambient space. We determine the Chow rank of generic cubics and quaternary forms by proving nondefectivity of all involved secant varieties. The main new ingredient in our proof is the generalization of a technique by [Brambilla and Ottaviani, On the Alexander–Hirschowitz theorem, J. Pure Appl. Algebra, 2008] that consists of employing Terracini's lemma and Newton's backward difference formula to compute the dimensions of secant varieties of arbitrary projective varieties. Via this inductive construction, the proof of nondefectivity ultimately reduces to proving a number of base cases. These are settled via a computer-assisted proof because of the large dimension of the spaces involved. The largest base case required in our proof consisted of computing the dimension of a vector space constructed from the 400th secant variety of a degree-82 Chow variety embedded in $\mathbb{P}^{98769}$. (Received July 23, 2020)

1161-14-84 **Salim Tayou*** (salim.tayou@ens.psl.eu). *Exceptional jumps of Picard rank of K3 surfaces over number fields.*

Given a K3 surface X over a number field K, we prove that the set of primes of K where the geometric Picard rank jumps is infinite, assuming that X has everywhere potentially good reduction. This result is formulated in the general framework of GSpin Shimura varieties and I will explain other applications to abelian surfaces. I

will also discuss applications to the existence of rational curves on K3 surfaces. The results in this talk are joint work with Ananth Shankar, Arul Shankar and Yunqing Tang. (Received August 11, 2020)

1161-14-90 **Mark Green**, **Yoon-Joo Kim*** (yoon-joo.kim@stonybrook.edu), **Radu Laza** and **Colleen Robles**. *Cohomology decomposition of hyperkähler manifolds.*

Looijenga, Lunts and Verbitsky showed the cohomology of a compact hyperkähler manifold admits additional symmetries than that of the usual compact Kähler manifolds. This symmetry structure on the cohomology dominates the Hodge structure, hard Lefschetz decomposition and hyperkähler rotations. In this talk, I will discuss the cohomology decomposition of hyperkähler manifolds with respect to this symmetry structure, and compute them explicitly for all currently known examples of hyperkähler manifolds. This is joint work with Mark Green, Radu Laza and Colleen Robles. (Received August 11, 2020)

1161-14-113 **Chunyi Li**, **Laura Pertusi*** (laura.pertusi@unimi.it) and **Xiaolei Zhao**. *Elliptic quintics on cubic fourfolds, O'Grady 10 and Lagrangian fibrations.*

In this talk we study certain moduli spaces of semistable objects in the Kuznetsov component of a cubic fourfold. We show that they admit a symplectic resolution $\tilde{M}$ which is a smooth projective hyperkaehler manifold deformation equivalent to the 10-dimensional example constructed by O'Grady. As a first application, we construct a birational model of $\tilde{M}$ which is a compactification of the twisted intermediate Jacobian of the cubic fourfold. Secondly, we show that $\tilde{M}$ is the MRC quotient of the main component of the Hilbert scheme of elliptic quintic curves in the cubic fourfold, as conjectured by Castravet. This is a joint work with Chunyi Li and Xiaolei Zhao. (Received August 13, 2020)

1161-14-160 **Emma Brakkee*** (e.l.brakkee@uva.nl). *Moduli of twisted K3 surfaces.*

Motivated by the relation between (twisted) K3 surfaces and special cubic fourfolds, we construct moduli spaces of polarized twisted K3 surfaces of fixed degree and order. We do this by mimicking the construction of the moduli space of untwisted polarized K3 surfaces as a quotient of a bounded symmetric domain. (Received August 16, 2020)

1161-14-181 **Spencer Tamagni*** (stamagni8@knights.ucf.edu), Orlando, FL 32816, and **Costas Efthimiou**. *Electric Fields from Geometry.*

We explain, in a manner accessible to undergraduates and non-experts, how techniques from the study of compact Riemann surfaces may be employed to efficiently solve for electric fields on surfaces of non-trivial topology. A byproduct of this analysis is an intuitive visualization of elliptic functions when their argument is complex-valued. Our goal is to provide students and instructors a quick reference article for an extraordinary topic that is not included in the standard books. (Received August 16, 2020)

1161-14-187 **Jack Petok*** (jack.petok@dartmouth.edu). *Kodaira dimensions of some moduli spaces of hyperkähler fourfolds.*

The Noether-Lefschetz locus in a moduli space of $K3^{[2]}$-fourfolds parametrizes fourfolds with Picard rank at least 2. Following Hassett's work on cubic fourfolds, Debarre, Iliev, and Manivel showed that the Noether-Lefschetz locus in the moduli space of degree 2 $K3^{[2]}$-fourfolds is a countable union of special divisors $\mathcal{M}_d$, where the discriminant d is a positive integer congruent to 0, 2, or 4 modulo 8. In this talk, we compute the Kodaira dimensions of these special divisors for all but finitely many discriminants; in particular, we show that, for $d > 176$ and certain other small values of d, the space $\mathcal{M}_d$ is a variety of general type. (Received August 17, 2020)

1161-14-191 **Chiara Camere*** (chiara.camere@unimi.it). *Mirror Symmetry and Lagrangian Fibrations on Hyperkähler Manifolds.*

In this talk I will explain a generalization of a lattice-theoretical mirror symmetry, as known for $K3$ surfaces, to higher dimensional hyperkähler manifolds, and give an application to the study of Lagrangian fibrations. Then I will discuss an example in the case of fourfolds of $K3^{[2]}$-type. Finally, if time allows, I will report on further developments about Lagrangian fibrations, joint work in progress with G. Saccà. (Received August 17, 2020)

1161-14-192 **Gretchen L Matthews*** (gmatthews@vt.edu). *Three-point Hermitian codes.*

One-point Hermitian codes are the best understood algebraic geometry codes from higher genus curves, meaning beyond the ubiquitous Reed-Solomon codes constructed from the projective line. In the last two decades, two-point Hermitian codes have been studied extensively, and their exact parameters (meaning dimension and minimum distance) have been established. Multipoint codes on the Hermitian curve supported by m points,

where $m \geq 3$, are more intricate in nature. In this talk, we share recent results on three-point Hermitian codes. This is joint work with Michael Wills. (Received August 17, 2020)

1161-14-204 **Justin Sawon*** (sawon@email.unc.edu), Department of Mathematics, University of North Carolina, Chapel Hill, NC 27599-3250. *Lagrangian fibrations by Prym varieties.*

Lagrangian fibrations on holomorphic symplectic manifolds and orbifolds are higher-dimensional generalizations of elliptic K3 surfaces. They are fibrations whose general fibres are abelian varieties that are Lagrangian with respect to the symplectic form. Markushevich and Tikhomirov described the first example whose fibres are Prym varieties, and their construction was further developed by Arbarello, Ferretti, and Sacca and by Matteini to yield more examples. In this talk we describe the general framework, and consider a new example. We describe its singularities and show that it is a 'primitive' symplectic variety. We also construct the dual fibration, using ideas of Menet. This is joint work with Chen Shen. (Received August 17, 2020)

1161-14-208 **Asher Auel*** (asher.auel@dartmouth.edu), Department of Mathematics, Kemeny Hall, Hanover, NH 05055. *Brill-Noether special cubic fourfolds.*

The study of special cubic fourfolds leads to beautiful connections between algebraic cycles, Hodge theory, the rationality problem, K3 surfaces, hyperkaehler varieties, and derived categories. In this talk, I'll explain a connection to the theory of algebraic curves, via the notion of Brill–Noether special cubic fourfolds. Specifically, I'll discuss the open problem of establishing lower bounds on the Clifford index of Brill–Noether special cubic fourfolds. (Received August 17, 2020)

1161-14-221 **Daniel C Smith-Tone*** (daniel-c.smith@louisville.edu). *Algebraic Models for the MinRank Problem and How Dramatically Our World is Changing.*

The MinRank Problem is a natural complexity theoretic problem closely related to cryptanalytic attacks on code- and multivariate-based schemes. Various methods of modelling the computational version of this problem exist, including two historically dominant algebraic techniques. For over a decade the so-called minors method has been supposedly proven to be superior to the Kipnis-Shamir (KS) method and any other method by way of a simple but trivially false argument.

We demonstrate the fallacy of the above claim by providing a tight analysis of the KS method in "superdetermined" MinRank instances as well as by introducing a new algebraic model that dramatically reduces the complexity of MinRank over a large parameter space. We then discuss the rather dramatic impact this new technique has on post-quantum cryptography. (Received August 17, 2020)

1161-14-257 **B. Bakker***, UIC, Dept. of Math, 851 S. Morgan St., Chicago, IL 60607, and **H. Guenancia** and **C. Lehn**. *Towards a BBDGGHKP decomposition theorem for nonprojective Calabi–Yau varieties.*

Calabi-Yau manifolds are built out of simple pieces in a precise sense by the Beauville–Bogomolov decomposition theorem: any Calabi–Yau Kahler manifold up to an etale cover is a product of complex tori, irreducible holomorphic symplectic manifolds, and strict Calabi-Yau manifolds (ones with no holomorphic forms except a unique top form). Recent work of Druel–Guenancia–Greb–Horing–Kebekus–Peternell extends this result to mildly singular projective Calabi–Yau varieties, and the proofs heavily use algebraic methods. In joint work with C. Lehn and H. Guenancia, we extend the decomposition to the nonprojective case for varieties admitting a (nonunique) symplectic form. We will also discuss progress towards the general case. (Received August 18, 2020)

1161-14-264 **Nicolas Addington**, **Brendan Hassett**, **Yuri Tschinkel** and **Anthony Várilly-Alvarado*** (varilly@rice.edu), Department of Mathematics MS 136, Rice University, 6100 S. Main St., Houston, TX 77007. *Recent progress on the rationality of some special cubic fourfolds.*

In this talk I will survey recent progress on the rationality of certain special cubic fourfolds, including work of Auel, Bolognesi, Russo, Stagliano, as well as collaborative work of Addington, Hassett, Tschinkel and myself. I will pay close attention to the case of cubic fourfolds of discriminant 18, which form a Noether-Lefshetz divisor $\mathcal{C}_{18}$ in the coarse moduli space of cubic fourfolds $\mathcal{C}$. I will explain how to produce a countably infinite union of codimension-one loci in $\mathcal{C}_{18}$ parametrizing rational cubic fourfolds. (Received August 18, 2020)

1161-14-274 **Jintai Ding**, **Joshua Deaton*** (deatonju@mail.uc.edu), **Vishakha** and **Bo-Yin Yang**.
The Nested Subset Differential Attack: A Practical Direct Attack Against LUOV which Forges a Signature within 210 Minutes.

In 2017, Ward Beullens et al. submitted Lifted Unbalanced Oil and Vinegar, which is a modification to the Unbalanced Oil and Vinegar Scheme by Patarin. Previously, Ding et al. proposed the Subfield Differential Attack which prompted a change of parameters by the authors of LUOV for the second round of the NIST post-quantum standardization competition. We have developed a refinement to the Subfield Differential Attack called the Nested Subset Differential Attack which fully breaks half of the parameter sets put forward. We also show by experimentation that this attack is practically possible to do in under 210 minutes for the level I security parameters and not just a theoretical attack. The Nested Subset Differential attack is a large improvement of the Subfield differential attack which can be used in real-world circumstances. Moreover, we will only use what is called the "lifted" structure of LUOV, and our attack can be thought of as a development of solving "lifted" quadratic systems. (Received August 18, 2020)

1161-14-278 **Sarah Frei*** (sarah.frei@rice.edu). *Rational points and derived equivalence.*

In this talk, I will consider the question: is the existence of a rational point preserved under derived equivalence? In recent joint work with Nicolas Addington, Ben Antieau, and Katrina Honigs, we show that over Q, the answer is no. We give two counterexamples: an abelian variety and a torsor over it, and a pair of hyperkaehler fourfolds. The hyperhaehler fourfold without rational points, which is a moduli space of sheaves on a K3 surface, has a Brauer-Manin obstruction. I will also report on ongoing work with Ryan Takahashi, in which we explore Brauer obstructions to rational points on other hyperkaehler fourfolds. (Received August 18, 2020)

15 ► Linear and multilinear algebra; matrix theory

1161-15-134 **Han Huang*** (hhuang421@gatech.edu). *Singularity of sparse Bernoulli matrices when* $1 \leq \liminf pn/\log(n) \leq \limsup pn/\log(n) < +\infty$.

It has been conjectured that for a sufficiently large n, and $p = p_n \in [\log(n)/n, 1/2)$, the probability that a $n \times n$ Bernoulli(p) matrix A is singular equals to the probability that A contains of a zero row or zero column up to a negligible error.

This conjecture has been recently proved by Litvak-Tikhomirov in the regime $C\log(n)/n < p < 1/C$ for some universal constant $C > 1$ with their new tool. While for $p = (1 + o(1))\log(n)/n$, it also holds due to a result of Basak-Rudelson. In this talk, we will discuss how to extend their results to fill the gap between these two regions. ($1 \leq pn/\log(n) < \infty$) (Received August 14, 2020)

1161-15-282 **Yi-Chyun Wong***, RISE-CRG, Cresskill, NJ , and **Richard Kyung**, RISE-CRG, Cresskill, NJ. *Study on the Spectrum Modification Algorithms to Enhance the Resolution of Old Photos Images.*

In this paper, spatial domain technique was employed to original images to have different effects on output images by changing the pixel intensity. Using mathematical functions and transformations such as b=T(a), where, a and b are the pixel values before and after processing and T is the transformation that maps pixel value a into b. A logarithmic transformation: b = T(a) = A*log(1+a), and power law (gamma) transformations: b= T(a)= A*a*exp(gamma) were used for the image process, where the 'A' and gamma are the positive constants. For the gamma transformation, we checked how the power law curves change as gamma varies. The cases such as gamma is greater than 1, gamma is equal to 1, and gamma is less than 1 are tested to map a range of input values into different output values. By performing the different transformation techniques and using various factors or constants, some enhancements and modifications were made. This paper showed the use of spectrum modification algorithms which can be applied to enhance the resolution of the images of old photos or historical relics using spatial domain technique. (Received August 19, 2020)

16 ► Associative rings and algebras

1161-16-7 **Soumyadip Acharyya*** (soumyadip.acharyya@gmail.com), **Sudip Acharyya**, **Sagarmoy Bag** and **Joshua Sack**. *Topologies on the rings of measurable functions.*

Let $(X, \mathcal{A})$ stand for a nonempty set X equipped with a σ-algebra $\mathcal{A}$ over X. The set of all real-valued $\mathcal{A}$-measurable functions on X forms a commutative lattice ordered ring with unity if the relevant operations are defined pointwise. My talk will focus on the so-called m-topology on this ring $\mathcal{M}(X, \mathcal{A})$ and its measure-theoretic

generalization. Important topological properties including first countability and connectedness will be discussed. (Received July 02, 2020)

1161-16-189 **Luigi Ferraro*** (lferraro@ttu.edu), **Frank Moore** and **Desiree Martin**. *The Taylor Resolution over a Skew Polynomial Ring.*

Abstract: Let I be a monomial ideal in the polynomial ring $R = k[x_1, \ldots, x_n]$ over a field k. In her thesis, Taylor introduced a complex which provides a multi-graded free resolution for R/I as an R-module. Later, Gemeda provided a differential graded structure on Taylor's complex. We generalize each of these results to monomial ideals in a skew polynomial ring. (Received August 17, 2020)

18 ► Category theory; homological algebra

1161-18-8 **Alina Iacob*** (aiacob@georgiasouthern.edu). *Generalized Gorenstein injective modules.*

We introduce a generalization of the Gorenstein injective modules - the Gorenstein FP_n-injective modules, denoted $\mathcal{GI}_n$. They are the cycles of the exact complexes of injective modules that remain exact when applying a functor $Hom(A, -)$, with A an FP_n-injective module. Thus $\mathcal{GI}_0$ is the class of classical Gorenstein injective modules, and $\mathcal{GI}_1$ is the class of Ding injective modules. We prove that when $n \geq 2$, the class $\mathcal{GI}_n$ is the right half of a perfect cotorsion pair, and therefore it is an enveloping class. For $n = 1$ we show that $\mathcal{GI}_1$ (i.e. the Ding injectives) forms the right half of a hereditary cotorsion pair. (Received July 03, 2020)

1161-18-9 **Ettore Aldrovandi*** (ealdrov@math.fsu.edu), Department of Mathematics, Florida State University, 208 Love Building, 1017 Academic Way, Tallahassee, FL 32306-5410. *Categorical groups, their morphisms, and higher algebraic structures.*

I will review the main notions about categorical groups, in particular their morphisms, presentations, and classification via cohomological invariants. I will use them as a springboard to discuss other types of higher algebraic structures, such as categorical rings and (presheaves of) ∞-groups. These latter are motivated by some recent applications to intersection theory and determinant functors, which, if time permits, I will shortly illustrate. (Received July 06, 2020)

1161-18-130 **Hannah Altmann*** (hannah.altmann@dsu.edu) and **Sean Sather-Wagstaff**. *On the Number of Semidualizing Modules over a Local Ring.* Preliminary report.

Let R be a commutative local noetherian ring with identity. A finitely generated R-module C is *semidualizing* if the homothety map $\chi_C^R : R \to \mathrm{Hom}_R(C, C)$ is an isomorphism and $\mathrm{Ext}^i_R(C, C) = 0$ for all $i > 0$. It has been proven recently that the number of isomorphism classes of semidualizing modules over R is finite, but little is known about the actual number. Gerko gives some bounds in the artinian case (which are easily extended to the Cohen Macaulay case). Using differential graded algebra techniques, we discuss extensions of this to the non Cohen Macaulay case and, time permitting, also for semidualizing complexes. (Received August 14, 2020)

20 ► Group theory and generalizations

1161-20-14 **Nham Ngo*** (nvngo@ung.edu), University of North Georgia - Gainesville, Department of Mathematics, Oakwood, GA 30566, and **Jerry Magana** (jmaga0242@ung.edu), University of North Georgia - Gainesville, Department of Mathematics, Oakwood, GA 30566. *Module varieties of quotients of polynomial algebras.*

In this talk, we give an explicit description of all irreducible components in the varieties of 3-dimensional modules for the algebras

$$k\left[x_1, \ldots, x_r\right] / \left\langle x_i^{a_i} : 1 \leq i \leq r \right\rangle$$

where k is an algebraically closed field, r is a positive integer and $a_i \geq 2$ for all $1 \leq i \leq r$.
This is joint work with J. Magana. (Received July 17, 2020)

1161-20-161 **Christopher Bendel**, **Daniel Nakano**, **Cornelius Pillen*** (pillen@southalabama.edu) and **Paul Sobaje**. *On Donkin's Tilting Module Conjecture and groups of type G_2.*

Tilting modules for reductive algebraic groups over fields with positive characteristics feature prominently in new approaches towards finding character formulas for simple modules. A long-standing conjecture by Donkin says that certain tilting modules are isomorphic to indecomposable injective modules when restricted to the Frobenius kernel of the algebraic group. It was recently shown that the conjecture does not hold for a group of

type G_2 and characteristic 2. In this talk we verify Donkin's conjecture for G_2 and characteristic 3. (Received August 16, 2020)

1161-20-230 **Irfan Bagci*** (ibagci@ung.edu). *Cohomology and Support Varieties.*
In this talk, we will briefly define cohomology and support varieties for finite groups. After that will show several uses of cohomology and support varieties for other algebraic structures including Lie superalgebras. (Received August 17, 2020)

1161-20-234 **Amrita Acharyya***, amrita.acharyya@utoledo.edu, and **Jon M Corson** and **Das C Bikash**. *Action of cofinite groups on cofinite graphs.*
Our idea of cofinite graph is motivated by cofinite groups due to B. Hartley. In this work, we define cofinite connectedness of a cofinite graph. Many of the properties of connectedness of topological spaces are analogous to cofinite connectedness. We define Group Actions on a cofinite graph to characterize a unique way of uniformly topologizing an abstract group with profinite topology, induced by the cofinite graph, so that the aforesaid action becomes uniformly equicontinuous. Example of such actions via the group of integers and it's profinite completion is given. (Received August 17, 2020)

22 ► Topological groups, Lie groups

1161-22-156 **Carmen Galaz-García***, carmengg@math.ucsb.edu. *Zariski dense surface subgroups in $SL(n,\mathbb{Q})$.*
In this talk I will explain how to construct a path $\rho_t\colon \pi_1(S) \to SL(n,\mathbb{R})$ for odd n of discrete, faithful and Zariski dense representations of a surface group such that $\rho_t(\pi_1(S)) \subset SL(n,\mathbb{Q})$ for every $t \in \mathbb{Q}$. As exemplified in the work of Long, Reid and Thistlethwaite, constructing Zariski dense surface subgroups in $SL(n,\mathbb{Q})$ has close connections to finding *thin groups*, which are infinite index subgroups of a lattice in $SL(n,\mathbb{R})$ that are Zariski dense. (Received August 16, 2020)

30 ► Functions of a complex variable

1161-30-13 **Garrett S. Moseley*** (920112221@student.ccga.edu), **Marianela Landi**, **Kayla Johnson** and **Aaron M. Yeager**. *Zeros of Complex Random Polynomials Spanned by Bergman Polynomials.*
We study the expected number of zeros of

$$P_n(z) = \sum_{k=0}^{n} \eta_k p_k(z),$$

where $\{\eta_k\}$ are complex-valued i.i.d standard Gaussian random variables, and $\{p_k(z)\}$ are polynomials orthogonal on the unit disk. When $p_k(z) = \sqrt{(k+1)/\pi}z^k$, $k \in \{0,1,\ldots,n\}$, we give an explicit formula for the expected number of zeros of $P_n(z)$ in a disk of radius $r \in (0,1)$ centered at the origin. From our formula we establish the limiting value of the expected number of zeros, the expected number of zeros in a radially expanding disk, and show that the expected number of zeros in the unit disk is $2n/3$. Generalizing our basis functions $\{p_k(z)\}$ to be regular in the sense of Ullman–Stahl–Totik, and that the measure of orthogonality associated to polynomials is absolutely continuous with respect to planar Lebesgue measure, we give the limiting value of the expected number of zeros of $P_n(z)$ in a disk of radius $r \in (0,1)$ centered at the origin, and show that asymptotically the expected number of zeros in the unit disk is $2n/3$. (Received July 17, 2020)

1161-30-65 **Dima Khavinson*** (dkhavins@usf.edu). *Some Questions on L^1-approximation in Bergman Spaces.*
Suppose a continuous on the closed unit disk function $\omega, ||\omega||_{L^\infty} = 1$, can be approximated by analytic functions in $L^1(dA)$-norm within ϵ, here dA stands for the area measure.

Question. Can we approximate it (in $L^1(dA)$) within $C\epsilon$ by analytic functions with the L^∞ norm at most 1? (The constant C is , of course, independent of ω.) The answer is NO, Moreover. the asymptotics, when ϵ, tends to 0 can be arbitrarily bad (DK, 2020). **History**. The problem is the ϵ - version of the celebrated Hoffman-Wermer Theorem for Uniform Algebras and was posed by J. Wermer in 1980. The answer of the analogous question in $L^1(d\theta)$, where $d\theta$ is the Lebesgue measure on the unit circle, or, in most generality, in the hyperdirichlet algebras content, is also NO (DK-H.Shapiro-F. Perez-Gonzalez, 1998, DK-2019), and "Yes", if we

replace $O(\epsilon)$ by $O(\epsilon)\log\frac{1}{\epsilon}$ for the disk , and $O(\epsilon)^{1/2}\log\frac{1}{\epsilon}$ for the hypodirichlet case (DK, 2020). (Received August 08, 2020)

1161-30-214 **Stephen J. Gardiner** and **Myrto Manolaki*** (myrto.manolaki@ucd.ie). *A strong form of Plessner's theorem.*

Let f be a meromorphic function on the unit disc. Plessner's theorem says that, for almost every point ζ on the unit circle, either (i) f has a finite nontangential limit at ζ, or (ii) the image $f(S)$ of any Stolz angle S at ζ is dense in the complex plane. In this talk, we will see that condition (ii) can be replaced by a much stronger assertion. Our new theorem and its harmonic analogue on halfspaces also strengthen classical results of Spencer, Stein and Carleson. Finally, we will discuss an application to the theory of universal approximation. (Received August 17, 2020)

1161-30-250 **Gou Nakamura*** (gou@aitech.ac.jp), 1247 Yakusa-cho, Toyota, 470-0392, Japan. *Automorphism groups of non-orientable extremal surfaces of genus 6.*

Let $\mathcal{K}_g$ be the moduli space of compact non-orientable hyperbolic surfaces of genus $g \geq 3$. A surface in $\mathcal{K}_g$ is called an extremal surface if it contains discs with maximum radius determined by g. The discs are called extremal discs. In this talk we consider the groups of automorphisms of extremal surfaces in $\mathcal{K}_6$, where 6 is the maximum genus of non-orientable extremal surfaces admitting more than one extremal discs. Considering those non-orientable extremal surfaces with automorphisms of the maximum order 10, we obtain two families of non-orientable extremal surfaces of even genus $g \geq 4$ with automorphisms of the maximum order $2(g-1)$. (Received August 18, 2020)

1161-30-258 **Catherine Beneteau*** (cbenetea@usf.edu). *A survey of optimal polynomial approximants and connections to digital filters.* Preliminary report.

In the last few years, the notion of optimal polynomial approximant has appeared in the mathematics literature in connection with Hilbert spaces of analytic functions of one or more variables. In the 70s, researchers in engineering and applied mathematics introduced least squares inverses in the context of digital filters in signal processing. It turns out that in the Hardy space H^2 these objects are identical. In this talk, I will survey results related to zeros and convergence of optimal polynomial approximants and implications for the design of ideal digital filters. This talk is based on a survey paper that is joint with Ray Centner. (Received August 18, 2020)

31 ► Potential theory

1161-31-210 **Koushik Ramachandran*** (koushik.math@gmail.com). *On the number of critical points of the localization landscape.*

Let $\Omega \subset \mathbb{C}$ be a bounded domain. The localization landscape of Ω is a function v which satisfies $\Delta v = -2$ in Ω with boundary data $v(z) = 0$, for $z \in \partial\Omega$. In this talk, we will present an upper bound for the number of critical points of v in various domains where we can make sense of some notion of "order" of the domain. We will talk about connections to eigenvalue problems and conclude the talk with some open problems. Based on joint work with Erik Lundberg. (Received August 17, 2020)

34 ► Ordinary differential equations

1161-34-4 **Paul W Eloe*** (peloe1@udayton.edu) and **Johnny Henderson**. *Two-point boundary value problems for ordinary differential equations, uniqueness implies existence.*

We consider a family of two-point boundary value problems for higher order nonlinear ordinary differential equations and obtain conditions in terms of uniqueness of solutions that imply existence of solutions. A standard hypothesis that has proved effective in uniqueness implies existence type results is to assume uniqueness of solutions of a large family of n point boundary value problems. Here, we shall replace that standard hypothesis with one in which we assume uniqueness of solutions of a large family of two-point boundary value problems. We then obtain readily verifiable conditions on the nonlinear term that in fact imply the uniqueness of solutions of the large family of two-point boundary value problems. (Received June 25, 2020)

1161-34-5 **Joseph Paullet*** (jep7@psu.edu), School of Science, Penn State Behrend, 4205 College Drive, Erie, PA 16509, and **Joseph Previte**. *Analysis of Flow Past a Permeable Stretching/Shrinking Sheet.*

We analyze a recently proposed model for boundary layer flow of a nanofluid past a permeable stretching/shrinking sheet. The boundary value problem (BVP) governing fluid velocity reduces to:

$$f'''(\eta) + f(\eta)f''(\eta) - f'(\eta)^2 = 0, \qquad 0 < \eta < \infty,$$

subject to

$$f(0) = S, \quad f'(0) = \lambda, \quad f'(\infty) = 0,$$

where λ measures the stretching ($\lambda > 0$) or shrinking ($\lambda < 0$) of the sheet, and S controls the suction ($S > 0$) or injection ($S < 0$) of fluid through the sheet.

For $\lambda \geq 0$ and $S \in \mathbb{R}$, we present a closed-form solution to the BVP and prove that this solution is unique. For $\lambda < 0$ and $S < 2\sqrt{-\lambda}$ we prove no solution exists. For $\lambda < 0$ and $S = 2\sqrt{-\lambda}$ we present a closed-form solution to the BVP and prove that it is unique. For $\lambda < 0$ and $S > 2\sqrt{-\lambda}$ we present two closed-form solutions to the BVP and prove the existence of an infinite number of solutions in this parameter range. The analytical results proved here differ from the numerical results reported in the literature. We discuss the mathematical aspects of the problem that lead to the difficulty in obtaining accurate numerical approximations to the solutions. (Received June 26, 2020)

1161-34-31 **Rudi Weikard*** (weikard@uab.edu). *On the spectral theory for first-order systems without the unique continuation property.*

We consider the differential equation $Ju' + qu = wf$ on the real interval (a, b) when J is a constant, invertible skew-Hermitian matrix and q and w are matrices whose entries are distributions of order zero with q Hermitian and w non-negative. In this situation, it may happen that there is no existence and uniqueness theorem for balanced solutions of a given initial value problem. After describing the set of solutions the equation does have we establish that the adjoint of the minimal operator is the maximal operator, investigate deficiency indices, describe all self-adjoint extensions of the minimal operator, and establish the existence of Green's function.

This is joint work with Kevin Campbell, Minh Nguyen, and Steven Redolfi. (Received August 12, 2020)

1161-34-41 **John R. Graef*** (john-graef@utc.edu), Department of Mathematics, University of Tennessee at Chattanooga, 615 McCallie ave, Chattanooga, TN 37403-2598. *Boundary value inclusions involving Hadamard fractional derivatives.*

The existence of solutions to a boundary value inclusion involving a Hadamard type fractional derivative is established. The proof is based on Mönch's fixed point theorem and the Kuratowski measure of noncompacteness. An example illustrating the results is also presented. (Received August 03, 2020)

1161-34-51 **Jeffrey W Lyons*** (jlyons3@citadel.edu) and **Jeffrey T. Neugebauer**. *Existence of a Positive Solution for a Singular Fractional Boundary Value Problem with Fractional Boundary Conditions using Convolution and Lower Order Problems.*

Existence of a positive solution is shown for two singular two-point fractional boundary value problems with fractional boundary conditions using fixed point theory, lower order problems, and convolution of Green's functions. A nontrivial example is included. (Received August 18, 2020)

1161-34-52 **Benjamin L Jeffers*** (bjeffers@trinity.edu), 8121 SW 47th ave, Portland, OR 97219, and **Jeffery W Lyons**. *On the solutions of the variational equation for an nth order boundary value problem with an integral boundary condition.*

In this paper, we discuss differentiation of solutions for the boundary value problem $y^{(n)} = f\left(x, y, y', y'', \ldots, y^{(n-1)}\right)$, $a < x < b$, $y^{(i)}(x_j) = y_{ij}$, $0 \leq i \leq m_j$, $1 \leq j \leq k-1$, and $y^{(i)}(x_k) + \int_c^d py(x)\,dx = y_{ik}$, $0 \leq i \leq m_k$ with respect to the boundary data. We show that under certain conditions partial derivatives of the solution $y(x)$ of the boundary value problem with respect to various boundary condition parameters exist and solve the associated variational equation along $y(x)$. (Received August 04, 2020)

1161-34-59 **Ahmed Ghatasheh*** (med.ghatasheh@gmail.com). *Sign-changing points of solutions of Sturm-Liouville equations with measure-valued coefficients.*

In this talk we investigate sign-changing points of nontrivial real-valued solutions of homogeneous Sturm-Liouville differential equations of the form $-d(du/d\alpha) + ud\beta = 0$, where $d\alpha$ is a positive Borel measure supported everywhere on (a, b) and $d\beta$ is a locally finite real Borel measure on (a, b). Since solutions for such equations are functions of locally bounded variation, sign-changing points are the natural generalization of zeros. We prove

that sign-changing points for each nontrivial real-valued solution are isolated in (a, b). We also prove a Sturm-type separation theorem for two nontrivial linearly independent solutions, and conclude the paper by proving a Sturm-type comparison theorem for two differential equations with distinct potentials.

This is joint work with Rudi Weikard. (Received August 06, 2020)

1161-34-119 **Pavel Kurasov*** (kurasov@math.su.se), Dept. of Mathematics, Stockholm Univ., 106 91 Stockholm, Sweden. *Inverse problems for quantum graphs.* Preliminary report.

To solve the inverse spectral problem for the Schrödinger equation on a metric graph one needs to determine: the metric graph; the potential in the Schrödinger equation; the vertex conditions (connecting the edges together).

The inverse problem is solved completely in the case of trees under mild restrictions on the vertex conditions. The main tool is a combination of the boundary control and M-function approaches to inverse problems. These two approaches are essentially equivalent in the case of single interval, but their different features may be effectively exploited to solve different partial inverse problems for trees. The bunch cutting procedure allows one to reduce the tree step-by-step by removing edges and vertices close to the boundary.

To solve the inverse problem for graphs with cycles we propose to use magnetic boundary control and magnetic M-functions where spectral data for a fixed potential are considered as functions of the magnetic fluxes through graph cycles. To solve the inverse problem we use **cycle opening procedure** mapping spectral data for arbitrary graphs with cycles to spectral data for trees on the same edge set. The graph and potential are reconstructed assuming so far standard vertex conditions. (Received August 14, 2020)

1161-34-122 **Gloria Huang*** (gloriahuang24@gmail.com), 206 Grand Teton Way, Bowling Green, KY 42104. *Using the Geometry of Parameter Space to Find the Most Influential Factors in the Healing of Diabetic Foot Ulcers.*

The treatment of chronic wounds has long been a challenge to wound care professionals and presents a substantial economic burden to healthcare systems globally. Over 50 billion is spent annually on the treatment of chronic wounds. To combat this issue, a mathematical model describing the interactions between matrix metalloproteinases (MMPs), their regulators (TIMPs), fibroblasts, and the extracellular matrix (ECM) was developed and analyzed to find the most influential factors, or parameters, in the healing process of diabetic foot ulcers. Using the differential equation model with de-identified patient data, the three-dimensional geometry of parameter space was visualized for all combinations of the twelve parameters in the model to see how these parameters affect the biological system. Knowledge of the identifiability of parameters can streamline treatment by allowing us to individualize treatment. This approach plots two parameters against the sum of squares errors to generate a three-dimensional graph. By analyzing the minimum of the graph, we can conclude if a parameter is identifiable and thus important in the healing response. This research shows that the parameters associated with the regulators of MMPs (TIMPs) are the most influential in a wound-healing model. (Received August 14, 2020)

1161-34-228 **Diksha Satish*** (diksha.satish627@topper.wku.edu), 136 Waterford Circle, Frankfort, KY 40601. *Calculating Individual and Population Parameter Values in the Healing of Chronic Wounds through Mixed-Effects Modeling.* Preliminary report.

In previous work, four differential equations were used to model the relationships between matrix metalloproteinases (MMP-1), their inhibitors (TIMP-1), and the extracellular matrix (ECM) using averaged patient data during the diabetic foot ulcer healing process. The patient data was acquired from a study that collected data on the concentration of these three factors from sixteen patients over twelve weeks. The aim of this study is to curve-fit individual patient data using mixed-modeling effects. Mixed-effects modeling splits each parameter into two parameters. The population parameter is the same for each patient representing an average patient response, while an individual parameter represents the individual patient response and varies across patients. Mixed-effects modeling is implemented in MONOLIX, where its Stochastic Approximation Expectation-Maximization Algorithm is used to calculate the population parameters. The resulting values were used as a starting point to calculate individual parameters. Through MONOLIX, all sixteen patients were curve fit generating both population and individual parameter values. Future work will include applying the parameter values in a sensitivity analysis to identify those that most affect an individual patient's healing response. (Received August 17, 2020)

1161-34-277 **Paul W. Eloe** and **Jeffrey T. Neugebauer***, jeffrey.neugebauer@eku.edu. *Smallest Eigenvalues for Boundary Value Problems of Two Term Fractional Differential Operators Depending on Fractional Boundary Conditions.*

Let $n \geq 2$ be an integer. Let $n-1 < \alpha \leq n$. We consider eigenvalue problems for two point $n-1,1$ boundary value problems of the form

$$D^{\alpha}_{0+}u + a(t)u + \lambda p(t)u = 0, \quad 0 < t < 1,$$
$$u^{(i)}(0) = 0,\ i = 0, 1, \ldots, n-2, \quad D^{\beta}_{0+}u(1) = 0,$$

where $0 \leq \beta \leq n-1$ and D^{α}_{0+} denotes the standard Riemann-Liouville differential operator. We prove the existence of smallest positive eigenvalues and then obtain comparisons of these smallest eigenvalues as functions of p and of β. (Received August 18, 2020)

35 ► Partial differential equations

1161-35-11 **Suzanne Lenhart*** (slenhart@utk.edu), University of Tennessee, Department of Mathematics, Knoxville, TN 37996-1320. *Optimal Control of Advection in a Parabolic PDE Model of an Invasive Species in a River.*

Invasive species in rivers may be managed by appropriate adjustment of flow rates. Using a parabolic PDE model representing an invasive population in a river, we investigate optimal control of the water discharge rate to force the invasive population downstream. We illustrate some numerical simulations in which parameters are varied to determine how far upstream the invasive population reaches. (Received July 10, 2020)

1161-35-19 **Irena Lasiecka*** (lasiecka@memphis.edu), Department of Mathematical Sciences, University of Memphis, Memphis, TN 38120, and **Marcelo Bongarti**. *Vanishing time relaxation in Jordan-Moore-Gibson-Thompson [JMGT] dynamics.*

The (third-order in time) MGT equation is a nonlinear (quasi-linear) Partial Differential Equation (PDE) describing a nonlinear propagation of sound in an acoustic medium.The interest in studying this type of problems is motivated by a large array of applications arising in engineering and medical sciences-including high intensity focused utrasound [HIFU] technologies. The important feature is that the model avoids the infinite speed of propagation paradox associated with a classical second order in time equation referred to as Westervelt equation. Replacing Fourier's law by Maxwell-Cattaneo's law gives rise to the third order in time derivative scaled by a small parameter $\tau > 0$, the latter represents the thermal relaxation time parameter and is intrinsic to the medium where the dynamics occurs. This talk will discuss an asymptotic analysis of the third order model when $\tau \to 0$. It is shown that the corresponding solutions converge *in a strong topology of the phase space* to a limit which is the solution of Westervelt equation. This addresses the open question raised recently in the context of JMGT dynamics.
Joint work with Marcelo Bongarti and Sutthirut Charoenphon. (Received July 21, 2020)

1161-35-20 **Roberto Triggiani*** (rtrggani@memphis.edu), 373 Dunn Hall, University of Memphis, Memphis, TN 38152. *Uniform Stabilization of Navier-Stokes Equations and Boussinesq Systems via Finite-dimensional Feedback Controls.*

Two main results will be given (joint with I.Lasiecka and B.Priyasad). First, uniform stabilization of the N-S Eqts in d=3, by a control strategy that includes a finite-dimensional tangential boundary feedback control acting on an arbitrarily small part of the boundary. Finite dimensionality was known for d=2 (I.Lasiecka-R.Triggiani, NLA 2015), but was an outstanding open problem for d=3. It required moving into a radically different functional setting of a Besov space with tight indeces, which is "close" to L3 in d=3. Second, uniform stabilization of a Boussinesq systems, any dimension, on the same Besov setting of low regularity as above, by means of localized interior finite dimensional feedback controls. In both cases the construction of feedback controllers is explicit. Solution to each problem relies - as a starting point - on suitable Unique Continuation Properties of suitable overdetermined conditions. Particularly the ones for the Boussinesq systems are new and offer interesting features (R.Triggiani-X.Wan, AMO, 2020) (Received July 22, 2020)

1161-35-26 **Tatiana Savina*** (savin@ohio.edu), Ohio University, Athens, OH 45701, and **Murdhy Aldawsary**, Ohio University, Athens, OH 45701. *Robin to Neumann mapping and Robin reflection of harmonic functions.*

In this talk, we review the celebrated Schwarz symmetry principle for harmonic functions. Using Robin to Neumann and Dirichlet to Neumann mappings, we discuss the reflection of harmonic functions satisfying Robin conditions on a real-analytic curve. (Received July 23, 2020)

1161-35-28 **Leonid Koralov*** (koralov@umd.edu). *Large Time Behavior of Diffusion Processes and a Class of Non-Standard Boundary Value Problems.*

We will discuss several asymptotic problems for diffusion processes (and the corresponding PDEs) that lead to a new class of elliptic and parabolic problems with non-standard boundary conditions. (Received July 24, 2020)

1161-35-29 **Heng Li***, Department of Mathematics, Governors State University, University Park, IL 60484, **Yongzhi Xu**, Department of Mathematics, University of Louisville, Louisville, KY 40292, and **Jianrong Zhou**, Department of Mathematics, Foshan University, Foshan, Guangdong 52800, Peoples Rep of China. *Some mathematical problems related to DCIS mathematical model.*

We discuss some mathematical problems related to Ductal carcinoma in situ (DCIS) model including direct problem, inverse problem and free boundary problem. DCIS is considered the earliest form of breast cancer. Theoretical and numerical results are presented. (Received July 27, 2020)

1161-35-35 **K.D. Chu**, **D.D. Hai** and **Ratnasingham Shivaji*** (r_shivaj@uncg.edu), Dept of Mathematics and Statistics, UNCG, 116 Petty, 317 College Ave, Greensboro, NC 27412. *Uniqueness of positive radial solutions for a class of infinite semipositone p-Laplacian problems in a ball.*

We prove uniqueness of positive radial solutions to the p-Laplacian problem

$$\begin{cases} -\Delta_p u = \lambda f(u) \text{ in } \Omega, \\ u = 0 \text{ on } \partial\Omega, \end{cases}$$

where $\Delta_p u = div(|\nabla u|^{p-2}\nabla u),\ p \geq 2,\ \Omega$ is the open unit ball in $R^N, N > 1,\ f : (0,\infty) \to \mathbb{R}$ is concave, $p-$ sublinear at ∞ with infinite semipositone structure at 0, and λ is a large parameter. (Received July 31, 2020)

1161-35-38 **Gampola Waduge Nalin Fonseka*** (fonsekan@carolinau.edu), 1825 Sherwood St, Apt C, Greensboro, NC NC 27403, **Ratnasingham Shivaji** (jgoddard@aum.edu), Montgomery, AL 36117, and **Jerome Goddard** and **Byungjae Son**. *A diffusive weak Allee effect model with U-shaped emigration and matrix hostility.*

We study positive solutions to steady-state reaction-diffusion equations of the form:

$$-\Delta u = \lambda f(u);\ \Omega$$
$$\alpha(u)\frac{\partial u}{\partial \eta} + \gamma\sqrt{\lambda}[1-\alpha(u)]u = 0;\ \partial\Omega$$

where u is the population density, $f(u) = \frac{1}{a}u(u+a)(1-u)$ represents a weak Allee effect type growth of the population with $a \in (0,1)$, $\alpha(u)$ is the probability of the population staying in the habitat Ω when it reaches the boundary, and positive parameters λ and γ represent the domain scaling and effective exterior matrix hostility, respectively. In particular, we analyze the case when $\alpha(s) = \frac{1}{[1+(A-s)^2+\epsilon]}$ for all $s \in [0,1]$, where $A \in (0,1)$ and $\epsilon \geq 0$. In this case $1-\alpha(s)$ represents a U-shaped relationship between density and emigration. Existence, nonexistence, and multiplicity results for this model are established via the method of sub-super solutions. (Received August 02, 2020)

1161-35-42 **Boris Vainberg***, 123 N Brackenbury Ln, Charlotte, NC 28270. *On the Near-Critical Behavior of Continuous Polymers.*

We will consider a mean-field model of polymers described in terms of solutions to a parabolic equation with a positive potential and a coupling constant proportional to the inverse temperature. At the critical value of the temperature, polymers exhibit a transition between folded (globular) and unfolded states. In a variety of chemical and biological publications, the molecule in its folded state is modeled by a ball, whose radius depends on the temperature and on the number of monomers comprising the molecule (which is usually very large). The radius is an important characteristic responsible for the diffusivity of polymers in a liquid.

We will describe the asymptotic distribution of polymers and the behavior of the radius when the temperature approaches to the critical value, and, simultaneously, the number of monomers goes to infinity. We also will discuss the critical value of elliptic exterior problems. Most of the presented results are joint with M. Cranston (UC Irvine), L. Koralov (UMD) and S. Molchanov (UNCC). (Received August 03, 2020)

1161-35-44 **Pelin Guven Geredeli*** (peling@iastate.edu), Department of Mathematics, 492 Carver Hall, Ames, IA 50011. *Alternate approach for the exponential stability of a linearized compressible flow-structure PDE Model.*

We consider a linearized compressible flow structure interaction (FSI) PDE model with a view of analyzing the stability properties of both the compressible flow and plate solution components. We operate in the time domain by way of obtaining the necessary energy estimates which culminate in an alternative proof for the uniform

stability of finite energy compressible flow-structure solutions via a certain multiplier method. In particular, our main (gradient type) multiplier is based upon the solution of a certain Neumann problem, a solution which is sufficiently smooth, even considering the unavoidable boundary interface singularities. (Received August 03, 2020)

1161-35-45 **J. T. Cronin** (jcronin@lsu.edu), Department of Biological Sciences, Louisiana State University, Baton Rouge, LA 70803, **J Goddard II** (jgoddard@aum.edu), Department of Math. & Stat., C. Comp., Auburn University Montgomery, Montgomery, AL 36124-4023, **A Muthunayake*** (akmuthun@uncg.edu), Department of Mathematics and Statistics, University of North Carolina at Greensboro, Greensboro, NC 27412, and **R Shivaji** (r_shivaj@uncg.edu), Department of Mathematics and Statistics, University of North Carolina at Greensboro, Greensboro, NC 27412. *Modeling the effects of trait-mediated dispersal on coexistence of mutualists.*

We analyse positive solutions (u, v) to the steady state reaction diffusion system:

$$\begin{cases} -\Delta u = \lambda u(1-u); \ \Omega \\ -\Delta v = \lambda r v(1-v); \ \Omega \\ \frac{\partial u}{\partial \eta} + \sqrt{\lambda} g(v) u = 0; \ \partial\Omega \\ \frac{\partial v}{\partial \eta} + \sqrt{\lambda} h(u) v = 0; \ \partial\Omega \end{cases}$$

where $\lambda > 0$, $r > 0$ are parameters and $g, h \in C^1([0,\infty),(0,\infty))$ are decreasing functions. This system models the steady states of two species living in a habitat where the interaction is limited to the boundary. Here, λ is directly proportional to the size of the habitat and we will study the ranges of λ where coexistence and nonexistence occurs. Namely, we will consider three cases: (a) $E_1(1, g(0)) = E_1(r, h(0))$, (b) $E_1(1, g(0)) > E_1(r, h(0))$, (c) $E_1(1, g(0)) < E_1(r, h(0))$. Here $E_1(r, K)$ denotes the principal eigenvalue of: $-\Delta z = rEz$; Ω, $\frac{\partial z}{\partial \eta} + K\sqrt{E} z = 0$; $\partial\Omega$. (Received August 03, 2020)

1161-35-62 **George Avalos*** (gavalos@math.unl.edu), Department of Mathematics, University of Nebraska-Lincoln, university of nebraska-lincoln, department of, Lincoln, NE 68588. *Qualitative Properties of Multi-Layered Structural Flow PDE.*

In this talk we will discuss a nonstandard Babuška-Brezzi variational formulation and subsequent stability analysis for a multilayered fluid-structure interaction (FSI) system. The composite multilayer PDE system under consideration constitutes a coupling of 3D fluid-2D elastic-3D elastic dynamics, where the coupling of the 3D fluid and the 3D (elastic) wave equation is realized via an additional 2D wave equation on the boundary interface. Inasmuch as the variational formulation is driven by the thick layer-thin layer structure component of the FSI dynamics, and not the fluid component, the argument proposed here for the necessary inf-sup estimate is quite different than that given for uncoupled fluid flows. This is joint work with Pelin Guven Geredeli. (Received August 07, 2020)

1161-35-69 **Marcelo M Disconzi*** (marcelo.disconzi@vanderbilt.edu), **Mihaela Ifrim** and **Daniel Tataru**. *The relativistic Euler equations with a physical vacuum boundary.*

We consider the relativistic Euler equations with a physical vacuum boundary and an equation of state $p(\varrho) = \varrho^\gamma$, $\gamma > 1$. We establish the following results. (i) local well-posedness in the Hadamard sense, i.e., local existence, uniqueness, and continuous dependence on the data; (ii) low regularity solutions: our uniqueness result holds at the level of Lipschitz velocity and density, while our rough solutions, obtained as unique limits of smooth solutions, have regularity only a half derivative above scaling; (iii) stability: our uniqueness in fact follows from a more general result, namely, we show that a certain nonlinear functional that tracks the distance between two solutions (in part by measuring the distance between their respective boundaries) is propagated by the flow; (iv) we establish sharp, essentially scale invariant energy estimates for solutions; (v) a sharp continuation criterion, at the level of scaling, showing that solutions can be continued as long as the the velocity is in $L^1_t Lip$ and a suitable weighted version of the density is at the same regularity level. (Received August 09, 2020)

1161-35-70 **Yongjin Lu*** (ylu@vsu.edu). *Pullback dynamics of non-autonomous Navier-Stokes equation on 2D Lipschitz-like domains.*

We study the tempered pullback dynamics of non-autonomous Navier-Stokes equation with a non-homogeneous boundary condition on 2D Lipschitz-like domains. We establish the existence of a minimal pullback attractor with respect to a universe of tempered sets with the presence of a pullback translation bounded time-dependent external force. We then give estimates on the finite fractal dimension of the pullback attractor expressed in terms of the Reynold and Grashof numbers. The last result we establish pertains to the upper semi-continuity of the pullback attractors when the external force is perturbed from a stationary force by a time-dependent

perturbation. Some of the key methods that we used to establish the results include trace formula and the construction of a background flow function. (Received August 09, 2020)

1161-35-91 **Parisa Fatheddin*** (fatheddin.1@osu.edu), 1465 Mt Vernon Ave, Marion, OH 43302, and **Zhaoyang Qiu** and **Yanbin Tang**. *Large deviation principle for two-dimensional stochastic Navier-Stokes and stochastic Boussinesq equations.*

We consider the large deviation principle by the classical Azencott method for two-dimensional stochastic Navier-Stokes equation and as applications derive the law of the iterated logarithm and exit problem. Large deviations by weak convergence approach is also presented for two-dimensional stochastic Boussinesq equation. (Received August 11, 2020)

1161-35-135 **Lorena Bociu**, **Suncica Canic**, **Boris Muha** and **Justin Webster*** (websterj@umbc.edu). *Multilayered Poro-elasticity Interacting with Stokes Flow.*

The problem of the filtration of slow fluid flow near an adjacent poro-elastic region is classical and arises in geological and biological applications. Mathematical results are largely numerical. We are concerned with the challenging problem of a 3-D Stokes flow adjacent to a 3-D poro-elastic region, modeled by a nonlinear Biot dynamics and accounting for the Beavers-Joseph-Saffman slip coupling condition. From a rigorous point of view, such coupled systems have been rarely considered. Indeed, there are technical trace issues with respect to velocity coupling between Stokes and Biot dynamics.

In this talk, we address an intermediate model on a periodic box, where a 2-D linear, poro-elastic plate mitigates the interaction between the two dynamics. We provide an existence result for the nonlinear, quasi-static case via a spatial semi-discretization that relies on the recent theory developed in the work of Bociu et al. [2016]. After limit passage, we address uniqueness through a regularity criterion for weak solutions. As an intermediate step, we obtain existence for the linear case with time-dependent permeability, and this result is robust to Biot compressibility as well as inertial terms. (Received August 14, 2020)

1161-35-150 **Thomas Hagen*** (thagen@memphis.edu), The University of Memphis, Deaprtment of Mathematical Sciences, Memphis, TN 38152. *Linearized free liquid films: Semigroup regularity from asymptotics.*

Free liquid films are formed during common plastics engineering processes, including cast film extrusion. A polymeric melt is fed through a die, stretched downward into a fluid sheet and solidified. For a highly viscous, thin material the mathematical model of this free liquid film was first given in the pioneering work of Yeow and Pearson in the 1970s. This model takes the form of a transport equation coupled to an elliptic system of momentum equations in two-dimensional space. The governing equations simplify to the one-dimensional "infinite width" model if dependence on the lateral direction is omitted.

In this presentation I will report about recent results for the linearization about steady state of the full two-dimensional model under the assumption of lateral periodicity. Specifically, I will show that the linearized equations satisfy the semigroup property and that the semigroup, being non-analytic, has distinct smoothing properties. This latter result is based on high-frequency asymptotic estimates for the momentum equations. These estimates prove that the semigroup is generated by a linear transport operator, perturbed by a compact operator which - surprisingly and in contrast to the one-dimensional case - is not Hilbert-Schmidt. (Received August 15, 2020)

1161-35-157 **Chongsheng Cao**, **Yanqiu Guo*** (yanguo@fiu.edu) and **Edriss S Titi**. *Analysis of a rapidly rotating convection model of tall columnar structure.*

This presentation is based on our analysis of a three-dimensional fluid model describing rapidly rotating convection that takes place in tall columnar structures. Global regularity is shown provided the model is regularized by a weak dissipation term. The main difficulty lies in the fact that the physical domain is three-dimensional, whereas the regularizing viscosity acts only on the horizontal variables. I will also discuss the global well-posedness of this model in the limit of infinite Prandtl number. (Received August 16, 2020)

1161-35-171 **Sanwar Uddin Ahmad*** (suahmad@colostate.edu), Department of Mathematics, Fort Collins, CO 80523, and **Jennifer L. Mueller**. *A Fast Approach for Solving Electrical Impedance Tomography Using Iteratively Regularized Kaczmarz Method.* Preliminary report.

Electrical impedance tomography (EIT) is an imaging modality that determines the internal conductivity and permittivity distribution based on the voltage measurements made on an object's surface when currents are applied. Due to its non-invasiveness, non-ionizing characteristics and cost-effectiveness, EIT is gaining a lot of attention in recent years. In this presentation, we propose a fast approach for solving 2D EIT inverse problems

for the complete electrode model using Iteratively Regularized Kaczmarz method. We also present a comparative analysis of the proposed method with the regular Kaczmarz method. (Received August 16, 2020)

1161-35-186 **Lorena Bociu***, lvbociu@ncsu.edu, and **Giovanna Guidoboni**, **Daniele Prada** and **Riccardo Sacco**. *Multiscale interface coupling between a poroelastic medium and a lumped circuit.*

Many applications in life science and engineering call for the use of multiscale interface coupling between PDEs and ODEs. The present work is focused on the case where the PDEs describe a poroelastic medium coupled with a lumped circuit including resistances, inductances and capacitances (RLC), in addition to sources of voltage and currents, and it is motivated by ocular tissue perfusion. To address the challenges brought by the multiscale coupling at the PDE/ODE interface, an energy-based operator splitting approach is proposed and compared with the fixed-point iterations methods. (Received August 16, 2020)

1161-35-219 **Jason A Kurz*** (jkurz@clemson.edu). *The Mindlin Timoshenko Plate Model and Related Carleman Estimates.*

In this talk we consider the Mindlin-Timoshenko (MT) model which is a system of three strongly coupled second-order hyperbolic partial differential equations that describe the motion of a thin plate under certain considerations. In particular, we provide Carleman type estimates for the MT system by properly diagonalizing the principle part of the equations and applying Carleman estimates for coupled systems of wave equations. Such estimates may lead to applications ranging from inverse problems for determining certain coefficients in the system to presenting a more tangible path for establishing observability inequalities for the system, and therefore the corresponding exact boundary controllability results. (Received August 18, 2020)

1161-35-269 **Shyla Rae Kupis*** (skupis@g.clemson.edu), 220 Parkway Dr, Clemson University, Martin Hall M-306, Clemson, SC 29634, and **Taufiquar Khan** (taufiquar.khan@uncc.edu), 9201 University City Blvd., Fretwell 360E, Charlotte, NC 28223. *A Wavelet-Based Solution to the Electrical Impedance Tomography Inverse Problem.* Preliminary report.

Electrical impedance tomography (EIT) is a non-invasive and radiation-free method for imaging the internal electrical conductivity distribution of a body using a set of electrodes to measure voltages on the boundary. There are many applications of EIT, such as imaging malignant tumors in the medical field, damage detection of industrial concrete, and geophysical monitoring of groundwater resources. However, the downside to this imaging modality is that the EIT inverse problem is exponentially ill-posed and its solutions are highly sensitive to perturbations in the data, especially from noise. To address this inherently ill-posed inverse problem, we project the EIT minimization problem to a smaller subspace in the wavelet domain to reduce reconstruction errors and provide better reconstructed images. We solve the EIT forward problem using the complete electrode model (CEM) to account for the effect of the electrodes, and then we solve the EIT inverse problem using a modified iteratively regularized Gauss-Newton (IRGN) method in the wavelet domain. We present our findings and compare recovered solutions produced from both the original domain and wavelet domain. (Received August 18, 2020)

37 ► Dynamical systems and ergodic theory

1161-37-58 **Cara Sulyok***, csulyok@vols.utk.edu, and **Judy Day** and **Suzanne Lenhart**. *A Mathematical Framework to Augment the Q-MARSH Score in the Diagnosis of Celiac Disease.* Preliminary report.

This work provides a mathematical framework to better understand the effects of immune activation on gut health. This mathematical model uses a system of ordinary differential equations to track changes in villus and crypt cell densities as well as intraepithelial lymphocytes to better understand the dynamics of small intestinal destruction. The model will be used to investigate and analyze various theories behind the progression of celiac disease by focusing on understanding the dynamics of the small intestine in situations mirroring healthy function, celiac disease, and refractory celiac disease. By doing so, we can assist in further quantifying and augmenting diagnostic measures and investigate potential therapies to mitigate the negative effects of celiac disease. (Received August 06, 2020)

1161-37-217 **Erik Lundberg*** (elundber@fau.edu). *Limit cycles in random polynomial systems.*
We study the number and distribution of the limit cycles of a planar vector field whose component functions are random polynomials. We present estimates for the number of limit cycles in both perturbative and non-perturbative settings. The proofs use novel combinations of techniques from dynamical systems and random analytic functions. (Received August 17, 2020)

39 ► Difference and functional equations

1161-39-17 **Johnny Henderson*** (johnny_henderson@baylor.edu), Department of Mathematics, Baylor University, Waco, TX 76798-7328, and **Jeffrey T. Neugebauer** (jeffrey.neugebauer@eku.edu), Department of Mathematics and Statistics, Eastern Kentucky University, Richmond, KY 40475-3133. *Existence of local solutions for fractional difference equations with left focal boundary conditions.*
For $1 < \nu \leq 2$ a real number and $T \geq 3$ a natural number, conditions are given for the existence of solutions of the νth order Atıcı-Eloe fractional difference equation, $\Delta^{\nu} y(t) + f(t+\nu-1, y(t+\nu-1)) = 0$, $t \in \{0, 1, \ldots, T\}$, and satisfying the left focal boundary conditions $\Delta y(\nu-2) = y(\nu+T) = 0$. (Received July 20, 2020)

1161-39-34 **Min Wang*** (mwang23@kennesaw.edu). *A discrete boundary value problem with mixed periodic boundary conditions.*
In this talk, a second order discrete boundary value problem with mixed periodic boundary conditions is studied. Sufficient conditions on the multiplicity of solutions in a weak sense are obtained by using the critical point theory. An example is given to demonstrate the applications of the results as well. (Received July 30, 2020)

1161-39-43 **John R. Graef**, **Shapour Heidarkhani**, **Lingju Kong*** (lingju-kong@utc.edu) and **Shahin Moradi**. *On an anisotropic discrete boundary value problem of Kirchhoff type.*
We prove the existence of at least one nontrivial solution of an anisotropic discrete boundary value problem of $p(k)$-Kirchhoff type and its associated parametric version. We provide one example to demonstrate the applicability of our main results. Our analysis mainly relies on variational methods and Ricceri's variational principle. (Received August 03, 2020)

1161-39-100 **Ferhan M. Atici***, Department of Matehmatics, Western Kentucky University, 1906 College Heights Blvd., Bowling Green, KY 42101, **Ngoc Nguyen**, Department of Mathematics, 1906 College Heights Blvd., Bowling Green, KY 42101, **Kamala Dadashova** (kamala.dadashova642@topper.wku.edu), Department of Mathematics, Western Kentucky University, Bowling Green, KY 42101, **Sarah Pedersen** (sarah.pedersen122@topper.wku.edu), Gatton Academy of Science and Mathematics, Western Kentucky University, Bowling Green, KY 42101, and **Gilbert Koch** (gilbert.koch@ukbb.ch), Pediatric Clinical Pharmacology, University Children's Hospital, Basel, Switzerland. *Pharmacokinetics and Pharmacodynamics Models of Tumor Growth and Anticancer Effects in Discrete Time.*
We study the h-discrete and h-discrete fractional representation of a pharmacokinetics-pharmacodynamics (PK-PD) model describing tumor growth and anticancer effects in continuous time considering a time scale $h\mathbb{N}_0$, where $h > 0$. Since the measurements of the drug concentration in plasma were taken hourly, we consider $h = 1/24$ and obtain the model in discrete time (i.e. hourly). We then continue with fractionalizing the h-discrete nabla operator in the h-discrete model to obtain the model as a system of nabla h-fractional difference equations. In order to solve the fractional h-discrete system analytically, we state and prove some theorems in the theory of discrete fractional calculus. We compare residual squared sum values of the newly introduced models in discrete time with the existing models in continuous time in one table. (Received August 12, 2020)

1161-39-294 **Sougata Dhar*** (sougata.dhar@uconn.edu), 341 Mansfield Road, Room 406, Department of Mathematics, University of Conn, Storrs, CT 06269, and **Jessica Stewart Kelly** and **Qingkai Kong**. *Lyapunov-type Inequalities for Third Order Linear and Half-Linear Difference Equations.*
We derive Lyapunov-type inequalities for third order linear and half-linear difference equations. The results are significant as the existing work on Lyapunov-type inequalities for difference equations is limited only to even order cases. As the techniques for even order difference equations are not applicable to odd order difference equations, the methods used here are novel. The resulting inequalities utilize $q_+(t)$ and $q_-(t)$. (Received August 18, 2020)

41 ► Approximations and expansions

1161-41-73 **Meredith Sargent**, Department of Mathematics, University of Arkansas, and **Alan A Sola***, Department of Mathematics, Stockholm University, 10691 Stockholm, Sweden. *Optimal approximants and orthogonal polynomials in several variables.*

We discuss the notion of optimal polynomial approximants in multivariable reproducing kernel Hilbert spaces. In particular, we analyze difficulties that arise in the multivariable case which are not present in one variable, for example, a more complicated relationship between optimal approximants and orthogonal polynomials in weighted spaces. Weakly inner functions, whose optimal approximants are all constant, provide extreme cases where nontrivial orthogonal polynomials cannot be recovered from the optimal approximants. Concrete examples are presented to illustrate the general theory and are used to disprove certain natural conjectures regarding zeros of optimal approximants in several variables. (Received August 10, 2020)

1161-41-131 **Brian Simanek*** (brian_simanek@baylor.edu), Baylor Math Department, One Bear Place #97328, Waco, TX 76798, and **Markus Hunziker**, **Andrei Martinez-Finkelshtein** and **Taylor Poe**. *Orthogonal Polynomials and Poncelet Ellipses.*

There is an extensive literature relating numerical ranges of special matrices, Blaschke products, and Poncelet curves in the unit disk. We will see how orthogonal polynomials on the unit circle (OPUC) are an ideal tool for understanding how these objects are connected and demonstrate how OPUC provide constructive tools for finding ellipses with special properties. (Received August 14, 2020)

1161-41-206 **Seungyeop Lee*** (lees3@usf.edu) and **Meng Yang**. *Asymptotics of the averaged characteristic polynomial for Ginibre Ensemble with finite number of fixed point charges.*

We consider the Ginibre Ensemble with a finite number of fixed point charges. The corresponding averaged characteristic polynomial of the Coulomb system has been identified with a multiple orthogonal polynomial (Type II). We obtain the asymptotic of the polynomials using the matrix Riemann-Hilbert analysis where the size of the matrix is determined by the number of fixed point charges. Such asymptotics are known only for a few (one or two) number of integer valued point charges (by Alfredo Deaño and Nick Simm). Our method yields the asymptotics for an arbitrary number of real valued point charges. The work is a joint project with Meng Yang (UC Louvain). (Received August 17, 2020)

46 ► Functional analysis

1161-46-190 **Ben (Li*** (bxl292@case.edu), **Castern Schuett** and **Elisabeth M Werner**. *Affine invariant maps for log-concave functions.*

Affine invariant points and maps for sets were introduced by Gruenbaum to study the symmetry structure of convex sets. We extend these notions to a functional setting. The role of symmetry of the set is now taken by evenness of the function. We show that among the examples for affine invariant points are the classical center of gravity of a log-concave function and its Santalo point. We also show that the recently introduced floating functions and the John- and Loewner-functions are examples of affine invariant maps. Their centers provide new examples of affine invariant points for log-concave functions. (Received August 17, 2020)

47 ► Operator theory

1161-47-18 **Fritz Gesztesy*** (fritz_gesztesy@baylor.edu), Department of Mathematics, Baylor University, Waco, TX 76798. *The limiting absorption principle and continuity properties of the spectral shift function for multi-dimensional, massless Dirac-type operators.* Preliminary report.

We report on recent results regarding the limiting absorption principle for multi-dimensional, massless Dirac-type operators and continuity properties of the associated spectral shift function.

This is based on various joint work with A. Carey, J. Kaad, G. Levitina, R. Nichols, D. Potapov, F. Sukochev, and D. Zanin. (Received July 20, 2020)

1161-47-85 **Boris P. Belinskiy**, **Don B. Hinton** and **Roger A. Nichols*** (roger-nichols@utc.edu), University of Tennessee at Chattanooga, Department of Mathematics, Dept. 6956, 615 McCallie Avenue, Chattanooga, TN 37403. *Spectral Analysis of Acoustic Black Holes.* Preliminary report.

A spectral analysis is performed for two-term Sturm–Liouville operators of the form $-\frac{1}{r}\frac{d}{dx}p\frac{d}{dx}$ on the interval $(0, 1]$, where the positive coefficients p and r satisfy power law decay to zero as $x \downarrow 0$. Specifically, we classify the endpoints, compute the essential spectrum, and determine conditions for the absence of positive eigenvalues of the corresponding self-adjoint extensions. Problems of this type arise, for example, in the study of the longitudinal vibrations of an elastic rod with cross sectional area that tapers to zero at one end and the transversal oscillations of an elastic string where the string tension and density taper to zero at one end. For certain powers of the decay, the time of propagation of a wave from $x = 1$ to $x = 0$ is infinite, so a wave does not reach the origin in finite time—a phenomenon known as an *acoustic black hole*. This talk is based on joint work with Boris P. Belinskiy (UTC) and Don B. Hinton (UTK). (Received August 11, 2020)

1161-47-145 **Jonathan Stanfill*** (jonathan_stanfill@baylor.edu), **Guglielmo Fucci**, **Fritz Gesztesy** and **Klaus Kirsten**. *Traces, Determinants, and Spectral Zeta Functions for Sturm–Liouville Operators.* Preliminary report.

Under appropriate hypotheses on regular as well as singular Sturm–Liouville operators associated with differential expressions of the type $\tau = r^{-1}[-(d/dx)p(d/dx) + q]$, the associated spectral ζ-function can be found through complex contour integration techniques to equal the residue of explicit functions, F (essentially, Fredholm determinants), involving a canonical system of fundamental solutions of $\tau y = zy$. The asymptotic behavior of F allows for a deformation of contours to find explicit expressions for the spectral ζ-function for $n \in \mathbb{N}$. We end by providing various concrete examples.

This is based on joint work with Guglielmo Fucci, Fritz Gesztesy, and Klaus Kirsten. (Received August 15, 2020)

49 ► Calculus of variations and optimal control; optimization

1161-49-105 **Liming Cai**, **Lanjing Bao**, **Logan Rose**, **Jeffery Summers** and **Wandi Ding***, 1301 E. Main Street, MTSU Box 34, Murfreesboro, TN 37132. *Mathematical Modeling and Optimal Control for Malaria Transmission Using Sterile Insect Technique (SIT) and Insecticide-Treated Nets (ITN).*

We consider a malaria transmission model with SEIR (susceptible-exposed-infected-recovered) classes for the human population, SEI (susceptible-exposed-infected) classes for the wild mosquitoes and an additional class for sterile mosquitoes. We derive the basic reproduction number of infection. We formulate an optimal control problem in which the goal is to minimize both the infected human populations and the cost to implement two control strategies: the release of sterile mosquitoes and the usage of insecticide-treated nets to reduce the malaria transmission. Adjoint equations are derived and the characterization of the optimal controls is established. Finally, we quantify the effectiveness of the two interventions aimed at limiting the spread of Malaria. A combination of both strategies leads to a more rapid elimination of the wild mosquito population that can suppress Malaria transmission. Numerical simulations are provided to illustrate the results. (Received August 13, 2020)

1161-49-120 **Sören Bartels**, Albert Ludwigs University of Freiburg, **Philipp Reiter***, Chemnitz University of Technology, and **Heiko von der Mosel**, RWTH Aachen University. *Elastic knots and rods.*

Minimizing the bending energy within knot classes leads to the concept of elastic knots. In relatively few cases we can predict the resulting shapes. In general, one faces a complicated energy landscape. The situation becomes even more involved by incorporating twist or prescribing symmetry. (Received August 14, 2020)

1161-49-308 **Lakmali Pradeepa Weerasena*** (lakmali-weerasena@utc.edu), 615 McCallie Avenue, Chattanooga, TN 37403. *A multi-objective mathematical model for optimal design of a compact-connected nature reserve system.*

Conservation biologists and wildlife managers are challenged with designing protected reserves for biodiversity conservation. In this study, we present a multi-objective mathematical model to obtain a compact shape for the reserve system. The model aims to include direct connections between reserve sites to increase the likelihood

of species persistence while maintaining a compact shape for the reserve system. For the proposed model, we introduce two conflicting objective functions that allow us to utilize the true purpose of multi-objective mathematical techniques and enhance the decision-making process with a trade-off analysis. We used synthetic reserve systems to demonstrate the working of the proposed model. An empirical application to a data set shows that the model produces quite impressive designs. (Received August 19, 2020)

51 ▶ Geometry

1161-51-30 **Charles Ouyang** and **Andrea Tamburelli*** (at52@rice.edu), TX. *Length spectrum compactification of the* $\mathrm{SL}(3,\mathbb{R})$*-Hitchin component.*

Higher Teichmüller theory studies geometric and dynamical properties of surface groups representations into higher rank Lie groups. One of these higher Teichmüller spaces is the $\mathrm{SL}(3,\mathbb{R})$-Hitchin component, a connected component in the $\mathrm{SL}(3,\mathbb{R})$-character variety that entirely consists of faithful and discrete representations that are the holonomies of convex real projective structures on a surface. In a joint work with Charles Ouyang, inspired by Bonahon's interpretation of Thurston's compactification of Teichmüller space by means of geodesic currents, we describe the length spectrum compactification of the $\mathrm{SL}(3,\mathbb{R})$-Hitchin component. We interpret the boundary points as hybrid geometric structures on a surface that are in part flat and in part laminar. (Received July 27, 2020)

1161-51-89 **Max Riestenberg*** (riestenberg@math.utexas.edu). *Explicit perturbation neighborhoods for Anosov representations.*

It has been known since their definition due to Labourie that Anosov representations are stable- any sufficiently small perturbation of an Anosov representation is still Anosov. In 2014, Kapovich, Leeb and Porti characterized Anosov represenations as "Morse actions" on the associated symmetric space and generalized the Morse lemma from hyperbolic geometry to higher rank symmetric spaces. Supplementing their proof with explicit estimates in the symmetric space allows us to describe explicit neighborhoods of Anosov representations. (Received August 11, 2020)

1161-51-267 **Jane Wang*** (wangjan@iu.edu) and **Sunrose Shrestha**. *Statistics of Square-Tiled Surfaces.*

Square-tiled surfaces are a class of translation surfaces that are of particular interest in geometry and dynamics because, as covers of the square torus, they share some of its simplicity and structure. One nice way to think about square-tiled surfaces is as surfaces created by gluing together unit squares along their edges. In this talk, we'll discuss various properties that square-tiled surfaces can have that give them extra symmetry. After drawing insights from experimental evidence, we will consider the implications between these properties and their frequency in each stratum of translation surfaces. (Received August 18, 2020)

1161-51-271 **Gil Kur*** (gilkur@mit.edu), Cambridge, MA 02139. *Connections between communication complexity and spherical geometry.*

In this talk, we are going to present an extremal problem in spherical geometry that is mainly motivated from Communication complexity. This problem was studied by Gao, Hug and Schneider in a restricted case.

Communication complexity is an area in theoretical computer science that quantifies the minimal amount of communication (in bits) required for multiple agents to solve a given task.

We show that proving such bounds for fundamental tasks reduces to extremal problems in spherical geometry which are of independent interest. (Received August 18, 2020)

1161-51-273 **Xian Dai** and **Giuseppe Martone***, martone@umich.edu. *Renormalized Hilbert length and convex projective surfaces.*

Convex projective surfaces, a generalization of hyperbolic surfaces, are naturally equipped with a geometric length function: the Hilbert length. We study the dynamics of the Hilbert length renormalized by the topological entropy, i.e. the exponential growth rate of closed curves. We estimate the number of closed curves which have roughly the same renormalized Hilbert length for two different convex projective surfaces. Furthermore, we describe the asymptotics of the renormalized Hilbert length along certain rays in the space of convex projective surfaces defined in complex analytic terms. This is joint work with Xian Dai. (Received August 18, 2020)

1161-51-307 **Sunrose Shrestha***, Wesleyan University. *The topology and geometry of random square-tiled surfaces.*

Square-tiled surfaces (STSs) are branched covers of the standard square torus with branching over exactly one point. They are concrete examples of translation surfaces which are an important class of singular flat metrics on 2-manifolds with applications in Teichmüller theory and polygonal billiards. In this talk we will consider a randomizing model for STSs based on permutation pairs and use it to compute the genus distribution. We also study holonomy vectors (Euclidean displacement vectors between cone points) on a random STS. Holonomy vectors of translation surfaces provide coordinates on the space of translation surfaces and their enumeration up to a fixed length has been studied by various authors such as Eskin and Masur. In this talk, we obtain finer information about the set of holonomy vectors, Hol(S), of a random STS. In particular, we will see how often Hol(S) contains the set of primitive integer vectors and find how often these sets are exactly equal. (Received August 19, 2020)

52 ► Convex and discrete geometry

1161-52-55 **Alexander Koldobsky*** (koldobskiya@missouri.edu), Department of Mathematics, University of Missouri, Columbia, MO 65211, and **Artem Zvavitch**. *Inequalities for the Radon transform on convex sets.*

We extend several results from convex geometry to general functions in place of volume. In particular, we prove a mean value inequality for the Radon transform. (Received August 05, 2020)

1161-52-66 **Alina Stancu*** (alina.stancu@concordia.ca), Department of Mathematics and Statistics, Concordia University, 1455 Blvd. de Maisonneuve Ouest, Montreal, Quebec H3G 1M8, Canada. *When convex body meets its polar.*

Convex sets whose polars are orthogonal transformations of themselves have been considered before in different mathematical contexts. We attempted a comprehensive study of planar convex bodies whose polars are the same convex bodies up to a rotation by pi. Roughly speaking, for any line passing through the origin, on one side we have part of the convex body and, on the other side, the mirror image of part of its polar, hence the title. We will show that the set of such bodies, called negatively self-polar, is very rich and has nice properties. (This is joint work with John-Mark Fortier.) (Received August 09, 2020)

1161-52-71 **Kaviita Ramanan*** (kavita_ramanan@brown.edu) and **Yin-Ting Liao**. *Random projections of high-dimensional measures: beyond universality.* Preliminary report.

The celebrated central limit theorem for convex sets states that most projections of any high-dimensional isotropic log-concave measure are close to Gaussian. More generally, it is known that if a high-dimensional measure satisfies the so-called thin shell condition, then most of its marginals are Gaussian. While these are beautiful universality results, they indiicate that the fluctuations of lower-dimensional random projections do not provide much information about the high-dimensional measures. Motivated by this, we study non-universal large deviation properties of the. projections, Specifically, we introduce a refined asymptotic thin shell condition for a sequence of measures and show that this implies that the corresponding sequence of random projections satisfies refined or sharp large deviations estimates. We also identify geometric information contained in these estimates, and construct importance sampling algorithms that allow one to more accurately estimate probabilities of interest. Finally, we verify the refined asymptotic thin shell condition for many high-dimensional measures of interest. (Received August 10, 2020)

1161-52-88 **Galyna Livshyts***, School of Math, 686 Cherry st NW, Atlanta, GA 30332. *On the maximal perimeter of convex sets with respect to probability measures.*

We will discuss upper and lower estimates for the maximal perimeter of convex sets with respect to probability measures. In particular, polynomial estimates are given for the case of isotropic log-concave measures. (Received August 11, 2020)

1161-52-93 **Sergii Myroshnychenko*** (myroshny@ualberta.ca). *On visual recognition and non-central sections.*

Assume that Earth is made out of a transparent glass and contains a convex body K in its interior. Let K be seen as a disk from every point on the planet's surface, possibly of different radii. Can one conclude that K is a Euclidean ball? What if it is always seen as an ellipse or a polygon?

We provide known and most recent results that answer **all** of the questions above, as well as their dual counterparts for non-central sections of convex bodies. (Received August 11, 2020)

1161-52-97 **Dmitry Ryabogin*** (ryabogin@math.kent.edu), Kent. *On bodies floating in equilibrium in every direction.*

Ulam asked if a solid of uniform density which floats in water in every position is necessarily a sphere? We will discuss known results related to this problem. (Received August 12, 2020)

1161-52-102 **Matthieu Fradelizi**, **Mokshay Madiman** and **Artem Zvavitch*** (zvavitch@math.kent.edu), The Department of Mathematical Sciences, Kent State University, Kent, OH 44242. *Links between different inequalities on mixed volumes of convex bodies.* Preliminary report.

The notion of mixed volumes and the inequalities involving them play a central role in the modern convex geometry, and have many connections to various other areas of mathematics. The most classical inequalities includes Brunn-Minkowski inequality and more general Alexandrov-Fenchel inequality.

In this talk we will discuss a local version of Alexandrov-Fenchel inequality and its connection to the search for the best constant in a number of geometric inequalities including inequality on the volume of orthogonal projections of convex bodies; isomorphic version of Bezout inequality; approximate submodularity of Minkowski sum; as well as to the property of distribution of roots of Steiner polynomial. (Received August 12, 2020)

1161-52-107 **Sergii Myroshnychenko**, **Kateryna Tatarko*** (tatarko@ualberta.ca) and **Vladyslav Yaskin**. *On the unique determination of ellipsoids by dual intrinsic volumes.*

In this talk, we show that an ellipsoid is uniquely determined up to an isometry by its dual Steiner polynomial. We reduce this result to the moment problem, and as a by-product obtain an alternative proof of the analogous known result for classical Steiner polynomials in $\mathbb{R}^3$. This is joint work with S. Myroshnychenko and V. Yaskin. (Received August 13, 2020)

1161-52-109 **Ferenc A. Bartha**, **Ferenc Fodor*** (fodorf@math.u-szeged.hu) and **Bernardo Gonzalez Merino**. *Central diagonal sections of the n-cube.*

We prove that the $(n-1)$-volume of central hyperplane sections of a unit cube in $\mathbb{R}^n$ orthogonal to a diameter is a strictly monotonically increasing function of n for $n \geq 3$. The proof is based on an integral formula for central sections of the cube, and on Laplace's method which we use to estimate the asymptotic behaviour of the integral. We treat small n by numerical methods. (Received August 13, 2020)

1161-52-124 **Yong Huang**, **Dongmeng Xi** and **Yiming Zhao*** (yimingzh@mit.edu). *The Minkowski problem in the Gaussian probability space.*

In this talk, we will discuss the Minkowski problem in the Gaussian probability space. Some partial results will be presented and we will also talk about difficulties in fully addressing the problem. (Received August 14, 2020)

1161-52-133 **Michael Roysdon*** (mroysdon@kent.edu), 233 MSB, 1300 Lefton Esplanade, Kent, OH 44242, and **Sudan Xing**. *L_p-Brunn-Minkowski type inequaliteis and a L_p-Borell-Brascamp-Lieb inequality.*

This talk concerns an L_p-extension of the famous Borell-Brascamp-Lieb inequality, which asserts that for any triple of measurable which satisfy reasonable averaging conditions, their integrals exhibit concavity.

As a consequence of this generalization, we extend the L_p-Brunn-Minkowski inequality to the class of β-concave measures on $\mathbb{R}^n$ and prove a related isoperimetric inequality.

This is a joint work with Sudan Xing. (Received August 14, 2020)

1161-52-143 **Giorgos Chasapis***, gchasapi@andrew.cmu.edu, and **Apostolos Giannopoulos** and **Nikos Skarmogiannis**. *Norms of weighted sums of log-concave random vectors.*

Given two centrally symmetric convex bodies C and K of volume 1 in $\mathbb{R}^n$, we study the multi-integral expression

$$\|\mathbf{t}\|_{C^s,K} := \int_C \dots \int_C \left\| \sum_{j=1}^s t_j x_j \right\|_K dx_1 \dots dx_s,$$

where $\mathbf{t} = (t_1, \dots, t_s) \in \mathbb{R}^s$. An old question by V. Milman in the pursuit of a generalized Khintchine-type inequality is whether $\|\mathbf{t}\|_{K^s,K}$ is actually equivalent to the Euclidean norm $\|\mathbf{t}\|_2$, up to a term which is logarithmic in the dimension. We discuss general upper bounds for $\|\mathbf{t}\|_{C^s,K}$ in the case that C is isotropic, that can be further tightened when $C = K$ under certain uniform convexity or cotype assumptions on $(\mathbb{R}^n, \|\cdot\|_K)$. Our approach also yields a new proof of the sharp lower bound on $\|\mathbf{t}\|_{C^s,K}$ provided in an earlier work of Gluskin and Milman. Based on joint work with A. Giannopoulos and N. Skarmogiannis. (Received August 15, 2020)

1161-52-154 **Niufa Fang**, **Sudan Xing** and **Deping Ye*** (deping.ye@gmail.com), Department of Mathematics and Statistics, Memorial University of Newfoundland, St. John's, NL A1C5S7, Canada. *The L_p Asplund sum of log-concave functions.*

Log-concave functions have properties very similar to those for convex bodies, and many fundamental results and concepts for convex bodies have found their counterparts for log-concave functions.

In this talk, I will present some progress on the L_p Asplund sum of log-concave functions, an L_p addition analogue to the L_p addition of convex bodies. Related results and properties will be discussed as well. This is a joint work with N. Fang and S. Xing. (Received August 15, 2020)

1161-52-159 **Han Huang**, **Boaz A Slomka**, **Tomasz Tkocz** and **Beatrice-Helen Vritsiou***, University of Alberta, Dpt. of Mathematical and Statistical Sciences, Edmonton, Alberta T6G 2G1, Canada. *On Hadwiger's covering conjecture.*

A central problem in Discrete Geometry, Hadwiger's covering conjecture, asks to find the smallest integer $N(n)$ with the property that every convex body in $\mathbb{R}^n$ can be covered by at most $N(n)$ translates of its interior.

We will discuss connections with Asymptotic Convex Geometry and measure concentration, as well as with entropic methods, that allow for at least a subexponential improvement to the long-standing general upper bounds of Rogers for $N(n)$. (Received August 16, 2020)

1161-52-193 **Peter Pivovarov*** (pivovarovp@missouri.edu) and **Jesus Rebollo Bueno**. *Stochastic forms of Brunn's principle.*

A number of geometric inequalities for convex sets arising from Brunn's concavity principle have recently been shown to yield local stochastic formulations. Comparatively, there has been much less progress towards stochastic forms of related functional inequalities. We work towards a stochastic geometry of concave functions to establish local versions of dimensional forms of Brunn's principle a la Borell, Brascamp-Lieb, and Rinott. To do so, we define shadow systems of convex epigraphs and hypographs, and revisit Rinott's approach in the context of multiple integral rearrangement inequalities. (Received August 17, 2020)

54 ► General topology

1161-54-6 **Sudip Kumar Acharyya** and **Rakesh Bharati*** (bharti.rakesh292@gmail.com), Vill. and P.O.- Tentulberia, P.S- Gaighata, Dist- North 24 Parganas, Bongaon, W.B 743245, and **A. Deb Ray**. *On the Intermediate rings of functionally countable subalgebra of $C(X)$.*

Let X be a Hausdorff zero-dimensional topological space. Suppose $C_c(X)$ is the ring of all continuous real valued functions on X which has countable range and $C_c^*(X)$ is the subring of $C_c(X)$ consisting of all bounded functions lying in $C_c(X)$. If $A_c(X)$ is an intermediate ring meaning that it is a ring lying between $C_c^*(X)$ and $C_c(X)$ then it is proved that the set of all maximal ideals of $A_c(X)$ with the well known hull-kernel topology is $\beta_0 X$, the Banaschewski compactification (the largest zero-dimensional compactification of a zero-dimensional Hausdorff space) of X. By defining the m_c-topology on $C_c(X)$ which is a countable analogue of the well known m-topology on $C(X)$, it is further proved that X is a P-space if and only if each ideal in $C_c(X)$ is closed in m_c-topology. There are quite a few open questions related to these rings versus the topological structure of X. (Received June 27, 2020)

1161-54-202 **Yuanan Diao**, **Michael Lee Finney** and **Dawn Ray***, Fretwell 376, 9201 University City Blvd., Charlotte, NC 28223. *The number of oriented rational links with a given deficiency number.*

Let $\mathcal{U}_n$ be the set of un-oriented and rational links with crossing number n, a precise formula for $|\mathcal{U}_n|$ was obtained by Ernst and Sumners in 1987. In this paper, we study the enumeration problem of oriented rational links. Let $\mathcal{M}_n$ be the set of oriented rational links with crossing number n and let $\mathcal{M}_n(d)$ be the set of oriented rational links with crossing number n ($n \geq 2$) and deficiency d. In this paper, we derive precise formulas for $|\mathcal{M}_n|$ and $|\mathcal{M}_n(d)|$ for any given n and d and show that

$$|\mathcal{M}_n(d)| = F^{(d)}_{n-d-1} + \frac{1+(-1)^{nd}}{2} F^{(\lfloor \frac{d}{2} \rfloor)}_{\lfloor \frac{n}{2} \rfloor - \lfloor \frac{d+1}{2} \rfloor},$$

where $F_n^{(d)}$ is the convolved Fibonacci sequence. (Received August 17, 2020)

55 ► Algebraic topology

1161-55-141 **Radmila Sazdanovic*** (rsazdan@ncsu.edu), Department of Mathematics NC State University, Raleigh, NC 27695, and **Pawel Dlotko**, **Jesse Levitt** and **Mustafa Hajij**. *Applied topology methods in knot theory.*

A multitude of knot invariants, including quantum invariants and their categorifications, have been introduced to aid with characterizing and classifying knots and their topological properties. Relations between knot invariants and their relative strengths at distinguishing knots are still mostly elusive. We use Principal Component Analysis (PCA), Ball Mapper, and machine learning to examine the structure of data consisting of various polynomial knot invariants and the relations between them. Although of different origins, these methods confirm and illuminate similar substructures in knot data. These approaches also enable comparison between numerical invariants of knots such as the signature and s-invariant via their distribution within the Alexander and Jones polynomial data. (Received August 15, 2020)

57 ► Manifolds and cell complexes

1161-57-24 **Elizabeth Denne***, Chavis Hall, Washington & Lee University, Lexington, VA 24450. *Crossing Number and Ribbonlength of Folded Ribbon Knots.*

We study Kauffman's model of a folded ribbon knot: a knot made from a thin strip of paper folded flat in the plane. The folded ribbonlength is the length to width ratio of such a ribbon. We give upper bounds on ribbonlength and relate these to crossing number for various families of knots. We also give a construction showing ribbonlength is sublinear in crossing number for torus knot families. This is joint work with undergraduate students. (Received July 23, 2020)

1161-57-46 **Tye Lidman** and **Allison H Moore***, Virginia Commonwealth University, Richmond, VA , and **Claudius Zibrowius**. *Essential Conway spheres and Floer homology via immersed curves.*

We will consider the problem of whether Dehn surgery along a knot in the three-sphere produces an L-space, which is a Floer-theoretic generalization of a lens space. The geometric characterization of these manifolds remains a difficult outstanding problem, and it is natural to ask whether the existence of certain essential surfaces in the complement of a knot can obstruct non-trivial surgeries yielding L-spaces. We will prove any knot in the three-sphere with a nontrivial L-space surgery admits no essential Conway spheres. As a corollary, we recover a classic result of Wu that states that if a knot K has an essential Conway sphere, then the fundamental group of rational Dehn surgery along K is always infinite. Our proof uses the technology of peculiar modules, a Floer theoretic invariant for tangles due to Zibrowius, and in particular, the geometric realization of these modules as certain decorated immersed curves on the four-punctured sphere. (Received August 03, 2020)

1161-57-75 **Feng Luo*** (fluo@math.rutgers.edu), Briarwood Drive, Warren, NJ 07059, and **Tianqi Wu**. *Koebe conjecture, Weyl problem and discrete uniformization.*

We will discuss our recent joint work with Tianqi Wu on the relationships among the discrete uniformization of polyhedral surfaces, the Koebe's circle domain conjecture and the Weyl problem on convex surfaces in the hyperbolic 3-space. (Received August 10, 2020)

1161-57-81 **Colin Adams*** (cadams@williams.edu), Bascom Hall, Williams College, 33 Stetson Ct., Williamstown, MA 01267, and **Judah Devadoss**, **Mohamed Elhamdadi** and **Alireza Mashaghi**. *Gauss Codes, Quandles and Bondles for Proteins.*

Proteins are linear molecules which when considered as such, make topology irrelevant. However, if intra-chain bonds are included, a rich structure ensues. We discuss representations of bonded proteins in terms of Gauss codes. Then we explore the possibility of applying the algebraic structure of quandles to distinguish the topologies of proteins. Because of the presence of bonds, we extend the theory to define bondles, a type of quandle particularly adapted to distinguishing the topological types of bonded proteins. (Received August 10, 2020)

1161-57-106 **Kenneth C Millett*** (millett@math.ucsb.edu), Department of Mathematics, University of California, Santa Barbara, Santa Barbara, CA 93106. *Knots in Proteins.*

Since 2000, knots in proteins have attracted the attention of biologists, biophysicists, and mathematicians. Following a brief historical inventory, we will describe the application of algebraic and topological methods to identify and quantify the presence of knotting in macromolecules, we will describe a new application of the

HOMFY-PT polynomial, the superposition of the DMS spectrum. We will employ this to describe important instances of knots and slipknots in proteins, their mathematical character, and explore what they tell us about the consequences of knotting entanglement. (Received August 13, 2020)

1161-57-112 **Eric J Rawdon*** (ejrawdon@stthomas.edu), 2042 Fairmount Av. *Open linking.* Preliminary report.

In the last 10 years, there has been increasing interest in measuring entanglement in open curves, e.g. knotting in proteins. We present a simple technique for measuring linking between two or more open curves. (Received August 13, 2020)

1161-57-115 **Thomas D. Eddy** and **Clayton Shonkwiler***, Department of Mathematics, Colorado State University, clay@shonkwiler.org. *New Stick Number Bounds from Random Sampling of Confined Polygons.*

The stick number of a knot is the minimum number of segments needed to build a polygonal version of the knot. Despite its elementary definition and relevance to physical knots, the stick number is poorly understood: for most knots we only know bounds on the stick number. We adopt a Monte Carlo approach to finding better bounds, producing very large ensembles of random polygons in tight confinement to look for new examples of knots constructed from few segments. We generated a total of 220 billion random polygons, yielding either the exact stick number or an improved upper bound for more than 40% of the knots with 10 or fewer crossings for which the stick number was not previously known, as well as determining for the first time the exact stick number of a non-torus prime knot with more than 9 crossings. (Received August 13, 2020)

1161-57-127 **Emily Shinkle*** (emilyss2@illinois.edu). *Finite Rigid Sets for Simplicial Complexes Associated to Surfaces.*

The arc complex and flip graph are simplicial complexes associated to a given surface, each describing combinatorial information about the surface. These complexes were used by Harer in his study of homological properties of the mapping class group, and have connections to Penner's decorated Teichmuller theory and Fomin-Shapiro-Thurston's theory of cluster algebras, among others. Irmak-McCarthy, Korkmaz-Papadopoulos, and Aramayona-Koberda-Parlier have shown correspondences between simplicial maps of these complexes and homeomorphisms of the associated surface. The complexes are infinite, but I show that these results can be extended for simplicial maps of certain finite subcomplexes, which we call finite rigid sets. (Received August 14, 2020)

1161-57-137 **Robert C. Haraway III*** (bobbycyiii@fastmail.com), **Neil R. Hoffman**, **Eric Sedgwick** and **Saul Schleimer**. *Knot diagram recovery in exponential time.* Preliminary report.

The most common combinatorial representation of knots is the diagram, a 4-regular planar graph with crossing information at the vertices. However, one may also represent a knot by a triangulation of its exterior. The problem of constructing a triangulation from a diagram of a knot can be solved in polynomial time. The same is not true for the reverse problem of knot diagram recovery. A family of torus knots shows that knot diagram recovery is not solvable in polynomial space, let alone polynomial time.

In joint work with Neil Hoffman, Eric Sedgwick, and Saul Schleimer, we show that knot diagram recovery is solvable in exponential time. (Received August 14, 2020)

1161-57-140 **Koya Shimokawa*** (kshimoka@rimath.saitama-u.ac.jp). *Handlebody decompositions of 3-manifolds and poly-continuous patterns.* Preliminary report.

In this talk, we consider handlebody decompositions of 3-manifolds. We will show that each closed orientable 3-manifold admits such a decomposition, and if there are two handlebody decompositions with some property, they are "stably equivalent". We will also discuss a characterization of handlebody decompositions of the 3-dimensional torus.

One motivation for this study comes from the material sciences. We are interested in the characterization of poly-continuous patterns of microphase structures of block copolymers. As an application, we discuss the characterization of poly-continuous patterns. We will also show that two poly-continuous patterns can be related by a finite sequence of moves. (Received August 15, 2020)

1161-57-199 **Claus Ernst*** (claus.ernst@wku.edu), Department of Mathematics, Western Kentucky University, 1906 College Heights Blvd., Bowling Green, KY 42101. *Generating all Minimal Diagrams of Alternating Knots.*

By the Taite Flyping Conjecture all all possible minimal diagrams of an alternating knot are related by a finite sequence of flypes. We we use this fact to generate all possible minimal diagrams of an alternating knot. Starting with any given minimal diagram our algorithm performs flypes exhaustively until all possible diagrams have been produced. We display our results for all distinct alternating knot diagrams up to 16 crossings. We discuss knots families that have only a small number of distinct minimal diagrams and knot families that have a large number of minimal diagrams. (Received August 17, 2020)

1161-57-227 **Sam Nelson*** (sam.nelson@cmc.edu). *Quiver Enhancements of Quandle Counting Invariants.*

In several recent papers, quivers have been used to enhance quandle counting invariant. In this talk we will survey these enhancements, starting with the original quandle coloring quivers and moving on to quandle cocycle quivers, quandle module quivers, and quandle coloring quivers for oriented surface-links. (Received August 17, 2020)

1161-57-229 **Uta Ziegler*** (uta.ziegler@wku.edu), 1906 College Heights Blvd, School of Engineering and Applied Sciences, Bowling Green, KY 42101. *Topological and geometric measures of knots in extreme confinement.*

Nearly-equilateral polygons are generated under extreme confinement. The knot types and some geometric quantities are computed for each polygon. The objective of this investigation is to compare these topological and geometric measures from these extremely confined polygons with previously reported measures of equilateral polygons in more moderate confinement. This work is in collaboration with Claus Ernst, Eric Rawdon, and Yuanan Diao. (Received August 17, 2020)

1161-57-235 **Florian Stecker*** (stecker@utexas.edu), **Gye-Seon Lee** and **Jaejeong Lee**. *Non-positive triangle reflection groups in SL(3,R).* Preliminary report.

We are interested in discrete subgroups of SL(3,R) which are isomorphic to fundamental groups of closed surfaces or 2-orbifolds. Two varieties of these are known: the "positive" ones correspond to convex projective structures and are stable under arbitrary continuous deformations. The non-positive ones are stable only under small deformations and their limit curves are not convex, and not even differentiable. In the case of a triangle reflection group, we can show exactly when how far we can deform until the subgroup ceases to be discrete. Furthermore, we show that at the transition point there is still a continuous, but non-transverse, limit curve. (Received August 17, 2020)

1161-57-248 **Jay Leach** and **Kate Petersen*** (petersen@math.fsu.edu). *Symmetries and SL(2,C) character varieties of knot groups.*

I'll discuss the SL(2,C) character variety of a group, and specifically how symmetries of a 3-manifold (or alternately of it's fundamental group) affect its character variety. (Received August 18, 2020)

1161-57-251 **Louis H Kauffman*** (kauffman@uic.edu), 5530 South Shore Drive, Apt 7C, Chicacgo, IL 60637-1946. *Representations of the Artin Braid Group Associated with Anyons.*

This talk will discuss the structure of unitary braid group representations associated with Fibonacci anyons and with anyons associated with Majorana Fermions, and how these representations are related with quantum algorithms and topological quantum computing. (Received August 18, 2020)

1161-57-254 **Hugh N Howards***, Math Department, Wake Forest University, Winston Salem, NC 27109, and **Jiong Li** and **Xiotian Liu**. *Systematically Detecting Flypes and Hexagonal Mosaics.*

We introduce a new tool which makes it easier to systematically recognize when two knots differ by a sequence of Flypes. We then apply it to hexagonal mosaics to find an infinite family of knots which do not achieve their hexagonal mosaic number while also in a projection which achieves their crossing number, extending a result of Lew Ludwig et al. (Received August 18, 2020)

1161-57-293 **Jason Parsley*** (parslerj@wfu.edu) and **Eric Rawdon**. *Census of Petal Knots.* Preliminary report.

This is joint work with Eric Rawdon (and many students). A petal diagram for a knot resembles a flower with n petals, symmetrically placed around a central point. Petal diagrams were introduced by Colin Adams and his students, who showed that every knot has a petal diagram, indexed by a permutation in S_n. Our work builds a census of all petal knots with 13 or fewer petals – over 480 million diagrams, classified by knot type. We

study the properties of these petal knots and draw connections to both classical and modern knot invariants. (Received August 18, 2020)

1161-57-303 **Siddhi Krishna*** (krishna@math.gatech.edu). *Taut Foliations and Dehn surgery along positive braid knots.* Preliminary report.

The L-Space Conjecture is taking the low-dimensional topology community by storm. It aims to relate seemingly distinct Floer homological, algebraic, and geometric properties of a closed 3-manifold Y. In particular, it predicts a 3-manifold Y isn't "simple" from the perspective of Heegaard-Floer homology if and only if Y admits a taut foliation. The reverse implication was proved by Ozsváth and Szabó. In this talk, we'll present some new results supporting the forward implication. Namely, we'll build taut foliations in manifolds obtained by Dehn surgery along positive braid closures. Our construction is concrete and combinatorial in nature. No background in Floer homology or foliation theories will be assumed. (Received August 18, 2020)

58 ► Global analysis, analysis on manifolds

1161-58-245 **Andrew Sungwan Park***, CRG-NJ, Cresskill, NJ , and **Richard Kyung**, CRG-NJ, Cresskill, NJ. *Study on the Economic Contributions and Social Statuses of Immigrants.*

A clear understanding of changes in the modern labor force and social interaction is crucial in determining trends in the United States economy. To review immigrants' contributions to the United States' economy from a general perspective, the following factors are needed to be considered: demographics, labor force participation, and social positions or statuses of immigrants in the United States. This paper includes qualitative and quantitative studies such as statistical analysis of gathered data to study the key characteristics of contributions made by immigrants as well as social interactions between immigrants and host country populations. As social skills are important in the modern labor market, such characteristics were then compared to those of the native-born population. Comparisons of historical immigrants' unemployment rate and those of natives were also made.

By analyzing the statistical data such as a seasonally adjusted series and extreme values of samples, it was possible to determine trends and probabilities for extreme events. For the analysis, a distribution method using an extreme value theory was employed to find distributions by quantifying the trends. (Received August 19, 2020)

60 ► Probability theory and stochastic processes

1161-60-63 **Ali Pirhadi*** (pirhadi@okstate.edu), 938 N. Main St., Apt. A, Davidson, NC 28036. *Random trigonometric polynomials with (strongly) dependent coefficient.*

It is well known that a random cosine polynomial $V_n(x) = \sum_{j=0}^{n} a_j \cos(jx),\ x \in (0, 2\pi)$, with the coefficients being independent and identically distributed (i.i.d.) real-valued standard Gaussian random variables (asymptotically) has $2n/\sqrt{3}$ expected real roots. On the other hand, out of many ways to construct a dependent random polynomial, one is to force *blocks* of the coefficients to be identified. We discuss four different cases of random cosine polynomials with dependent coefficients–three of which employ strongly dependent coefficients, namely the expectation of the number of real zeros is not universal any longer. We plan to discuss two examples of random cosine polynomials with contiguous blocks of coefficients as well as one with palindromic blocks, and at last random cosine polynomials with periodic blocks. (Received August 07, 2020)

1161-60-64 **Mokshay Madiman*** (madiman@udel.edu). *The regularization effect of certain Markov semigroups on integrable functions.*

In 1989, Talagrand proposed a conjecture regarding the regularization effect on integrable functions of a natural Markov semigroup on the Boolean hypercube. While this conjecture remains unresolved, the analogous conjecture for the Ornstein-Uhlenbeck semigroup was recently resolved by Eldan-Lee and Lehec, by combining an inequality for the log-Hessian of this semigroup with a new deviation inequality for log-semiconvex functions under Gaussian measure. Our first goal is to explore the validity of both these ingredients for a number of semigroups (the $M/M/\infty$ queue on $\mathbb{Z}_+$, the Laguerre semigroups on $\mathbb{R}_+$, and certain diffusions in $\mathbb{R}^n$). Our second goal is to prove an analogue of Talagrand's conjecture for these settings, even in those cases where these ingredients are not valid. This is joint work with Nathael Gozlan, Xue-Mei Li, Cyril Roberto, and Paul-Marie Samson. (Received August 08, 2020)

1161-60-79 **Arnaud Marsiglietti*** (a.marsiglietti@ufl.edu) and **Sergey Bobkov**. *Entropic CLT for Smoothed Convolutions and Associated Entropy Bounds.*

We explore an asymptotic behavior of entropies for sums of independent random variables that are convolved with a small continuous noise. Convergence along the central limit theorem for strong distances and connections to the slicing problem are discussed. (Received August 10, 2020)

1161-60-86 **Sean O'Rourke*** (sean.d.orourke@colorado.edu), Department of Mathematics, University of Colorado, Campus Box 395, Boulder, CO 80309-0395, and **Tulasi Ram Reddy**. *Sums of random polynomials with independent roots.*

We consider the sum of independent random polynomials as their degrees tend to infinity. Namely, let p and q be two independent random polynomials of degree n, whose roots are chosen independently in the complex plane. We compute the limiting distribution for the zeros of the sum $p + q$ as n tends to infinity by analyzing its logarithmic potential. We will also discuss a generalization for more than two polynomials. These results can be viewed as describing a version of the free additive convolution from free probability theory for zeros of polynomials. (Received August 11, 2020)

1161-60-96 **Fabrice Baudoin**, **Maria Gordina** and **Tai Melcher*** (melcher@virginia.edu). *Quasi-invariance for generalized Kolmogorov diffusions in infinite dimensions.*

The Kolmogorov diffusion stands as one of the simplest and most natural examples of hypoelliptic diffusions, that is diffusions whose end point distribution is smooth with respect to Lebesgue measure. We consider different infinite-dimensional versions of these diffusions and show that they retain smoothness properties. These results are attained by establishing uniform control over coefficients appearing in certain functional analytic inequalities for approximating finite-dimensional diffusions.

This is joint work with Fabrice Baudoin and Maria Gordina. (Received August 12, 2020)

1161-60-101 **Marcus Michelen*** (michelen.math@gmail.com) and **Will Perkins**. *Analyticity for classical gasses via recursion.*

Classical gasses are a central object of study in statistical mechanics. We will describe analogies between these continuous objects and more familiar discrete random models such as the hardcore model. For the case of repulsive pair potentials, we give a new criterion for uniqueness of the infinite volume Gibbs measure and analyticity of the pressure. Our improvement on the bound for analyticity is by a factor e^2 over the classical cluster expansion approach and a factor e over the known limit of cluster expansion convergence. Based on joint work with Will Perkins. (Received August 12, 2020)

1161-60-144 **Souvik Dhara***, sdhara@mit.edu, and **Debankur Mukherjee** and **Kavita Ramanan**. *The r-to-p norm of non-negative random matrices.*

For an $n \times n$ matrix A_n, the $r \to p$ operator norm is defined as

$$\|A_n\|_{r\to p} := \sup_{x\in\mathbb{R}^n : \|x\|_r \le 1} \|A_n x\|_p \quad \text{for} \quad r, p \ge 1.$$

This talk considers $r \to p$ norms of symmetric random matrices with nonnegative entries, including adjacency matrices of Erdős-Rényi random graphs, matrices with positive sub-Gaussian entries, and certain sparse matrices. For $1 < p \le r < \infty$, the asymptotic normality of the appropriately centered and scaled norm $\|A_n\|_{r\to p}$ is established. Furthermore, a sharp ℓ_∞-approximation for the unique maximizing vector in the definition of $\|A_n\|_{r\to p}$ is obtained, which may be of independent interest.

The results obtained can be viewed as a generalization of the seminal results of Füredi and Komlós (1981) on asymptotic normality of the largest singular value. In the general case with $1 < p \le r < \infty$, the spectral methods are no longer applicable, which requires a new approach, involving a refined convergence analysis of a nonlinear power method and establishing a perturbation bound on the maximizing vector. (Received August 15, 2020)

1161-60-151 **Grigorios Paouris**, **Konstantin Tikhomirov** and **Petros Valettas*** (valettasp@missouri.edu). *Hypercontractivity, and lower deviation estimates in normed spaces.*

We consider the problem of estimating probabilities of lower deviation $\mathbb{P}\{\|G\| \leqslant \delta\mathbb{E}\|G\|\}$ in normed spaces with respect to the Gaussian measure. These estimates occupy central role in the probabilistic study of high-dimensional structures. It has been confirmed in several concrete situations, using ad hoc methods, that lower deviations exhibit very different and more complex behavior than the corresponding upper estimates. In this work we develop a general method for proving small ball estimates for norms. In the case of 1–unconditional norms our bounds are best possible up to numerical constants. We also study the lower small deviation estimates

for both 1-unconditional and general norms and we obtain optimal results. In all regimes, $\|G\|_\infty$ arises (at the computational level) as an extremal case in the problem. The proofs exploit the convexity and hypercontractive properties of the Gaussian measure. (Received August 15, 2020)

1161-60-152 **Keshav Pokhrel*** (kpokhrel@umich.edu), MI 48128. *Composite Generalizers of Weibull Distribution: Order of Compositions in the light of Reliability Models.*

We study the composite generalizers of Weibull distribution using exponentiated, Kumaraswamy, transmuted, and beta distributions. The composite generalizers are constructed using both forward and reverse order of each of these distributions. The usefulness and effectiveness of the composite generalizers and their order of composition is investigated by studying the reliability behavior of the resulting distributions. Two sets of real-world data are analyzed using the proposed generalized Weibull distributions. (Received August 16, 2020)

1161-60-179 **Yen Do**, **Hoi Nguyen** and **Oanh Nguyen***, Department of Mathematics, University of Illinois at Urbana-Champaign, Urbana, IL 61802. *Universality and non-universality of random trigonometric polynomials.*

In this talk, we discuss the number of real roots of random trigonometric polynomials with iid coefficients. We show that the variance of the number of real roots is asymptotically linear in terms of the expectation; furthermore, the multiplicative constant in this linear relationship depends only on the kurtosis of the common distribution of the polynomial's coefficients. Our method gives a fine comparison framework throughout Edgeworth expansion, asymptotic Kac-Rice formula and a detailed analysis of characteristic functions. (Received August 16, 2020)

1161-60-180 **Boris Hanin** and **Grigorios Paouris***, Texas A&M University, College Station, TX 77843. *Non-asymptotic results for singular values of Gaussian Matrix products.*

This article concerns the non-asymptotic analysis of the singular values (and Lyapunov exponents) of Gaussian matrix products in the regime where N, the number of term in the product, is large and n, the size of the matrices, may be large or small and may depend on N. We obtain concentration estimates for sums of Lyapunov exponents, a quantitative rate of convergence of the empirical measure of singular values to the Triangle Law, and results on the joint normality of Lyapunov exponents when N is sufficiently large as a function of n. (Received August 16, 2020)

1161-60-203 **Vishesh Jain*** (visheshj@stanford.edu), **Ashwin Sah** and **Mehtaab Sawhney**. *Perfectly sampling $k \geq (8/3 + o(1))\Delta$-colorings in graphs.*

Efficiently sampling a uniformly random (proper) k-coloring of a graph G with maximum degree Δ is an intensely studied algorithmic problem. Traditionally, the interest has been in obtaining *approximately uniform* samples, and the goal is to do this efficiently for k as close to Δ as possible.

In this talk, we will discuss the problem of efficiently obtaining *perfectly uniform* samples. This was first studied in 1998 by Huber, who provided an efficient perfect sampling algorithm provided that $k > \Delta(\Delta + 2)$. Recently, interest in this problem has been revived due to works on partial rejection sampling and deterministic approximate counting, and especially the very recent work of Bhandari and Chakraborty, who provided a perfect sampler for $k > 3\Delta$.

We will present a perfect sampler which breaks this barrier at $k = 3\Delta$, and works efficiently for all $k \geq (8/3 + o_\Delta(1))\Delta$.

Joint work with Ashwin Sah (MIT) and Mehtaab Sawhney (MIT). (Received August 17, 2020)

1161-60-233 **Gokarna R. Aryal**, **Sher B. Chhetri** and **Hongwei Long*** (hlong@fau.edu), Department of Mathematical Sciences, Florida Atlantic University, 777 Glades Road, Boca Raton, FL 33431, and **Alfred A. Akinsete**. *On the Beta-G Poisson Family.*

In this talk, we present a new family of distributions which is defined by using the genesis of the truncated Poisson distribution and the beta distribution. Some mathematical properties of the new family will be discussed. We also discuss the parameter estimation procedures and potential applications of such a generalized family of distributions. (Received August 17, 2020)

1161-60-244 **Andrew Ledoan*** (andrew-ledoan@utc.edu), University of Tennessee at Chattanooga, Department of Mathematics, 615 McCallie Avenue, Chattanooga, TN 37403, and **Christopher Corley** (christopher-corley@utc.edu), University of Tennessee at Chattanooga, Department of Mathematics, 615 McCallie Avenue, Chattanooga, TN 37403.
The density of complex zeros of random sums.

Let $\{\eta_j\}_{j=0}^N$ be a sequence of independent and identically distributed random complex Gaussian variables, and let $\{f_j(z)\}_{j=0}^N$ be a sequence of given analytic functions that are real-valued on the real line. I will present an exact formula for the expected density of the distribution of complex zeros of the random equation $\sum_{j=0}^N \eta_j f_j(z) = \mathbf{K}$, where $\mathbf{K} \in \mathbb{C}$. The method of proof employs a formula for the expected absolute value of quadratic forms of Gaussian random variables. I will also discuss the limiting behavior of the density function as N tends to infinity and provide some numerical computations for the density function and empirical distributions for random sums with various choices of the functions $f_j(z)$, including polynomials orthogonal on the real line and the unit circle. (Received August 17, 2020)

1161-60-253 **Indranil Ghosh*** (ghoshi@uncw.edu), 601 S College Road, Wilmington, NC 28403, and **Hon Keung Tony Ng** (ngh@mail.smu.edu), Southern Methodist University, Dallas, TX.
A class of skewed distributions with applications in environmental data.

In environmental studies, many data are typically skewed and it is desired to have a flexible statistical model for this kind of data. In this article, we study a class of skewed distributions by invoking skewness into some known distributions. In particular, we consider using the logistic kernel to derive a class of univariate distribution called the truncated-logistic skew-symmetric (TLSS) distribution. We provide some structural properties of the proposed distribution and develop the statistical inference for the TLSS distribution. A simulation study is conducted to investigate the efficacy of the maximum likelihood method. For illustrative purposes, two real data sets from environmental studies are used to exhibit the applicability of such a model. (Received August 18, 2020)

1161-60-290 **Cory Ball***, ballcbh@gmail.com. *An Introduction to Stable Random Variables with an Application.*

In this talk, we will look at stable random variables. We will see the definition and a characterization. Finally, we will look briefly at the change point problem in regards to the drift parameter of Ornstein-Uhlenbeck processes driven by α-stable Lévy motions. (Received August 18, 2020)

1161-60-295 **Eyal Lubetzky**, **Itai Benjamini** and **Yuval Peled***, yp24@nyu.edu. *Minimum Weight Disk Triangulations and Fillings.*

We study the minimum total weight of a disk triangulation using vertices out of $\{1, \ldots, n\}$, where the boundary is labeled (123) and the $\binom{n}{3}$ triangles have independent rate-1 exponential weights.

We show that, with high probability, the minimum weight is $(c_1 + o(1))] \log n/\sqrt{n}$ for an explicit constant c_1, and that it is attained by a triangulation that consists of $(1/4 + o(1)) \log n$ vertices with distinct labels.

In addition, we prove that, with high probability, the minimum weights of a homological filling and a homotopical filling of the cycle (123) are both attained by the minimum weight disk triangulation. (Received August 18, 2020)

1161-60-299 **Duval Zephirin*** (dzephiri@fau.edu), 363 New York Dr., Fort Myers, FL 33905, and **Hongwei Long** (hlong@fau.edu). *Portfolio maximization for investors in Fads models driven by Lévy distributions.*

In recent years, there has been a growing interest in the use of Lévy distributions to model stock market behavior. Empirical studies of stock prices indicate distributions with heavy tails, which are incompatible with a Gaussian distribution and a more realistic approach is one that allows small jumps in small time intervals. Lévy distributions are used to model stock prices for informed and uninformed investors. We derive the optimal portfolio for both investors in a Fads model driven by Lévy distributions under asymmetric information and stochastic volatility of the stock price. Buckley's (2015) results correspond to a special case of the utility function and under deterministic variance. (Received August 18, 2020)

1161-60-300 **Andrew Thomack*** (andrew.thomack@greenville.edu). *The Asymptotic Growth of the Expected Number of Zeros of Random Harmonic Polynomials.*

A harmonic polynomial is a complex function of the form $h(z) = p(z) + \overline{q(z)}$ where p and q are complex analytic polynomials. We build off of work by Li and Wei, using the Kac-Rice formula to study the expected number of zeros for different models of random harmonic polynomials. Our main focus is the asymptotic growth of the number of zeros of these polynomials as the degree increases, but in this talk we will also look at the shape of

the first intensity functions of several models of random harmonic polynomials and note how they compare to their analytic analogs. (Received August 18, 2020)

1161-60-302 **Sher B. Chhetri*** (schhetri@mailbox.sc.edu), **Hongwei Long** (hlong@fau.edu) and **Cory Ball** (ballcbh@gmail.com). *Parameter Estimation for Geometric Lèvy Processes with Constant Volatility.* Preliminary report.

The alpha-stable distributions introduced by Paul Lèvy play an important role in many areas including finance, economics, and life sciences. In particular, they are important models for data sets reporting extreme events, such as earthquakes or stock market crashes. In finance, various stochastic models are used to model the price movements of financial instruments. After Robert Merton's (1976) seminal work, several jump-diffusion models for option pricing and risk management have been proposed. In this work, we introduce the alpha-stable Geometric Lèvy processes where we add alpha-stable Lèvy motion to the process related to the dynamics of log-returns in the Black-Scholes model. We use sample characteristic functions approach and study parameter estimation for discretely observed stochastic differential equations driven by Lèvy noises. We also discuss the consistency and asymptotic properties of the proposed estimators and establish a Central Limit Theorem. (Received August 18, 2020)

62 ► Statistics

1161-62-47 **Binod Rimal*** (brimal2014@fau.edu) and **William Hahn**. *Compressed Topological Data Analysis.*

The Johnson-Lindenstrauss lemma concerns the existence of embeddings such that the geometry of a point cloud is approximately preserved when mapping from a high dimensional space to a lower-dimensional space. Random matrices, which w.h.p. satisfy the restricted isometry property, allow data to be mapped to a lower dimension before computing persistent topology. In this talk, we discuss techniques for computing persistence diagrams directly from lower dimensional random projections. We analyze a synthetic classification problem and examine the performance of topological data analysis on compressed persistence diagrams through the machine learning technique. (Received August 03, 2020)

1161-62-78 **Scott Smith*** (sesmith@uiwtx.edu), San Antonio, TX 78209. *Log-Weibull X Generalized Distributions.*

We propose a new category of distribution families, called Log-Weibull-X families (LWX). This method of distribution construction applies the log function in the place of the popular odds function. The log function is particularly suited to use with the Weibull transformer, as it produces a true generalization of the transformed family. Focusing on the application of the Exponential and Burr(c,1) models, we will present probability weighted moments, reliability results, properties of maximum likelihood estimation, and an application to real data with comparison to similar transform constructions. (Received August 10, 2020)

1161-62-108 **Chad M Schafer*** (cschafer@cmu.edu). *Astrostatistics in the Era of LSST.* Preliminary report.

The Legacy Survey of Space and Time (LSST), to be conducted at the under-construction Vera C. Rubin Observatory, will yield 15 Terabytes of data each evening over a ten-year period, revolutionizing our understanding of the Universe. The complexity of the statistical challenges demands increased involvement from the data science community. In this talk I will describe some of the opportunities, focusing on the recurring challenges when working with high-dimensional and noisy astronomical data. In their raw form, these data are difficult to model, and assumptions that may have been reasonable at small sample sizes could be revealed to be inadequate by LSST-scale data sets. Such inference challenges provide statisticians with opportunities to both contribute to groundbreaking science, and to advance statistical methodology. (Received August 13, 2020)

1161-62-116 **Nonhle C Mdziniso*** (nmdziniso@bloomu.edu), Bloomsburg University, 400 E. Second St., Bloomsburg, PA 17815, and **Kahadawala Cooray** (coora1k@cmich.edu), Central Michigan University, Mount Pleasant, MI 48858. *Parametric Analysis of Renal Failure Data using the Exponentiated Odd Weibull Distribution.*

In this work, we analyze renal failure data from patients with mesangioproliferative glomerulonephritis (MPGN) which was modeled by Vikse et al. (2002) non-parametrically using the Kaplan-Meier curve. In their work, they showed that clinical variables were part of the variables which indicated progression to end-stage renal failure (ESRF). Though survival curves associated with these variables may be difficult to model by existing parametric distributions in literature. Therefore, we introduce a four-parameter Odd Weibull extension, the

exponentiated Odd Weibull (EOW) distribution which is very versatile in modeling lifetime data that its hazard function exhibits ten different hazard shapes as well as various density shapes. Basic properties of the EOW distribution are presented. In the presence of random censoring, a small simulation study is conducted to assess the coverage probabilities of the estimated parameters of the EOW distribution using the maximum likelihood method. Our results show that the EOW distribution is very convenient and reliable to analyze the MPGN data since it provides an excellent fit for the clinical variables. Furthermore, advantages of using the EOW distribution over the Kaplan-Meier curve are discussed. (Received August 13, 2020)

1161-62-129 **Karl B Gregory*** (gregorkb@stat.sc.edu), LeConte College, 1523 Greene St, Columbia, SC 29201, and **Dewei Wang** and **Chris McMahan**. *Penalized regression methods for group testing data.*

Regression methods for the binary outcome of disease status when individuals are tested in groups will be discussed. Group testing can be used to reduce the number of chemical assays required to screen a population for a disease, but, given assays that are subject to error, group testing induces complicated likelihood functions. We discuss, in particular, adaptive elastic net penalization for fitting logistic regression models with group testing data. In addition, we introduce a semi-parametric additive model with sparsity/smoothness penalization for modeling nonlinear effects; this model makes use of group lasso penalties to achieve sparsity in the nonparametric effects. We consider also how data from multiple studies of a single disease can be combined such that the analyses share information. (Received August 14, 2020)

1161-62-149 **Dewei Wang*** (deweiwang@stat.sc.edu), **Yan Liu** and **Xichen Mou**. *Varying-coefficient regression with pooled biomonitoring data.* Preliminary report.

Human biomonitoring involves measuring the accumulation of contaminants in biological specimens (such as blood or urine) to assess individuals' exposure to environmental contamination. Due to the expensive cost of a single assay and possible unreliable data caused by the assay's limit of detection, the use of pooling has become increasingly common in environmental studies. The implementation of pooling starts by physically mixing specimens into pools, and then measures pooled specimens for the level of contaminants. An important task is to reconstruct individual-level statistical characteristics based on pooled measurements. In this talk, we will use the varying-coefficient regression model for the individual-level biomonitoring and provide methods to estimate the varying-coefficients based on different types of pooled data. Asymptotic properties of the estimators will be presented. We will illustrate our methodology via simulation and with application to pooled biomonitoring of a brominated flame retardant provided by the National Health and Nutrition Examination Survey. (Received August 15, 2020)

1161-62-158 **Suvra Pal*** (suvra.pal@uta.edu), 411 S Nedderman Drive, 478 Pickard Hall, Arlington, TX 76019. *On the Flexibility of COM-Poisson distribution in Cure Rate Modeling.*

In this talk, I will first introduce a cure rate model. Then, assuming a latent competing risks scenario, I will show the advantages of using the flexible COM-Poisson distribution in capturing the latent number of competing risks that are related to the occurrence of an event of interest. Considering right censored data, I will present the steps of the EM algorithm for the maximum likelihood estimation of the model parameters. To demonstrate the performance of the EM algorithm, I will present the results of an extensive Monte Carlo simulation study. Results of a sensitivity analysis with respect to model mis-specification will also be presented. Finally, to illustrate the flexibility of the COM-Poisson distribution in the context of cure rate modeling, I will analyze a well-known melanoma data. (Received August 16, 2020)

1161-62-188 **Jinwung Jung***, CRG-NJ, Cresskill, NJ , and **Richard Kyung**, CRG-NJ, Cresskill, NJ. *Study on the Forecasting of Natural Gas Economy Using Statistical and Computational Analysis.*

In this paper, statistical and computational analyses of the forecasting of the natural gas economy were performed. As evident in our observations, lower gas price values have much shorter return periods. In other words, they are more likely to reoccur; however, as the values increase for higher gas price values, the length of the return periods increase exponentially. Therefore, there is a tendency for gas price values to remain in lower ranges. The higher the gas price value, the more over-speculation and overestimation in the stock market. As a result, it is natural for the gas price values to eventually decrease. Since stock market bubbles are bound to pop at any given point, this trend is expected to repeat in the future and bring high gas price values down to normalcy. For the analysis, the extreme value theory was utilized to assess extreme events within probability distributions by quantifying tail behavior. By analyzing the minimum and maximum values of samples, it was possible to determine probabilities

for extreme events. A comparison was made with events previously observed and analyzed for authenticity. (Received August 19, 2020)

1161-62-212 **Ryan Pittman**, **John M Grego** and **David B Hitchcock*** (hitchcock@stat.sc.edu), 216 Leconte College, Columbia, SC 29208-0001. *Concurrent Functional Regression to Reconstruct River Stage Data during Flood Events.*

On October 4, 2015, the Cedar Creek gage at Congaree National Park stopped reporting stages, and readings did not resume until around two weeks later due to record-breaking rainfall that led to some of the worst flooding in South Carolina history. Our goal is to reconstruct the Cedar Creek stage during the missing two weeks, using a sample of ten historical flood events from the last 25 years. The Congaree River gage in Congaree National Park remained functioning throughout the October 2015 flood. The stages from the two gages are directly related during floods. We introduce a new method to objectively determine the start and end points of each flood event in the sample and then use these events to predict the missing Cedar Creek stage. We treat the stage as functional data and use a concurrent functional regression model to establish the relationship between the two locations during each timepoint of prior flood events. Once this relationship is found, the known Congaree stage is used to predict the missing Cedar Creek stage during the 2015 flood. The results show that there is a strong functional relationship between the two locations, and that the crest of Cedar Creek is predicted to have been a historic high of above 17 feet, compared to a previous high of just over 16 feet. (Received August 17, 2020)

1161-62-262 **Xiaonan Zhu*** (xzhu7@una.edu), UNA Box 5051, Department of Mathematics, University of North Alabama, Florence, AL 35632, and **Tonghui Wang**. *The variance-mean mixture of closed skew-normal distributions.*

In this talk, the variance-mean mixture of the closed skew-normal distribution with the generalized inverse Gaussian distribution as the mixing distribution is studied. The new distribution includes many distributions as special cases such as the variance-mean mixture of multivariate (skew) normal distributions. Explicit forms of its moment generating function and density function are derived. Some statistical properties, such as linear transformations, conditional distributions, and EM estimation, are discussed. (Received August 18, 2020)

1161-62-268 **Gokarna Raj Aryal*** (aryalg@pnw.edu), Purdue University Northwest, Hammond, Hammond, IN USA. *On the Transmuted-G Poisson Family of Distributions with Applications.*

In recent years, extensive efforts have been made to develop new models in the area of distribution theory and related statistical applications. These studies are mainly motivated due to the modeling of various data sources and find out the probabilistic structure. In this study, we present a new family of distributions called the Transmuted-G Poisson (TGP) family. This family of distributions is constructed by using the transmuted map and the genesis of the zero truncated Poisson (ZTP) distribution. Some mathematical and statistical properties of TGP family are provided. The parameter estimation and simulation procedures are also discussed. Usefulness of TGP family is illustrated by modeling couple of real-life data. (Received August 18, 2020)

1161-62-288 **Ping-Shou Zhong*** (pszhong@uic.edu), Chicago, IL. *Homogeneity Tests of Covariance and Change-Points Identification for High-Dimensional Functional Data.*

We consider inference problems for high-dimensional functional data with a dense number (T) of repeated measurements taken for a large number of p variables from a small number of n experimental units. Our first aim is to solve the theoretical and computational challenges in detecting and identifying change points among covariance matrices from high-dimensional functional data. The second aim is to provide computationally efficient and tuning-free tools with a guaranteed stochastic error control. The change point detection procedure is developed in the form of testing the homogeneity of covariance matrices. The weak convergence of the stochastic process formed by the test statistics is established under the "large p, large T and small n" setting. Our change point identification estimator is proven to be consistent for change points at any location of a sequence. Its rate of convergence is shown to depend on the data dimension, sample size, number of repeated measurements, and signal-to-noise ratio. Simulation results demonstrate both finite sample performance and computational effectiveness of our proposed procedures. An application to fMRI data demonstrates that our proposed methods can identify event boundaries in the preface of the movie *Sherlock*. (Received August 18, 2020)

65 ► Numerical analysis

1161-65-21 **Leo Rebholz*** (rebholz@clemson.edu). *Accelerating nonlinear solvers for PDEs from fluids.* Preliminary report.

After reviewing some recent results for Anderson acceleration applied to fixed point iterations, we extend the methodology to several application problems including Navier-Stokes, Boussinesq, and Bose-Einstein condensates. We formulate solvers for these problems as fixed point iterations, and show the fixed point functions satisfy the assumptions of the general theory. Results of several numerical tests will be given. (Received July 22, 2020)

1161-65-98 **Xiaobing Feng*** (xfeng@math.utk.edu), Department of Mathematics, 1403 Circle Drive, Knoxville, TN 37996. *Recent developments in numerical methods for stochastic Stokes equations.*

Besides the mathematical interests, stochastic Stokes (and Navier-Stokes) equations have been proposed to study turbulence flow under random forcing. Even in the simplest setting, their PDE solutions have very low regularity in time, which then poses a significant challenge for developing efficient and convergent numerical methods for the stochastic Stokes and (Navier-Stokes) equations. In particular, the most natural and popular class of numerical methods for those equations, namely mixed finite element methods, had not been proven to work. In this talk we shall present some recent developments in mixed finite element and splitting methods for the Stokes equations with multiplicative noise. We shall highlight the continuous and discrete stochastic inf-sup conditions and the strong convergence for both the velocity and pressure approximations, as well as the new analysis techniques used to obtain these results. Numerical experiments will also be presented to validate the theoretical results. (Received August 12, 2020)

1161-65-110 **Arvind K Saibaba***, 2311 Sullivan Drive, Campus Box 8205, Raleigh, NC 27518, and **Pranjal Pranjal**, **Eric de Sturler**, **Eric Miller** and **Misha Kilmer**. *Randomized approaches to accelerate MCMC algorithms for Bayesian Inverse Problems.* Preliminary report.

Markov chain Monte Carlo (MCMC) approaches are traditionally used for uncertainty quantification in inverse problems where the forward problem is described by a partial differential equation (PDE). However, the use of MCMC algorithms is prohibitively expensive in certain applications where each log-likelihood evaluation may require hundreds to thousands of PDE solves. We show how to mitigate this computational cost by using several randomized techniques and discuss various ways in which the randomized approximations can be embedded within MCMC algorithms. We demonstrate the accuracy and computational benefits of our proposed algorithms on a model application from diffuse optical tomography. (Received August 13, 2020)

1161-65-114 **Jennifer L Mueller*** (mueller@math.colostate.edu), 101 Weber Hall, Department of Mathematics, Colorado State University, Fort Collins, CO 80523, and **Diego A C Cardenas** and **Sergio S Furuie**. *Regularized reconstructions by the Distorted Born Iterative Method for low frequency ultrasound computed tomography.*

While conventional ultrasound is generally considered unsuitable for pulmonary imaging, low frequency (10 – 750 kHz) ultrasound has been shown to penetrate the lung, motivating our work in developing a tomographic pulmonary ultrasound imaging system. In this work, tomographic reconstructions of sound speed from numerically simulated low frequency ultrasound data on a ring array of transducers are computed using the distorted Born iterative method (DBIM) with total variation (TV) regularization with an automatic choice of regularization parameter. (Received August 13, 2020)

1161-65-177 **Lara Kassab*** (lara.kassab@colostate.edu). *Nonnegative CP Tensor Decomposition for Dynamic Topic Modeling.*

Dynamic topic modeling has been widely used in social and data sciences with the goal of extracting latent themes in the data along with their temporal evolution. Previous work on dynamic topic modeling primarily employ the method of Nonnegative Matrix Factorization (NMF), where slices of the data tensor are each factorized into the product of lower-dimensional nonnegative matrices. With this approach, however, information contained in the temporal dimension of the data is often neglected or underutilized. We propose instead adopting the method of nonnegative CANDECOMP/PARAPAC (CP) tensor decomposition (NNCPD), where the data tensor is directly decomposed into a minimal sum of outer products of nonnegative vectors, thereby preserving the temporal information. The viability of NNCPD is demonstrated through application to both synthetic and real data, where improved results are obtained compared to those of typical NMF-based methods. This is joint work with

Miju Ahn, Nicole Eikmeier, Jamie Haddock, Alona Kryshchenko, Kathryn Leonard, Deanna Needell, R. W. M. A. Madushani, Elena Sizikova, and Chuntian Wang. (Received August 16, 2020)

1161-65-249 **Mirjeta Pasha*** (mpasha3@asu.edu), Arizona State University, Arizona, AZ 85212, and **Lothar Reichel** and **Alessandro Buccini**. *Modulus-based iterative methods for constrained ℓ_p-ℓ_q minimization.*

The need to solve discrete ill-posed problems arises in many areas of science and engineering. Solutions of these problems, if they exist, are very sensitive to perturbations in the available data. Regularization replaces the original problem by a nearby regularized problem, whose solution is less sensitive to the error in the data. The regularized problem contains a fidelity term and a regularization term. Recently, the use of a p-norm to measure the fidelity term and a q-norm to measure the regularization term has received considerable attention. The balance between these terms is determined by a regularization parameter. In many applications, such as in image restoration, the desired solution is known to live in a convex set, such as the nonnegative orthant. It is natural to require the computed solution of the regularized problem to satisfy the same constraint(s). In this talk we will discuss that this procedure induces a regularization method and describes a modulus-based iterative method for computing a constrained approximate solution of a smoothed version of the regularized problem. Convergence of the iterative method is shown, and numerical examples that illustrate the performance of the proposed method are presented. (Received August 18, 2020)

1161-65-255 **Lin Mu***, University of Georgia, University of Georgia. *The Recent Development of Weak Galerkin Finite Element Methods.*

In this talk, we shall introduce the weak Galerkin finite element method (FEM), a new polygonal FEM, for solving partial differential equations. The key concept of discrete weak derivatives will be introduced and discussed for the development of numerical schemes. Weak Galerkin Method is a natural extension of the classical Galerkin finite element method with advantages in many aspects. For example, due to its high structural flexibility, the weak Galerkin finite element method is well suited to most partial differential equations on the general meshing by providing the needed stability and accuracy in approximation. In this talk, the speaker shall discuss the basics of weak Galerkin finite element methods (WG), including a priori error estimate and a posteriori error estimate. We start with second-order elliptic equations, for which WG shall be applied and explained in detail. Later, we shall extend our schemes to other CFD models, including Stokes equations, Brinkman equations, and Navier-Stokes equations. We shall discuss the new divergence preserving schemes and upwind stabilization techniques in designing the robust numerical schemes. (Received August 18, 2020)

68 ► Computer science

1161-68-53 **Yongge Wang*** (yonwang@uncc.edu), Matthews, NC 28105. *Byzantine Fault Tolerance Based Consensus for Blockchains.* Preliminary report.

The Internet communication channel can be considered as a partial synchronous network. The problem of Byzantine Fault Tolerance (BFT) in partial synchronous or asynchronous networks has received a lot of attention in the last 30 years. Our analysis shows that most existing (widely deployed) BFT protocols for blockchains (e.g., Tendermint BFT, Ethereum Casper FFG etc) may not achieve their goals in the open networks such as Internet. Based on these analysis, we introduce BDLS protocol which is the BFT protocol BDLS which, up to our knowledge, is the most efficient BFT protocol for open networks. Furthermore, we will briefly discuss the algebraic description based BFT protocol Casper CBC (Correct by Construction) for Ethereum 2.0 by Ethereum research lab and point out the potential challenges there. (Received August 05, 2020)

1161-68-148 **Susan Hohenberger*** (susan@cs.jhu.edu), **Venkata Koppula** and **Brent Waters**. *Building Secure Encryption from a Simple Function.*

Chosen-ciphertext attack (CCA) security is the de facto standard for public key encryption. The cryptographic community has had success in realizing CCA secure encryption from a variety of number-theoretic assumptions. This success encouraged a strong drive to realize CCA secure encryption from a general assumption, such as injective trapdoor functions. A trapdoor function is a primitive in which any user given a public key PK can evaluate an input x by calling $F_{eval}(PK, x) = y$. And a user with the secret key SK can recover x from y as $F_{inverse}(SK, y) = x$. However, a polynomial-time attacker without the secret key should not be able to output x given $F_{eval}(PK, x) = y$ for a randomly chosen x. By injective, we require a one-to-one mapping of the function input and evaluation spaces.

In this talk, I will present a construction of CCA secure public-key encryption from injective trapdoor functions. The construction is black box and assumes no special properties (e.g. "lossy", "correlated product secure") of the trapdoor function. This result is joint work with Venkata Koppula and Brent Waters. It was recognized with a Best Paper Award at CRYPTO 2020. (Received August 15, 2020)

1161-68-218 **Jonathan Mosheiff**, **Nicolas Resch**, **Noga Ron-Zewi**, **Shashwat Silas** and **Mary Wootters***, marykw@stanford.edu. *LDPC Codes Achieve List Decoding Capacity.*

We show that Gallager's ensemble of Low-Density Parity Check (LDPC) codes achieve list decoding capacity. These are the first graph-based codes shown to have this property. Previously, the only codes known to achieve list-decoding capacity were completely random codes, random linear codes, and codes constructed by algebraic (rather than combinatorial) techniques. This result opens up a potential avenue towards truly linear-time list-decodable codes which achieve list-decoding capacity.

Our result on list decoding follows from a much more general result: any local property satisfied with high probability by a random linear code is also satisfied with high probability by a random LDPC code from Gallager's distribution. Local properties are properties characterized by the exclusion of small sets of codewords, and include list-decoding, list-recovery and averageradius list-decoding. Along the way, we give a characterization of sets of codewords that are likely to appear in a random linear code, which may be of independent interest.

This is joint work with Jonathan Mosheiff, Nicolas Resch, Noga Ron-Zewi, and Shashwat Silas. (Received August 17, 2020)

1161-68-301 **Daniel C Apon*** (daniel.apon@nist.gov). *Passing the final checkpoint! NIST PQC's 3rd Round begins.*

In recent years, there has been a substantial amount of research on quantum computers – machines that exploit quantum mechanical phenomena to solve mathematical problems that are difficult or intractable for conventional computers. If large-scale quantum computers are ever built, they will be able to break many of the public-key cryptosystems currently in use. This would seriously compromise the confidentiality and integrity of digital communications on the Internet and elsewhere.

In response, NIST initiated a process in late 2017 to solicit, evaluate, and standardize one or more PKEs/KEMs and digital signature schemes. The Round 3 candidates were announced on July 22, 2020. This third round is expected to continue for another 12-18 months, at which point the initial post-quantum cryptographic standards are expected to be finalized and announced.

In this talk, I will survey the decision-making process of NIST in choosing its third round candidates – in particular, I will aim to explain NIST's thinking in choosing its finalists for near-term standardization.

Further, I will highlight a variety of issues and open questions that NIST believes are the most important to answer in this next year and a half before the first post-quantum cryptographic standards are chosen. (Received August 18, 2020)

1161-68-305 **Josh Vekther**, **Jiacheng Zhuo**, **Luisa Gil Fandino**, **Qixing Huang** and **Etienne Vouga*** (evouga@cs.utexas.edu). *Weaving as Geodesic Foliations.*

We then connect the physics governing surfaces woven out of thin ribbons to the geometry of geodesic foliations—foliations whose leaves are all approximately geodesic curves—and develop several new variational algorithms for computing such foliations. Our key insight is a relaxation of vector field integrability in the discrete setting, which allows us to optimize for curl-free unit vector fields that remain well-defined near singularities and robustly recover a scalar function whose gradient is well aligned to these fields. We present a design and fabrication pipeline for approximating surfaces of arbitrary geometry and topology by triaxially-woven structures, where the ribbon layout is determined by a geodesic foliation on a sixfold branched cover of the input surface. We validate the effectiveness of our pipeline on a variety of simulated and fabricated woven designs. (Received August 19, 2020)

70 ► Mechanics of particles and systems

1161-70-281 **Tom Herschberg*** (mgr992@mocs.utc.edu) and **Eleni Panagiotou**. *A Topology User Package for LAMMPS.*

We created an extension of LAMMPS, a popular molecular dynamics simulator. Our extension is a part of the creation of a larger user-package that enables the measurement of topological entanglement through rigorous mathematical measures from topology. In this part of the user-package we created a method to measure the linking in a system of macromolecules in Molecular Dynamics (MD) simulations. We present the software and

explanatory material which are easily accessible to the engineering and bio-engineering community. (Received August 18, 2020)

76 ► Fluid mechanics

1161-76-74 **De Witt L. Sumners*** (sumners@math.fsu.edu), Department of Mathematics, Florida State University, Tallahassee, FL 32304, and **Renzo L. Ricca** and **Irma I. Cruz-White**. *Helicity of Seifert Framed Defects.*

We provide a topological proof that under Seifert framing the helicity of line defects that arise in optics, condensates and excitable media is zero. We prove that when each of the individual helicities of the components of a defect link is zero, then the link is Seifert-framed. As corollary the linking exclusion principle is proven, and that total twist remains conserved under change of topology due to anti-parallel reconnection of defect strands. (Received August 10, 2020)

1161-76-123 **John Gounley*** (gounleyjp@ornl.gov). *Reducing time-to-solution in lattice Boltzmann simulations via regularization.*

The lattice Boltzmann method (LBM) is a popular numerical method for solving the Navier-Stokes equations in fluid and fluid-structure interaction simulations. The method differs from traditional solvers by tracking the evolution of a particle distribution function instead of hydrodynamic variables. On high performance computing systems, lattice Boltzmann methods are bandwidth-bound, with computational performance largely determined by the time required to transfer the LBM particle distribution function to and from memory at each timestep. This presents a particular challenge for high-order, multi-speed LBM lattices which involve significantly more particle distribution components per lattice site. In this talk, we address this challenge by applying a regularization process that allows for the LBM particle distribution to be stored in an effectively lossless compressed format. While LBM regularization was originally developed to improve simulation stability, we show that regularized-based compression can be used to improve simulation time-to-solution as well. This is joint work with Madhurima Vardhan and Amanda Randles (Duke University), Luiz Hegele (Santa Catarina State University), Erik Draeger (LLNL), and Shirley Moore (ORNL). (Received August 14, 2020)

1161-76-142 **Luoding Zhu*** (luozhu@iupui.edu), 402 N Blackford Street, Indianapolis, IN 46202, and **Kaoru Sakai**. *Modeling and simulation of blood flow past the distal arteriovenous graft with intimal hyperplasia.* Preliminary report.

Late-stage-kidney-disease patients rely on hemodialysis for the maintenance of their regular lives. Arteriovenous graft (AVG) is one of the commonly used devices for dialysis. However, this artificially created shunt may get clotted and eventually causes the dialysis to fail. The culprit behind the AVG clotting and failure is the intimal hyperplasia (IH), the gradual thickening of vein-wall in the vicinity of the vein-graft conjunction, which is not yet well understood. In this work we investigate the effects of the IH development, including its location and severity on the flow and force fields in the AVG anastomosis. The stenosis is modelled in the shape of a Gaussian function with different height, spread, and location. The blood is modelled as a viscous incompressible fluid, and the blood flow (pulsatile) is modelled by the lattice Boltzmann equations (D3Q19). The fluid-structure-interaction is modelled by the immersed boundary (IB) framework. Our computational results show that the IH severity has the most influence on the WSS, WNS, and the axial OSI. The stenosis location and flow pulsatility have little effect on flow and force fields. Our results indicate that the progression of the IH tends to accelerate the closure of the vein lumen. (Received August 15, 2020)

1161-76-153 **Haoxiang Luo*** (haoxiang.luo@vanderbilt.edu), VU Station B 351592, 2301 Vanderbilt Pl, Nashville, TN 37235. *Large-scale simulation of fluid-structure interaction for biological flows using an immersed-boundary method.*

Fluid–structure interaction (FSI) can be found in many biological tissues and organs. Among the examples are insect wings, fish fins, heart valves, blood vessels, and vocal folds. These applications typically involve complex geometries and large deformations and are therefore difficult for the boundary-conformal methods in computational fluid dynamics to handle. In our approach, we developed an immersed-boundary method based on the Cartesian grid to solve the viscous, incompressible Navier-Stokes equation. The nonlinear tissue mechanics is solved using a finite-element method using an independent, unstructured mesh discretizing the solid domain. The two solvers are coupled in a partitioned manner and are iterated at each time step to ensure numerical stability. Large-scale parallel computing is adopted for simulations involving hundreds of millions of mesh points

in the flow domain. I will present several applications using this approach, including animal flight, fish swimming, vocal cord vibration, and heart valves. (Received August 15, 2020)

1161-76-246 **Pengtao Yue*** (ptyue@vt.edu), 460 McBryde, 225 Stanger Street, Blacksbuerg, VA 24061, and **Jiaqi Zhang**. *Phase-field modeling of contact angle hysteresis and its application in drop impact dynamics.* Preliminary report.

Abstract: In reality, most solid surfaces are intrinsically rough or chemically heterogeneous. As a consequence, the contact line may get pinned at topological or chemical defects, which is known as contact angle hysteresis. In this talk, we introduce a thermodynamically consistent phase-field model that accurately captures contact angle hysteresis. The formulation satisfies a dissipative energy law, based on which we develop a finite-element discretization that is unconditionally energy-stable in some simple cases. In the end, we use this new method to simulate the transient process of a drop impacting a solid surface, where the drop may deposit, rebound, or partially rebound depending on the drop impact velocity and the surface wetting property. By tuning the hysteresis window, our numerical results can achieve very good agreement with the experimental measurements, which also manifests the importance of hysteresis in contact line dynamics. (Received August 18, 2020)

1161-76-261 **Ehssan Nazockdast*** (ehssan@email.unc.edu). *Cell nucleus as a microrheological probe to study the rheology of the cytoskeleton.*

Mechanical properties of the cell are important biomarkers for probing its architectural changes caused by cellular processes and/or pathologies. Recent advancements in microfluidic and high-speed imaging technologies have enabled probing the cell cortex deformations in microchannels, while also tracking different intracellular components in high-throughputs. Most previous studies of cell mechanics using microchannels only measure the cell stiffness, and do not disentangle the rheology of different cellular components, including the cortex, the cytoplasm and the nucleus. We present a novel method that utilizes the correlation between the cortical deformations that are induced by external microfluidic flows, and the nucleus displacements, induced by those cortical deformations, to decouple the cell cortex and the cytoplasm mechanics. As a proof of concept, we consider a rigid spherical nucleus centered in a spherical cell. We obtain analytical expressions for the time-dependent nucleus velocity vs the cell deformations, when the interior cytoplasm is modeled as a viscous, viscoelastic, porous and poroelastic material, and demonstrate how the nucleus velocity can be used to characterize the linear rheology of the cytoplasm over a wide range of forces and timescales/frequencies. (Received August 18, 2020)

1161-76-272 **Evan M Gildernew*** (nmx363@mocs.utc.edu) and **Eleni Panagiotou** (eleni-pangiotou@utc.edu). *Molecular Dynamic Study of Polymer Chains to Viscoelastic Response.* Preliminary report.

We used topological methods to measure the entanglement of varying molecular weight polymer melts alongside the development of molecular dynamic simulations to predict the polymer melt system's mechanical response to uni-axial elongational shear flow. The melts studied show characterization consistent with the transition region with some crossover into the rubbery plateau region and the polymer melt systems simulated were topologically analyzed for entanglement characterization, using the Gauss linking integral. (Received August 18, 2020)

82 ► Statistical mechanics, structure of matter

1161-82-60 **EJ Janse van Rensburg*** (rensburg@yorku.ca), Mathematics & Statistics, York University, Toronto, Ontario M3J 1P3, Canada. *Phase diagrams of linked and unlinked polygons in a square.*

The effect of entanglements on the physical properties of polymers and strings remains an important topic in biology and related areas. In this talk I will give account of a simple two dimensional model examining phase behaviour in a statistical mechanics model of linked and unlinked polygons (as models of ring polymers). Our model is composed of two square lattice polygons confined to a square, in one case linked to each other (by one polygon being placed inside the other), and in the other case not linked (by their interiors being disjoint). I will discuss briefly the numerical approach to this problem using a parallel implementation of the GARM algorithm, and then discuss the phase diagram, and how it depends on the underlying topology of the two polygons. Generally, the phase diagram includes a multicritical point where lines of first order and continuous transitions meet. I will discuss the properties of these transitions, and in some cases determine scaling exponents characterising them. This study is joint work with Enzo Orlandini (Padua, Italy). (Received August 07, 2020)

1161-82-121 **Michael Lang*** (lang@ipfdd.de), Leibniz Institut für Polymerforschung, Dresden, Hohe Straße 6, 01069 Dresden, Germany, and **Jakob Fischer**, **Ron Dockhorn**, **Benjamin Köhler**, **Marco Werner** and **Jens-Uwe Sommer**. *Olympic Gels.*

We use large scale computer simulations to create topologically linked gels, so-called "Olympic gels". We discuss briefly several different options to prepare these gels either using cyclic DNA and topoisomerase, a multi-batch ring-chain equilibrium - also called "DNA-Origami", a progressive construction method, or a reversible ring-opening polymerization at extremely low reaction rates. The structure of the gels is analyzed based upon a distribution of pairwise linkages that we determine using the HOMFLY polynomial. We also report briefly on the physical properties of Olympic gels regarding deformation behavior and swelling. These properties are rather unique, as the initial regime of both deformation and swelling costs almost no conformational entropy. (Received August 14, 2020)

1161-82-173 **Cristian Micheletti*** (michelet@sissa.it), SISSA, Trieste, Italy. *Knots and links in channel and slit confinement: static and dynamics.*

I will report on a series of studies where we looked at how the static and dynamics of entangled polymers is affected by confinement. Specifically, I will first by consider the knotting of semi-flexible chains inside channels of different size and discuss how the size and complexity evolves during the free or externally-driven dynamics of the chain[1,2]. Next, I will turn to the case of linked rings inside channels and slits and discuss how the size and dynamics of their linked portion responds to different types of confinement[3,4].

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(Received August 16, 2020)

1161-82-183 **Tetsuo Deguchi*** (deguchi@phys.ocha.ac.jp), Ohtsuka 2-1-1, Bunkyo-ku, Tokyo 112-8610, Japan. *Exact viscoelasticity of topological polymers and polymer networks via homology.* Preliminary report.

We show exact results on dynamical and viscoelastic properties of topological polymers and polymer networks through the Rouse dynamics by making an extensive use of their homological properties, which are part of topological properties. In particular, we rigorously express the stress of the polymers or networks in terms of the edge (or bond) vectors. The stress expression should be consistent with many polymer experiments such as those supporting the stress optical law. Here, let us explain about the background of this study: Recently complex polymers with their chemical connectivity expressed by graphs have been synthesized in chemistry. We call them topological polymers or graph-shaped polymers. We regard polymer networks as an extension of topological polymers. The edge vector is defined by the displacement or the difference of the position vectors of neighboring vertices (monomers) in a given topological polymer or polymer network. This presentation is in collaboration with Jason Cantarella, Clayton Shonkwiler, and Erica Uehara. (Received August 16, 2020)

1161-82-275 **Wei Li** (wli52@utk.edu), Chemical and Biomolecular Engineering, University of Tennessee, Knoxville, Knoxville, TN 37996, and **Manolis Doxastakis*** (edoxasta@utk.edu), Chemical and Biomolecular Engineering, University of Tennessee, Knoxville, Knoxville, TN 37996. *Crossover from Rouse to entangled polyisoprene dynamics: a multiscale simulation approach.*

Topological constraints govern the dynamics of entangled macromolecules when polymers exceed a critical molecular weight. The crossover from the low molecular limit described by the Rouse model to dynamics prescribed by entanglements has long been the subject of extensive research with significant challenges present. Constraint release and contour length fluctuations are often introduced into phenomenological descriptions to capture dynamics that depart from the original Doi-Edwards tube theory. However, significant recent studies find that many of these features present chemically specific characteristics rather than a ubiquitous behavior questioning our comprehensive understanding of the fundamental underlying mechanisms. Polyisoprene melts have long served

as a prototype experimental system to study the evolution of the end-to-end distance along with mechanical relaxation. In this study we present new results by a systematic investigation of the transition of polyisoprene molecules from the Rouse regime to dynamics dictated by topological constraints. We report findings on mean correlation times as well as a complete description of full relaxation spectra and contrast our data to experimental studies. (Received August 18, 2020)

91 ► Game theory, economics, social and behavioral sciences

1161-91-298 **Kuanseok Song***, CRG-NJ, Cresskill, NJ , and **Richard Kyung** (nycrick@gmail.com), CRG-NJ, Cresskill, NJ. *Study on the Economic Growth in Bangladesh and Laos: A Quantitative and Qualitative Analysis.*

Although beset by poverty in the past, new developments in Bangladesh and Laos have brought about steady economic growth. Advances in employment and technology, accompanied by new opportunities, investments, and the opening of the borders, have provided the two countries with an opportunity for increased economic development. Until recently, extreme poverty in Bangladesh and Laos resulted in the United Nations declaring the country as one of the most undeveloped in the world. However, the countries are beginning to see positive signs of improvement in employment, technology, and education. Agricultural and textile industries have produced surpluses, and wages have been rising for much of the country's poor. As the economic situation continues to improve, many of the issues associated with poverty, such as hunger, disease, and inadequate housing are being addressed, as well. This paper also aims to use statistical analysis to follow the current trends in the country's unemployment rates - a common indicator of economic stability. By analyzing their economic data, policy makers are better able to forecast potential economic development. (Received August 19, 2020)

92 ► Biology and other natural sciences

1161-92-10 **Maeve L McCarthy*** (mmccarthy@murraystate.edu), Mathematics & Statistics, 6C Faculty Hall, Murray, KY 42071, and **K. Renee Fister** and **Seth Oppenheimer**. *Optimal control of mosquito populations.*

We develop an optimal control framework to investigate the introduction of sterile type mosquitoes to reduce the overall mosquito populations. The goal is to establish the existence of a solution given an optimal sterilization protocol as well as to develop the corresponding optimal control representation to minimize the infiltrating mosquito population while minimizing fecundity and the number of sterile type mosquitoes introduced into the environment per unit time. (Received July 09, 2020)

1161-92-12 **Orit Peleg*** (orit.peleg@colorado.edu), Boulder, CO 80303. *Flow-Mediated Olfactory Communication in Honey Bee Swarms.*

Honey bee swarms are a landmark example of collective behavior. To become a coherent swarm, bees locate their queen by tracking her pheromones, but how can distant individuals exploit these chemical signals which decay rapidly in space and time? We find that bees collectively create a communication network to propagate pheromone signals, by arranging in a specific spatial distribution where there is a characteristic distance between individuals and a characteristic direction in which individuals broadcast the signals. To better understand such a flow-mediated directional communication strategy, we connect our experimental results to an agent-based model where virtual bees with simple, local behavioral rules, exist in a flow environment. Our model shows that increased directional bias leads to a more efficient aggregation process that avoids local equilibrium configurations of isotropic communication, such as small bee clusters that persist throughout the simulation. Our results highlight a novel example of extended classical stigmergy: rather than depositing static information in the environment, individual bees locally sense and globally manipulate the physical fields of chemical concentration and airflow. (Received July 17, 2020)

1161-92-61 **Neil Osheroff*** (neil.osheroff@vanderbilt.edu), Department of Biochemistry, 654 Robinson Research Building, Nashville, TN 37232-0146. *Recognition of DNA Topology by Topoisomerases.*

The double helical structure, length, and compaction of DNA, along with normal nucleic acid processes, generate a number of topological problems that the cell must be able to resolve in order to survive. Examples are DNA under- and overwinding (negative and positive supercoiling), knotting, and tangling. Levels of DNA supercoiling

strongly affect processes such as replication and transcription and DNA knots and tangles must be removed in order to open the double helix and segregate chromosomes during mitosis. Topological issues in DNA are resolved in human and bacterial cells by enzymes called topoisomerases. Type I topoisomerases regulate DNA supercoiling by generating transient single-stranded breaks in the genetic material. Type II topoisomerases regulate DNA supercoiling and remove knots and tangles by generating transient double-stranded breaks in the double helix. This presentation will familiarize the audience with DNA topoisomerases, how they resolve topological issues, and how they discern the geometry of DNA supercoils. (Received August 07, 2020)

1161-92-259 **Aarini P Panzade*** (aarinippanzade@gmail.com), 421 Rockcrest Way, 421 Rockcrest Way, Louisville, KY 40245-3214. *The Effect of TIMP-1 levels, MMP-1 Levels, and Healing Times on the Wound Surface Area.*

Matrix metalloproteinases (MMPs) are enzymes that degrade all kinds of extracellular matrix proteins during the wound-healing process. TIMP-1 is a tissue inhibitor of metalloproteinases (MMPs). In this work, we are investigating how MMPs and TIMPs, and the ratio of MMPs to TIMPs affect the wound surface area and the healing time. The data that was used for this work was of sixteen patients with diabetic foot ulcers in which measurements were taken of MMPs and TIMPs during a 12-week period. A multilinear regression analysis was performed on the interaction terms of different variables and compared to the wound surface area and compared to the regression analysis done in the spring semester on just the individual variables. It was concluded that the linear regression analysis done by the interaction terms was more beneficial to the wound surface area than the one done on individual variables. (Received August 18, 2020)

93 ► Systems theory; control

1161-93-111 **Ahmet Ozkan Ozer*** (ozkan.ozer@wku.edu), 1906 College Heights Hill Blvd, Department of Mathematics, Bowling Green, KY 42101. *Investigation of the uniform boundary observability of finite difference approximations of several beam equations with only one boundary observation.* Preliminary report.

First, a space-discretized Finite Difference approximation for the beam equations is introduced. These equations mainly describe transverse vibrations for single or mutli-layer beams. With only one boundary controller, even though these equations are known to be exactly observable, their Finite Difference approximations are not able to mimic the exact observability with respect to the discretization parameter. This is mainly due to the loss of the uniform gap among the eigenvalues of the approximated finite dimensional model. Lack of observability is proved. To obtain a uniform gap, and therefore, an exact observability result, we consider a direct filtering technique. Current and future research will also be discussed. (Received August 13, 2020)

1161-93-139 **Scott Hansen*** (shansen@iastate.edu). *Boundary controllability of Schrodinger and Euler Bernoulli equations with an internal point mass.*

We analyze the space of exact controllability of a Schrodinger equation with a singular internal dynamic boundary condition which can be thought of as an internal point mass. Surprisingly, the controllability space (with control active at one end point) can be identical on both sides of the point mass, or differ by one Sobolev order, or contain subspaces with both types of behavior. These results lead to some new characterizations of controllability spaces for the Euler-Bernoulli beam with an internal point mass with a single control active at an endpoint. (Received August 14, 2020)

94 ► Information and communication, circuits

1161-94-68 **Heide Gluesing-Luerssen*** (heide.gl@uky.edu). *Distance distribution of cyclic orbit codes.*

Cyclic orbit codes are subspace codes that arise as the orbit of an $\mathbb{F}_q$-subspace in $\mathbb{F}_{q^n}$ under the natural action of the cyclic group $\mathbb{F}_{q^n}^*$. We discuss the distance distribution of cyclic orbit codes, where the distance is defined as the subspace distance. We show that for optimal full-length orbit codes (i.e., the cardinality is maximal and distance is maximal for that cardinality) the distance distribution depends only on q, n and the dimension of the generating subspace. For full-length codes with near-maximal distance, the distance distribution depends on an additional parameter, which is closely related to the number of fractions in $\mathbb{F}_{q^n}$ that can be obtained from elements of the generating subspace. This is joint work with Hunter Lehmann. (Received August 09, 2020)

1161-94-92 **Tovohery H Randrianarisoa*** (trandrianarisoa@fau.edu), Florida Atlantic University, FL. *Coding theory using Linear Complexity of Finite Sequences.*

In this work, we present two metrics coming from the linear complexity of sequences over finite field. The first metric is when linear complexity is used on finite sequences viewed as vectors over a finite field. The second metric is the linear complexity where we consider periodic sequences. We show that the problems of decoding random codes embedded with these metrics are hard problems. This opens the possibility of using these metrics for code based cryptosystems. We give various properties of codes with these metrics. (Received August 11, 2020)

1161-94-225 **Allison Beemer*** (beemera@uwec.edu), **Eric Graves**, **Joerg Kliewer**, **Oliver Kosut** and **Paul Yu**. *Throwing out the bathwater: authentication over multiple-access channels.*

In this talk, we present results on authenticated communication over adversarial multiple-access channels (MACs). We first consider a standard definition of authentication, in which a receiver may either decode all messages correctly or discard all messages if a malicious adversary is detected. For this setting, we see that an extension of the arbitrarily-varying channel condition overwritability characterizes the two-user authentication capacity region. We then define gamma-correcting authentication, where we require that at least a gamma fraction of the users' messages be recovered, even if an adversary forces the receiver to throw out other messages. We give necessary conditions for the two-user gamma-correcting authentication capacity region to have nonempty interior, and show that positive rate pairs are achievable over a particular adversarial MAC that satisfies these conditions. We conclude by exploring coding schemes that will allow for partial message correction over such channels. (Received August 17, 2020)

1161-94-241 **Dalton Seth Gannon*** (dalton.gannon@louisville.edu), 1124 Reutlinger Ave, Apt 204, Louisville, KY 40204, and **Hamid Kulosman**. *LCD codes and the condition for cyclic codes over $\mathbb{Z}_4$ to be LCD.*

A linear code with complementary dual (LCD code) is a linear code C whose intersection with its dual code, $C^{\perp}$, is only the zero codeword (i.e. $C \cap C^{\perp} = \{0\}$). These codes are of importance for not only theory, but for application as well. LCD codes give an optimum linear coding solution for a two user binary adder channel, they have applications to data storage, they simplify the maximum-likelihood decoding problem, and these codes have recently been shown to have applications to cryptography. It was first shown by James Massey a condition of linear codes to have complementary dual. Later, Yang & Massey produced a condition for a cyclic code over a Galios Field to have a complementary dual. It will be shown that a necessary and sufficient condition for a cyclic code C over $\mathbb{Z}_4$ of odd length N to be an LCD code is that $C = (f(x))$, where f is a self-reciporocal monic divisor of $X^N - 1 \in \mathbb{Z}_4[X]$. This result provides an interesting likeness between LCD cyclic codes over Galios Fields and LCD cyclic codes over the ring $\mathbb{Z}_4$. (Received August 17, 2020)

00 ► General

1162-00-7 **Andrei Okounkov*** (okounkov.columbia@gmail.com), Columbia University. *Lie theory without groups.*

Lie theory, which deals with smooth groups of transformations, is one of the cornerstones of mathematics and has great importance for both thoery and appliactions. It is also visibly limited, as the globe of Lie groups has been explored and inhabited. However, in recent years, some new geometric and algebraic structures have been recognized as being as good as Lie groups in every respect, including e.g. their contribution to the supply of special functions. My goal in this talk will be to explain where these new avenues of Lie theory lead. (Received June 29, 2020)

1162-00-158 **Kevin Seo***, CRG-NJ, Cresskill, NJ , and **Richard Kyung**, CRG-NJ, Cresskill, NJ. *Study on Bioimaging Using Mathematical Transformation and Digital Image Processing.*

The original MRI image of a brain with a tumor showing pseudo-progression and Alzheimer's was analyzed with the linear algebra based transformation method. This method was performed by adjusting the contrast by a specific factor. The structures inside the brain and the tumor are easily distinguished and identifiable using the presented method. In comparison to using linear transformation, quadratic transformation allowed for us to observe the bright regions becoming brighter and the dark regions becoming darker. As a result, an image of high contrast was present; this is desirable when attempting to view tumors and small structures. Although the quadratic transformation factor is significantly less than the linear transformation factor for the MRI images, the tumor analyzed in the quadratic transformation showed more of a contrast against the rest of the brain than the tumor analyzed in the linear transformation. When the factors are too small, certain regions are significantly brighter than others. This not only renders the image but also is ineffective for observations, for a balance between overly-bright areas and detectable structures is ideal. (Received August 31, 2020)

1162-00-175 **Istvan G Lauko*** (iglauko@uwm.edu), **George Berzsenyi** and **Gabriella A Pinter**. *Problems from The Mathematics Student Competition Corner (1979-1981).*

The Competition Corner, edited by George Berzsenyi, ran from 1979 to 1981 in the NCTM publication The Mathematics Student. It provided interested students motivation to test their creativity and problem solving skills and recognized students for well-formulated solutions, extensions and generalizations to problems. Each year, the 3-4 winners of the year-round competition were recognized with the Pólya Prize for Problem Solving and were awarded programmable calculators by Hewlett-Packard; they were among the first on the market. The competition also built a community of problem solvers who still remember the excitement and challenge that each new problem set generated. In this talk we present a few memorable problems and emphasize the open-ended nature of problem solving, the creation of new problems, extensions and generalizations. Our aim is to highlight a few great problems that could be developed into Math Circle activities or used in other programs aimed at exposing students to creative problem solving. (Received August 31, 2020)

03 ► Mathematical logic and foundations

1162-03-8 **Can Başkent*** (c.baskent@mdx.ac.uk). *Learning from and Working with Contradictions in Heuristics.*

Contradictions (or *monsters*) are essential part of Lakatosian heuristics. Depending on the characteristics of monsters, the theory is revised. Similarly, Hintikka "brackets" abnormalities in order to obtain a classical, contradiction-free system for epistemic inquiries.

But, if contradictions are such central elements of dynamic heuristics, why are they excluded? If we choose to work with them, how can we achieve it?

In this talk, I will argue that the underlying logic of reasoning for Lakatosian and even Hintikkan heuristics is non-classical. I will also maintain that it is a kind of logic that is inconsistency-tolerant.

Particularly, I will give an example of how Lakatosian heuristics works and how it is sympathetic to contradictions. Next, I will do the same for Hintikka's interrogative models of inquiry. I will then conclude that

meta-mathematical heuristic reasoning must be inconsistency- and incompleteness-tolerant based on the aforementioned cases. This will also substantiate the thesis that inquiry and heuristics are indeed "games" between rational players who may have inconsistent knowledge or incomplete strategies. (Received July 07, 2020)

1162-03-60 **Andras Benedek*** (benedek.andras@btk.mta.hu), Research Centre for the Humanities, 4 Tóth Kálmán str., Budapest, Bp. IX 1094, Hungary. *Models and Logics in the Practice of Mathematical Heuristics: Tools for concept development and discovery or just for retroactive reconstruction?*

I survey some Models of Building Theories and History Based Logics of concept formation both from the point of view of conceptual and historical reconstructions of heuristics. Discussing Inquisitive Semantics and Interrogative Models of Inquiry in terms of Dynamic Epistemic Logic I raise three questions. 1) Is it necessary that "the owl of Minerva spreads its wings only with the falling of the dusk", i.e., we obtain logic models of reasoning only after the historical event of discovery, or 2) can we also use the Dynamic Epistemic Logical and Game Theoretical tools for concept development in mathematics education and actual scientific research? 3) Could we blend these approaches with the toolbox of Model Theory? My answer to the first question is "No"; "Yes" to the second; and "Should" for the third one. I argue my case w.r.t (1) by pointing out that reconstructions of thoughts overlap with the field of discovery; show (2) by logical formalizations of Polya-type and Lakatosian heuristics; and support (3) by use-cases of Model Theory and Reverse Mathematics. The main point of the talk is that the challenge of describing real-world practice of mathematical heuristics is to combine modeling mathematical structures with the dynamics of epistemic logics of reasoning about them. (Received August 22, 2020)

1162-03-70 **Valentina Harizanov*** (harizanv@gwu.edu), Department of Mathematics, Washington, DC 20052. *Orderable quandles.*

We investigate algebraic and computability-theoretic properties of orderable quandles and, more generally, magmas. A magma is an algebraic structure with a single binary operation. Important examples of magmas include groups and semigroups, but also those that come from knot theory and do not necessarily have an associative operation, such as quandles. A magma M is right-orderable (left-orderable) if there is a linear ordering of the domain of M, which is right-invariant (left-invariant) with respect to the magma operation. If M has a right order that is also a left order, then M is bi-orderable. A computable orderable magma does not necessarily have a computable order. There is a natural topology on the set of all (left, right, bi-) orders of M. These spaces are compact, and in some cases homeomorphic to the Cantor set. (Received August 23, 2020)

1162-03-115 **Dario Verta***, Department of Mathematics, Washington, DC 20052. *Computability-theoretic complexity of properties of magmas.* Preliminary report.

We investigate computability-theoretic complexity of determining whether an algebraic structure has a certain property, relative to the algorithmic description that presents a structure. We study computable structures, but also, taking examples from algebraic categories, consider structures admitting a recursive presentation in terms of generators and relators. We establish that certain properties (such as Markov properties) are hard in a given class of structures. We can apply general results to various specific structures including various magmas such as semigroups, racks and quandles. (Received August 28, 2020)

05 ► Combinatorics

1162-05-16 **Xavier Perez-Gimenez** and **Abigail Raz*** (araz2@unl.edu). *Perfect matchings in the random bipartite geometric graph.* Preliminary report.

We consider the standard random bipartite geometric graph process in which n red vertices and n blue vertices are placed at random on the unit d-dimensional cube and edges are added sequentially, between vertices of different colors, in increasing order of edge-length. A natural question is to ask whether the first edge in the process that results in the minimum degree being at least one coincides, with high probability, with the first edge that creates a perfect matching. While this was already known to be false when $d = 2$, as the thresholds are not even of the same order, we are able to positively answer it for $d \geq 3$.

Joint work with Xavier Pérez-Giménez (Received August 10, 2020)

1162-05-19 **Jeff Kahn** and **Jinyoung Park*** (jinyoungpark0@gmail.com). *Tuza's Conjecture for random graphs.*

A celebrated conjecture of Zs. Tuza says that in any (finite) graph, the minimum size of a cover of triangles by edges is at most twice the maximum size of a set of edge-disjoint triangles. Resolving a recent question of

Bennett, Dudek, and Zerbib, we show that this is true for random graphs; more precisely:

$$\text{for any } p = p(n),\ \mathbb{P}(G_{n,p} \text{ satisfies Tuza's Conjecture}) \to 1 \text{ (as } n \to \infty).$$

(Received August 12, 2020)

1162-05-20 **Mark Kempton*** (mkempton@mathematics.byu.edu). *Cospectral Vertices in Graphs.* Preliminary report.

The basic question of spectral graph theory is: "What can the spectrum of a matrix tell us about a graph?" Fundamental to answering this question is an understanding of when two graphs can be "cospectral," that is, when can two non-isomorphic graphs have the same (adjacency matrix) spectrum. A related question concerns cospectral pairs of vertices, namely two vertices of a graph whose respective deletion yields a cospectral pair of graphs. The study of cospectral vertices shows up in various applications of spectral graph theory. We will discuss various characterizations of cospectral vertices and methods for constructing graphs with cospectral pairs. (Received August 12, 2020)

1162-05-24 **Steve Butler**, **Elizabeth Cooper**, **Aaron Li** and **Kate Lorenzen*** (lorenkj@iastate.edu), 411 Morrill Rd, Carver Hall 421, Ames, IA 50011, and **Zoe Schopick**. *Spectral properties of the exponential distance matrix.*

Given a graph G, the exponential distance matrix is defined entry-wise by letting the (u, v)-entry be $q^{dist(u,v)}$, where $dist(u, v)$ is the distance between the vertices u and v with the convention that if vertices are in different components, then $q^{dist(u,v)} = 0$. We establish several properties of the characteristic polynomial (spectrum) for this matrix and inertia of some graph families. (Received August 12, 2020)

1162-05-25 **Jennifer Morse**, **Jianping Pan**, **Wencin Poh*** (wpoh@ucdavis.edu) and **Anne Schilling**. *Crystal for stable Grothendieck polynomials.*

We construct a type A crystal whose character is the stable Grothendieck polynomials for fully-commutative permutations. This crystal is a K-theoretic generalization of Morse-Schilling crystal on decreasing factorizations. We prove, among other things, that this crystal is isomorphic to that of pairs of semistandard Young tableaux using a newly defined insertion. This offers a combinatorial interpretation to the Schur positivity of the stable Grothendieck polynomials. This is joint work with Jennifer Morse, Jianping Pan and Anne Schilling. (Received August 12, 2020)

1162-05-28 **Leslie Hogben*** (hogben@aimath.org). *Product throttling for zero forcing, power domination, and Cops and Robbers.*

The graph processes of power domination and zero forcing are involved in the analysis of electrical network monitoring and other applications, and pursuit-evasion games such as Cops and Robbers can be used to analyze intrusion detection. For each of these and other related parameters, the number of rounds that it takes to finish the process (time cost, called propagation or capture time) is also studied. When the process may use more than the minimum number of resources, the cost trade-off between time and resources is called throttling. Throttling the sum of time and resources has been studied for nine years. Recently product throttling, i.e., minimizing the product of time and resources, has been studied for a few parameters. This talk will survey results on product throttling. (Received August 14, 2020)

1162-05-47 **Cory Glover*** (cglover@mathematics.byu.edu) and **Mark Kempton**. *Understanding the Non-Backtracking Spectrum of Graphs.*

The non-backtracking matrix of a graph models a non-backtracking random walk as a Markov chain across the directed edges of a graph. It is commonly known that the spectrum of said matrix can be related to the spectrum of the adjacency matrix by Ihara's Theorem. In this talk, we examine specific graphs to find the explicit values of the non-backtracking spectrum that satisfy Ihara's Theorem. These graphs include bipartite graphs, cycles, trees and others. In addition, we identify properties of the non-backtracking spectrum for graphs with degree at least 2. Our method uses an invariant subspace, in particular its eigenvectors, to identify the values of and bounds on the non-backtracking spectrum. In addition, we create a bound on the spectral radius of the non-backtracking matrix in terms of the spectral radius of the adjacency matrix. (Received August 19, 2020)

1162-05-51 **Katherine F Benson**, **Daniela Ferrero**, **Mary Flagg**, **Veronica Furst**, **Leslie Hogben** and **Violeta Vasilevska*** (violeta.vasilevska@uvu.edu), 800 W University Parkway, Orem, UT 84058. *Nordhaus-Gaddum problems for power domination.*

Power domination is a coloring process on a graph that consists of finding a set of vertices that can 'observe' (color) the entire graph by following certain rules. The power domination number of a graph is the smallest number of vertices needed to observe (color) the whole graph. The study of power domination was motivated

by the placement problem in the electrical engineering that consists of finding network locations where phase measurement units (PMUs) can be placed to monitor the entire network with minimum number of PMUs possible. The Nordhaus-Gaddum problems for power domination consist of finding a (tight) lower and upper bounds for the sum or product of the power domination number of a graph and of its complement. In this talk, we first discuss the Nordhaus-Gaddum problems for power domination. Then we present results about the improved tight Nordhaus-Gaddum sum upper bound for graphs that are connected and have connected complements, as well as for graphs whose components and the components of their complements have order at least 3. (Received August 20, 2020)

1162-05-54 **Christopher Duffy*** (christopher.duffy@usask.ca). *Colourings, Homomorphisms, Brooks' Theorem, and Oriented Graphs.*

The definition of graph colouring by way of graph homomorphism allows us to give reasonable definitions for colouring for other types of graph objects. In this talk we consider oriented graph colouring — proper vertex colourings of anti-symmetric digraphs that, in some sense, respect the orientation of the arcs. We see how our intuition for graph colouring both helps and hinders us when we examine colourings of oriented graphs. In particular we explore the state-of-the-art for the oriented graph analogue to Brooks' Theorem for graphs with bounded maximum degree. (Received August 20, 2020)

1162-05-66 **Eric Sundberg*** (sundberg@oxy.edu) and **Ramin Naimi**. *Bin-based pairing strategies for the Maker–Breaker game on the boolean hypercube with subcubes as winning sets.* Preliminary report.

We consider the Maker–Breaker positional game on the vertices of the n-dimensional hypercube $\{0,1\}^n$ with k-dimensional subcubes as winning sets. We describe a pairing strategy which allows Breaker to win when $k = n/4+1$ in the case where n is a power of 4. Our results also imply the general result that there is a Breaker's win pairing strategy for any $n \geq 3$ if $k = \left\lfloor \frac{3}{7} n \right\rfloor + 1$. (Received August 22, 2020)

1162-05-68 **Eva Czabarka*** (czabarka@math.sc.edu), **Stephen J Smith** (sjs8@email.sc.edu) and **Laszlo A Szekely**. *An infinite antichain of planar tanglegrams.*

A tanglegram is a specific kind of graph, consisting of two rooted binary trees of the same size and a perfect matching joining their leaves. A tanglegram layouts is a straight line drawing of the two trees as plane trees such that the leaves are on two parallel lines, and the matching edges are the only ones that can cross.The tangle crossing number of a tanglegram is the minimum crossing number among its layouts. The tanglegram is planar, if it has a layout without crossings.

Tanglegrams play a major role in phylogenetics, especially in the theory of cospeciation. The first binary tree is the phylogenetic tree of hosts, while the second binary tree is the phylogenetic tree of their parasites, e.g. gopher and louse. The matching connects the host with its parasite. The tanglegram crossing number has been related to the number of times parasites switched hosts, or, working with gene trees instead of phylogenetic trees, to the number of horizontal gene transfers.

Here we show that he induced subtanglegram partial order is not a well-quasi-ordering by exhibiting an infinite antichain of planar tanglegrams. (Received August 23, 2020)

1162-05-69 **Eva Czabarka** (czabarka@math.sc.edu), **Inne Singgih** (inne.singgih@uc.edu) and **Laszlo A. Szekely*** (szekely@math.sc.edu). *On the diameter of k-colorable graphs.*

Erdős, Pach, Pollack and Tuza [*J. Combin. Theory* **B 47** (1989), 279–285] conjectured that the diameter of a K_{2r}-free connected graph of order n and minimum degree $\delta \geq 2$ is at most $\frac{2(r-1)(3r+2)}{(2r^2-1)} \cdot \frac{n}{\delta} + O(1)$ for every $r \geq 2$, if δ is a multiple of $(r-1)(3r+2)$. For every $r > 1$ and $\delta \geq 2(r-1)$, we create K_{2r}-free graphs with minimum degree δ and diameter $\frac{(6r-5)n}{(2r-1)\delta+2r-3} + O(1)$, which are counterexamples to the conjecture for every $r > 1$ and $\delta > 2(r-1)(3r+2)(2r-3)$. The rest of the paper proves positive results under a stronger hypothesis, k-colorability, instead of being K_{k+1}-free. We show that the diameter of connected k-colorable graphs with minimum degree $\geq \delta$ and order n is at most $\left(3 - \frac{1}{k-1}\right) \frac{n}{\delta} + O(1)$, while for $k = 3$, it is at most $\frac{57n}{23\delta} + O(1)$. (Received August 23, 2020)

1162-05-77 **Ron Aharoni**, **Eli Berger**, **Joseph Briggs**, **Erel Segal-Halevi** and **Shira Zerbib***, zerbib@iastate.edu. *Fractionally balanced hypergraphs and fair division.* Preliminary report.

An r-partite hypergraph is fractionally balanced if there exists a non-negative function on its edge set, that has constant degrees in each vertex side. Using a topological version of Hall's theorem due to Aharoni and Haxell (2000), we prove lower bounds on the matching number of such hypergraphs. This yields results in questions concerning fair division of cakes among hungry players. (Received August 24, 2020)

1162-05-87 **Michael Tait***, michael.tait@villanova.edu. *A meta-conjecture in extremal spectral graph theory.*

Let $ex(n, F)$ be the maximum number of edges in an F-free graph on n vertices and $spex(n, F)$ be the maximum spectral radius (of the adjacency matrix) of an F-free graph on n vertices. In this talk we discuss a conjecture regarding for which graphs F the extremal graphs for these two problems are the same. This is motivated by a recent theorem of ours which determines $spex(n, F)$ when F is a friendship graph. This is joint work with Sebi Cioaba, Lihua Feng, and Xiao-Dong Zhang (Received August 25, 2020)

1162-05-91 **Colin Defant***, 18434 Hancock Bluff Rd., Dade City, FL 08540, and **Kai Zheng**. *Stack-Sorting with Consecutive-Pattern-Avoiding Stacks.*

We introduce consecutive-pattern-avoiding stack-sorting maps SC_σ, which are natural generalizations of West's stack-sorting map s and natural analogues of the classical-pattern-avoiding stack-sorting maps s_σ recently introduced by Cerbai, Claesson, and Ferrari. We characterize the patterns σ such that $\mathrm{Sort}(\mathrm{SC}_\sigma)$, the set of permutations that are sortable via the map $s \circ \mathrm{SC}_\sigma$, is a permutation class, and we enumerate the sets $\mathrm{Sort}(\mathrm{SC}_\sigma)$ for $\sigma \in \{123, 132, 321\}$. We also study the maps SC_σ from a dynamical point of view, characterizing the periodic points of SC_σ for all $\sigma \in S_3$ and computing $\max_{\pi \in S_n} |\mathrm{SC}_\sigma^{-1}(\pi)|$ for all $\sigma \in \{132, 213, 231, 312\}$. In addition, we characterize the periodic points of the classical-pattern-avoiding stack-sorting map s_{132}, and we show that the maximum number of iterations of s_{132} needed to send a permutation in S_n to a periodic point is $n-1$. The new concepts introduced in this paper lead to several open problems and conjectures that we will discuss. (Received August 25, 2020)

1162-05-92 **Emily Yi Zhang*** (eyzhang@mit.edu), Madison, MS 39110. *On the Broadcast Dimension of a Graph.*

A function $f : V(G) \to \mathbb{Z}^+ \cup \{0\}$ is a *resolving broadcast* of a graph G if, for any distinct $x, y \in V(G)$, there exists a vertex $z \in V(G)$ with $f(z) > 0$ such that $\min\{d(x, z), f(z) + 1\} \neq \min\{d(y, z), f(z) + 1\}$. The *broadcast dimension* of G is the minimum of $\sum_{v \in V(G)} f(v)$ over all resolving broadcasts f of G. The concept of broadcast dimension was introduced by Geneson and Yi as a variant of metric dimension and has applications in areas such as network discovery and robot navigation.

In this paper, we derive an asymptotically tight lower bound on the broadcast dimension of an acyclic graph in the number of vertices, and we show that a lower bound by Geneson and Yi on the broadcast dimension of a general graph in the adjacency dimension is asymptotically tight. We also study the change in the broadcast dimension of a graph under a single edge deletion. We show that both the additive increase and decrease of the broadcast dimension of a graph under edge deletion is unbounded. Moreover, we show that under edge deletion, the broadcast dimension of any graph increases by a multiplicative factor of at most 3. These results fully answer three questions asked by Geneson and Yi. (Received August 25, 2020)

1162-05-101 **Karen Gunderson*** (karen.gunderson@umanitoba.ca), Department of Mathematics, Winnipeg, MB R3T 2N2, Canada. *Longest cycles in a preferential attachment random graph model.* Preliminary report.

In preferential attachment graph models, a graph is built vertex by vertex and when each new vertex arrives, it sends m new edges to older vertices, randomly, with vertices of high degree being more likely to be selected as a neighbour. One such example is the Buckley–Osthus random graph model which is a generalisation of the Barabási–Albert preferential attachment graph model. I will discuss new bounds on the size of the longest cycles and the largest matchings in the Buckley–Osthus random graphs. These results can be used to answer a question of Frieze, Pérez-Giménez, Prałat, and Reiniger on the Hamiltonicity of the Barabási–Albert preferential attachment graph model with parameter $m = 3$. Based on joint work with Michał Przykucki. (Received August 27, 2020)

1162-05-102 **Mihir Singhal*** (mihirs@mit.edu). *Families with no perfect matchings.*

We consider families of k-subsets of $\{1, \ldots, n\}$, where n is a multiple of k, which have no perfect matching. An equivalent condition for a family $\mathcal{F}$ to have no perfect matching is for there to be a *blocking set*, which is a set of b elements of $\{1, \ldots, n\}$ that cannot be covered by b disjoint sets in $\mathcal{F}$. We are specifically interested in the largest possible size of a family $\mathcal{F}$ with no perfect matching and no blocking set of size less than b. Frankl resolved the case of families with no singleton blocking set (in other words, the $b = 2$ case) for sufficiently large n and conjectured an optimal construction for general b. Though Frankl's construction fails to be optimal for $k = 2, 3$, we show that the construction is optimal whenever $k \geq 100$ and n is sufficiently large. (Received August 27, 2020)

1162-05-116 **Stephen J Young***, stephen.young@pnnl.gov. *Ramanujan Graphs and Supercomputing Topologies.* Preliminary report.

In modern supercomputing systems a significant fraction of the overall computation time is dedicated to interprocess communication. As a consequence, significant amount of hardware and algorithmic resources are devoted to decreasing the impact of communication congestion on process runtime. In this work we evaluate the potential applicability of Ramanujan graphs to design supercomputing topologies which are more robust to communication congestion. We specifically consider the ability of supercomputing topologies inspired by the explicit Ramanujan graph constructions given by Margulis as well as Lubotzky, Phillips, and Sarnak, to minimizing the communications overhead of supercomputing topologies. Joint work with Sinan Aksoy, Kevin Barker, Paul Bruillard, Tobias Hagge, Mark Kempton, Joseph Manzano, Carlos Ortiz Marrero, Mark Raugas, and Joshua Suetterlein. (Received August 28, 2020)

1162-05-119 **Michelle Delcourt*** (michelledelcourt@gmail.com) and **Luke Postle**. *Progress towards Nash-Williams' Conjecture on Triangle Decompositions.*

Partitioning the edges of a graph into edge disjoint triangles forms a triangle decomposition of the graph. A famous conjecture by Nash-Williams from 1970 asserts that any sufficiently large, triangle divisible graph on n vertices with minimum degree at least 0.75 n admits a triangle decomposition. In the light of recent results, the fractional version of this problem is of central importance. A fractional triangle decomposition is an assignment of non-negative weights to each triangle in a graph such that the sum of the weights along each edge is precisely one.

We show that for any graph on n vertices with minimum degree at least 0.827327 n admits a fractional triangle decomposition. Combined with results of Barber, Kühn, Lo, and Osthus, this implies that for every sufficiently large triangle divisible graph on n vertices with minimum degree at least 0.82733 n admits a triangle decomposition. This is a significant improvement over the previous asymptotic result of Dross showing the existence of fractional triangle decompositions of sufficiently large graphs with minimum degree more than 0.9 n. This is joint work with Luke Postle. (Received August 28, 2020)

1162-05-124 **Sinan Aksoy***, sinan.aksoy@pnnl.gov, and **Koby Hayashi**, **Cheonghee Park** and **Haesun Park**. *Hypergraph random walks, Laplacians, and clustering.*

We discuss several approaches for constructing Laplacian matrices for non-uniform hypergraphs. Recent work has suggested utilizing edge-dependent vertex weights to define non-reversible hypergraph random walks. Based on these random walks, we suggest several ways of constructing Laplacians and design clustering algorithms that utilize them as inputs. Using several data sets from real-life applications, we compare the performance of these clustering algorithms experimentally against a variety of existing hypergraph clustering methods. We conclude by highlighting avenues for future work. (Received August 28, 2020)

1162-05-125 **Debarun Ghosh**, **Ervin Győri** and **Ryan R Martin*** (rymartin@iastate.edu), Department of Mathematics, 428 Carver Hall, Iowa State University, Ames, IA 50011, and **Addisu Paulos** and **Chuanqi Xiao**. *Planar Turán number of the 6-cycle.*

Let $\mathrm{ex}_{\mathcal{P}}(n, T, H)$ denote the maximum number of copies of T in an n-vertex planar graph which does not contain H as a subgraph. When $T = K_2$, $\mathrm{ex}_{\mathcal{P}}(n, T, H)$ is the well studied function, the planar Turán number of H, denoted by $\mathrm{ex}_{\mathcal{P}}(n, H)$. The topic of extremal planar graphs was initiated by Dowden (2016). He obtained sharp upper bound for both $\mathrm{ex}_{\mathcal{P}}(n, C_4)$ and $\mathrm{ex}_{\mathcal{P}}(n, C_5)$. Later on, Y. Lan, et al. (2019) continued this topic and proved that $\mathrm{ex}_{\mathcal{P}}(n, C_6) \leq \frac{18(n-2)}{7}$. In this talk, we give a sharp upper bound $\mathrm{ex}_{\mathcal{P}}(n, C_6) \leq \frac{5}{2}n - 7$, for all $n \geq 18$, which improves Lan's result. (Received August 28, 2020)

1162-05-140 **P. Mark Kayll*** (mark.kayll@umontana.edu). *On coloring and D-coloring large-girth digraphs.*

Erdős' 1959 article establishing the existence of graphs with arbitrarily large girth and chromatic number not only exposed a facet of a diamond —the then-nascent probabilistic method—but also sparked a cottage industry refining and polishing his gem. Partly historical, partly contemporary, this talk traces a thread analogizing graphs to digraphs, generalizing coloring to homomorphing, and adding uniqueness to the mix. (Received August 30, 2020)

1162-05-153 **Steve Butler*** (butler@iastate.edu). *Hadamard diagonalizable graphs of order at most 36.*

If the Laplacian matrix of a graph has a full set of orthogonal eigenvectors with entries ± 1, then the matrix formed by taking the columns as the eigenvectors is a Hadamard matrix and the graph is said to be Hadamard

diagonalizable. By combination of theoretical tools and computational search all Hadamard diagaonalizable graphs through order 36 have been determined. We give an overview of the approach used.

This is joint work with Jane Breen, Melissa Fuentes, Bernard Lidický, Michael Phillips, Alexander W. N. Riasanovsky, Sung-Yell Song, Ralihe R. Villagrán, Cedar Wiseman, and Xiaohong Zhang. (Received August 30, 2020)

1162-05-159 **Gregory P. Wene*** (gpwene2011@hotmail.com), 7342 Oak Manor Dr, Apt 7109, San Antonio, TX 78229. *The Hypercube Semigroups.* Preliminary report.

An m-dimensional hypercube A, $m \geq 1$, is an array of n^m elements belonging to a field; the elements are denoted by $A_{ij\ldots u}$ where there are m subscripts, and each varies from 1 to n. Knuth (J. Algebra 2 (1965), 182-217) introduced hypercubes as a generalization of the display of the structure constants of a finite semifield. We examine the semigroups formed by these under binary compositions suggested by Knuth. (Received August 31, 2020)

1162-05-161 **Kai Zheng*** (kzheng@princeton.edu). *On the e-positivity of trees and spiders.*

We prove that the chromatic symmetric function of any tree with a vertex of degree at least six is not e-positive, that is, it cannot be written as a nonnegative linear combination of elementary symmetric functions. This makes significant progress towards a recent conjecture of Dahlberg, She, and van Willigenburg, who conjectured the result for the chromatic symmetric functions of all trees with a vertex of degree at least four. We also provide a series of conditions that can identify when the chromatic symmetric function of a spider, a tree consisting of multiple paths identified at an end, is not e-positive. These conditions generalize to trees and graphs with cut vertices as well. Finally, by applying a result of Orellana and Scott, we provide a method to inductively calculate certain coefficients in the elementary symmetric function expansion of the chromatic symmetric function of a spider, leading to further e-positivity conditions for spiders. (Received August 31, 2020)

1162-05-165 **Marthe Bonamy**, **Michelle Delcourt**, **Richard Lang** and **Luke Postle*** (lpostle@uwaterloo.ca). *Edge Coloring with Local List Sizes.*

The well-known List Colouring Conjecture from the 1970s states that for every graph G the chromatic index of G is equal to its list chromatic index. In a seminal paper in 1996, Kahn proved that the List Colouring Conjecture holds asymptotically. Our main result is a local generalization of Kahn's theorem. More precisely, we show that, for a graph G with sufficiently large maximum degree Δ and minimum degree $\delta \geq \ln^{25} \Delta$, the following holds: For every assignment of lists of colours to the edges of G, such that $|L(e)| \geq (1 + o(1)) \cdot \max\{\deg(u), \deg(v)\}$ for each edge $e = uv$, there is an L-edge-colouring of G. Furthermore, Kahn showed that the List Colouring Conjecture holds asymptotically for linear, k-uniform hypergraphs, and recently Molloy generalized this to correspondence colouring. We also prove a local version of Molloy's result. In fact, we prove a weighted version that simultaneously implies all of our results. (Received August 31, 2020)

1162-05-172 **Michael Ferrara*** (mferrara@nsf.gov), **Catherine Erbes**, **Nathan Graber** and **Paul Wenger**. *Stability Theorems for Degree Sequences of Graphs and Hypergraphs.* Preliminary report.

A graphic sequence π is *potentially H-graphic* if there is some realization of π that contains H as a subgraph. The Erdős-Jacobson-Lehel problem asks to determine $\sigma(H, n)$, the minimum even integer such that any n-term graphic sequence π with sum at least $\sigma(H, n)$ is potentially H-graphic. The parameter $\sigma(H, n)$ is known as the *potential function* of H, and can be viewed as a degree sequence variant of the classical extremal function $\mathrm{ex}(n, H)$.

In this talk, we investigate a stability concept for the potential number, inspired by Simonovits' classical result on the stability of the extremal function. We first define a notion of stability for the potential number that is a natural analogue to the stability given by Simonovits. However, under this definition, many families of graphs are not σ-stable, establishing a stark contrast between the extremal and potential functions. We then extend this notion to the degree sequences of uniform hypergraphs and provide some new results that demonstrate some interesting differences between the graph case and that for r-graphs with $r \geq 3$. (Received August 31, 2020)

1162-05-174 **Stephen G Hartke*** (stephen.hartke@ucdenver.edu) and **Luke Nelsen**. *Subcubic planar graphs are 7-square-choosable.*

The square of a graph G is the graph with vertex set V(G) where two vertices are adjacent if their distance in G is at most 2. Wegner (1977) conjectured that the square of any subcubic planar graph is 7-colorable, which was proved independently by Thomassen (2018) and Hartke, Jahanbekam, and Thomas (2019+). Cranston and

Kim (2008) asked whether the list chromatic number of the square of any subcubic planar graph is also at most 7. We answer this question in the affirmative, proving that all subcubic planar graphs are 7-square-choosable. This is joint work with Luke Nelsen. (Received August 31, 2020)

1162-05-180 **Blair D Sullivan*** (sullivan@cs.utah.edu). *Approximation Algorithms for Graphs Near Structured Classes.*

We describe a framework for generalizing approximation algorithms for structural graph classes so that they apply to graphs "close" to a class (a scenario we expect is common when working with real-world networks) while still guaranteeing approximation ratios. The "structural rounding" framework edits graphs into an algorithmically tractable class, applies a class-specific approximation algorithm, then lifts the partial solution to the original graph. We give a general characterization of when an optimization problem is amenable to this approach, and show that it includes many well-studied graph problems, such as Independent Set, Vertex Cover, Feedback Vertex Set, Minimum Maximal Matching, Chromatic Number, (ℓ-)Dominating Set, Edge (ℓ-)Dominating Set, and Connected Dominating Set. Further, we describe recent experimental evaluation of this framework and benchmark it against standard heuristics and approximation techniques. (Received August 31, 2020)

1162-05-185 **Michael Ren*** (mren36@mit.edu). *Wilf Equivalences and Stanley-Wilf Limits for Patterns in Rooted Labeled Forests.*

Building off recent work of Garg and Peng, we continue the investigation into classical and consecutive pattern avoidance in rooted forests, resolving some of their conjectures and questions and proving generalizations whenever possible. Through extensions of the forest Simion-Schmidt bijection introduced by Anders and Archer, we demonstrate a new family of forest-Wilf equivalences, completing the classification of forest-Wilf equivalence classes for sets consisting of a pattern of length 3 and a pattern of length at most 5. We also find a new family of nontrivial c-forest-Wilf equivalences between single patterns using the forest analogue of the Goulden-Jackson cluster method, showing that a $(1-o(1))^n$-fraction of patterns of length n satisfy a nontrivial c-forest-Wilf equivalence and that there are c-forest-Wilf equivalence classes of patterns of length n of exponential size. Finally, we prove a forest analogue of the Stanley-Wilf conjecture for avoiding a single pattern as well as certain other sets of patterns. Our techniques are analytic, easily generalizing to different types of pattern avoidance and allowing for computations of convergent lower bounds of the forest Stanley-Wilf limit in the cases covered by our result. (Received August 31, 2020)

1162-05-193 **Sebastian M Cioaba***, Department of Mathematical Sciences, Ewing Hall, Newark, DE 19716-2553. *Eigenvalues and linear programming bounds for regular graphs and hypergraphs.* Preliminary report.

The spectrum of a graph is closely related to many graph parameters. In particular, the spectral gap of a regular graph which is the difference between its valency and second eigenvalue, is widely seen an algebraic measure of connectivity and plays a key role in the theory of expander graphs. In this paper, we extend previous work done for graphs and bipartite graphs and present a linear programming method for obtaining an upper bound on the order of a regular uniform hypergraph with prescribed distinct eigenvalues. Furthermore, we obtain a general upper bound on the order of a regular uniform hypergraph whose second eigenvalue is bounded by a given value. Our results improve or generalize some results obtained by Feng–Li (1996) in the context of Alon-Boppana theorems for regular hypergraphs and by Dinitz–Schapira–Shahaf (2020) in the context of the Moore or degree-diameter problem. We also determine the largest order of an r-regular u-uniform hypergraph with second eigenvalue at most θ for several parameters (r, u, θ). In particular, orthogonal arrays give the structure of the largest hypergraphs with second eigenvalue at most 1 for every sufficiently large r. This is joint work with Jack Koolen, Masato Mimura, Hiroshi Nozaki and Takayuki Okuda. (Received August 31, 2020)

1162-05-196 **Alexander Riasanovsky*** (awnr@iastate.edu) and **Ryan Martin** (rymartin@iastate.edu). *On the edit distance function of random graphs.*

Given a hereditary property of graphs $\mathcal{H}$ and a $p \in [0,1]$, the edit distance function $\mathrm{ed}_{\mathcal{H}}(p)$ is asymptotically the maximum proportion of edge-additions plus edge-deletions applied to a graph of edge density p sufficient to ensure that the resulting graph satisfies $\mathcal{H}$. The edit distance function is directly related to other well-studied quantities such as the speed function for $\mathcal{H}$ and the $\mathcal{H}$-chromatic number of a random graph.

Let $\mathcal{H}$ be the property of forbidding an Erdős-Rényi random graph $F \sim \mathbb{G}(n_0, p_0)$, and let φ represent the golden ratio. In this paper, we show that if $p_0 \in [1 - 1/\varphi, 1/\varphi]$, then **a.a.s.** as $n_0 \to \infty$,

$$\mathrm{ed}_{\mathcal{H}}(p) = (1 + o(1)) \frac{2 \log n_0}{n_0} \cdot \min \left\{ \frac{p}{-\log(1-p_0)}, \frac{1-p}{-\log p_0} \right\}.$$

Moreover, this holds for $p \in [1/3, 2/3]$ for any $p_0 \in (0,1)$. (Received August 31, 2020)

1162-05-201 **Ivan Contreras** and **Andrew Tawfeek*** (atawfeek21@amherst.edu), 0401 Keefe Campus Center, Amherst College, Amherst, MA 01002. *Enumeration of Forman Equivalence Classes on Graphs.*

Discrete Morse theory has been developed over the past few decades, since its original formulation by Forman in 1998. We provide a novel approach to enumerating the Forman equivalence classes of discrete Morse functions on finite simple graphs and explicitly show that their generating function is given by the characteristic polynomial of the graph Laplacian Δ – and illustrate what this connection says about acyclic matching occurring on the Hasse diagram of the graph. Furthermore, we provide a discussion of our current research on generalizing our results to Forman equivalence classes on higher-dimensional simplicial complexes – and what can be said about the generating function in larger dimensions. (Received August 31, 2020)

1162-05-202 **Shannon Dillman** and **Franklin Kenter*** (kenter@usna.edu). *Leaky Forcing: A New Variation of Zero Forcing.*

Zero forcing is a one-player game played on a graph. The player chooses some set of vertices to color, then iteratively applies a color change rule: If all but one of a colored vertex's neighbors are colored, color (i.e. "force") the remaining uncolored neighbor. The goal is to find the minimum number of vertices to initially color so that all vertices eventually become colored. Equivalent formations of zero forcing have been developed including sensor allocation to solve linear systems (K.-Lin 2018), controllability in follower-leader dynamics (Monshizadeh-Zhang-Camlibel 2014), and edge covering in specific hypergraphs (Brimkov-Fast-Hicks 2016).

We ask what if there is a juncture in a network that has a leak? In the context of zero forcing this corresponds to the following variation we call ℓ-forcing: Given ℓ, find a set of vertices such that for *any* set of ℓ vertices that are unable to force, all vertices will still be colored.

We compute the ℓ-forcing number for selected families of graphs including grid graphs and find examples where adding edges make the graph more "resilient" to leaks. Further, we implement known computational methods for our new leaky forcing variation. (Received August 31, 2020)

1162-05-208 **Juergen Kritschgau***, Jkritsch@iastate.edu, and **Abhishek Methuku**, **Michael Tait** and **Craig Timmons**. *Generalized Saturation Problems.*

A graph G is F-saturated if G is F-free and the addition of any non-edge to G creates a copy of F. Generalized saturation problems consider minimizing the number of copies of some graph H in an F-saturated graph on n vertices. This talk will discuss some results and open problems in generalized saturation. (Received August 31, 2020)

1162-05-210 **Katalin Berlow***, kberlow@andrew.cmu.edu. *Restricted Stacks as Functions.*

The stack sort algorithm has been the subject of extensive study over the years. In this paper we explore a generalized version of this algorithm where instead of avoiding a single decrease, the stack avoids a set T of permutations. We let s_T denote this map. We classify for which sets T the map s_T is bijective. A corollary to this answers a question of Baril, Khalil, and Vajnovszki about stack sort composed with $s_{\{\sigma,\tau\}}$, known as the (σ,τ)-machine. This fully classifies for which σ and τ the preimage of the identity under the (σ,τ)-machine is counted by the Catalan numbers. We also prove that the number of preimages of a permutation under the map s_T is bounded by the Catalan numbers, with a shift of indices. For T of size 1, we classify exactly when this bound is sharp. We also explore the periodic points and maximum number of preimages of various s_T for T containing two length 3 permutations. (Received August 31, 2020)

1162-05-211 **Fan Zhou*** (fanzhou@college.harvard.edu). *On Eventually Periodic Sets as Minimal Additive Complements.* Preliminary report.

For subsets C, W of $\mathbb{Z}$, C is said to be a minimal additive complement to W if $C+W = \{c+w : c \in C, w \in W\} = \mathbb{Z}$ and $C' + W \neq \mathbb{Z}$ for any proper subset $C' \subset C$. In their recent paper, Burcroff and Luntzlara studied the conditions under which "eventually periodic sets", which are of form $(m\mathbb{N}+A)\cup B\cup F$ (for finite A, B, F, where B is in the same congruence classes mod m as A and F is in different congruence classes than A), arose as minimal additive complements in $\mathbb{Z}$. In the present paper we shall study this question further. For example, we give, in the form of bounds on the period m, some sufficient conditions for an eventually periodic set to be a minimal additive complement; in particular we show that "all eventually periodic sets are eventually minimal additive complements". Moreover, we generalize this to a framework in which "patterns" of points are projected down to $\mathbb{Z}$, and we show that all sets which arise this way are eventually minimal additive complements. We also introduce a formalism of formal power series, which serves purely as a bookkeeper in writing down proofs. (Received August 31, 2020)

1162-05-214 **Jianping Pan*** (jppan@math.ucdavis.edu), One Shields Ave, Davis, CA 95616, and **Joseph Pappe**, **Wencin Poh** and **Anne Schilling**. *Uncrowding algorithm for hook-valued tableaux.* Preliminary report.

Whereas set-valued tableaux are the combinatorial objects associated to stable Grothendieck polynomials, hook-valued tableaux are associated to stable canonical Grothendieck polynomials. In this paper, we define a novel uncrowding algorithm for hook-valued tableaux. The algorithm "uncrowds" the entries in the arm of the hooks and yields a set-valued tableau. We prove that our uncrowding algorithm intertwines with crystal operators. (Received August 31, 2020)

1162-05-221 **Rachel Kirsch** (kirsch@iastate.edu) and **Jamie Radcliffe*** (jamie.radcliffe@unl.edu). *Counting cliques in degree-bounded hypergraphs.* Preliminary report.

There has been a lot of recent progress on the problem of giving an upper bound on the number of k-cliques in a graph with maximum degree at most r. I'll describe preliminary work on the same problem for hypergraphs. (Received September 01, 2020)

1162-05-225 **Abdul Basit*** (basit.abdul@gmail.com), **Artem Chernikov**, **Sergei Starchenko**, **Terence Tao** and **Chieu-Minh Tran**. *Zarankiewicz's problem for semilinear hypergraphs.*

A bipartite graph $H = (V_1, V_2; E)$ with $|V_1| + |V_2| = n$ is *semilinear* if $V_i \subseteq \mathbb{R}^{d_i}$ for some d_i and the edge relation E consists of the pairs of points $(x_1, x_2) \in V_1 \times V_2$ satisfying a fixed Boolean combination of q linear equalities and inequalities in $d_1 + d_2$ variables for some q. We show that for a fixed k, the number of edges in a $K_{k,k}$-free semilinear H is almost linear in n, namely $|E| = O_{q,k}(n^{1+\varepsilon})$ for any $\varepsilon > 0$; and more generally, $|E| = O_{q,k,r}(n^{r-1+\varepsilon})$ for a $K_{k,\dots,k}$-free semilinear r-partite r-uniform hypergraph.

As an application, we obtain the following incidence bound: given n points and m open boxes with axis parallel sides in $\mathbb{R}^d$ such that their incidence graph is $K_{k,k}$-free, there can be at most $O_k((n+m)^{1+\varepsilon})$ incidences. Instead of boxes one can take polytopes cut out by the translates of an arbitrary fixed finite set of halfspaces. (Received September 01, 2020)

1162-05-234 **Ilani S Axelrod-Freed*** (ilani_af@mit.edu). *312-Avoiding Reduced Valid Hook Configurations and Duck Words.*

Valid hook configurations are combinatorial objects used to understand West's stack sorting map as well as cumulants in noncommutative probability theory. We show a bijection between reduced valid hook configurations on 312-avoiding permutations with the maximal allowed number of points and 3D-Dyck words, proving a conjecture of Sankar's. We extend to a bijection between all 312-avoiding reduced valid hook configurations with 3D-Dyck words with specified modifications. We show how these can be counted in terms of the number of 3D-Dyck words of length $3k$ in which exactly i Y's do not have an X immediately before them, the (k, i) Duck words, and use this relationship to prove several properties about sums of 312-avoiding reduced valid hook configurations, including two more of Sankar's conjectures. We also show that the number of $(k, 1)$ Duck words is given by a variant of the tennis ball numbers. (Received September 01, 2020)

1162-05-241 **József Balogh**, **Nathan Lemons** and **Cory Palmer*** (cory.palmer@umontana.edu). *Maximum size intersecting families of bounded minimum positive co-degree.*

Let $\mathcal{H}$ be an r-uniform hypergraph. The *minimum positive co-degree* of $\mathcal{H}$, denoted $\delta^+_{r-1}(\mathcal{H})$, is the minimum k such that if S is an $(r-1)$-set contained in a hyperedge of $\mathcal{H}$, then S is contained in at least k distinct hyperedges of $\mathcal{H}$.

In this talk we determine the maximum possible size of an intersecting r-uniform n-vertex hypergraph with minimum positive co-degree $\delta^+_{r-1}(\mathcal{H}) \geq k$ and characterize the unique hypergraph attaining this maximum, for n sufficiently large. We will also discuss motivations for this new hypergraph degree condition and its connection to existing hypergraph parameters. (Received September 02, 2020)

14 ► Algebraic geometry

1162-14-59 **Owen Finn Barrett*** (ofb@uchicago.edu), Univ of Chicago, 5734 S University Ave, Chicago, IL 60637. *The derived category of the abelian category of constructible sheaves.*

Nori proved in 2002 that given a complex algebraic variety X, the bounded derived category of the abelian category of constructible sheaves on X is equivalent to the usual triangulated category $D(X)$ of bounded constructible complexes on X. He moreover showed that given any constructible sheaf $\mathcal{F}$ on $\mathbf{A}^n$, there is an injection $\mathcal{F} \hookrightarrow \mathcal{G}$ with $\mathcal{G}$ constructible and $H^i(\mathbf{A}^n, \mathcal{G}) = 0$ for $i > 0$.

In this talk, I will explain how to extend Nori's theorem to the case of a variety over an algebraically closed field of positive characteristic, with Betti constructible sheaves replaced by ℓ-adic sheaves. This is the case $p = 0$ of the general problem which asks whether the bounded derived category of p-perverse sheaves is equivalent to $D(X)$, resolved affirmatively for the middle perversity by Beilinson, and has possible applications to a future motivic sheaf theory and to defining a six-functor formalism for algebraic stacks. (Received August 21, 2020)

1162-14-73 **Grégoire Naisse***, gregoire.naisse@gmail.com, **Jens Niklas Eberhardt**, mail@jenseberhardt.com, and **Arik Wilbert**, arik.wilbert@uga.edu. *Real Springer fibers and odd Khovanov homology.*

The main ingredient for Khovanov link homology is a certain TQFT associated to the Frobenius algebra $\mathbb{Z}[X]/(X^2)$. It is known that this TQFT can be obtained from the geometry of the complex projective line $\mathbb{C}P(1)$. Furthermore, Khovanov homology can be extended to tangles through the use of arc algebras. These algebras can be constructed as convolution algebras over the components of the complex two-row Springer fibers. In another direction, Khovanov homology can be deformed to give an odd version (basically replacing tensors product with exterior ones). I'll explain how this odd Khovanov homology is related to the geometry of real spaces instead of complex ones. This is a joint work with Jens Niklas Eberhardt and Arik Wilbert. (Received August 24, 2020)

15 ► Linear and multilinear algebra; matrix theory

1162-15-22 **Bryan Curtis**, **Bryan Shader*** (bshader@uwyo.edu) and **Jason Williford**. *Sign patterns of orthogonal matrices.*

At one of the earliest conferences on combinatorics, Miroslav Fielder put forth the challenge of characterizing the sign-patterns of real orthogonal matrices. Today, over 70 years later, that challenge remains largely unmet. This talk will discuss some recent advances that utilize a mix of analysis, matrix theory, and combinatorics. In particular, it is shown that if A is the adjacency matrix of a doubly regular tournament, then each sign-pattern obtained from the $I - A$ by replacing some of its zeros by $+1$ or -1 is the sign-pattern of an orthogonal matrix. (Received August 12, 2020)

1162-15-46 **Shaun M Fallat*** (shaun.fallat@uregina.ca), Department of Mathematics and Statistics, University of Regina, Regina, Sask S4N5C3, Canada. *Variants of Maximum Nullity and Zero Forcing on Trees.*

For a graph G and a nonnegative integer q, we let $S_q(G)$ denote the subset of all real symmetric matrices in $S(G)$ having q negative eigenvalues. We define $M_q(G)$ to be the maximum nullity over the set $S_q(G)$. As with the standard maximum nullity parameter for G, a q-analogue of zero forcing exists, known as $Z_q(G)$, and satisfies $M_q(G) \leq Z_q(G)$. The Z_q game on a graph is an adaption of conventional zero forcing involving a second player and a third forcing option.

In this talk, we present a suite of results and properties about these parameters in the case of trees with particular attention when $q = 1$. (Received August 19, 2020)

1162-15-50 **Christina Pospisil*** (pospisil.christina@gmx.de), pospisil.christina@gmx.de. *Generalization Theory of Linear Algebra II.*

This talk continues Generalization Theory of Linear Algebra I (first part was presented at JMM 2019) and provides inverses for non-injective mappings in multiple dimensions, inverses for non-surjective mappings in one and multiple dimensions and introduces a general determinant theory (second part was presented at JMM 2020). In future work there will be further operations and applications to physics and other natural sciences be explored. (Received August 20, 2020)

1162-15-150 **Jane Breen*** (janebreen@ontariotechu.ca). *Kemeny's constant and random walks on graphs.*

Kemeny's constant is an interesting and useful quantifier of how well-connected the states of a Markov chain are, and is calculated using the eigenvalues of the transition matrix. By considering the random walk on a simple, undirected graph, and the eigenvalues of the normalized Laplacian matrix of the graph, we can compute Kemeny's constant and regard this value as a graph parameter with a concrete interpretation in terms of the expected length of a random trip in the graph. In this talk we give a survey of known results, consider extremal graphs where Kemeny's constant is largest-possible, and present new techniques in spectral graph theory which facilitate the computation of Kemeny's constant for these graphs. (Received August 30, 2020)

16 ► Associative rings and algebras

1162-16-17 **Xin Tang***, Math & Computer Science, Fayetteville State U, 1200 Murchison Road, and **James J. Zhang** and **Xiangui Zhao**. *Cancellation Problems of Morita and Skew Types.*

In this talk, we will present a couple of results on the problem concerning the Morita and skew cancellation for non-commutative algebras. In particular, we will explore what classes of non-commutative algebras are indeed Morita cancellative (respectively, skew cancellative). Several new results concerning these two types of cancellations, as well as the classical cancellation, will be proved. This is joint work with James J. Zhang and Xiangui Zhao. (Received August 10, 2020)

1162-16-35 **Ramy Yammine***, ramy.yammine@temple.edu. *On actions of connected Hopf algebras.*

Let H be a connected Hopf algebra acting on an algebra A. Working over a base field having characteristic 0, we show that for a given prime (semi-prime, completely prime) ideal I of A, the largest H-stable ideal of A contained in I is also prime (semi-prime, completely prime). We also prove a similar result for certain subrings of convolution algebras. (Received August 17, 2020)

1162-16-37 **Daniel Rogalski**, **Robert Won** and **James Zhang*** (zhang@math.washington.edu), Department Of Mathematics, Box 354350, Seattle, WA 98195-0001. *Residue complexes and the Brown-Goodearl conjecture.*

Brown and Goodearl conjectured that every noetherian Hopf algebra is Artin-Schelter Gorenstein. This conjecture is known to be true for many cases, in particular, for affine polynomial identity Hopf algebras. Weak Hopf algebras are an important generalization of Hopf algebras, and the category of modules over a weak Hopf algebra has a monoidal structure. Let W be a weak Hopf algebra that is a finitely generated module over its affine center. We prove that W has finite self-injective dimension and is a direct sum of Artin-Schelter Gorenstein algebras. Therefore Brown-Goodearl conjecture holds in this special weak Hopf setting. The main tool for the proof is the residue complex introduced by Yekutieli. We will also give some motivations and consequences of Brown-Goodearl conjecture. (Received August 17, 2020)

1162-16-41 **K. R. Goodearl*** (goodearl@math.ucsb.edu) and **S. Launois** (s.launois@kent.ac.uk). *Catenarity in Quantum Nilpotent Algebras.*

A guiding principle in the study of quantized coordinate rings is the expectation that these algebras display properties which parallel those of their classical counterparts. We will discuss the important property of *catenarity*, namely that all saturated chains of prime ideals between any two fixed primes have the same length. It holds in the classical coordinate rings of affine varieties and has been extended to many quantized coordinate rings. Recently, catenarity has been established for the large class of quantum nilpotent algebras. (Received August 18, 2020)

1162-16-45 **Nicholas Davidson**, **Jonathan Kujawa*** (kujawa@math.ou.edu) and **Robert Muth**. *Webs and Howe duality in type P.*

For the type P Lie superalgebras we introduce combinatorial web categories which completely describe the monoidal categories generated by the symmetric powers of the natural representation and their duals. Using these categories we also establish a Howe duality between pairs of type P Lie superalgebras. (Received August 19, 2020)

1162-16-49 **Chelsea Walton**, **Elizabeth Wicks** and **Robert Won*** (robwon@uw.edu). *Algebraic structures in comodule categories over weak bialgebras.*

A weak bialgebra is both an algebra and a coalgebra satisfying compatibility conditions which are weaker than those defining a bialgebra. These weakened compatibility conditions still ensure that the categories of modules and comodules over a weak bialgebra admit a monoidal structure. We will discuss algebra objects and coalgebra objects in comodule categories over weak bialgebras, and their relationships to comodule algebras and comodule coalgebras. This work is joint with Chelsea Walton and Elizabeth Wicks. (Received August 19, 2020)

1162-16-55 **Joshua Sussan*** (joshuasussan@gmail.com). *p-DG structures in higher representation theory.*

The study of p-DG algebras arises naturally when trying to category quantum groups at prime roots of unity. We will review this framework and construct some categorical representations for the case of sl(2). (Received August 21, 2020)

1162-16-56 **Izuru Mori** and **Adam Nyman*** (adam.nyman@wwu.edu). *An abstract characterization of noncommutative projective spaces.* Preliminary report.

Let k denote a base field and let C denote a k-linear abelian category. I. Mori and K. Ueyama give a homological characterization for C to be a noncommutative projective space, i.e. a category of the form $\mathsf{cohproj}A$ where A is an AS-regular right coherent $\mathbb{Z}$-graded algebra over k. We describe a $\mathbb{Z}$-algebra generalization of this result that is broad enough to encompass the noncommutative deformations of $\mathbb{P}^1 \times \mathbb{P}^1$ discovered by M. Van den Bergh. We apply our result to prove that various noncommutative smooth quadrics are noncommutative deformations of $\mathbb{P}^1 \times \mathbb{P}^1$. (Received August 21, 2020)

1162-16-61 **Jason Gaddis**, **Padmini Veerapen** and **Xingting Wang*** (xingting.wang@howard.edu). *Reflection groups and rigidity of quadratic Poisson algebras.*

In this talk, we discuss the invariant theory of quadratic Poisson algebras. Let G be a finite group of the graded Poisson automorphisms of a quadratic Poisson algebra A. When the Poisson bracket of A is skew-symmetric, a Poisson version of the Shephard-Todd-Chevalley theorem is proved stating that the fixed Poisson subring A^G is skew-symmetric if and only if G is generated by reflections. For many other well-known families of quadratic Poisson algebras, we show that G contains limited or even no reflections. This kind of Poisson rigidity result ensures that the corresponding fixed Poisson subring A^G is not isomorphic to A as Poisson algebras unless G is trivial. This is a joint work with Jason Gaddis and Padmini Veerapen. (Received August 22, 2020)

1162-16-80 **Kenneth Chan**, **Jason Gaddis*** (jdgaddis@gmail.com), **Robert Won** and **James Zhang**. *Reflexive hull discriminants of quantum generalized Weyl algebras.* Preliminary report.

We introduce the reflexive hull discriminant as a tool to study invariant theory questions for algebras that are finitely generated, but not necessarily free, over their centers. We compute this discriminant for quantum generalized Weyl algebras and use this to determine their automorphism groups. This is joint work with Kenneth Chan, Robert Won, and James Zhang. (Received August 24, 2020)

1162-16-85 **Van C. Nguyen***, U.S. Naval Academy, Department of Mathematics, Annapolis, MD 21402. *McKay matrices for finite-dimensional Hopf algebras.*

In 1980s, McKay showed that for the finite subgroups G of SU_2, the (so-called McKay) matrices that record the result of tensoring the simple G-modules with the natural G-module $V = \mathbb{C}^2$ exactly correspond to the affine Cartan matrices of type A, D, E. This result, subsequently referred to as the *McKay correspondence*, has been an inspiration for much work on a host of topics in singularity theory, group theory, orbifolds, and many other subjects.

Generalizing from the group algebra case, the purpose of this talk is to study the McKay matrices defined over an arbitrary finite-dimensional Hopf algebra. For a finite-dimensional Hopf algebra A, the McKay matrix M_V of an A-module V encodes the relations for tensoring the simple A-modules with V. We prove general results about McKay matrices, their eigenvalues, and their (left and right) eigenvectors by using the coproduct and the characters of simple and projective A-modules. We illustrate these results for the Drinfeld double D_n of the Taft algebra by deriving expressions for the eigenvalues and eigenvectors of M_V in terms of several kinds of Chebyshev polynomials.

This is joint work with Georgia Benkart, Rekha Biswal, Ellen Kirkman, and Jieru Zhu. (Received August 25, 2020)

1162-16-94 **Mee Seong Im** and **Chun-Ju Lai***, 1, Sec. 4, Roosevelt Rd, Taipei City, 10617, Taiwan, and **Arik Wilbert**. *Springer fibers via quiver varieties using Maffei-Nakajima isomorphism.*

It is a remarkable theorem by Maffei-Nakajima that the Slodowy variety, which consists of certain complete flags, can be realized as certain Nakajima quiver variety of type A. The isomorphism is known to be rather implicit as it takes to solve a system of equations in which variables are linear maps. In this talk, we will talk about an explicit and efficient way to realize these quiver varieties in terms of complete flags in the corresponding Slodowy varieties. As an application, we provide an explicit description of irreducible components of two-row Springer fibers in terms of a family of kernel relations via quiver representations, which allows us to formulate a characterization of irreducible components of Springer fibers of classical type. This is a joint work with Mee Seong Im and Arik Wilbert. (Received August 26, 2020)

1162-16-106 **Daniel Yee*** (yeedo@uwec.edu) and **Jason Gaddis**. *Examples of Congenial Algebras.*
If an algebra satisfies the congeniality property, then the algebra must satisfy the noncommutative version of Auslander's Theorem: an isomorphism between the skew group algebra and the endomorphism ring over the invariant subalgebra. Since Auslander's Theorem is a well known condition in noncommutative invariant theory, our goal was to identify algebras with congeniality. Thus, we study congeniality defined in the work of Bao, He, and Zhang (2016) and discovered several examples of algebras that satisfy this property. One of the examples we will explore is the quantized Weyl algebra. (Received August 28, 2020)

1162-16-111 **Bach Nguyen**, **Kurt Trey Trampel*** (ktrampel@nd.edu) and **Milen Yakimov**. *Quantum cluster algebras at roots of unity.*
Quantum cluster algebras at roots of unity are introduced, and standard cluster theory results such as the quantum Laurent phenomenon are obtained. Examples of these algebras include quantum cluster algebras specialized to a root of unity. For roots of unity with nice compatibility, we present a canonical central subalgebra. We also present a general theorem on the structure of discriminants of these quantum algebras. These discriminants encode representation theoretic information about these algebras. The discriminants of quantum Schubert cells for any symmetrizable Kac-Moody algebra are found as an example of the general theorem. (Received August 28, 2020)

1162-16-131 **Jonas T Hartwig** and **Daniele Rosso*** (drosso@iu.edu). *Grothendieck rings of towers of twisted generalized Weyl algebras.*
Twisted generalized Weyl algebras (TGWAs) are a family of algebras that includes as special cases many algebras of interest in representation theory, like Weyl algebras and certain quotients of universal enveloping algebras. Their definition depends on the choice of a base ring R and certain parameters σ, t. In general TGWAs do not have a natural bialgebra nor Hopf algebra structure, however we show that, if we fix R and σ, there is an algebra map $A(R, \sigma, tt') \to A(R, \sigma, t) \otimes_R A(R, \sigma, t')$. This allows us to define a tensor product operation for modules of different TGWAs and an algebra structure on the direct sum (over all t) of the Grothendieck groups of their categories of weight modules. We discuss examples of this construction and describe the resulting algebras. This is joint work with Jonas Hartwig. (Received August 29, 2020)

1162-16-152 **Benjamin Briggs*** (briggs@math.utah.edu) and **Sarah Witherspoon**. *The Hochschild cohomology of twisted tensor products.*
The bicharacter twisted tensor product of graded algebras is an interesting source of quantum behaviour, often producing especially computable examples, such as the quantum complete intersections.

I will explain how to compute the Hochschild cohomology of a twisted tensor product in terms of the Hochschild cohomology of the two component algebras. This description generalises a result of Bergh and Oppermann. I'll try to give lots of examples, showing how this can be used to give simple calculations of Hochschild cohomology in cases where the computations were previous quite complicated.

This is joint work with Sarah Witherspoon. (Received August 30, 2020)

1162-16-189 **Zajj B Daugherty*** (zdaugherty@gc.cuny.edu), The City College of New York, Department of Mathematics, 160 Convent Ave, New York, NY 10031, and **Nicolle Gonzáles**, **Lisa Schneider** and **Louise Sutton**. *Degenerate two boundary BMW algebras.* Preliminary report.
The Birman-Murakami-Wenzl (BMW) algebras and their generalizations arise both as diagram algebras and as algebras of operators that preserve symmetry in tensor products of simple modules for symplectic and orthogonal Lie algebras and quantum groups. Their diagrammatic presentations are that of tangles, potentially in spaces with one or more puncture. They are fundamental to realizing certain knot and link invariants, but also have ties into lattice models in Statistical Mechanics. In this talk, we will introduce specifically the degenerate two-boundary BMW algebras and explore some of their algebraic and combinatorial structure. (Received August 31, 2020)

1162-16-231 **Arkady Berenstein** and **Jacob Greenstein*** (jacobg@ucr.edu). *Quantum Grothendieck resolutions and their representations.* Preliminary report.
In this talk we discuss classical, quantum and representation theoretic aspects of Grothendieck resolutions of a semisimple Lie group and its Poisson dual group. The link between the algebraic (respectively, quantum) versions of these resolutions is provided by the algebraic (respectively, quantum version) of the celebrated Semenov-Tian-Shansky map which is the core of our constructions. (Received September 01, 2020)

1162-16-233 **Nicolas Andruskiewitsch**, **Ivan Angiono** and **Milen Yakimov*** (yakimov@math.lsu.edu), Department of Mathematics, Louisiana State University, Baton Rouge, LA 70808. *Poisson geometry of large quantum groups.*

In this talk we will address the classification of irreducible representations of the Drinfeld doubles of the distinguished pre-Nichols algebras associated to finite dimensional Nichols algebras of diagonal type (large quantum groups in short). This is a large class of Hopf algebras which are module finite over their centers. It includes all quantum supergroups at roots of unity as special cases. On each of them we construct a structure of Poisson order in the sense of De Concini-Kac-Procesi and Brown-Gordon. Unlike the previous arguments for quantum groups at roots of unity which used low rank computations, our constructions are based on general arguments with dualities between specializations of integral forms of Hopf algebras and Poisson geometric arguments. (Received September 01, 2020)

17 ► Nonassociative rings and algebras

1162-17-120 **Cody A Tipton*** (cat1184@uw.edu). *Frobenius A_∞ Algebras.*

We will present the definition of a Frobenius A_∞- algebra that extends the classical notation of Frobenius algebras and is a generalization of cyclic A_∞-algebras. The latter being symmetric Frobenius A_∞-algebras that are already extensively studied because of their connection with Fukaya categories. Then we will extend some classical results of Frobenius algebras to Frobenius A_∞-algebras. (Received August 28, 2020)

1162-17-148 **David E. V. Rose*** (davidrose@unc.edu) and **Logan Tatham**. *Webs in type C.*

Web categories are generators-and-relations descriptions of categories of quantum group representations. We will review known descriptions of such categories (in low rank and type A) before discussing a conjectural description of the web category in type C, rank 3. We will present a number of results in support of this conjecture, and (time-permitting) will touch on work in progress that should lead to its resolution. (Received August 30, 2020)

1162-17-157 **Lee A Jenkins*** (lee.jenkins25@uga.edu), 200 D.W. Brooks Drive, Boyd Graduate Studies Research Center, Athens, GA 30602, and **Daniel K Nakano**. *The Nilpotent Cone for Classical Lie Superalgebras.*

In this talk, the speaker will discuss an analogue of the nilpotent cone, $\mathcal{N}$, for a classical Lie superalgebra, $\mathfrak{g}$, that generalizes the definition for the nilpotent cone for semisimple Lie algebras. For a classical simple Lie superalgebra, $\mathfrak{g} = \mathfrak{g}_{\bar{0}} \oplus \mathfrak{g}_{\bar{1}}$ with Lie $G_{\bar{0}} = \mathfrak{g}_{\bar{0}}$, we will show that there are finitely many $G_{\bar{0}}$-orbits on $\mathcal{N}$. It will also be shown that the Duflo-Serganova commuting variety, $\mathcal{X}$, is contained in $\mathcal{N}$ for any classical simple Lie superalgebra. Consequently, the finiteness result we discuss generalizes and extends the work of Duflo-Serganova on the commuting variety. (Received August 31, 2020)

1162-17-192 **Vera Serganova*** (serganov@math.berkeley.edu), 2318 Tulare ave, El Cerrito, CA, CA 94530. *The second real form of the category of vector superspaces.* Preliminary report.

Symmetric monoidal category of superspaces (over complex numbers) has two real forms in the sense of Etingof–Gelaki. One of these forms is the category of superspaces over R. We will discuss the second real form, which we call HVect. In particular, we classify simple Lie algebra objects in HVect.

We show that studying HVect has interesting consequences in the theory of Lie superalgebras. The existence of a compact form for all basic superalgebras (missing in the standard theory) is one of these advantages. The theory of symmetric superspaces gives another application of the category HVect.

The talk is a part of joined project with S. Sahi and H. Salmasian. (Received August 31, 2020)

18 ► Category theory; homological algebra

1162-18-33 **Cris Negron*** (cnegron@email.unc.edu). *Cohomology for Drinfeld doubles of finite group schemes.*

Recall that the Drinfeld double of a finite group scheme G is a finite-dimensional Hopf algebra which integrates, in a certain way, both the group ring of G and the algebra of global functions on G. I will discuss our recent proof of the fact that the Drinfeld double of an arbitrary finite group scheme has finitely generated cohomology. (One should view this result in light of Etingof and Ostrik's conjecture, which proposes that any finite tensor categories has finitely generated cohomology.) In this talk I will focus on how deformation theory informs, and

contributes to, our analysis of cohomology for such doubles. This talk is based in part on joint work with Eric Friedlander. (Received August 17, 2020)

1162-18-40 **Eugene Gorsky*** (egorskiy@math.ucdavis.edu), **Matthew Hogancamp** and **Paul Wedrich**. *Derived traces of Soergel categories.*

We study two kinds of categorical traces of (monoidal) dg categories, with particular interest in categories of Soergel bimodules. First, we explicitly compute the usual Hochschild homology, or derived vertical trace, of the category of Soergel bimodules in arbitrary types. Secondly, we introduce the notion of derived horizontal trace of a monoidal dg category and compute the derived horizontal trace of Soergel bimodules in type A. As an application we obtain a derived annular Khovanov-Rozansky link invariant with an action of full twist insertion, and thus a categorification of the HOMFLY-PT skein module of the solid torus. (Received August 18, 2020)

1162-18-44 **Alistair Savage*** (alistair.savage@uottawa.ca). *Affinization of monoidal categories.*

We will describe a general procedure for affinizing monoidal categories. When the original category is braided, its affinization is again a monoidal category. We will discuss examples coming from representation theory, skein theory, and the theory of braids and tangles. (Received August 19, 2020)

1162-18-65 **Mee Seong Im*** (meeseongim@gmail.com), Chauvenet Hall, Office 342, Department of Mathematics, United States Naval Academy, Annapolis, MD 21402, and **Can Ozan Oguz** (canozanoguz@gmail.com), Department of Mathematics, Galatasaray University, Istanbul, Ortaköy, Turkey. *Natural transformations between induction and restriction on iterated wreath product of symmetric group of order* 2. Preliminary report.

Considering the group algebra $\mathbb{C}[A_n]$ of n-step iterated wreath product of S_2, we describe the category of (A_m, A_n)-bimodules, by explicitly describing the natural transformations between induction and restriction functors on the representations of the iterated wreath product tower. This is joint work in progress with Can Ozan Oğuz. (Received August 23, 2020)

1162-18-74 **Rhea Palak Bakshi** and **Dionne Kunkel***, dfkunkel@gwu.edu, and **Sujoy Mukherjee**, **Takefumi Nosaka** and **Jozef Przytycki**. *The second quandle homology of quasigroup Alexander quandles.*

For X, an Alexander quandle with invertible $(1-t)$, we express the second quandle homology of quasigroup Alexander quandles in terms of the exterior algebra of X and present a short self-contained proof of its structure and provide some computational examples. (Received August 24, 2020)

1162-18-82 **Daniel Rogalski*** (drogalski@ucsd.edu) and **Jason Gaddis** (jdgaddis@gmail.com). *On the Local Structure at a Point on a Noncommutative Projective Variety.* Preliminary report.

Noncommutative Projective Varieties arise from graded rings through a generalization of the Proj construction from commutative algebraic geometry. Then a point in the variety comes from a graded module of dimension 1, and to study the local structure at a point one wants to understand the category of graded modules of dimension 1 over the algebra. This was done by Van Gastel and Van den Bergh for the case of surfaces. In this talk we describe some generalizations of these results to certain threefolds. In some nice cases the graded modules of dimension 1 can be described using the representation theory of infinite quivers. The result has applications to defining blowups at points on such varieties. (Received August 25, 2020)

1162-18-84 **Daniel K. Nakano*** (nakano@math.uga.edu), **Kent B. Vashaw** and **Milen T. Yakimov**. *Noncommutative Tensor Triangular Geometry.*

In this talk, I will show how to develop a general noncommutative version of Balmer's tensor triangular geometry that is applicable to arbitrary monoidal triangulated categories (MΔC). Insights from noncommutative ring theory are used to obtain a framework for prime, semiprime, and completely prime (thick) ideals of an MΔC, **K**, and then to associate to **K** a topological space–the Balmer spectrum Spc **K**.

We develop a general framework for (noncommutative) support data, coming in three different flavors, and show that Spc **K** is a universal terminal object for the first two notions (support and weak support). The first two types of support data are then used in a theorem that gives a method for the explicit classification of the thick (two-sided) ideals and the Balmer spectrum of an MΔC. The third type (quasi support) is used in another theorem that provides a method for the explicit classification of the thick right ideals of **K**, which in turn can be applied to classify the thick two-sided ideals and Spc **K**.

This is joint and ongoing work with Milen Yakimov and Kent Vashaw (Received August 25, 2020)

1162-18-108 **Joshua R Edge*** (edgej@denison.edu). *Spin models for singly-generated Yang-Baxter planar algebras.*

Spin models for planar algebras are useful tools in building new and interesting fiber functors, which can be used to create new Hopf algebras. In this talk, we discuss the classification of all spin models for singly-generated Yang-Baxter planar algebras in terms of certain highly symmetric graphs. Using Liu's classification of singly-generated Yang-Baxter planar algebras, this classifies all spin models for the Jones polynomial, the Bisch-Jones planar algebras, and the Kauffman polynomial. This simplifies and clarifies Jaeger's classification of spin models for the Kauffman polynomial. We also explain the numerous exceptional cases in Jaeger's classification by demonstrating a new, discrete two-parameter family of spin models for the Bisch-Jones planar algebras. (Received August 28, 2020)

1162-18-128 **Nicolle Gonzalez*** (nicolle@math.ucla.edu) and **Matt Hogancamp**. *A skein theoretic Carlsson-Mellit algebra.* Preliminary report.

The Carlsson-Mellit algebra, or $A_{q,t}$ algebra, originally arose in the proof of the celebrated Shuffle conjecture. This algebra, built from Hecke algebra generators and a family of raising and lowering operators, has a particularly interesting representation, known as the polynomial representation, on which its action is given by complicated plethystic operations. In this talk I will discuss how this algebra (specialized at $t = q^{-1}$) and its polynomial representation can be formulated skein theoretically, as certain braids on an annulus, and consequently categorified. This is joint work with Matt Hogancamp. (Received August 29, 2020)

1162-18-135 **Max Vargas*** (mvargas2@uoregon.edu). *Okounkov-Vershik approach to representations of the partition category.*

We explain a new approach to the representation theory of the partition category based on a reformulation of Enyang's Jucys-Murphy elements. We establish branching rules in the style of Okounkov and Vershik, and use these to compute the Gelfand-Tsetlin characters of the standard modules. This leads to new proofs of results of Comes and Ostrik on blocks of Deligne's category $\operatorname{Rep} S_t$. (Received August 29, 2020)

1162-18-163 **Iva Halacheva*** (i.halacheva@northeastern.edu), **Anthony Licata**, **Ivan Losev** and **Oded Yacobi**. *The cactus group and perverse equivalences.* Preliminary report.

Given a categorical $\mathfrak{g}$-action on an abelian category $\mathcal{C}$, where $\mathfrak{g}$ is a semisimple Lie algebra, Chuang and Rouquier construct so called Rickard complexes, one for each simple root of $\mathfrak{g}$, which act as equivalences on the derived category $D^b(\mathcal{C})$. These complexes satisfy the braid relations for $\mathfrak{g}$, as shown by Cautis and Kamnitzer, and hence give an action of the braid group. We show that the complex corresponding to the positive lift of the longest Weyl group element (of any parabolic in $\mathfrak{g}$) is a perverse equivalence on $D^b(\mathcal{C})$. Hence, it induces a bijection on the irreducible objects of $\mathcal{C}$, and recovers the cactus group action on the corresponding $\mathfrak{g}$-crystal. This is joint work in progress with Tony Licata, Ivan Losev, and Oded Yacobi. (Received August 31, 2020)

1162-18-186 **Julia Plavnik*** (jplavnik@iu.edu). *Cohomology of finite tensor categories: duality and Drinfeld centers.*

In the talk, we consider the *finite generation property* for cohomology of a finite tensor category $\mathcal{C}$, which requires that the self-extension algebra of the unit $\operatorname{Ext}^*_{\mathcal{C}}(\mathbf{1},\mathbf{1})$ is a finitely generated algebra and that the graded extension group $\operatorname{Ext}^*_{\mathcal{C}}(\mathbf{1},V)$ is a finitely generated $\operatorname{Ext}^*_{\mathcal{C}}(\mathbf{1},\mathbf{1})$-module, for all V in $\mathcal{C}$. We explore how this cohomological finiteness property is affected by different constructions of finite tensor categories, such as duality (with respect to exact module categories) and taking the Drinfeld center. This talk is based on a join work with Cris Negron. (Received August 31, 2020)

1162-18-218 **Kent Vashaw*** (kvasha1@lsu.edu). *Balmer spectra and Drinfeld centers.* Preliminary report.

The Balmer spectrum of a monoidal triangulated category is an important geometric invariant which can be used in many cases to obtain a classification of the thick ideals of the category. Many examples of monoidal triangulated categories arise as stable module categories of finite-dimensional Hopf algebras. We will describe the relationship between the Balmer spectra of stable module categories associated to a Hopf algebra and its Drinfeld double. In particular, we show that there always exists a continuous map between them, and describe how this map relates to classifications of thick ideals. (Received September 01, 2020)

19 ► K-theory

1162-19-12 **Shanna Dobson*** (sdobson@calstatela.edu). *K-Theory of Diamonds.* Preliminary report.

A perfectoid space is an adic space covered by affinoid adic spaces of the form $Spa(R, R^+)$ with R perfectoid. Let $Perfd$ be the category of perfectoid spaces. Let $Perf \subset Perfd$ be the subcategory of perfectoid spaces in characteristic p. Let $\mathcal{Y}$ be a pro-étale sheaf on $Perf$. $\mathcal{Y}$ is a diamond if $\mathcal{Y}$ can be expressed as the quotient X/R of a perfectoid space X of characteristic p by a pro-étale equivalence relation $R \subset X \times X$ such that R is a perfectoid space with $s, t : R \to X$ pro-étale. We survey Efimov's definition of K-theory and give a preliminary outline of the K-theory of diamonds. (Received August 03, 2020)

20 ► Group theory and generalizations

1162-20-42 **Christopher Ryba*** (ryba@math.berkeley.edu). *Littlewood Complexes for Symmetric Groups.*

Certain categories of representations of GL_∞ and S_∞ have been studied by Sam and Snowden. They admit "specialisation" functors to their finite counterparts. We give a resolution of irreducible representations of S_n by representations restricted from GL_n (which contains S_n as permutation matrices), which yields injective resolutions in $Rep(S_\infty)$. This construction categorifies a family of symmetric functions studied by Orellana and Zabrocki as well as Assaf and Speyer. (Received August 18, 2020)

1162-20-71 **Elijah Bodish*** (ebodish@uoregon.edu), University of Oregon, Department of Math, 1222 Univ of Oregon, Eugene, OR 97403. *Web Calculus and Tilting Modules in Type C_2.*

Using Kuperberg's B_2/C_2 webs, and following Elias and Libedinsky, we describe a "light leaves" algorithm to construct a basis for the space of morphisms between arbitrary tensor products of fundamental representations for $\mathfrak{sp}_4$ (and the associated quantum group). Our argument has very little dependence on the base field, and we leverage this to interpret Kuperberg's webs as a diagrammatic presentation for the monoidal category of tilting modules. Finally, we define a new diagrammatic category which we argue is equivalent to both the monoidal category $\mathrm{Rep}(\mathbb{C}^\times \rtimes \mathbb{Z}/2)$ and the semisimplification of the category of tilting modules for $\mathcal{U}_q^{\mathbb{Z}}(\mathfrak{sp}_4)$ at a primitive third root of unity. (Received August 23, 2020)

1162-20-75 **Alexandru Chirvasitu***, Department of Mathematics, 244 Mathematics Building, Buffalo, NY 14260, and **Jacek Krajcok** and **Piotr Soltan**. *Quantum disks, balls and groups.*

The Toeplitz algebra attached to the unit disk is the universal C^*-algebra generated by an isometry, and is a non-commutative analogue of the unit disk. Similarly, one can attach algebras to "quantum spaces" that can be regarded as non-commutative versions of non-compact Hermitian symmetric spaces. I will discuss results to the effect that such quantum spaces (e.g. the non-commutative cousins of the unit balls in complex spaces $\mathbb{C}^n$) cannot admit quantum group structures, i.e. their attached non-commutative "function algebras" do not admit reasonable Hopf algebra structures. (Received August 24, 2020)

1162-20-93 **Cris Negron*** (cnegron@email.unc.edu), University of North Carolina Chapel Hill. *Quantum groups at even order roots of unity.*

I will discuss unique phenomena for quantum groups at even order roots of unity. Here the small quantum group is not a Hopf subalgebra in Lusztig's divided power algebra, but a quasi-Hopf algebra constructed as a de-equivariantization of the "big quantum group" via the quantum Frobenius. We will discuss relations with conformal field theory. (Received August 25, 2020)

1162-20-107 **Dongkwan Kim*** (kim00657@umn.edu), 206 Church St. SE, 203B, Minneapolis, MN 55455, and **Pavlo Pylyavskyy**. *Two-row W-graphs in affine type A.*

For a Coxeter group W, a W-graph is a graph satisfying certain combinatorial axioms. It yields a good basis of the corresponding representation and describes the action of W on the basis elements concretely. Therefore, understanding the structure of W-graphs is useful for investigating properties of the representations of W, even when W is finite and its irreducible characters are known. In this session, I will discuss W-graphs when W is an (extended) affine symmetric group (which is an infinite group) and these graphs are associated with "two-row partitions". Also I will briefly mention how it is related to Lusztig's periodic W-graphs. This work is joint with Pavlo Pylyavskyy. (Received August 28, 2020)

1162-20-169 **You Qi*** (yq2dw@virginia.edu), 141 Cabell Drive, Kerchof Hall, P.O. Box 400137, 141 Cabell Drive, Kerchof Hall, P.O. Box 4001, Charlottesville, VA 22904. *On some p-differential graded link homologies.*

We report on a categorification of Jones polynomial specialized at a pth root of unity, utilizing the monoidal category of Soergel bimodules under a p differential. (Received August 31, 2020)

1162-20-204 **Nate Harman*** (nateharman1234@gmail.com). *The Drinfeld center of Deligne's category $Rep(S_t)$.*

I will describe recent joint work with Flake and Laugwitz in which we characterize simple and indecomposable objects inside the Drinfeld center of $Rep(S_t)$. (Received August 31, 2020)

22 ► Topological groups, Lie groups

1162-22-76 **Gurbir Dhillon***, 10 Hillhouse Ave, DL 417, New Haven, CT 06511. *Singular Soergel bimodules and quantum Langlands duality for W-algebra representations.* Preliminary report.

The fundamental local equivalence is a partially conjectural analog of the Satake isomorphism for the quantum Langlands correspondence proposed by Gaitsgory and Lurie. Gaitsgory has further conjectured a compatibility of this equivalence with Feigin–Frenkel duality for affine W-algebras. Some important cases of the latter conjecture were established by Arakawa and Frenkel. Building on previous work of the author, joint in parts with Justin Campbell and Sam Raskin, which established the fundamental local equivalence non-factorizably, we will explain work in progress which similarly establishes Gaitsgory's latter conjecture. A key tool is a relationship between singular Soergel bimodules and representations of affine W-algebras previously conjectured by the author. (Received August 24, 2020)

1162-22-142 **Scott Larson*** (scott.larson@uga.edu), 160 Pine Ridge Trace, Athens, GA 30605. *Small Resolutions of Closures of K-Orbits in Flag Varieties.*

Let G be a connected complex reductive algebraic group and K the fixed points of an algebraic involution. The geometry of closures of K-orbits on the flag variety of G governs key properties in the representation theory of a corresponding real reductive group. We recall results from Barbasch-Evens on resolutions of singularities of K-orbit closures for the group $U(p,q)$. The group $Sp(2n,R)$ admits similar but more complicated resolutions, and we provide families of small resolutions in this case. We conclude with a general construction of resolutions of K-orbits. (Received August 30, 2020)

30 ► Functions of a complex variable

1162-30-32 **Yury Grabovsky** and **Narek Hovsepyan*** (hovnarek@yahoo.com). *On feasibility of extrapolation of complex electromagnetic permittivity functions.*

As a function of frequency, the complex electromagnetic permittivity has analytic extension into the upper half-plane. It has further physical properties such as positive imaginary part and reflection symmetry about the imaginary axis. Such a function can be measured in a band of frequencies and one would like to extrapolate to a wider band of frequencies, using its analyticity. How reliable such extrapolation procedures can possibly be? Given two analytic functions, representing extrapolants of the same experimental data, we give a sharp bound on the worst case error at an extrapolation point outside of the experimentally accessible frequency band. We show that this bound exhibits a power law precision deterioration as one moves further away from the frequency band containing measurement data. This quantifies the uncertainty of any extrapolation procedure. (Received August 16, 2020)

1162-30-67 **William Gryc**, **Loredana Lanzani**, **Jue Xiong*** (xiong.147@osu.edu) and **Yuan Zhang**. *Neumann boundary value problem for $\bar{\partial}$.* Preliminary report.

Let Ω be a bounded domain with Lipschitz boundary in the complex plane. There are several ways to define the Hardy space associated to Ω. For example, we can define the Hardy space of interior functions denoted by $H^p(\Omega)$ and the Hardy space of boundary values denoted by $h^p(\partial\Omega)$. By considering the Dirichlet boundary value problem for $\bar{\partial}$, we can identify these two Hardy spaces. Correspondingly, we are interested in the Neumann boundary value problem for $\bar{\partial}$: what are the natural data space and solution space associtated to such PDEs? We will answer this question and show that our choice of the data space is optimal for the Neumann BVP.

Furthermore, we obtain an integral representation for the solution with norm estimates. This is a joint work inspired by the SCV workshop the speaker attended at AIM. (Received August 22, 2020)

1162-30-170 **John Treuer***, 340 Rowland Hall (Bldg.# 400), University of California, Irvine, Irvine, CA 92697-3875, Los Angeles, CA. *Characterization of minimal domains of the Bergman kernel in the complex plane using complex analysis of one variable.*

Let $\Omega \subset \mathbb{C}$ be a domain, $K(\cdot,\cdot)$ equal the Bergman kernel of Ω, and $v(\Omega)$ denote the Euclidean volume of Ω. A minimal domain for the Bergman kernel is a bounded domain such that $K(z,z_0) = v(\Omega)^{-1}$ for all $z \in \Omega$. In 2020, Dr. Xin Dong and I characterized the minimal domains in the complex plane by showing that they are disks minus relatively closed polar sets. In our proof we used the Suita conjecture (now a theorem), which is a very deep result proved by Guan and Zhou using L^2-methods of several complex variables. Since my and Dr. Dong's theorem is about domains in the complex plane, it would be desirable to remove the Suita conjecture from our proof and only use techniques of complex analysis of one variable. In this talk, I will provide a new proof of the characterization of minimal domains in $\mathbb{C}$, which only uses complex analysis of one variable. I will also discuss generalizations of this result to higher dimensions. (Received August 31, 2020)

32 ► Several complex variables and analytic spaces

1162-32-38 **Jennifer Brooks** and **Kevin Palencia***, 98 E Meadow Dr, Cortland, IL 60112. *A Rank Question for Homogeneous Polynomials.*

The Ebenfelt sum of squares conjecture and the degree estimate conjecture for proper rational mappings between balls in complex Euclidean spaces are open problems in several complex variables that naturally lead to the study of the rank of the bihomogeneous polynomial $r(z,\bar{z})\|z\|^2$ on $\mathbb{C}^{n+1}$ under certain additional hypotheses. When r has a diagonal coefficient matrix, these questions reduce to questions about the rank of the real polynomial $P = SQ$, where Q is a homogeneous polynomial and $S(x) = \sum_{j=0}^{n} x_j$. In this talk, we describe joint work with Jennifer Brooks that uses techniques from commutative algebra to estimate the minimum rank of $P = SQ$ under the additional hypothesis that Q has maximum rank. The problem has already been solved for $n+1 \leq 3$, and so we consider $n+1 \geq 4$. We obtain a minimum rank estimate that is sharp when $n+1 = 4$, and we exhibit a family of polynomials having this minimum rank. We exhibit an estimate for $n+1 > 4$ that is not sharp, but is nonetheless nontrivial. (Received August 17, 2020)

1162-32-97 **Al Boggess** and **Andrew Raich*** (araich@uark.edu). *The Complex Green Operator on Quadric Submanifolds.*

The Kohn Laplacian $\Box_b$ is one of the fundamental objects in several complex variable, and its inverse (when it exists) is called the complex Green operator. Quadric submanifolds are one of the few classes of manifolds on which we can find an integral formula (or exact expression) for the complex Green operator. Al Boggess and I have been investigating the $\Box_b$-equation on quadrics for the past several years, and in this talk, I will present some surprising calculations in the very nice case that the eigenvalues of the directional Levi forms are nonvanishing. (Received August 26, 2020)

1162-32-99 **Emil J Straube*** (straube@math.tamu.edu), Department of Mathematics, Texas A&M University, College Station, TX 77843, and **Sönmez Şahutoğlu** and **Mehmet Çelik**. *Compact Hankel operators on convex domains.* Preliminary report.

This is a report on ongoing work with Mehmet Çelik and Sönmez Şahutoğlu. I will discuss compactness of Hankel operators on convex domains with affine varieties in their boundaries. (Received August 26, 2020)

1162-32-104 **Zhenghui Huo**, **Nathan Wagner*** (nathanawagner@wustl.edu) and **Brett Wick**. *A Békollè-Bonami Class of Weights for Certain Pseudoconvex Domains.*

We prove the weighted L^p regularity of the ordinary Bergman projection on certain pseudoconvex domains where the weight belongs to an appropriate generalization of the Békollè-Bonami class. The main tools used are estimates on the Bergman kernel obtained by McNeal and Békollè's original approach of proving a good-lambda inequality. In particular, we view the Bergman projection as a Calderón Zygmund singular integral operator with respect to a particular quasi-metric. This quasi-metric arises from a scaling approach employed by McNeal and others to obtain estimates on the Bergman kernel for various pseudoconvex domains of finite type. (Received August 27, 2020)

1162-32-122 **Yunus Zeytuncu*** (zeytuncu@umich.edu). *A Tauberian approach to an analog of Weyl's Law for the Kohn Laplacian.*

In this talk, we look at the eigenvalue counting function for the Kohn Laplacian on spheres $\mathbb{S}^{2n-1}$. We compute the leading coefficient in the asymptotic expansion of the counting function by a Tauberian argument. This statement (partially) generalizes a result by Stanton and Tartakoff, where they obtained an analog of Weyl's Law for eigenvalues of the Kohn Laplacian acting on $(0,q)$-forms ($q \geq 1$) on hypersurfaces in $\mathbb{C}^n$. (Received August 28, 2020)

1162-32-126 **David E Barrett*** (barrett@umich.edu). *Hearing domains in projective space.* Preliminary report.

A famous question of (Bers-Bochner-)Kac asks to what extent a domain in $\mathbb{R}^2$ can be recovered up to isometry from the unitary equivalence class of its (self-adjoint [compact]) Green's operator. This talk will explore the question of to what extent a (strongly $\mathbb{C}$-convex) domain in complex projective space can be recovered (up to projective automorphism) by the unitary equivalence class of its (non-self-adjoint) Leray integral (projection) operator. (Received August 28, 2020)

1162-32-151 **Alexander Nagel** and **Malabika Pramanik***, malabika@math.ubc.ca. *Bergman spaces under monomial maps.*

If a proper monomial map carries a complex domain $\Omega_1 \subseteq \mathbb{C}^n$ onto $\Omega_2 \subseteq \mathbb{C}^n$, how are the Bergman spaces of Ω_1 and Ω_2 related? We will discuss a structure theorem that addresses this question, with applications. (Received August 30, 2020)

1162-32-166 **Martino Fassina*** (martino.fassina@gmail.com) and **Yifei Pan** (pan@pfw.edu). *Some remarks on the global distribution of the points of finite D'Angelo type.* Preliminary report.

Let M be a smooth real hypersurface in $\mathbb{C}^n$. We present some examples and a stratification result for the set of Levi degenerate points of finite D'Angelo type of M. (Received August 31, 2020)

1162-32-167 **Jiri Lebl*** (lebl@okstate.edu), **Alan Noell** and **Sivaguru Ravisankar**. *Removable CR singularities.*

We study stability of a CR singularity of a real-analytic submanifold $MsubsetmathbbC^n$ whose CR structure extends across the CR singularity. In this case the manifold can be realized as an image under a holomorphic map of a CR submanifold M. We are interested in perturbing M without disturbing the CR structure, that is, perturbing the map. For instance, we find a sharp inequality of the dimensions is found for when the CR singularity can always be perturbed away. A related question that needs to be answered is when can a CR function be perturbed to remove its zeros. This is joint work with Alan Noell and Sivaguru Ravisankar. (Received August 31, 2020)

1162-32-171 **Yonatan Shelah*** (yonshel@umich.edu), Ann Arbor, MI. *An Inverse Spectral Problem for the Leray Transform in Two Settings.* Preliminary report.

I will investigate the inverse spectral problem for the Leray transform on a class of Reinhardt domains in $\mathbf{C}^2$, which are C^2 away from the axes and satisfy milder smoothness conditions on the axes. The spectrum was calculated by L. Lanzani and D. Barrett in a 2009 paper, where the eigenvalues were marked according to the Fourier decomposition. This gives rise to the notion of a marked spectrum, indexed by $\mathbf{Z}^2_{\geq 0}$. I will highlight the following topics:

1. To what extent does the marked spectrum determine the Reinhardt domain? There is a measure approach to this question by considering different limit distributions of eigenvalues. This recovery process can only be done up to dilations, variable swap and the duality map, but is that it? This is true at least in some cases.

2. I will show that there is a remarkable analogy with the setting of what's called rigid Hartogs domains in $\mathbf{C}^2$, which are given by $\mathrm{Im}(z_2) > f(|z_1|)$, where f is a C^2 real-valued function on $(0,\infty)$ with some regularity conditions at the endpoints. The marked spectrum is indexed by $\mathbf{Z}_{\geq 0} \times \mathbf{R}_{<0}$, but the general recovery algorithm is essentially unchanged. This is joint work with L. Edholm. (Received August 31, 2020)

1162-32-183 **Andrew Zimmer*** (amzimmer2@wisc.edu). *Compactness of the $\bar{\partial}$-Neumann problem on domains with bounded intrinsic geometry.*

By considering intrinsic geometric conditions, we introduce a new class of domains in complex Euclidean space. This class is invariant under biholomorphism and includes strongly pseudoconvex domains, finite type domains in dimension two, convex domains, $\mathbb{C}$-convex domains, and homogeneous domains. For this class of domains, we show that compactness of the $\bar{\partial}$-Neumann operator on $(0,q)$-forms is equivalent to the boundary not containing

any q-dimensional analytic varieties (assuming only that the boundary is a topological submanifold). (Received August 31, 2020)

1162-32-205 **Chase Bender**, **Debraj Chakrabarti**, **Luke Edholm*** (edholm@umich.edu) and **Meera Mainkar**. *Complexity of singular Reinhardt domains and the regularity of the Bergman projection.*

We consider a large class of singular Reinhardt domains in $\mathbb{C}^n$ and develop a general framework from which we may understand the L^p-regularity of their Bergman projections. The domains in this class may be realized as quotients of simpler domains under the action of a finite group of biholomorphic automorphisms. It is shown that the range of Bergman L^p-boundedness is closely connected to the "arithmetic complexity" of the original domain, a notion that will be made precise.

This talk combines various projects of the speaker with collaborators Chase Bender, Debraj Chakrabarti and Meera Mainkar. (Received August 31, 2020)

34 ► Ordinary differential equations

1162-34-136 **Vinodh Kumar Chellamuthu***, 225 South University Avenue, Saint George, UT 84770, and **Noelle West**. *Modeling the Effects of Passive Immunity in Birds for the Disease Dynamics of West Nile Virus.* Preliminary report.

West Nile Virus (WNV) is a mosquito-borne virus that circulates among birds but also affects humans. Migrating birds carry these viruses from one place to another each year. WNV has spread rapidly across the continental United States, resulting in numerous human infections and deaths. Several studies suggest that larval mosquito control measures should be taken as early as possible in a season to control the mosquito population size. Also, adult mosquito control measures are necessary to prevent the transmission of WNV from mosquitoes to birds and humans. To better understand the effective strategy for controlling the affected larvae mosquito population, we have developed a mathematical model using a system of first-order differential equations to investigate the transmission dynamics of WNV in a mosquito-bird-human community. We also incorporated vertical transmission in mosquitoes and passive immunity in birds to more accurately simulate the spread of the disease. (Received August 29, 2020)

1162-34-200 **Aseel Farhat*** (afarhat@fsu.edu), 1017 Academic Way, Tallahassee, FL 32304, and **Zoran Grujić**. *Local near-Beltrami structure of turbulence and regularity of the 3D Navier-Stokes flows.*

Computational simulations of turbulent flows indicate that the regions of low dissipation feature high degree of local alignment between the velocity and the vorticity. Hence, one could envision a geometric scenario in which the persistence of the local near-Beltrami property might be consistent with a (possible) finite-time singularity formation. We will show that this scenario is in fact prohibited if the sine of the angle between the velocity and the vorticity is small enough with respect to the local enstrophy. (Received August 31, 2020)

35 ► Partial differential equations

1162-35-10 **Hakima Bessaih*** (bessaih@uwyo.edu), Department of Mathematics & Statistics, Dept. 3036, 1000 E. University Ave., Laramie, WY 82071, and **Benedetta Ferrario**. *Invariant measures for stochastic damped 2D Euler equation.*

We study a two-dimensional Euler equation, damped by a linear term and driven by an additive noise. The existence of weak solutions has already been studied; pathwise uniqueness is known for solutions that have bounded vorticity. We prove the Markov property and the existence of an invariant measure in the space of bounded vorticity. Since this is not a Polish space, we cannot apply the classical tools found in the literature but we develop a Krylov-Bogoliubov's type method working with the weak star and the bounded weak star topologies. (Received July 15, 2020)

1162-35-14 **Kazuo Yamazaki*** (kyamazak@ttu.edu), Department of Mathematics & Statistics, Texas Tech University, Lubbock, TX 79409. *Remarks on the non-uniqueness in law of the Navier-Stokes equations up to the J.-L. Lions' exponent.* Preliminary report.

Lions in 1959 introduced the Navier-Stokes equations with a viscous diffusion in the form of a fractional Laplacian; subsequently, in 1969 he claimed the uniqueness of its solution when its exponent is not less than $1/2 + n/4$ where n is the spatial dimension. Following the stochastic framework of convex integration presented by Hofmanová,

Zhu, and Zhu (2019, arXiv:1912.11841 [math.PR]), we prove the non-uniqueness in law for the three-dimensional stochastic Navier-Stokes equations generalized via the viscous diffusion in the form of a fractional Laplacian with its exponent less than five quarters. Moreover, we prove non-uniqueness in law of the two-dimensional stochastic Navier-Stokes equations when the exponent of the fractional Laplacian is strictly less than one. It should be emphasized that the definitions of these solutions are rougher than that of Leray-Hopf type. (Received August 09, 2020)

1162-35-15 **Marcelo M. Disconzi*** (marcelo.disconzi@vanderbilt.edu), **Mihaela Ifrim** and **Daniel Tataru**. *The relativistic Euler equations with a physical vacuum boundary.*

We consider the relativistic Euler equations with a physical vacuum boundary and an equation of state $p(\varrho) = \varrho^\gamma$, $\gamma > 1$. We establish the following results. (i) local well-posedness in the Hadamard sense, i.e., local existence, uniqueness, and continuous dependence on the data; (ii) low regularity solutions: our uniqueness result holds at the level of Lipschitz velocity and density, while our rough solutions, obtained as unique limits of smooth solutions, have regularity only a half derivative above scaling; (iii) stability: our uniqueness in fact follows from a more general result, namely, we show that a certain nonlinear functional that tracks the distance between two solutions (in part by measuring the distance between their respective boundaries) is propagated by the flow; (iv) we establish sharp, essentially scale invariant energy estimates for solutions; (v) a sharp continuation criterion, at the level of scaling, showing that solutions can be continued as long as the the velocity is in $L^1_t Lip$ and a suitable weighted version of the density is at the same regularity level. (Received August 10, 2020)

1162-35-21 **leoncior rodriguez-quinones*** (leoncio.quinones@usu.edu), **Luis Gordillo** (luis.gordillo@usu.edu) and **Jia Zhao** (jia.zhao@usu.edu). *The effects of simple density-dependent prey diffusion and refuge in a predator-prey system.*

We study a spatial (two-dimensional) Rosenzweig-MacArthur model under the following assumptions: (1) prey spread follows a nonlinear diffusion rule, (2) preys have a refuge zone (sometimes called "protection zone") where predators cannot enter, (3) predators move following linear diffusion. We present a bifurcation analysis for the system that shows the existence of positive solutions at the steady state. We complement the theoretical results with numerical computations and compare our results with those obtained in the case of having linear diffusion for the prey movement. Our results show that both models, with linear and nonlinear diffusion for the prey, have the same bifurcation point and the positive solution curves are virtually the same in a neighborhood of this point, but they get drastically different as the bifurcation parameter approaches to zero. (Received August 12, 2020)

1162-35-23 **Vera Mikyoung Hur***, 1409 W Green Street, University of Illinois at Urbana-Champaign, Department of Mathematics, Urbana, IL 61801. *Stokes waves in a constant vorticity flow: theory and numerics.*

Stokes in the 1800s made many contributions about periodic waves at the surface of water, under the influence of gravity, propagating in permanent form a long distance at a practically constant velocity. In an irrotational flow, for instance, he observed that crests become sharper and troughs flatter as the amplitude increases, and that the so-called wave of greatest height, or extreme wave, possesses a 120 degree's angle at the crest. The irrotational flow assumption is justified in many situations, and facilitates rigorous analysis and numerical computation. However, rotational effects are significant in many others. I will review recent progress in a constant vorticity flow. Numerical findings include folds and gaps in the wave speed vs. amplitude plane, and a profile enclosing multiple bubbles of fluids. I will discuss analytical and numerical applications if time permits. (Received August 12, 2020)

1162-35-30 **Jeremy L Marzuola***, Department of Mathematics, CB #3250, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599. *Numerical algorithms for water waves with background flow over obstacles and topography.*

With David Ambrose, Jon Wilkening, Roberto Camassa, Rich McLaughlin and Quentin Robinson, we have developed fast and efficient algorithms for solving the incompressible, irrotational Euler equations with a free surface for a water layer that includes multiple stationary obstacles, resulting in non-simply connected fluid domains, as well variable bottom topography, in two-dimensions. One approach uses Layer Potentials and has the ability to be generalized to three dimensions. The other approach uses Cauchy Integrals and is fast and efficient for computing conserved quantities. We present convergence data, energy conservation, stabilization methods, as well as explore various dynamical regimes with non-simply connected fluid domains. The methods generalize to 3D as well. (Received August 14, 2020)

1162-35-52 **Anna L Mazzucato*** (alm24@psu.edu). *Enhanced diffusion and global existence for the Kuramoto-Sivashinsky equation.*

We consider the Kuramoto-Sivashinsky equation (KSE) on the two-dimensional torus in scalar form. We prove global existence for small data in the absence of growing modes. If growing modes are present, we show that global existence for arbitrary data holds for the advective KSE, provided the advecting flow field induces a sufficient small diffusion time for the linearized operator, for example if the flow is mixing with large amplitude. This is joint work with Yuanyuan Feng (PSU). (Received August 20, 2020)

1162-35-57 **Zachary Bradshaw*** (zb002@uark.edu), **Igor Kukavica** and **Tai-Peng Tsai**. *Global existence of infinite energy solutions to fluid equations.*

We discuss two sets of results. The first concerns global existence of solutions to the 3D Navier-Stokes equations with data that is not decaying in a local L^2-sense. Indeed, local L^2 values may be unbounded. Results in this direction have recently been independently established by Fernàndez-Dalga and Lemarié-Rieusset as well as Bradshaw, Kukavica and Tsai. The new solution classes are noteworthy as they allow for scaling symmetric data. The second concerns the existence and properties of global solutions which occupy a class strictly between Leray's original class of solutions and the local Leray solutions introduced by Lemarié-Rieusset at the turn of the century. These results due to Bradshaw and Tsai shed light on what properties of Leray's original solutions are inherited by the more general class of Lemarié-Rieusset. (Received August 21, 2020)

1162-35-58 **Theodore D Drivas***, Fine Hall – Washington Road, Princeton, NJ, NJ 08540. *Flexibility, rigidity and stability of steady fluid motion.*

We discuss the possibility and the impossibility of smooth deformations of equilibrium fluid states in the presence of symmetries. We will also make some remarks about the stability/instability of stationary states. (Received August 21, 2020)

1162-35-62 **Gampola Waduge Nalin Fonseka*** (fonsekan@carolinau.edu), 1825 Sherwood St, Apt C, Greensboro, NC 27403, and **Ananta Acharya** and **Ratnasingham Shivaji**. *Analysis of reaction-diffusion systems where a parameter influences both the reaction terms as well as the boundary.*

We study positive solutions to steady state reaction diffusion models of the form:

$$\begin{cases} -\Delta u = \lambda f(v);\ \Omega \\ -\Delta v = \lambda g(u);\ \Omega \\ \frac{\partial u}{\partial \eta} + \sqrt{\lambda} u = 0;\ \partial\Omega \\ \frac{\partial v}{\partial \eta} + \sqrt{\lambda} v = 0;\ \partial\Omega \end{cases}$$

where $\lambda > 0$ is a positive parameter, Ω is a bounded domain in $\mathbb{R}^N$; $N > 1$ with smooth boundary $\partial\Omega$ or $\Omega = (0,1)$, $\frac{\partial z}{\partial \eta}$ is the outward normal derivative of z. We assume that f and g are continuous increasing functions such that $f(0) = 0 = g(0)$ and $\lim_{s\to\infty} \frac{f(Mg(s))}{s} = 0$ for all $M > 0$. (Received August 22, 2020)

1162-35-64 **Marta Lewicka*** (lewicka@pitt.edu). *Expansions of averaging operators and applications.*

The following approach of finding solutions to a partial differential equation Lu=0, proved to be quite versatile and allow to deal with several elliptic and parabolic nonlinear operators L:

(i) find an asymptotic expansion of suitable averaging operator on u, in which the appropriate order coefficient matches Lu;

(ii) study the related mean value equation by removing higher order error terms in the expansion;

(iii) interpret the mean value equation as the dynamic programming principle of a two-player game incorporating deterministic and stochastic components;

(iv) pass to the limit in the radius of sampling/averaging, in order to recover solutions to Lu=0 from the values of the game process.

In my talk, I will explain this approach in the following contexts: p-Laplacian; non-local geometric p-Laplacian; Robin boundary conditions; and weighted Laplace-Beltrami operator on a manifold. In each case, finding the appropriate averaging principle is the key starting point in order to develop (i)-(iv). (Received August 22, 2020)

1162-35-78 **Yuan Pei*** (yuan.pei@wwu.edu), 516 High St., MS-9063, Bellingham, WA 98225, and **Adam Larios** and **Leo Rebholz**. *Velocity-vorticity-Voigt model for PDEs in fluid dynamics.*

In this talk, we propose the so-called 3D velocity-vorticity-Voigt (VVV) model for Navier-Stokes equations as well as Boussinesq equations. We add a Voigt regularization term to the momentum equation in velocity-vorticity formulation without regularizing the vorticity. We prove global well-posedness and regularity of this model along with an energy identity. We also show convergence of the model's velocity and vorticity to their counterparts in the 3D Navier-Stokes equations as the Voigt modeling parameter tends to zero. Similar discussion will be given for the Boussinesq model with thermal fluctuation. Part of the work is jointly with Adam Larios and Leo Rebholz. (Received August 24, 2020)

1162-35-100 **Animikh Biswas**, **Zachary Bradshaw** and **Michael Jolly*** (msjolly@indiana.edu), Department of Mathematics, Indiana University, Bloomington, IN 47405. *Data assimilation using local observables.*

We develop, analyze, and test an approximate, it global data assimilation/synchronization algorithm based on purely it local observations for the two-dimensional Navier-Stokes equations on the torus. We prove that, for any error threshold, if the reference flow is analytic with sufficiently large analyticity radius, then it can be recovered within that threshold. Numerical computations are included to demonstrate the effectiveness of this approach, as well as variants with data on moving subdomains. In particular, we demonstrate numerically that machine precision synchronization is achieved for it mobile data collected from a small fraction of the domain. (Received August 26, 2020)

1162-35-103 **ALVIS ZHAODH***, 1132 Saluda Ct., 5, 5, Chapel Hill, NC 27514, and **Mark Williams**. *Hyperbolic boundary problems with large oscillatory coefficients on small frequency region.*

We study weakly stable hyperbolic boundary problems with highly oscillatory coefficients that are large, $O(1)$, compared to the small wavelength ϵ of oscillations. Such problems arise, for example, in the study of classical questions concerning the stability of Mach stems and compressible vortex sheets. For such applications one seeks to prove energy estimates that are in an appropriate sense "uniform" with respect to the small wavelength ϵ, but the large oscillatory coefficients are a formidable obstacle to obtaining such estimates. In this paper we analyze a simplified form of the linearized problems that are relevant to the above stability questions, and obtain positive results in the small frequency region with the approach presented in [Wi19] under "unfavorable" conditions. We also construct geometric optics solutions for those systems as applications to the main estimation theorem. (Received August 27, 2020)

1162-35-113 **Adam Larios***, 203 Avery Hall, PO BOX 880130, Lincoln, NE 68588-0130, and **Kazuo Yamazaki**. *2D Kuramoto-Sivashinsky: A reduced model with comparisons to Navier-Stokes.*

The Kuramoto-Sivashinsky equation (KSE) is a highly chaotic dynamical system that arises in flame fronts, plasmas, crystal growth, and many other phenomena. Due to its lack of a maximum principle, the KSE is often studied as an analogue to the 3D Navier-Stokes equations (NSE). Much progress has been made on the 1D KSE since roughly 1984, but even for the 2D KSE, global well-posedness remains a major open question. In analogy with regularizations of the 3D NSE, we present modifications of the 2D KSE which allow for global well-posedness, while still retaiing many imporant features of the 2D KSE. However, as has been demonstrated recently by Kostianko, Titi, and Zelik, standard regularizations, that work well for Navier-Stokes, fail when applied to even the 1D KSE. Thus, we present entirely new types of modification for the 2D KSE. This talk will describe key ideas of the analysis, and also show many colorful movies of solutions. (Received August 28, 2020)

1162-35-118 **Jeff Borggaard**, **Nathan Glatt-Holtz** and **Justin Krometis*** (jkrometis@vt.edu). *A Bayesian Approach to Estimating Background Flows from a Passive Scalar.*

We consider the Bayesian inverse problem of estimating a background flow field from the partial and noisy observation of a passive scalar (e.g., a solute concentration) governed by advection and diffusion. We provide conditions under which the inference is consistent, i.e., the posterior converges to a Dirac measure on the true flow as the number of observations grows large. We also attack the problem computationally by leveraging MCMC methods adapted in recent years to infinite-dimensional settings. (Received August 28, 2020)

1162-35-129 **Cecilia Mondaini*** (cf823@drexel.edu) and **Nathan Glatt-Holtz**. *Mixing for Hamiltonian Monte Carlo in infinite dimensions.*

The analysis of convergence/mixing rates for testing sampling efficiency in Markov Chain Monte Carlo (MCMC) algorithms, a fundamental question for establishing their range of applicability, has gained increased attention

recently. Of particular interest are MCMC methods that are designed to be well-defined in infinite dimensions, a property that allows them to overcome the curse of dimensionality when applied to corresponding finite-dimensional approximations. We analyze such question for an infinite-dimensional version of the Hamiltonian/Hybrid Monte Carlo algorithm, for which mixing rates had been an open problem until being very recently addressed via an exact coupling approach. Our proof uses the weak Harris theorem together with a generalized coupling argument, providing a flexible methodology to establish mixing rates for other MCMC algorithms. Furthermore, as an application of our general result, we show that all required assumptions can be verified in the context of a Bayesian inversion approach to advection-diffusion type PDEs. This is a joint work with Nathan Glatt-Holtz (Tulane U). (Received August 29, 2020)

1162-35-145 **William M Feldman*** (feldman@math.utah.edu). *Contact angle hysteresis and free boundaries in heterogeneous media.* Preliminary report.

I will discuss some mathematical results related to the study of capillary droplets on rough surfaces. (Received August 30, 2020)

1162-35-146 **Joceline Lega***, Department of Mathematics, 617 N. Santa Riga Avenue, Tucson, AZ 85721. *Phase singularities and defects in the Swift-Hohenberg equation.* Preliminary report.

We are interested in understanding the structure of symmetric grain boundary solutions of a canonical pattern-forming model, the Swift-Hohenberg equation. We focus on the so-called strong bending limit (small angle between the roll patterns on each side of the defect), when dislocations are present along the core of the grain boundary. In this relatively simple situation, the latter takes the form of a straight line.

The geometry of this particular pattern allows one to approach the problem on the half plane bounded by the line of dislocations, and to view the solution as resulting from spatial dynamics in a direction perpendicular to the core of the grain boundary. Moreover, because the solution is periodic in the direction parallel to the line of dislocations, we consider its Poisson extension to the unit disc, together with its harmonic conjugate, which is built from the Hilbert transform of the real field. As a consequence, the appearance of phase singularities as one approaches the line of dislocations may be related to the dynamics of zeros of the resulting complex function defined on the unit disc.

This work is joint with Nick Ercolani. (Received August 30, 2020)

1162-35-149 **Inwon Kim***, Mathematical Science Building, 520 Portola Plaza, Los Angeles, CA 90095, and **Nestor Guillen** and **Antoine Mellet**. *Incompressible limit of degenerate diffusions with absorption.* Preliminary report.

We study the incompressible limit of the porous medium equation with a sink term. The model gives a simplified description of de-congestion phenomena in tumor growth and crowd motion. (Received August 30, 2020)

1162-35-155 **Dallas Albritton*** (dallas.albritton@nyu.edu). *Self-similar solutions of active scalars with critical dissipation.*

In PDE analyses of fluid models, often we may identify a so-called critical space that lives precisely at the borderline between well-posedness and ill-posedness. What happens at this borderline? We explore this question in two active scalar equations with critical dissipation. In the surface quasi-geostrophic equations, we investigate the connection between non-uniqueness and large self-similar solutions that was established by Jia, Sverak, and Guillod in the Navier-Stokes equations. This is joint work with Zachary Bradshaw. In the critical Burgers equation, and more generally in scalar conservation laws, the analogous self-similar solutions are unique, and we show that all front-like solutions converge to a self-similar solution at the diffusive rates. This is joint work with Raj Beekie. (Received August 30, 2020)

1162-35-164 **Lorena Bociu***, lvbociu@ncsu.edu. *Minimizing Drag in A Moving Boundary Fluid-Elasticity Interaction.*

We are interested in minimizing fluid vorticity in the case of an elastic body moving and deforming inside an inviscid fluid, using a distributed control. This translates into analyzing an optimal control problem subject to a moving boundary fluid-structure interaction (FSI). The FSI is described by the coupling of Navier-Stokes and wave equation. The control is inherently a nonlinear control, acting as feedback on the moving frame. Its action depends on the flow map of the domain, which is itself defined through the dynamics of the problem. A key ingredient in the optimal control problem is represented by the long time behavior of the forced dynamics, which was an open problem in the field. Our results include existence of solutions for all times with small distributed sources and small initial data, as well as existence of optimal control for the problem of minimization of drag in the fluid. (Received August 31, 2020)

1162-35-176 **Braden Balentine*** (braden.balentine@colorado.edu), University of Colorado Boulder, Department of Mathematics, Campus Box 395, Boulder, CO 80302. *Well-Posedness and Global in Time Behavior for Mild Solutions to the Navier-Stokes Equation on the Hyperbolic Space with Initial Data in L^p.*

We discuss mild solutions to the Navier-Stokes equation on the n-dimensional hyperbolic space $\mathbb{H}^n$, $n \geq 2$. Substantial results have been already obtained by Pierfelice. In this talk, we show how to extend the rest of the Fujita-Kato theory of mild solutions from $\mathbb{R}^n$ to $\mathbb{H}^n$. This includes well-posedness results for L^p initial data in the range $1 < p < \infty$, global in time results for small initial data, and L^p norm decay results for both u and ∇u. As part of this, we discuss extending to the hyperbolic space $\mathbb{H}^n$ known facts in Euclidean space concerning the strong continuity and contractivity of the semigroup generated by the Laplacian. Also, we discuss necessary boundedness and commutation properties for a certain projection operator in the setting of $\mathbb{H}^n$. This work, together with Pierfelice's, contributes to providing a full theory for mild solutions on $\mathbb{H}^n$. (Received August 31, 2020)

1162-35-177 **Elizabeth A Carlson***, elizabeth.carlson@huskers.unl.edu, and **Adam Larios**, **Joshua Hudson**, **Luke Van Roekel**, **Humberto C Godinez** and **Mark Petersen**. *AOT Data Assimilation Algorithm: Parameter Recovery, Applications, and a Nonlinear Algorithm.*

One of the challenges of the accurate simulation of turbulent flows is that initial data is often incomplete. Data assimilation circumvents this issue by continually incorporating the observed data into the model. A new approach to data assimilation known as the Azouani-Olson-Titi (AOT) algorithm introduced a feedback control term to the 2D incompressible Navier-Stokes equations (NSE) in order to incorporate sparse measurements. The solution to the AOT algorithm applied to the 2D NSE was proven to converge exponentially to the true solution of the 2D NSE with respect to the given initial data. In this talk, we present our previous work on how the AOT algorithm applied to a system of equations with an incorrect parameter still converges to the correct solution up to an error determined by the error in the parameters, and how this led to the development of a simple parameter recovery algorithm. We will also present how this work led to the implementation of the AOT algorithm in the Model for Prediction Across Scales - Ocean model. Finally, in a different direction we also present a proof of the convergence of a nonlinear version of the AOT algorithm in the setting of the 2D NSE, where for a portion of time the convergence rate is proven to be double exponential. (Received August 31, 2020)

1162-35-179 **Nathan Glatt-Holtz*** (negh@tulane.edu). *On the long-time statistical behavior of smooth solutions of the weakly damped stochastically-driven KdV Equation.*

This talk presents a collection of new results concerning the ergodic and regularity properties of the long time statistics of the weakly damped Korteweg-de Vries (KdV) equations stirred by a white in time but spatially smooth (degenerate) Gaussian source term. This is work carried out in collaboration with Vincent Martinez (Hunter College) and Geordie Richards (Utah State). (Received August 31, 2020)

1162-35-181 **Hyunju Kwon*** (hkwon@ias.edu). *Higher derivatives of suitable weak solutions of the hypodissipative Navier-Stokes equations.*

I'll consider the hypo-dissipative version of the Navier-Stokes equations (i.e., the dissipation is given as a fractional Laplacian $(-\Delta)^s$ for $s < 1$), and discuss higher derivative estimates for suitable Leray-Hopf weak solutions. This is joint work with Wojciech S. Ożański. (Received August 31, 2020)

1162-35-191 **Sarah S Hagen*** (hagenss@wou.edu) and **Radu Dascaliuc**. *Spectral Structure of Solutions to the Navier-Stokes Equations with Constant Energy and Enstrophy.*

Motivated by open questions in the long-time dynamics and fluid turbulence, we investigate the existence of nonstationary solutions to the Navier-Stokes equations (NSE) with constant energy profiles, with constant enstrophy profiles, or both. In this talk we make progress in this area by constructing a nonstationary solution to the incompressible NSE (on the 3D torus) whose energy remains constant. Similarly, we construct a nonstationary solution whose enstrophy remains constant. These constructions necessarily exist outside of the attractor and are supported on an infinite number of Fourier modes. On the 2D torus we show that nonstationary constant energy (resp. enstrophy) solutions (if they exist) must necessarily be supported on an infinite number of modes when the force is an eigenvector of the Stokes operator. (Received August 31, 2020)

1162-35-197 **Nathan Glatt-Holtz**, **Vincent Martinez** and **Geordie Richards*** (geordie.richards@usu.edu). *Asymptotic coupling and unique ergodicity for damped stochastic KdV.*

We will review a straightforward asymptotic coupling strategy for establishing uniqueness of the invariant measure for stochastic PDEs. As a novel application, we will provide a concise proof of unique ergodicity for damped

stochastic KdV with essentially elliptic stochastic forcing, employing a "nudging" strategy which has also been useful in data assimilation problems. This is a joint work with Nathan Glatt-Holtz (Tulane) and Vincent Martinez (Hunter). (Received August 31, 2020)

1162-35-198 **Nathan Glatt-Holtz**, **Vincent Martinez** and **Geordie Richards*** (geordie.richards@usu.edu). *Ergodic and mixing properties of damped stochastic KdV.*

We will present some new ergodic and mixing theorems for damped stochastic KdV: uniqueness and higher regularity of the invariant measure, and exponential mixing for large damping. This is a joint work with Nathan Glatt-Holtz (Tulane) and Vincent Martinez (Hunter). (Received August 31, 2020)

1162-35-199 **Aseel Farhat*** (afarhat@fsu.edu), 1017 Academic Way, Tallahassee, FL 32304, and **Hans Johnston**, **Michael S. Jolly** and **Edriss S. Titi**. *Assimilation of nearly turbulent Rayleigh-Bénard flow through local circulation measurements.*

We will discuss a continuous (downscaling) data assimilation algorithm for the 2D Bénard convection problem using vorticity or local circulation measurements. Our numerical results indicate that the approximate solution of the algorithm is converging to the unknown reference solution (vorticity and temperature) corresponding to the measurements of the 2D Bénard convection problem. Moreover, this convergence is realized using data which is much more coarse than the resolution needed to satisfy rigorous analytical estimates. (Received August 31, 2020)

1162-35-203 **Gautam Iyer*** (gautam@math.cmu.edu), Department of Mathematical Sciences, Carnegie Mellon University, PA 15213, and **Son Van**. *Bounds on the heat transfer rate via passive advection.*

In heat exchangers, an incompressible fluid is heated initially and cooled at the boundary. The goal is to transfer the heat to the boundary as efficiently as possible. In this talk we study a related steady version of this problem: Consider the steady state temperature of in a fluid that is stirred, uniformly heated and cooled on the boundary. For a given Péclet number is large, how should one stir to minimize the total heat? This problem was studied by Marcotte, Doering, Thiffeault and Young in '18, where the authors provided many heuristics and numerical simulations. In this talk we will show that when the Péclet number is large, one can always find a stirring velocity field so that the total heat is at most $O((\log \mathrm{Pe})^9 / \mathrm{Pe}^{1/2})$. We suspect this is optimal (up to a logarithmic correction), but are presently unable to prove a matching lower bound. (Received August 31, 2020)

1162-35-213 **Juraj Foldes*** (foldes@virginia.edu) and **Mouhamadou Sy**. *Invariant measures for Hamiltonian systems.*

Using a fluctuation dissipation method, we construct an invariant measure for the surface quasi-geostrophic equation (SQG) and 3D Euler equation. Since the support of the measure contains entire solutions, we obtain a manifold containing solutions that do not blow-up. This complements results in which a blow-up solutions for SQG and grow up solutions for Euler are constructed. The method of the proof relies on an addition of a stochastic forcing and a small dissipation to the equation. For such stochastic equation, one can construct an invariant measure and by passing the strength of the forcing and the dissipation to zero, we obtain the desired invariant measure. We also discuss the size of the support of the measure, which relies on the number of conservation laws for the particular equation.

This is a joint project with Mouhamadou Sy. (Received August 31, 2020)

1162-35-215 **Tuan Ngoc Pham*** (tuan.pham@mathematics.byu.edu). *Blowup solutions of a Navier-Stokes-like equation - A probabilistic perspective.*

Although it is not known if an L^3-initial datum can produce a blowup solution of the 3-dimensional Navier-Stokes equations, various results have been established based on the assumption that there is such a blowup-generating datum. One of the results is the existence of a blowup-generating datum with minimal norm (Jia-Sverak 2013, Gallagher et al 2013, etc). From a control theory perspective, the existence of minimal blowup data implies that there is a minimal cost one has to pay to generate blowup. The existence of minimal blowup data has been studied in several differential equations such as the harmonic map heatflow, the complex Ginzburg-Landau equation and the Schrodinger equation. In 1997, Le Jan and Sznitman constructed solutions of the Navier-Stokes equations via branching processes, similar to McKean's approach to the KPP-Fisher equation in 1975. We employ their idea to study the minimal blowup data of a Navier-Stokes-like equation called cheap NSE by Montgomery-Smith, 2002. Joint work with Radu Dascaliuc and Chris Orum. (Received September 01, 2020)

1162-35-240 **John C. Bowman*** (bowman@ualberta.ca) and **Pedram Emami**. *Bounds on the global attractor of 2D incompressible turbulence in the palenstrophy–enstrophy–energy space.* Preliminary report.

Analytic bounds on the projection of the global attractor of 2D incompressible turbulence in the palenstrophy–enstrophy plane [Dascaliuc, Foias, and Jolly 2005, 2010] are observed to vastly overestimate the values obtained from numerical simulations. This is due to the lack of a good estimate for the inner product of the advection term $(\mathcal{B}(u,u), A^2u)$ and the biLaplacian. Sobolev inequalities like Ladyzhenskaya or Agmon's inequalities yield an upper bound [Foias, Jolly, Manley, and Rosa, 2002] that we show is not sharp. In fact, for statistically isotropic turbulence, the expected value of this term is zero. The implications for estimates on the behaviour of the global attractor are discussed. (Received September 02, 2020)

41 ► Approximations and expansions

1162-41-79 **Vira Babenko*** (vira.babenko@drake.edu) and **Vladyslav Babenko**. *Function L-spaces. Optimal recovery and best approximation.*

We consider the problem of approximation of unbounded positively homogeneous operators in L-spaces using Lipschitz operators. In this talk, we discuss its connection to the problem of computing modulus of continuity of the unbounded operator on the class of elements, as well as, to the problem of optimal recovery of an unbounded operator by a Lipschitz one on the class of elements given with an error. As applications, we consider the problem of approximation of unbounded operator, that for functions with values in some L-space puts in a correspondence Hukuhara-type derivatives, by Lipschitz operators. Further, we discuss the solution of the problem of the optimal recovery of this operator on the class of functions that have Hukuhara-type derivative with the given majorant of the modulus of continuity. (Received August 24, 2020)

1162-41-187 **Elena Cherkaev*** (elena@math.utah.edu), Salt Lake City, UT 84112. *Matrix Pade approximations of Herglotz-Nevanlinna Functions and Applications to Model Order Reduction.*

Pade approximations of Stieltjes and Herglotz-Nevanlinna functions play a central role in the operator theory connecting the moment problem with continuous fractions, Jacobi matrices, orthogonal polynomials, the spectral theory, and approximation problems. The talk will discuss matrix Pade approximations and reduced order representations of spectral decompositions of operators arising in homogenization and inverse problems. (Received August 31, 2020)

1162-41-195 **Tatyana Sorokina*** (tsorokina@towson.edu), 8000 York Rd, Towson, MD 21093. *Conforming harmonic finite elements.*

Instead of using the full polynomial space, conforming finite element methods are designed where only harmonic polynomials (a much smaller space) are employed in the computation. The conforming quadratic harmonic polynomial finite element is defined only on a special triangular grid. The optimal order of convergence is proved, and confirmed by numerical computations. In addition, numerical comparisons with the standard conforming and nonconforming finite elements are presented. Joint work with S. Zhang. (Received August 31, 2020)

44 ► Integral transforms, operational calculus

1162-44-114 **Xuan Thinh Duong** and **Loredana Lanzani*** (llanzani@syr.edu), Department of Mathematics, 900 South Crouse Ave., Syracuse, NY, and **Ji Li** and **Brett D. Wick**. *The commutator of the Cauchy-Szegő projection for domains in $\mathbb{C}^n$ with minimal smoothness.*

Let $D \subset \mathbb{C}^n$ be a bounded, strongly pseudoconvex domain whose boundary bD satisfies the minimal regularity condition of class C^2. We characterize boundedness and compactness in $L^p(bD,\omega)$, for $1 < p < \infty$, of the commutator $[b, S_\omega]$ where S_ω is the Cauchy–Szegő (orthogonal) projection of $L^2(bD,\omega)$ onto the holomorphic Hardy space $H^2(bD,\omega)$ and the measure ω belongs to a family (the "Leray Levi-like" measures) that includes induced Lebesgue measure σ. We next consider a much larger family of measures $\{\Omega\}$ modeled after the Muckenhoupt weights for σ: we define the holomorphic Hardy spaces $H^p(bD,\Omega)$ for any A_p-like measure Ω and we characterize boundedness and compactness of $[b,_\Omega]$ in $L^2(bD,\Omega)$ for any A_2-like measure Ω. Earlier closely related results rely upon an asymptotic expansion, and subsequent pointwise estimates, of the Cauchy–Szegő kernel that are not available in the settings of minimal regularity of bD and/or A_p-like measures. (Received August 28, 2020)

46 ► Functional analysis

1162-46-89 **Daniele Venturi*** (venturi@ucsc.edu). *Spectral methods for nonlinear functionals and functional differential equations.*

We present a rigorous convergence analysis for cylindrical approximations of nonlinear functionals, functional derivatives, and functional differential equations (FDEs). The purpose of this analysis is twofold: first, we prove that continuous nonlinear functionals, functional derivatives and FDEs can be approximated uniformly on any compact subset of a real separable Hilbert space by high-dimensional multivariate functions and high-dimensional partial differential equations (PDEs), respectively. Second, we show that the convergence rate of such functional approximations can be exponential, depending on the regularity of the functional (in particular its Fréchet differentiability), and its domain. We also provide necessary and sufficient conditions for consistency, stability and convergence of functional approximations to linear FDEs. These results open the possibility to utilize numerical techniques for high-dimensional systems such as deep neural networks and numerical tensor methods to approximate nonlinear functionals in terms of high-dimensional functions, and compute approximate solutions to FDEs by solving high-dimensional PDEs. Numerical applications are presented for prototype nonlinear functionals, and for an initial value problem involving a linear FDE. (Received August 25, 2020)

49 ► Calculus of variations and optimal control; optimization

1162-49-95 **Benjamin Cowen***, 6 MetroTech Center, Brooklyn, NY 11201, and **Anna Choromanska** and **Apoorva Nandini Saridena**. *LSALSA: Accelerated Source Separation via Learned Sparse Coding.*

We propose an algorithm for generalized sparse coding (SC) inference. SC inference is the problem of sparsely encoding a data point with respect to the columns of a given dictionary, or in the generalized setting as given in morphological component analysis (MCA), across multiple dictionaries. In MCA, each dictionary provides a distinct sparse representation corresponding to a component of the data, forming an additive mixture model for the original data point. Both settings have been cast as ℓ_1-regularized least-squares optimization problems; these convex formulations and their solvers such as the Split Augmented Lagrangian and Shrinkage Algorithm (SALSA) are trusted by practitioners but can be computationally extensive. We propose a deep learning solution called Learned SALSA (LSALSA), and demonstrate vast acceleration on image vision tasks, in addition to increasing the quality of estimated sparse codes and visual clarity on both classic SC and MCA problems in computation-starved settings. Finally, we present a theoretical framework for analyzing LSALSA, concluding that it exactly implements a truncated instance of the alternating direction method of multipliers (ADMM) applied to a new, learned cost function with curvature modified by one of the learned matrices. (Received August 26, 2020)

51 ► Geometry

1162-51-123 **Jerzy Kocik*** (jkocik@siu.edu), Department of Mathematics, Carbondale, IL 62901. *Pauli spinors, Apollonian depth function, and fractals.* Preliminary report.

A surprising applications of Pauli spinors allows one to visualize the space of tricycles (triples of disks in contact) as a complex plane. The fractal of the Apollonian depth function assumes a particularly elegant form, with a symmetry group that extends the standard modular group. (Received August 28, 2020)

54 ► General topology

1162-54-11 **Robert Owczarek*** (rowczare@unm.edu), 59 Coryphodon Ln, Jemez Springs, NM 87025. *Graphs, knots, invariants.* Preliminary report.

Recent results on graph and knot invariants will be presented (Received July 29, 2020)

1162-54-34 **Vincent Longo***, vincent.longo@huskers.unl.edu. *An infinite family of counterexamples to Batson's conjecture.* Preliminary report.

Batson's conjecture is a non-orientable version of Milnor's conjecture, which states that the 4-ball genus of a torus knot $T(p, q)$ is equal to $\frac{(p-1)(q-1)}{2}$. Batson's conjecture asks if the nonorientable 4-ball genus is equal to

the pinch number of a torus knot, i.e. the number of a specific type of (nonorientable) band surgeries needed to obtain the unknot. The conjecture was recently proved to be false by Lobb, whose counterexample we will show fits into an infinite family of counterexamples. (Received August 17, 2020)

1162-54-105 **Hannah R. Schwartz*** (hs25@princeton.edu). *Isotopy vs. homotopy for disks with a common dual.*

Recent work of both Gabai and Schneiderman-Teichner on the smooth isotopy of homotopic surfaces with a common dual has reinvigorated the study of concordance invariants defined by Freedman and Quinn in the 90's, along with homotopy theoretic invariants of Dax from the 70's. We will show that in any smooth 4-manifold with boundary, pairs of homotopic properly embedded disks with a common dual are isotopic if and only if the 'Dax invariant' of the pair vanishes. (Received August 27, 2020)

55 ► Algebraic topology

1162-55-39 **Subhankar Dey*** (subhanka@buffalo.edu). *A note on a geography problem in knot Floer homology.* Preliminary report.

Using Legendrian knot invariant defined by Lisca-Ozasvath-Stipsicz-Szabo, we prove that knot Floer homology of a certain class of knots is non-trivial in next-to-top Alexander grading. This gives a partial affirmative answer to a question posed by Baldwin and Vela-Vick. (Received August 18, 2020)

1162-55-63 **Deborah J Weeks*** (deweeks@gwu.edu), 801 22nd St. Northwest, Washington, DC 20052. *Framing Changes of Links in 3-Manifolds.*

Extending the work of McCullough and Chernov, the conditions under which the framing of a link can be changed via ambient isotopy are examined. A change in framing can only be accomplished via the Dirac trick in a manifold with a properly embedded non-separating S^2. Due to the invariance of spin structure and the parallelizability of every compact oriented 3-manifold the framing of a knot can only change by an even power. (Received August 22, 2020)

1162-55-139 **Neil Saunders*** (n.saunders@greenwich.ac.uk), University of Greenwich, Park Row, London, United Kingdom. *Geometry and Combinatorics coming from the Exotic Nilpotent Cone.*

The exotic nilpotent cone as defined by Kato gives a 'Type A-like' Springer correspondence in Type C. In particular, there is a bijection between the symplectic group orbits on the exotic nilpotent cone and the irreducible representations of the Weyl group of Type C. In this talk, I will outline some various geometric and combinatorial consequences that follow from this, such as a parameterisation of the irreducible components of the 'exotic' springer fibres using standard Young bitableaux. These results are joint work with Vinoth Nanadakumar and Daniele Rosso, and Arik Wilbert. (Received August 30, 2020)

1162-55-173 **Henry Adams** (henry.adams@colostate.edu), **Johnathan Bush*** (johnathan.bush@colostate.edu) and **Florian Frick** (frick@cmu.edu). *Borsuk-Ulam theorems for maps into higher-dimensional codomains.*

We say a continuous map from a sphere to Euclidean space is odd if it respects the standard antipodal action. One formulation of the Borsuk-Ulam theorem states that the image of an odd map from an n-sphere to $\mathbb{R}^n$ must contain the origin. We generalize this result by increasing the dimension of the codomain, and by finding a small diameter subset of the sphere whose image contains the origin in its convex hull.

For example, given an arbitrary odd map from the circle of circumference 1 to $\mathbb{R}^{2k+1}$, there is some subset of the circle of diameter at most $\frac{k}{2k+1}$ whose image contains the origin in its convex hull. Furthermore, this diameter bound is sharp: there exists an odd map $S^1 \to \mathbb{R}^{2k+1}$ such that the convex hull of the image of $X \subset S^1$ does not contain the origin whenever $\operatorname{diam}(X) < \frac{k}{2k+1}$.

Interestingly, these Borsuk-Ulam type theorems follow from an understanding of the topology of certain simplicial complexes defined on spheres. We will outline this connection and summarize current work toward obtaining sharper bounds for arbitrary odd maps $S^n \to \mathbb{R}^k$. (Received August 31, 2020)

57 ► Manifolds and cell complexes

1162-57-29 **Rhea Palak Bakshi*** (rhea_palak@gwu.edu) and **Józef H. Przytycki**. *The Kauffman Bracket Skein Module of the Connected Sum of Two Handlebodies: A Counterexample.*

In this talk we disprove a twenty-two-year-old theorem about the structure of the Kauffman bracket skein module of the connected sum of two handlebodies. We achieve this by analysing handle slidings on compressing discs in a handlebody. We find more relations than previously predicted for the Kauffman bracket skein module of the connected sum of handlebodies, when one of them is not a solid torus. Additionally, we speculate on the structure of the Kauffman bracket skein module of the connected sum of two solid tori. This is joint work with Józef H. Przytycki. (Received August 14, 2020)

1162-57-31 **Jozef H Przytycki*** (przytyck@gwu.edu), George Washington Universuty, Phillips Hall, Phillips Hall, Washington, DC 20052-0001, and **Rhea Palak Bakshi** (rhea_palak@gwmail.gwu.edu). *Remarkable closed formula in Temperley-Lieb algebra.*

With the aim of computing the Kauffman bracket skein module of the connected sum of two solid tori, we prove the following in Temperley-Lieb algebra. Consider a positive pure braid $w_k = \sigma_{k-1}\sigma_{k-2}\cdots\sigma_2\sigma_1^2\sigma_2\cdots\sigma_{k-2}\sigma_{k-1}$ and project it onto Temperley-Lieb algebra by taking $\sigma_i = AId + A^{-1}e_i$. We write w_k in the classical (Catalan) basis of TL_k and obtain the following formula for the coefficients of E or its any partial permutation (notation is explained below):

$$A^{2k-2j_s-4}(A^{2i_1} - A^{-2i_1})(A^{2(i_2-j_1-1)} + A^{-2(i_2-j_1-1)})\cdots(A^{2(i_s-j_{s-1}-1)} + A^{-2(i_s-j_{s-1}-1)}).$$

Let $w_k = \sum_{C\in Cat} C(A)C$ where the sum is taken over all classical Catalan basis elements and $C(A)$ is a Laurent polynomial coefficient. Then $C(A) = 0$ if C written in TL_k generators (in a minimal form) has some letter appearing twice. Otherwise let $E_{i,j} = e_ie_{i+1}...e_j$ for any $1 \le i \le j \le k-1$. Define $E = E_{i_1,j_1}E_{i_2,j_2}...E_{i_s,j_s}$ where $j_{t-1} \le i_t - 2$. Our formula is useful in studying the effect of handle slidings on the KBSM of connected sums of arbitrary 3-manifolds. (Received August 14, 2020)

1162-57-36 **Olga Plamenevskaya** and **Laura Starkston*** (lstarkston@math.ucdavis.edu). *Unexpected Stein fillings and plane curve arrangements.*

We use singular braided symplectic surfaces in $\mathbb{C}^2$ in comparison with algebraic curves in $\mathbb{C}^2$ in order to exemplify differences between Stein fillings and algebraic deformations of a surface singularity. We work with Lefschetz fibrations and open book decompositions arising naturally from the projection of $\mathbb{C}^2$ to $\mathbb{C}$. We develop a symplectic analogue to results in the complex algebraic setting by de Jong and van Straten. (Received August 17, 2020)

1162-57-83 **Dave Auckly***, Math Department, Kansas State University, Manhattan, KS 66506, and **Daniel Ruberman**. *Exotic homotopy classes in diffeomorphism groups.*

The homotopy groups of the diffeomorphism/homeomorphism group of a four-dimensional manifold are in many ways analogous to the smooth/topological concordance group. Both reflect the large difference between embedded surfaces in the smooth and topological categories. This talk will present results demonstrating the large difference between these groups in the smooth and topological categories, describe how smoothly knotted surfaces lead to exotic homotopy classes of diffeomorphisms, and outline some parts of the computation of the relevant gauge-theoretic invariants that detect such exotic homotopy classes. (Received August 25, 2020)

1162-57-86 **Allison N. Miller*** (allison.miller@rice.edu) and **Mark Powell**. *Stabilization of knotted surfaces.*

Any two embedded surfaces representing the same homology class in a given 4-manifold become isotopic after finitely many 1-handle additions, as proved by Baykur-Sunukjian. Given two surfaces properly embedded in a fixed 4-dimensional manifold, their *stabilization distance* is the minimal number of 1-handle stabilizations necessary for the surfaces to become ambiently isotopic, and their *generalized stabilization distance* is the minimal number of 1-handle stabilizations if one is also allowed to take connected sum with arbitrary 2-knots and count that as distance zero. For every nonnegative integer m we give a knot J_m in the 3-sphere with two slice discs in the 4-ball whose generalized stabilization distance is exactly m, using the homology of cyclic covers. We then use metabelian twisted homology to derive 'higher-level' bounds on the generalized stabilization number, and use these bounds to prove that slicing discs for a knot coming from a fixed metabolizing link on a Seifert surface can have arbitrarily large generalized stabilization distance.

This is joint work with Mark Powell. (Received August 25, 2020)

1162-57-96 **Daniel Ruberman*** (ruberman@brandeis.edu), Brandeis University, Department of Mathematics, MS 050, 415 South Street, Waltham, MA 02454. *A Levine-Tristram invariant for knotted tori.*

In 1969, Tristram and Levine independently introduced an integer-valued function $\sigma_\alpha(K)$ of a knot, where α is a unit complex number. It gives rise to a concordance invariant that in turn shows that the the concordance group is infinitely generated. I will explain a generalization of this invariant to the setting where the 3-sphere is replaced by $X =$ a homology $S^1 \times S^3$, and the knot is replaced by an embedded torus (that carries the first homology). In the classical setting, $\frac{1}{2}\sigma_{\alpha^2}(K)$ is equal to a count of irreducible $SU(2)$ representations on the knot complement with holonomy α on the meridian. A similar count of representations, for embedded tori, was introduced last year by Echeverria. While our invariant is equal in some cases to (a normalized version of) Echeverria's, I will give examples to show that the two are not equal in general. (Received August 26, 2020)

1162-57-109 **Peter Feller** and **JungHwan Park***, jungpark0817@gmail.com. *Distances between torus knots.*

I will talk about several notions of distance between torus knots. More precisely, we consider distances of knots that are related to the slicing number, the 4-dimensional clasp number, and the 4-ball genus. This is joint work with Peter Feller. (Received August 28, 2020)

1162-57-121 **J Scott Carter*** (carter@southalabama.edu), 16003 Windroot St., Austin, TX 78728. *Codimension one properly embedded surfaces as a (higher) category.*

This is based upon joint work with Seiichi Kamada. A proof of the following (sort-of) classical result will be proven. A generic properly and smoothly embedding of a surface that has boundary corresponds to a globular composition of homogenous triple arrows in a 4-category that has two objects and back-and-forth non-identity arrows between them. Properly isotopic surfaces correspond to equivalent homogenous triple arrows in that 4-category. Moreover, given a globular composition of homogeneous triple arrows, a representative proper embedding can be constructed. Equivalent triple arrows give rise to properly isotopic surfaces. The construction of the category gives rise to a method of constructing higher dimensional versions thereof. In these relations among weakly equivalent (multi)-arrows correspond to embedded handles.

As one might expect, the talk will be replete with illustrations. (Received August 28, 2020)

1162-57-134 **Razvan Gelca*** (rgelca@gmail.com), Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409, and **Cezar Lupu** (cezar.lupu@ttu.edu), Department of Mathematics and Statistics, Texas Tech University, Lubbock, TX 79409. *Weil's action of the Heisenberg group on theta functions and the colored Jones polynomials of torus knots.*

We will explain how the action of the finite Heisenberg group on theta functions can be used for understanding the colored Jones polynomials of torus knots and give a quick proof of the formula for these invariants. (Received August 29, 2020)

1162-57-138 **Hongtaek Jung** (htjung@ibs.re.kr), Center for Geometry and Physics, Institute for Basic Science, 77 Cheongam-ro, Nam-gu, Pohang, Gyeongbuk 37673, South Korea, **Sungkyung Kang** (skkang@ibs.re.kr), Center for Geometry and Physics, Institute for Basic Science, 77 Cheongam-ro, Nam-gu, Pohang, Gyeongbuk 37673, South Korea, and **Seungwon Kim*** (math751@ibs.re.kr), Center for Geometry and Physics, Institute for Basic Science, 77 Cheongam-ro, Nam-gu, Pohang, Gyeongbuk 37673, South Korea. *Concordance invariants and the Turaev genus.* Preliminary report.

In this talk, we introduce new lower bounds of Turaev genus using various concordance invariant. As an application, we showed that for any positive integer g, we can find a quasi-alternating knot with Turaev genus g, which answers the question originally asked by Champanerkar and Kofman. This is a joint work with Hongtaek Jung and Sungkyung Kang. (Received August 30, 2020)

1162-57-141 **Micah Chrisman**, **Sujoy Mukherjee*** (mukherjee.166@osu.edu) and **Józef H. Przytycki**. *On multi-term distributive homology.* Preliminary report.

Two-term distributive (rack) homology, introduced in 1990, and its enhancement, twisted quandle homology, are very useful tools for constructing link invariants and for studying self-distributive algebraic structures. The third author proposed the theory of multi-term distributive homology in 2010 to study the homology of self-distributive algebraic structures under a general framework.

After a brief discussion about the monoid of all binary operations on a set X and multi-shelves, I will introduce multi-term distributive homology. Finally, I will talk about our present work on the topic. (Received August 30, 2020)

1162-57-143 **R. Bakshi**, **D. Ibarra** and **G. Montoya-Vega*** (gmontoyavega@gwu.edu), 2514 K ST NW, Apt 25, Washington, DC 20037, and **J. Przytycki** and **D. Weeks**. *Framing Changes of Links in* 3*-Manifolds in the language of skein modules.*

The framing of a knot K in an oriented 3-manifold remains unchanged unless there exists a properly embedded non-separating 2-sphere which intersects K exactly once; in which case, the change of framing is given by the Dirac trick. Since 1987, when Przytycki introduced skein modules, these have been extensively studied with the goal of building an algebraic topology based on knots. In this talk, we formulate the results on the framing of knots in the language of skein modules. (Received August 30, 2020)

1162-57-144 **Alexandra Kjuchukova*** (sashka@mpim-bonn.mpg.de), sashka@mpim-bonn.mpg.de, and **Kent Orr**. *Extending knot group quotients over surfaces in* B^4*.*

Let G be a group which arises as a quotient of a knot group. A "G-knot" $K \subset S^3$ is a knot equipped with a surjective homomorphism $\rho : \pi_1(S^3 - K) \to G$. When does a G-knot K bound a smooth G-surface $F \subset B^4$ extending ρ? We find a sharp obstruction, $\Theta_G(K, \rho)$, to the existence of such a surface F. When G is a dihedral group or a group of the form $Z_n \rtimes Z := \langle u, t | t^n, utut^{-1} \rangle$, the obstruction $\Theta_G(K, \rho)$ admits a description in terms of the Seifert form of K. Joint with Kent Orr. (Received August 31, 2020)

1162-57-156 **Jeffrey Meier**, **Abigail Thompson** and **Alexander Zupan*** (zupan@unl.edu). *Cubic graphs induced by bridge trisections.*

A bridge trisection of a knotted surface $\mathcal{K} \subset S^4$ induces a cell decomposition of the underlying surface whose 1-skeleton is a cubic Tait-colored graph Γ; that is, every vertex of Γ has valence three, and each edge of Γ is colored red, blue, or green, with the property that every vertex is incident to edges of all three colors. A natural question is whether this process can be reversed: Given a cubic Tait-colored graph Γ, does there exist a surface $\mathcal{K} \subset S^4$ and a bridge trisection of $\mathcal{K}$ inducing Γ? We answer the question in the affirmative, showing further that $\mathcal{K}$ can even chosen to be unknotted and with any possible normal Euler number (if $\mathcal{K}$ is nonorientable). (Received August 30, 2020)

1162-57-162 **Kyle Hayden*** (hayden@math.columbia.edu), MC 4403, RM 626, 2990 Broadway, New York, NY 10027. *Exotically knotted disks and complex curves.*

I will present a new construction of surfaces in 4-manifolds that are exotically knotted, i.e., isotopic through ambient homeomorphisms but not through diffeomorphisms. This gives rise to exotically knotted surfaces of all genera in the 4-ball. The construction is uniquely well-suited to the complex/symplectic setting, producing the first examples of exotically knotted complex curves and symplectic 2-spheres. (Received August 31, 2020)

1162-57-182 **Dave Auckly** (dav@math.ksu.edu), Department of Mathematics, Kansas State University, **Hee Jung Kim*** (heejungorama@gmail.com), Department of Mathematics, Western Washington University, **Paul Melvin** (pmelvin@brynmawr.edu), Department of Mathematics, Bryn Mawr College, and **Daniel Ruberman** (ruberman@brandeis.edu), Department of Mathematics, Brandeis University. *Homologous 2-spheres in 4-manifolds.*

It has been shown that there are infinite families of simply-connected 4-manifolds for which all primitive ordinary homology classes of square 1 are represented by topologically isotopic but smoothly distinct 2-spheres that become smoothly isotopic after one external stabilization. We will discuss examples of homologous 2-spheres of higher intersections that hold similar results and as an interesting feature, it will be seen that geometric intersections of any pair of these spheres must be larger than their algebraic intersections. This is joint work with Dave Auckly, Paul Melvin, and Daniel Ruberman. (Received August 31, 2020)

1162-57-190 **Clayton McDonald*** (mcdonafi@bc.edu), 48 Lanark Road Apt 2, Brighton, MA 02135, and **Duncan McCoy**. *Doubly Slice Links.*

A knot K in S^3 is slice if it is the cross section of an embedded sphere in S^4, and it is doubly slice if the sphere is unknotted. We will discuss various different generalizations of double slicing to links and the various constructions and obstructions involved in distinguishing them. (Received August 31, 2020)

1162-57-194 **Patrick Naylor*** (patrick.naylor@uwaterloo.ca). *Gluck twisting roll spun knots.*

Given an embedded 2-sphere in the 4-sphere, a Gluck twist is a surgery which produces a 4-manifold homeomorphic, but not obviously diffeomorphic to the 4-sphere. This operation has been well studied, but it remains an open question whether all Gluck twists are standard. Many 2-spheres arising from "spinning" constructions are known to have standard Gluck twists, but roll-spun knots are notably absent from the list of such spheres. In this talk, I will discuss some recent results showing that the Gluck twist of the roll spin of an unknotting number one knot is standard. This is joint work with Hannah Schwartz. (Received August 31, 2020)

1162-57-209 **Jason Joseph*** (jason.joseph@rice.edu). *Unknotting numbers of 2-spheres in the 4-sphere.*

In this talk we consider two unknotting operations on knotted spheres embedded in 4-space. One, the stabilization number, is the minimal number of 1-handle stabilizations needed to produce an unknotted surface. The other, which we call the Casson-Whitney number, counts the minimal number of pairs of finger and Whitney moves needed during a regular homotopy to the unknotted sphere. Using an algebraic lower bound coming from the knot group we will prove that these unknotting numbers are distinct in infinitely many examples. Time permitting we will show that both unknotting numbers can exhibit strongly nonadditive behavior under connected sum. This is joint work with Michael Klug, Benjamin Ruppik, and Hannah Schwartz. (Received August 31, 2020)

1162-57-223 **Inanc Baykur*** (baykur@math.umass.edu) and **Noriyuki Hamada**. *Lefschetz fibrations, symplectic geography and knotted surfaces.*

We will discuss how to build symplectic Lefschetz fibrations with prescribed signature and spin type, along with novel applications to the symplectic geography problem and speculations on exotically knotted orientable surfaces in the 4-sphere. This is joint work with N. Hamada. (Received September 01, 2020)

1162-57-239 **Samuel J Lomonaco*** (lomonaco@umbc.edu), 1000 Hilltop Circle, 10236 Little Brick House Court, Baltimore, MD 21250. *Four Dimensional Knot Theory Revisted.* Preliminary report.

We will talk about the second homotopy group as a ZPi1-module of the exterior of 2-knots in 4-space. (Received September 01, 2020)

60 ► Probability theory and stochastic processes

1162-60-127 **Anthony Sanchez*** (asanch33@uw.edu), Seattle, WA 98115. *Gaps of saddle connection directions for some branched covers of tori.* Preliminary report.

Consider the class of translation surfaces given by gluing two identical tori along a slit. Every such surface has genus two and two cone-type singularities of angle 4π. There is a distinguished set of geodesics called saddle connections that are the geodesics between cone points. We can recover a vector in the plane representing the saddle connection by keeping track of the amount that the saddle connection moves in the vertical and horizontal direction. How random is the set of saddle connections? We motivate the gap distribution of slopes as a measure of randomness and compute the gap distribution of slopes of saddle connections. (Received August 29, 2020)

1162-60-230 **James L Carroll***, jlcarroll@lanl.gov. *Hierarchical Bayesian Models for Radiographic Uncertainty Quantification.* Preliminary report.

Uncertainty estimation is an essential problem in many fields. For complex non-linear inverse problems, traditional uncertainty propagation can be difficult. Even when it is possible, the presence of unknown unknowns can further complicate the issue. For radiography, one possible approach is to perform the inverse of an image of a known object. The error from the known truth can then be considered a sample from the uncertainty distribution.

DARHT radiography of known static objects is expensive and rare. Therefore each experiment is rarely repeated exactly. The available data varies in machine tune, object thickness, endpoint energies, camera settings, the presence or absence of exit windows, phosphors, etc. and in many other details of fielding.

We propose that the best way to deal with uncertainty quantification from such rare, sparse data is with a Bayesian Hierarchical model that can use wide Bayesian priors to inform uncertainty where data is sparse, and can borrow support from more well attested regions of the domain to help fill in the gaps for other, less well attested regions of the domain. This approach has the advantage of illuminating the portions of the domain that would benefit most from additional testing using the Expected Value of Sample Information (EVSI). (Received September 01, 2020)

1162-60-236 **Steven T Morrow*** (morrows@wit.edu). *An L^p Multiplicative Coboundary Theorem for Sequences of Unitriangular Random Matrices.*

R. Bradley (1996) proved a "multiplicative coboundary" theorem for sequences of unitriangular random matrices with integer entries, requiring tightness of the family of distributions of the entries from the partial matrix products of the sequence. This was an analog of K. Schmidt's (1977) result for sequences of real-valued random variables with tightness of the family of partial sums. Here is an L^p moment analog of Bradley's result which also relaxes the restriction of entries being integers. (Received September 01, 2020)

62 ► Statistics

1162-62-5 **Hanna Makaruk*** (hanna_m@lanl.gov). *Inverse Problem of Experimental Data Analysis.* Preliminary report.

Astronomical observations, unusual medical cases, physics experiments too costly to repeat – all of them produce important information not achievable in other way. They should not be treated as qualitative, anecdotal evidence only. They should be analyzed in a mathematically rigorous way as inverse problems of a one-of-a-kind event to produce quantitative experimental data. Methods for analysis of data from one-of-a-kind event are needed, since this analysis differs in important ways from analysis of repeated experiments' data. For repeated experiments the experimental error includes a range of true values generated by repetitions of the experiment, and measurement uncertainty caused by detectors. They are always independent. Repetitions of any experiment, as similar as achievable, always have built-in differences resulting in a range of the true values rather than in a single value. Measurement uncertainty depends on the measurement system only. Digital measurements have very small uncertainty, frequently smaller than the range of true experimental values resulting from built-in differences in the experiment repetitions. When data from one–of–a kind experiment are analyzed, only the measurement uncertainty can be reported. (Received August 17, 2020)

1162-62-137 **Sky K Sjue***, P.O. Box 1663, Los Alamos, NM 87545, and **Elise Tang**. *Density reconstructions from proton radiographic images of detonation waves in high explosives.*

We will discuss the imaging techniques used in proton radiography at Los Alamos, then discuss some approximations for the non-stationary chromatic blur due to non-constant energy loss of the protons through a dynamic experiment. A low order form for the density profile of a detonation wave in the high explosive will be assumed based on the ZND model. The effect of motion blur will be included due to integration over proton micro-pulses 5 nanoseconds wide for a duration of 50 nanoseconds in a single image. The time averaging will be performed consistently through a series of images. These assumptions will be subjected to a constant mass constraint to estimate the most likely density of the explosives in the reaction zone, the width of the reaction zone, and the velocity of the detonation wave, plus confidence intervals for these quantities of interest. The results from this forward analysis will be compared to an alternative analysis based on a more direct Abel inversion of the images. (Received August 29, 2020)

1162-62-168 **Thilini V Mahanama*** (thilini.v.mahanama@ttu.edu) and **Dimitri Volchenkov**. *Tornado Property Loss Scale: Up to $8 Billion by 2025 (Classification, Dependence, and Prediction of Tornado Events in the U.S.).*

The National Oceanic and Atmospheric Administration (NOAA) reports that the U.S. annually sustains about 1,300 tornadoes. Currently, the Enhanced Fujita scale is used to categorize the severity of a tornado event based on a wind speed estimate. We propose the new Tornado Property Loss scale (TPL-Scale) to classify tornadoes based on associated damage costs. The dependence between the tornado affected area and the associated property losses vary strongly over time and location. The overall tornado damage costs forecasted by a trained long short-term memory network trained on historical data might reach $8 billion over the next five years although no systematic increase in the number and cost of disasters is observed over time. (Received August 31, 2020)

1162-62-229 **Francesco Zuniga*** (fzuniga@nevada.unr.edu), **Anna K Panorska** (ania@unr.edu) and **Tomasz J Kozubowski** (tkozubow@unr.edu). *A new trivariate model for stochastic episodes.*

We study the joint distribution of stochastic events described by (X, Y, N), where N has a 1-inflated geometric distribution and X, Y are the sum and the maximum of N exponential random variables. Models with similar structure have been used in several areas of applications, including actuarial science, finance, and weather and climate, where such events naturally arise. We provide basic properties of this class of multivariate distributions of mixed type, and discuss their applications. Our results include marginal and conditional distributions, joint integral transforms, moments and related parameters, stochastic representations, estimation and testing. An example from finance illustrates the modeling potential of this new model. (Received September 01, 2020)

65 ► Numerical analysis

1162-65-26 **Jia Zhao*** (jia.zhao@usu.edu), 3900 Old Main Hill Animal Science 108, Logan, UT 84322. *Arbitrarily High-order Unconditionally Energy Stable Schemes for Thermodynamically Consistent Gradient Flow Models.*

In this talk, I will present a systematical approach to developing arbitrarily high order, unconditionally energy stable numerical schemes for thermodynamically consistent gradient flow models that satisfy energy dissipation laws. Utilizing the energy quadratization (EQ) method, we formulate the gradient flow model into an equivalent form with a corresponding quadratic free energy functional. Based on the equivalent form with a quadratic energy, we adopt the Gaussian collocation method to discretize the equivalent form with a quadratic energy, arriving at an arbitrarily high-order scheme for gradient flow models. Schemes derived using both approaches are proved rigorously to be unconditionally energy stable. Numerical results will be shown to illustrate their accuracy and effectiveness. This is a joint work with Yuezheng Gong and Qi Wang. (Received August 13, 2020)

1162-65-27 **Jia Zhao*** (jia.zhao@usu.edu), 3900 Old Main Hill Animal Science 108, Logan, UT 84322. *Energy Stable Numerical Approximations for Hydrodynamic Liquid-Crystal Models.*

The hydrodynamic liquid crystal models have been used for studying flows of liquid crystals and liquid crystal polymers. In this talk, I will present a generic numerical technique that could develop full discrete, spatial-temporally second-order schemes, where only a linear system needs to be solved in each time step. These schemes are proved to be unconditionally energy stable so that a large time step is plausible. Then, we utilize these schemes to solve several widely-used liquid crystal models. Numerical examples will be presented to demonstrate their effectiveness. This is joint work with Qi Wang (University of South Carolina). (Received August 13, 2020)

1162-65-72 **Yekaterina Epshteyn*** (epshteyn@math.utah.edu), Department of Mathematics, University of Utah, Salt Lake City, UT 84112. *Central-upwind schemes for shallow water models.*

In this talk, we will present central-upwind schemes for shallow water models. Shallow water models are widely used in many scientific and engineering applications related to modeling of water flows in rivers, lakes and coastal areas. We will show that the designed central-upwind schemes deliver high-resolution, can handle complicated geometry, and satisfy necessary stability conditions. We will illustrate the performance of the developed methods on a number of challenging numerical tests. Current and future research will be discussed as well. (Received August 23, 2020)

1162-65-81 **Sergiy Borodachov*** (sborodachov@towson.edu), 7800 York Rd., Department of Mathematics, Towson University, Towson, MD 21252. *Optimal recovery of three times differentiable functions based on discrete data of smoothness two.*

We obtain an optimal multivariate spline recovery method for global reconstruction of functions from the Sobolev class $W^3_\infty(P)$ on a given d-dimensional convex polytope P. This optimal method uses as information functions values, gradients, and Hessians at a fixed finite point set $X \subset P$, which includes the vertices of P. We prove that this algorithm is optimal among all non-adaptive recovery methods. The recovery error is measured in the C-norm on P. This optimal method is a cubic continuous spline over a Delaunay triangulation X in P interpolating function values at points of X. We also construct a generalization of this recovery method for data of arbitrary smoothness $r-1$ and give its worst-case error bound over the Sobolev class $W^r_\infty(P)$. (Received August 24, 2020)

1162-65-90 **Mirjeta Pasha*** (mpasha3@asu.edu), Arizona State University, Tempe, AZ 85280, and **Lothar Reichel** and **Alessandro Buccini**. *Linearized Krylov subspace Bregman iteration with nonnegativity constraint.*

Bregman-type iterative methods have received considerable attention in recent years due to their ease of implementation and the high quality of the computed solutions they deliver. However, these iterative methods may require a large number of iterations and this reduces their usefulness. In this talk we will discuss a computationally attractive linearized Bregman algorithm by projecting the problem to be solved into an appropriately chosen low-dimensional Krylov subspace. The projection reduces the computational effort required for each iteration. A variant of this solution method, in which nonnegativity of each computed iterate is imposed, also is described. Extensive numerical examples illustrate the performance of the proposed methods. (Received August 25, 2020)

1162-65-110 **Henry Adams**, **Lara Kassab*** (lara.kassab@colostate.edu) and **Deanna Needell**. *An Adaptation for Iterative Structured Matrix Completion.*

The task of filling-in or predicting missing entries of a matrix, from a subset of known entries, is known as *matrix completion*. Structured matrix completion includes any setting in which data is not missing uniformly at random. In recent work, a modification to the standard nuclear norm minimization (NNM) for matrix completion has been developed to take into account *sparsity-based* structure in the missing entries. This notion of structure is motivated in many settings including recommender systems, where the probability that an entry is observed or not depends on the value of the entry. We propose adjusting an Iteratively Reweighted Least Squares (IRLS) algorithm for low-rank matrix completion to take into account sparsity-based structure in the missing entries. We also present an iterative gradient-projection-based implementation of the algorithm that can handle large-scale matrices. Finally, we present a robust array of numerical experiments on matrices of varying sizes, ranks, and level of structure. We show that our proposed method is comparable with the adjusted NNM on small structured matrices. Further, we show that our proposed method often outperforms the IRLS algorithm in structured settings on matrices up to size 1000×1000. (Received August 29, 2020)

1162-65-117 **Hakima Bessaih***, Department of Mathematics & Statistics, Dept. 3036, 1000 E. University Ave., Laramie, WY 82071, and **Annie Millet**. *Numerical schemes for the 2d Stochastic Navier-Stokes equations.* Preliminary report.

We consider space-time discretization schemes for the 2d stochastic Navier-Stokes equations on the torus. We prove a mean square rate of convergence. This refines previous results established with a rate of convergence in probability only. Using exponential moment estimates of the solution of the Navier-Stokes equations and a convergence of a localized scheme, we can prove strong convergence of fully implicit and semi-implicit time Euler discretization and also a splitting scheme. We finite elements for the space discretization. (Received August 28, 2020)

1162-65-132 **Md Masud Rana*** (md-masud.rana@ttu.edu), Texas Tech University, Box 1042, Lubbock, TX 79409, and **Victoria E. Howle**, **Katharine Long**, **Ashley Meek** and **William Milestone**. *A New Block Preconditioner for Implicit Runge–Kutta Methods for Parabolic PDE.*

Implicit Runge–Kutta (IRK) methods offer an appealing combination of stability and high order; however, these methods are not widely used for PDE because they lead to large, strongly coupled linear systems. An s-stage IRK system has s-times as many degrees of freedom as the systems resulting from backward Euler or implicit trapezoidal rule discretization applied to the same equation set. Here, we introduce a new block preconditioner for IRK methods, based on a block LDU factorization with algebraic multigrid subsolves for scalability. We demonstrate the effectiveness of this preconditioner on the heat equation as a simple test problem, and compare in condition number and eigenvalue distribution, and in numerical experiments with other preconditioners currently in the literature. Experiments are run with IRK stages up to $s = 7$, and it is found that the new preconditioner outperforms the others, with the improvement becoming more pronounced as spatial discretization is refined and as temporal order is increased. (Received August 29, 2020)

1162-65-154 **Varun Shankar***, shankar@cs.utah.edu, and **Grady B Wright** and **Aaron L Fogelson**. *An Efficient High-Order Meshless Method for Advection-Diffusion Equations on Time-Varying Irregular Domains.*

We present a high-order radial basis function finite difference (RBF-FD) framework for the solution of advection-diffusion equations on time-varying domains. Our framework is based on a generalization of the recently-developed Overlapped RBF-FD method, and utilizes a novel automatic procedure for computing RBF-FD weights on stencils in variable-sized regions around stencil centers. This procedure paves the way for tuning-free assembly of RBF-FD differentiation matrices on moving domains. In addition, our framework utilizes a simple and efficient procedure for *updating* RBF-FD differentiation matrices on moving domains. Finally, advection-diffusion in time-varying domains is handled through a high-order semi-Lagrangian method that utilizes the parameter-free overlapped RBF-FD method combined with rapid and localized node set modification. The resulting framework has no tuning parameters and allows for large time-steps. We demonstrate high-orders of convergence for advection-diffusion equations on time-varying 2D and 3D domains for both small and large Peclet numbers. Finally, we utilize our method to solve a coupled 3D problem motivated by models of platelet aggregation and coagulation, once again demonstrating high-order convergence rates in a refinement study. (Received August 30, 2020)

1162-65-188 **Kathryn P Drake**, **Edward J Fuselier** and **Grady B Wright*** (gradywright@boisestate.edu). *A meshfree partition of unity method for divergence-free/curl-free approximation.* Preliminary report.

Divergence-free/curl-free vector fields are pervasive in many areas of science and engineering, from fluid dynamics to electromagnetism. A common problem that arises in applications is that of reconstructing these vector fields based only on discrete samples. Additionally, it is often necessary that these reconstructions preserve the divergence-free or curl-free properties of the field to maintain certain physical constraints. Divergence-free/curl-free radial basis functions (RBFs) are a particularly good choice for this application as they are meshfree and the reconstructed fields analytically satisfy the divergence-free or curl-free property. A negative aspect of this approach is that the method is computationally expensive due to its global nature. In this talk, we discuss a new technique for bypassing this issue that combines divergence-free/curl-free RBFs in a partition of unity framework, where one solves local reconstruction problems and blends them together to a global reconstruction. A particular challenge with this approach that we explain how to overcome is maintaining the local divergence-free/curl-free properties in the global reconstruction. We present theoretical approximation results and demonstrate the effectiveness of the method on several problems. (Received August 31, 2020)

1162-65-227 **Bamdad Hosseini*** (bamdadh@caltech.edu), 1200 E California Blvd, MC 305-16, Pasadena, CA 91125. *Model reduction and neural networks for parametric PDEs.* Preliminary report.

I will discuss some ideas at the intersection of machine learning and uncertainty quantification with a particular focus on data-driven methods that do not require explicit knowledge of processes that generate the data. In particular we focus on supervised learning on Banach spaces for emulation of parametric PDE models and outline a method that combines principal component analysis with neural network regression for mesh-independent approximation of PDE solutions. Our method is supported by theoretical analysis and numerical experiments that elucidate the flexibility and mesh independence of our method. (Received September 01, 2020)

1162-65-228 **Seth Armstrong**, **Sarah Duffin*** (duffin@suu.edu) and **Jianlong Han**. *A Numerical Scheme for the Lotka-Volterra Competition Model with Nonlocal Interaction.*

The Lotka-Volterra competition model involving nonlocal diffusion terms can be used to model two species competing with each other for the same prey. We introduce a new semi-implicit numerical scheme that is shown to mimic the dynamical properties of the true solution. We prove that the scheme is uniquely solvable and unconditionally stable. The convergence rate of the numerical solution to the true solution of the system is also given. (Received September 01, 2020)

68 ► Computer science

1162-68-216 **Simina Branzei*** (simina.branzei@gmail.com), West Lafayette, IN 47906, and **Dimitris Paparas** and **Nicholas Recker**. *Sorting and cake cutting in rounds.* Preliminary report.

We study the problem of sorting in rounds in a query model where we are given a vector x and each question is of the form: "Is rank(x[i]) ¡= k?", where $k \in \{1, ..., n\}$. We study sorting and selection in this model and show the bounds for sorting imply bounds for cake cutting in rounds. (Received September 01, 2020)

76 ► Fluid mechanics

1162-76-130 **Cecilia Mondaini*** (cf823@drexel.edu) and **Nathan Glatt-Holtz**. *Numerical approximation for invariant measures of the 2D Navier-Stokes equations.*

We consider the problem of approximating statistically steady states of the 2D stochastic Navier-Stokes equations (SNSE) via an approximating sequence of measures generated from a space-time discretization of the 2D SNSE. More specifically, we consider a spectral Galerkin spatial discretization and a semi-implicit Euler time scheme. We show that successive iterations of the Markov semigroup associated to the discretized system, starting from any initial probability distribution, converge to the invariant measure of the continuous system. The proof is obtained with two main steps: a spectral gap result for the discretized system which is independent of the discretization parameters, and finite time L^2-convergence of the discretized system towards the continuous one. Most importantly, this approach allows us to obtain explicit rates of convergence with respect to the number of iterations in the Markov chain, up to (also explicit) numerical discretization error. This is a joint work with Nathan Glatt-Holtz (Tulane U). (Received August 29, 2020)

1162-76-133 **Gung-Min Gie***, Department of mathematics, University of Louisville, Louisville, KY 40292, and **Chang-Yeol Jung** and **Hoyeon Lee**. *Enriched Finite Volume approximations of the plane-parallel flows at a small viscosity.*

We discuss the viscous boundary layers of the plane-parallel flow, governed by the stationary Navier-Stokes equations under a certain symmetry. Performing boundary layer analysis, we first construct the so-called corrector, which is an analytic approximation of the velocity vector field near the boundary. Then, by embedding the corrector function into the classical Finite Volume schemes, we construct the semi-analytic enriched Finite Volume schemes for the plane-parallel flow, and numerically verify that our new enriched schemes reduce significantly the computational error of classical schemes especially near the boundary. (Received August 29, 2020)

1162-76-147 **James P Kelliher*** (kelliher@math.ucr.edu), **Anna L Mazzucato** and **Gung-Min Gie**. *Well-posedness and regularity of solutions to the 3D Euler equations with inflow, outflow.* Preliminary report.

The 3D incompressible Euler equations in a bounded domain are most often supplemented with impermeable boundary conditions, which constrain the fluid to neither enter nor leave the domain. In 1983, Antontsev, Kazhikhov, and Monakhov published a proof of the existence and uniqueness of solutions to the Euler equations in which on certain inflow boundary components fluid is forced into the domain while on other outflow components fluid is drawn out of the domain. We extend their result to multiply connected domains and establish compatibility conditions on the initial data that allow higher regularity solutions, addressing an open issue in the literature. (Received August 30, 2020)

1162-76-212 **Tuan Ngoc Pham*** (tuan.pham@mathematics.byu.edu), UT. *Stochastic models of the Navier-Stokes equations and the problem of stochastic explosion.*

In 1997, Le Jan and Sznitmann introduced a probabilistic method for the deterministic Navier-Stokes equations (later relaxed to permit the possibility of stochastic explosion) in which the solution is constructed from a (possibly explosive) stochastic branching process. This opened an interesting approach to study the existence and uniqueness of solutions alternative to many well-known analytic approaches, e.g. those of Leray (1930s), Kato, Fujita (1960s, 80s) and others. Central to Le Jan-Sznitmann's approach is the so-called majorizing kernel. The choice of majorizing kernels is not unique. Two known majorizing kernels are: the self-similar kernel and the Bessel kernel. Each choice gives a different branching distribution. The question of whether a branching process terminates eventually is known as the "stochastic explosion" problem. This question is of its own interest in probability theory, whether or not the branching process is tied to a PDE. A classical method to treat such a problem is perhaps using large deviation estimates. In this talk, I will briefly explain an alternative method called "cutset argument", and apply it to show that the Bessel kernel results in a non-explosive branching process. Joint work with Radu Dascaliuc, Enrique Thomann, and Edward Waymire. (Received August 31, 2020)

1162-76-219 **Animikh Biswas*** (abiswas@umbc.edu) and **Randy Price** (randyp1@umbc.edu). *Data Assimilation for the Three Dimensional Navier-Stokes Equations.*

We provide conditions, *based solely on the observed data*, for the global well-posedness, regularity and convergence of the Azouni-Olson-Titi (AOT) data assimilation algorithm for a Leray-Hopf weak solutions of the three dimensional Navier-Stokes equations (3D NSE). The aforementioned conditions on the observations, which in this case comprise either of *modal* or *volume element observations*, are automatically satisfied for solutions that are globally regular and are uniformly bounded in the H^1-norm. However, neither regularity nor uniqueness is necessary for the efficacy of the AOT algorithm. To the best of our knowledge, this is the first such rigorous analysis of the AOT data assimilation algorithm for the 3D NSE. (Received September 01, 2020)

1162-76-220 **Vincent R Martinez*** (vrmartinez@hunter.cuny.edu), 695 Park Ave, New York, NY 10065. *Higher-order synchronization for a nudging algorithm for 2D Navier-Stokes with nodal observables.* Preliminary report.

The analytical study of a nudging algorithm in the infinite-dimensional setting of PDEs was initially carried out by Azouani, Olson, and Titi for the two-dimensional (2D) incompressible Navier-Stokes equations (NSE). In their seminal work, convergence of the approximating solution to the true solution was shown to take place at least in the topology of the Sobolev space H^1. However, this does not include uniform convergence. This talk will discuss convergence in stronger topologies, including the uniform topology, of this nudging based algorithm for data assimilation in the context of the 2D NSE when observations are given as on nodal values of the velocity field. This is joint work with Animikh Biswas (University of Maryland-Baltimore County) and Ken Brown (Hunter College). (Received September 01, 2020)

81 ► Quantum theory

1162-81-43 **Louis H Kauffman*** (kauffman@uic.edu), 5530 South Shore Drive, Apt 7C, Chicago, IL 60637-1946. *Spacetime Algebra and the Majorana Dirac Equation.*

(Joint work with Peter Rowlands) We formulate the Dirac equation in terms of spacetime algebra, a Clifford algebra generated by e_1, e_2, e_3, e_4 , all pariwise anti-commuting and such that the first three have square 1 and e_4 has square -1. In this formulation we classify those solutions that give rise to Majorana Dirac equations (with all real solutions) and we discuss properties of Majorana Fermions from this point of view, including representations of the Artin Braid group and possible applications to quantum computing. (Received August 19, 2020)

90 ► Operations research, mathematical programming

1162-90-184 **Mark A Abramson*** (mark.abramson@uvu.edu), 800 West University Parkway, LA 121U, Orem, UT 84058, **Griffin D. Kent** (10764442@my.uvu.edu), Dept of Industrial and Systems Engineering, 19 Memorial Drive West, Bethlehem, PA 18015, and **Gavin W. Smith** (gavin.w.smith@intel.com), Intel Corporation, 2501 NE Century Blvd, Hillsboro, OR 97124. *A global optimization approach to computing penetration depth between two convex polyhedra.*

During the detailed design phase of an aerospace program, one of the most important consistency checks is to ensure that no two distinct objects occupy the same physical space. Since exact geometrical modeling is usually intractable, geometry models are discretized, which often introduces small interferences not present in the fully detailed model. We focus on computing the depth of the interference, so that these false positive interferences can be removed, and attention can be properly focused on the actual design. Specifically, we focus on efficiently computing the penetration depth between two polyhedra, which is a well-studied problem in the computer graphics community. We formulate the problem as a five-variable constrained global optimization problem, and then derive an equivalent unconstrained, 2-variable nonsmooth problem. To solve the optimization problem, we apply a popular stochastic multistart optimization algorithm in a novel way, which exploits the advantages of each problem formulation simultaneously. We apply our algorithm to 14 randomly generated pairs of penetrating polytopes and show very promising results. (Received August 31, 2020)

91 ► Game theory, economics, social and behavioral sciences

1162-91-9 **Lorraine Marie Arangno** and **Deborah C Arangno*** (darangno@hcc-nd.edu), 54515 S.R. 933 North, P.O. Box 308, Notre Dame, IN 46556-0308. *AN APPLICATION OF MATHEMATICAL STRUCTURES TO A PHILOSOPHICAL DEBATE.*

In contemporary metaethics, realist-antirealist debates centering on the ontology of moral properties continue to generate significant and exciting research. We have explored the use of methods from Graph Theory to clarify a debated position of Meta-Ethics, previously encumbered by intrinsic vagueness and ambiguity. We employ rigorous mathematical formalism to symbolize, parse, and thus disambiguate, particular philosophical questions regarding ethical ontological materialism of the reductivist variety. In our research, we seek to revisit the once vexed question regarding the multiple-realizability of moral properties by employing the mathematical machinery of hypergraphs. The utilization of hypergraphs offers explanatory flexibility to the hypothesis that there are, potentially, an infinite number of unrelated subvening facts (viz., the subvening sets of natural non-moral properties) which have the possibility of coming together as subsets (hyperedges) to form supervening moral properties. We hope our research will be of interest to both the Analytic Philosophy and the applied Mathematics communities, and will help facilitate discussions of metaethics, particularly pertaining to ethical naturalism. We believe there is much research yet to be done (Received July 11, 2020)

1162-91-217 **Seokhyun Ryu***, RISE-CRG, Cresskill, NJ , and **Richard Kyung**, RISE-CRG, Cresskill. *Study on the Housing Start and Mortgage Rate Trends Using Statistical and Computational Analysis.*

According to the data that details the past ten years in Housing Stars in America, there has been a steady increasing trend. Nearing the year 2020, there was a significant increase in the trend where the prices rose to around 1600 thousand, however due to the COVID-19 pandemic, affecting businesses worldwide, prices dropped immensely all the way down to around 900 thousand.

In this paper, correlation between housing starts, mortgages rates and other economic indicators were studied using statistical and computational analysis. For the analysis, we assumed the mortgage rates will be getting lower since they have reached their lowest since the late 1900s averaging 3.06 percent in August of 2020. At the beginning of 2020, when the coronavirus was affecting companies most severely, there was a slight increase in mortgage rates that reached up to around 3.8 percent.

As for the stock market, indicators such as NASDAQ and Dow Jones were investigated if they have shown similar trends as with the Housing Starts and the Mortgage Rates. When the Coronavirus was at its peak, the stock market decreased significantly. However, as the pandemic's effects lessened, the stock market started to come back to its original state. (Received September 01, 2020)

1162-91-242 **Emily Dai***, RISE-CRG, Cresskill, NJ , and **Richard Kyung**, RISE-CRG, Cresskill, NJ. *Study on Qualitative and Quantitative Changes in Social Behavior During Economic Crises.*

The COVID-19 pandemic has altered not only how people behave but also how people shop. As more consumers turn to online shopping to avoid contracting the illness, small businesses are experiencing heavy losses in both sales and customers. Statistics indicate that low-wage workers are at the highest risk of facing the repercussions of the COVID-19 virus. Throughout the presented research, qualitative and quantitative analysis for the economic and social factors is performed to figure out the symptoms of behavioral and societal impacts. Given such a situation, this paper describes the responses of stress exhibited by customers, evident in the significant changes in motivations. As a result, quantitative and qualitative changes in behavior are analyzed. Furthermore, this paper discusses the high unemployment rates and emotional distress caused by depressions and stock market crashes. Such factors were studied using the Volatility Index, commonly referred to as the fear index or fear gauge. (Received September 02, 2020)

92 ► Biology and other natural sciences

1162-92-178 **Michael I Ham***, P.O. Box 1663, Los Alamos, NM 87545. *An Exploration of Novel Filter Material and Mask Performance For Preventing the Spread of COVID-19.* Preliminary report.

Mask wearing by the public has been found to reduce the spread of COVID-19. Masks allow some percentage of small droplets to pass in and out of the material and through gaps where the mask is not sealed to the face. Here we show that the primary benefit of masks is that they significantly reduce the velocity of air moving in and out of the mask while acting as filters that protect both the wearer and those around them from droplets. We demonstrate mask filter efficiency approaches 100% as droplet size increases, a result that becomes important for preventing larger droplets generated while speaking or coughing and sneezing. Our results show that the conventional assessment of mask filtration efficiency in the 300nm-10um range is insufficient for fully assessing mask performance. Combined with statistical inference of droplet virion levels our results demonstrate why masks are able to reduce the spread of COVID-19 because not every droplet from an infected person is likely to have a virion, and every droplet you prevent from entering an uninfected person matters. We apply our results to suggest a metrics that would help better inform consumers when purchasing masks. (Received August 31, 2020)

1162-92-222 **Churni Gupta*** (churnibidisha@ufl.edu), **Necibe Tuncer** and **Maia Martcheva**. *A multi-scale Network model on HIV.* Preliminary report.

In this paper we formulate a multi-scale nested immuno-epidemiological model of HIV on complex networks. The system is described by ordinary differential equations coupled with partial differential equations. First, existence and uniqueness results of the immunological system is proved. Next, wellposedness of the multi-scale system is established. We derive an explicit expression of the basic reproduction number of the immuno-epidemiological model. Existence and stability of disease free equilibrium and endemic equilibrium is established. Numerical simulations suggest that with all things held fixed, if the number of nodes in the network is increased, so is the basic reproduction number. Further, we find that for a scale free network the number of infected individuals at equilibrium is a hump-like function of the within-host reproduction number; however, the dependence becomes monotone if the network has predominantly low connectivity nodes or high connectivity nodes. (Received September 01, 2020)

97 ► Mathematics education

1162-97-13 **Zoe C Ashton*** (ashton.95@osu.edu), 230 N Oval Mall OSU, Dept of Philosophy, Columbus, OH 43210. *Linking IBL and Research through Audience-Based Heuristics.*

Hadamard, in "Thoughts on the Heuristic Method," wrote that "I tried, in other words to discern the rules, that all mathematicians unconsciously follow when they reason, at least the main ones ... They are on the other hand common sense rules – truisms so to speak" (pg. 40). In this talk, I explore what exactly this 'truisms' claim amounts to and how it relates the way mathematicians solve problems and how students should solve problems. In part, I will argue that these truisms are reflected in a certain way that mathematicians conceive of their audiences. Namely that these audiences are universal – the way we solve problems in mathematics is not completely removed from the way we solve problems in dialectical milieus. By viewing our problem-solving practices through the dialectical lens, this can help the expert construct designed tasks to implement in IBL. The core idea is to reveal (a) how mathematical problem solving is dialectical, (b) recognize what the dialectical moves look like, (c) identify what kind of audience is involved in that dialectic, (d) recreate that dialectic in IBL tasks. (Received August 05, 2020)

1162-97-18 **Péter Juhász*** (juhasz.peter@renyi.hu), Budapest, Hungary. *Unusual chessboard tasks for gifted learners.*

Lajos Pósa has been organizing and leading weekend math camps for highly gifted students for more than 30 years. A special form of gifted education is being implemented in these camps, in which the focus is laid not on preparing for competitions, but on intense and autonomous thinking.

In one of the long term sequences of tasks, the colors of the squares of a chessboard, or some numbers written on the squares can be changed, and various related questions are to be answered. Students are also encouraged to pose connected questions, which may expand the set of problems to be solved by the group. The presentation offers a sample of this sequence of tasks. (Received August 12, 2020)

1162-97-48 **Erhan Selcuk Haciomeroglu*** (erhansh@tuj.temple.edu), **Janet B Andreasen** (janet.andreasen@ucf.edu) and **Brianna A Kurtz** (brianna.a.kurtz@gmail.com). *Exploring the Stages of Polya's Problem-Solving Strategy.*

How is the graph of a race between two runners related to the stages of Polya's problem-solving strategy? This talk explores the potential effectiveness of Polya's problem-solving strategy and illustrates how dynamic activities can be used to help students to make meaningful connections between different representations of the same concept. Attendees will discuss how to create their own tasks to elicit students' thinking and deepen their understanding of mathematics, followed by discussions of implications for teaching and learning mathematics. (Received August 19, 2020)

1162-97-53 **Robert G Stein*** (bstein@csusb.edu), 4 Beacon Way Apartment 503, Jersey City, NJ 0730-4. *Let's Bring Polya Back to School.*

It is time to bring Polya's educational approach to the entire school mathematics curriculum. This talk presents examples that are appropriate for elementary grades and above and challenges listeners to design an entire school curriculum around inquiry based learning. (Received August 20, 2020)

1162-97-98 **Jennifer Elyse Clinkenbeard*** (jclinkenbeard@csumb.edu), 100 Campus Center, Seaside, CA 93955. *Food For Thought: Exploring Euclidean and non-Euclidean Geometries.* Preliminary report.

We present a series of inquiry-oriented tasks to visualize and model different geometries using common food items. We explore: (1) Euclid's parallel postulate and its consequences in the first book of Elements with spaghetti; (2) spherical geometry on the surface of an orange; and (3) hyperbolic geometry on Pringles chips. This activity series has been implemented in an inquiry-oriented, upper-division geometry course at CSU Monterey Bay. We discuss samples of the tasks above that have been implemented in the course, as well as samples of student models and responses to metacognitive writing prompts. We discuss the strengths and limitations of this approach with respect to heuristic reasoning. As Pólya observed in How to Solve It, heuristic reasoning supports the development of provisional reasoning. This does not constitute formal proof, nor should it be presented as such. We conclude with a discussion of these considerations in the context of Euclidean and non-Euclidean geometries, and how students' exploration with the models may contribute to readiness for more formal argument. (Received August 26, 2020)

1162-97-112 **Mario Banuelos*** (mbanuelos22@csufresno.edu). *Investigating Validation Metrics for Statistical and Mathematical Modeling within Upper Division Courses.* Preliminary report.

In the context of supervised learning and assessing a mathematical framework, researchers rely on a variety of practices to validate models. In statistics, resampling methods provide a means to assess trained models' performance on unseen data, known as test data. In mathematical modeling, the comparison to data is essential to challenge previous assumptions. Rather than providing a set of procedures for students to follow after training or implementing their models, students will investigate the effects of different validation metrics with respect to model testing.

Specifically, I will discuss two different collaborative approaches in upper division math courses to explore validation set splits and model accuracy. Students explore the fraction of testing and training splits in regression problems and how that affects the test mean squared error. For the model accuracy activity, students work in groups to answer specific questions about model success, bias, and accuracy. Activities are implemented in RStudio and Python. Preliminary student feedback will also be discussed. (Received August 28, 2020)

1162-97-224 **Ravi Vakil*** (rvakil@stanford.edu). *Learning through problem-solving from high school through postgraduate.*

"Problem-solving" has been central to learning and investigation in mathematics throughout human history, and it is striking how much consensus there is within the mathematical community on this. This has been elucidated in many ways, including through math circles, Polya's teachings (including his volumes with Szego), and inquiry-based learning. I will discuss some common themes and some attempts to stretch the envelope, with examples from high school (including "Proof School" in the San Francisco Bay Area) through advanced graduate "courses". (Received September 01, 2020)

1162-97-226 **Megan E Selbach-Allen*** (mselbach@stanford.edu). *The What and Why Behind IBL Communities: Beginning to Build in California, Nevada and Beyond.*

As evidence has mounted for the effectiveness of various active learning techniques, the use of Inquiry Based Learning (IBL) in college mathematics classes has grown rapidly. IBL is now a special interest group in the MAA and several national conferences have been held in recent years along with a series of practitioner workshops across multiple summers to help instructors change their practice and understand the teaching methodology. While these national efforts have sown the seeds, the initial funding streams are ending and the need for more localized, regional communities is emerging. These smaller regional groups have the potential to serve as communities of practice for mathematics faculty who desire to continuously improve their instruction and use of IBL techniques. In this talk I will share some of the short history of IBL communities and how they have been enacted in different regions around the United States. I will also share recent efforts to establish an IBL community in California and Nevada and solicit feedback from participants to think collaboratively about what this community might become for our region and how other communities might begin. (Received September 01, 2020)

1162-97-232 **Angie Hodge*** (angie.hodge@nau.edu), 4000 East Summer Run Drive, 4000 East Summer Run Drive, Flagstaff, AZ 86004. *IBL in a remote learning environment: Tips and trips from the field.*

This session will be used to share lessons learned (both good and bad) from employing inquiry-based learning (IBL) in remote mathematics classes. Examples will be shared from a variety of courses including calculus and mathematics for elementary teachers. (Received September 01, 2020)

1162-97-235 **Agnes Tuska*** (agnest@csufresno.edu). *The Way George Pólya Solved It.* Preliminary report.

George Pólya published his seminal book of "How to Solve It: A New Aspect of Mathematical Method" in the USA in English in 1945. The book had several editions and was translated into about 20 languages at various times since then. What cultural and societal factors influenced the times of the publications? How did the book influence the goals and the effectiveness of mathematics education in various cultures? (Received September 01, 2020)

1162-97-238 **Lance D Burger*** (lburger@csufresno.edu). *LACAN EXTENDED: THE TWENTY FOUR DISCOURSES.* Preliminary report.

This presentation discussed an article in progress, written from the viewpoint of a mathematician, which extends the well-known four discourses of Jacques Lacan by constructing a generalized model based on the symmetric group S4. After discussing preliminaries on the original four discourses and some elementary group theory, an Unconscious discourse is introduced which functions as a group identity element inducing a table of interactions, inverse elements and subgroups rich in structure and interpretations. Some immediate results of the model

include an inverse element to the Capitalist discourse theorized to correspond to a Socialist discourse. Also, subgroups are produced which organize the original four discourses differently than Lacan's original serial rotation of the master's discourse, leading to interesting arrangements lending credence to the general model, yet invite speculation. (Received September 01, 2020)

2010 MATHEMATICS SUBJECT CLASSIFICATION

Compiled in the Editorial Offices of MATHEMATICAL REVIEWS and ZENTRALBLATT MATH

00 General
01 History and biography
03 Mathematical logic and foundations
05 Combinatorics
06 Order, lattices, ordered algebraic structures
08 General algebraic systems
11 Number theory
12 Field theory and polynomials
13 Commutative rings and algebras
14 Algebraic geometry
15 Linear and multilinear algebra; matrix theory
16 Associative rings and algebras
17 Nonassociative rings and algebras
18 Category theory; homological algebra
19 *K*-theory
20 Group theory and generalizations
22 Topological groups, Lie groups
26 Real functions
28 Measure and integration
30 Functions of a complex variable
31 Potential theory
32 Several complex variables and analytic spaces
33 Special functions
34 Ordinary differential equations
35 Partial differential equations
37 Dynamical systems and ergodic theory
39 Difference and functional equations
40 Sequences, series, summability
41 Approximations and expansions
42 Fourier analysis
43 Abstract harmonic analysis
44 Integral transforms, operational calculus
45 Integral equations
46 Functional analysis
47 Operator theory
49 Calculus of variations and optimal control; optimization
51 Geometry
52 Convex and discrete geometry
53 Differential geometry
54 General topology
55 Algebraic topology
57 Manifolds and cell complexes
58 Global analysis, analysis on manifolds
60 Probability theory and stochastic processes
62 Statistics
65 Numerical analysis
68 Computer science
70 Mechanics of particles and systems
74 Mechanics of deformable solids
76 Fluid mechanics
78 Optics, electromagnetic theory
80 Classical thermodynamics, heat transfer
81 Quantum theory
82 Statistical mechanics, structure of matter
83 Relativity and gravitational theory
85 Astronomy and astrophysics
86 Geophysics
90 Operations research, mathematical programming
91 Game theory, economics, social and behavioral sciences
92 Biology and other natural sciences
93 Systems theory; control
94 Information and communication, circuits
97 Mathematics education

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